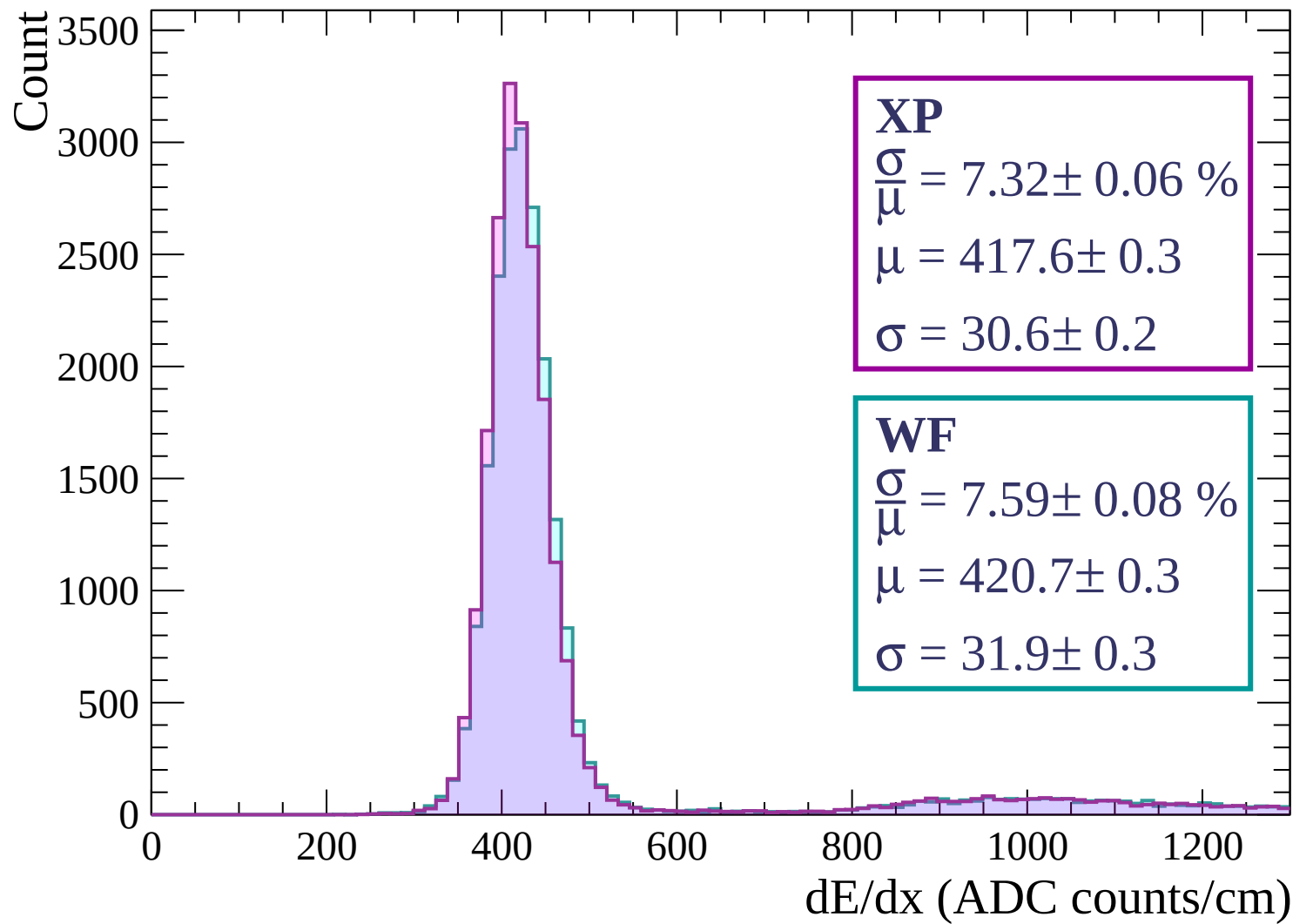
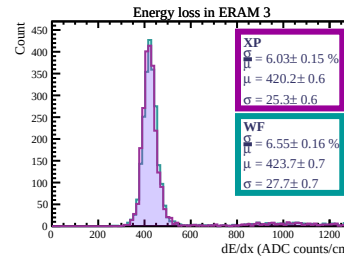
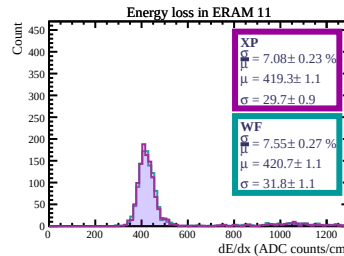
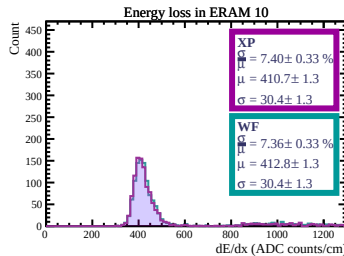
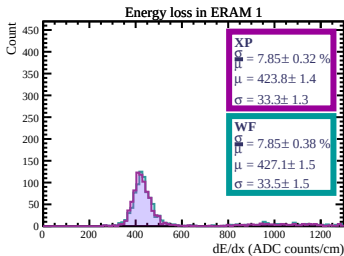
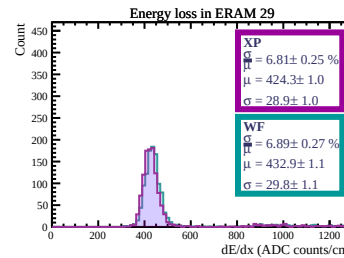
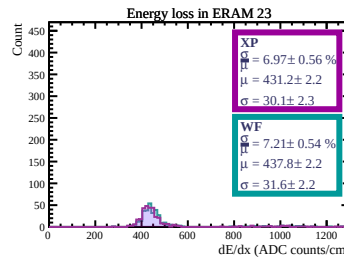
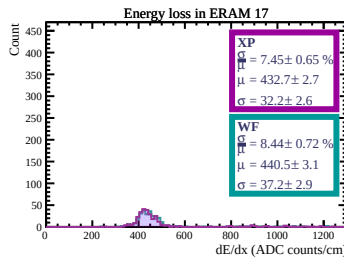
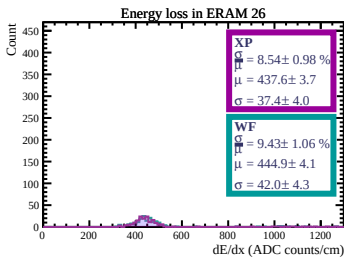
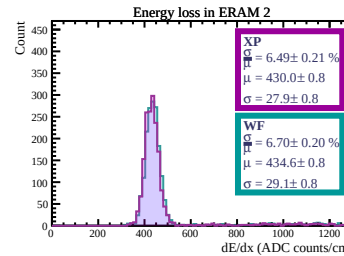
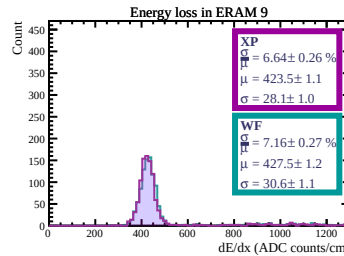
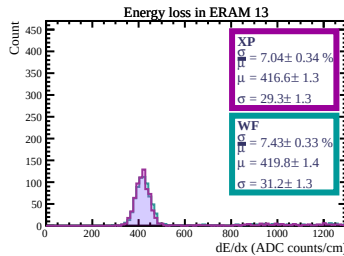
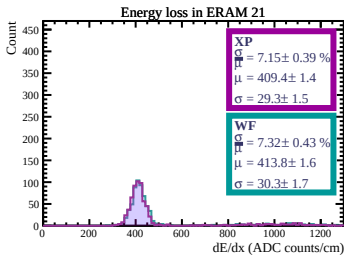
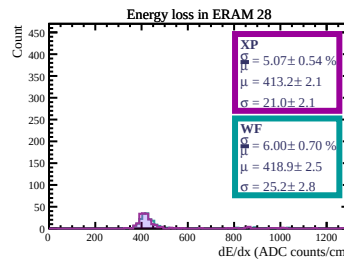
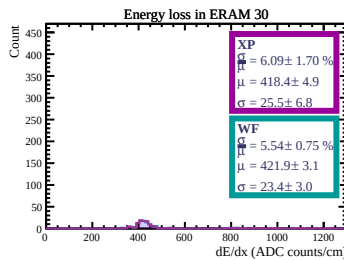
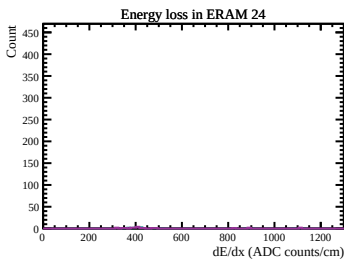
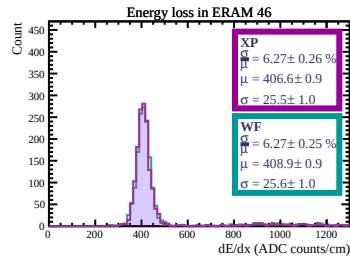
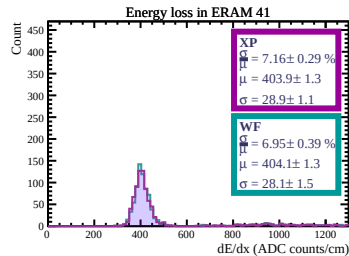
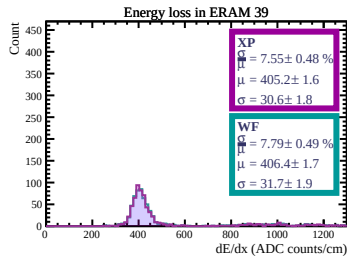
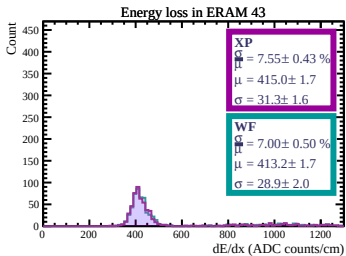
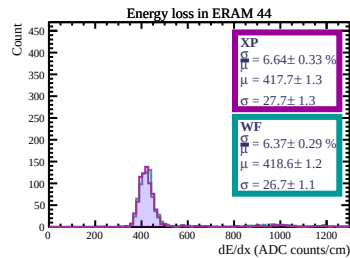
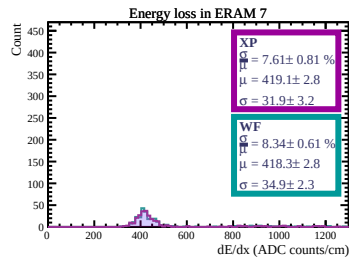
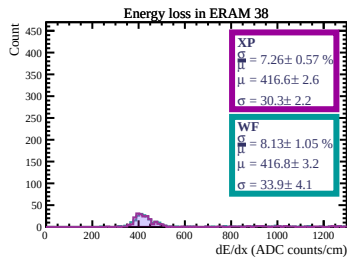
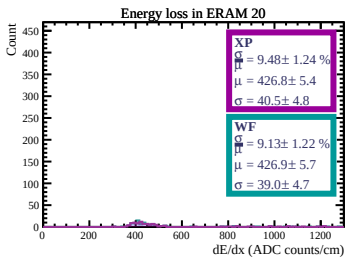
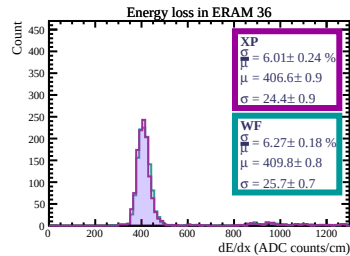
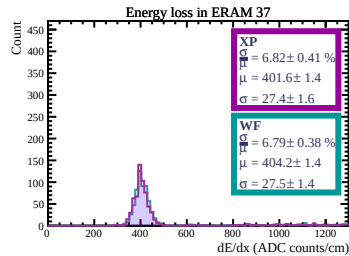
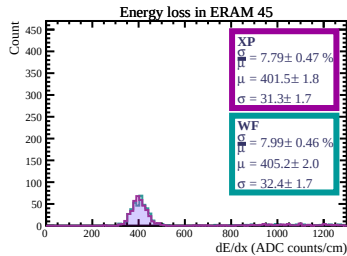
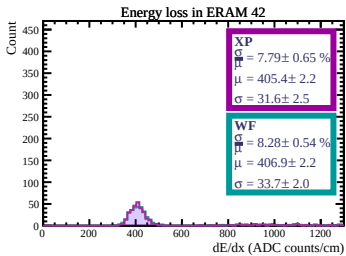
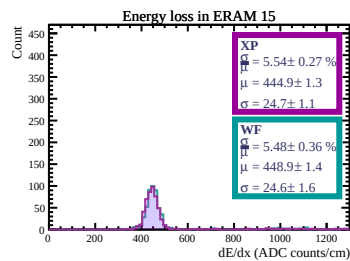
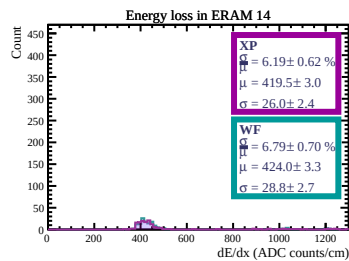
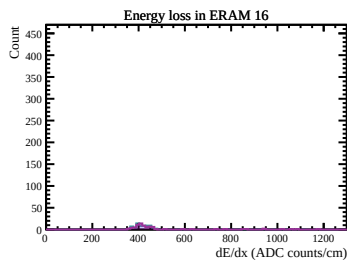
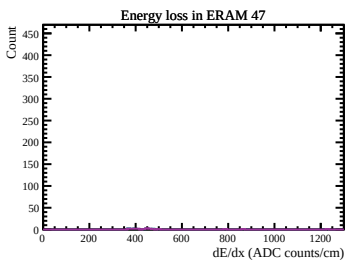
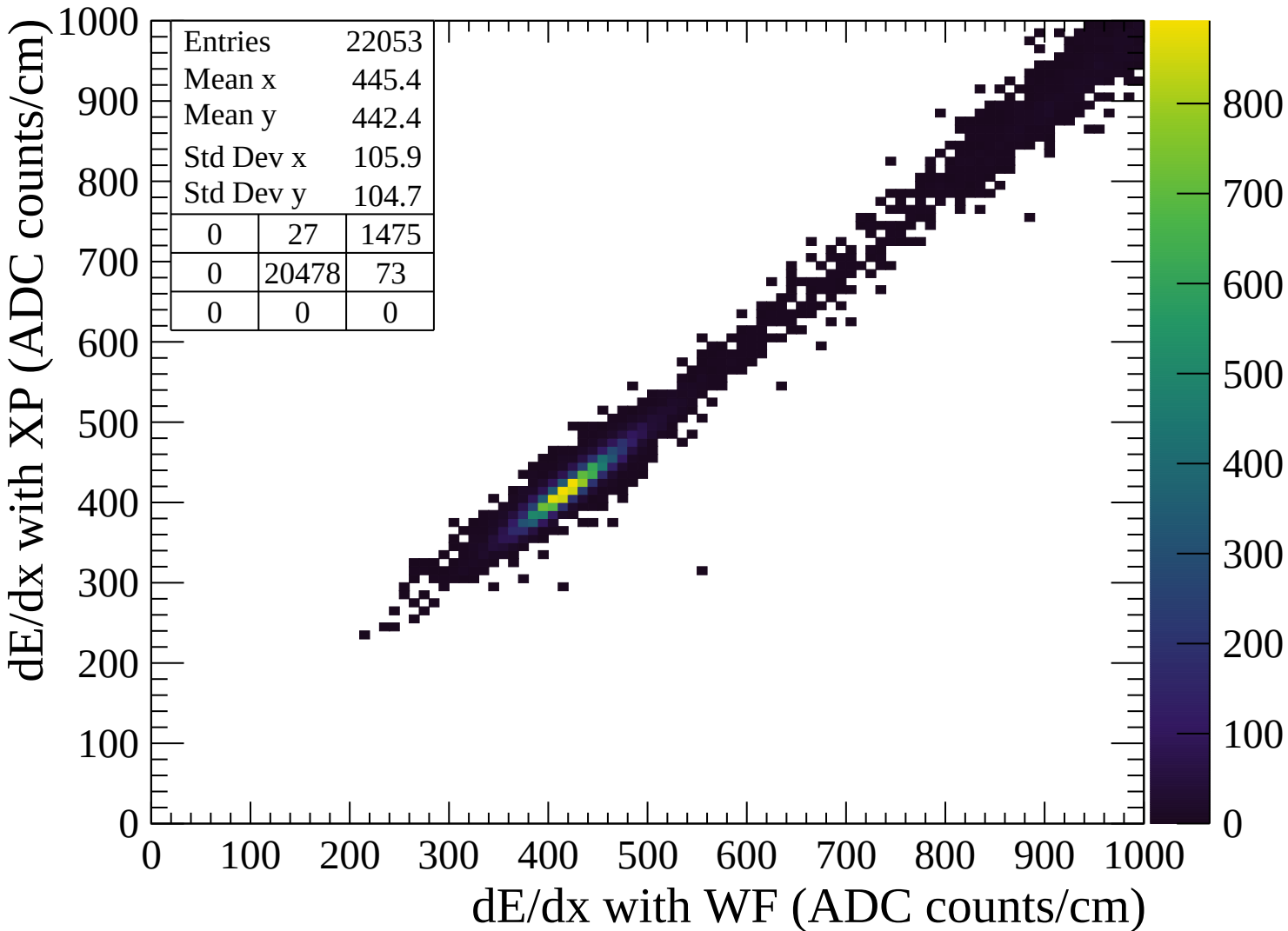
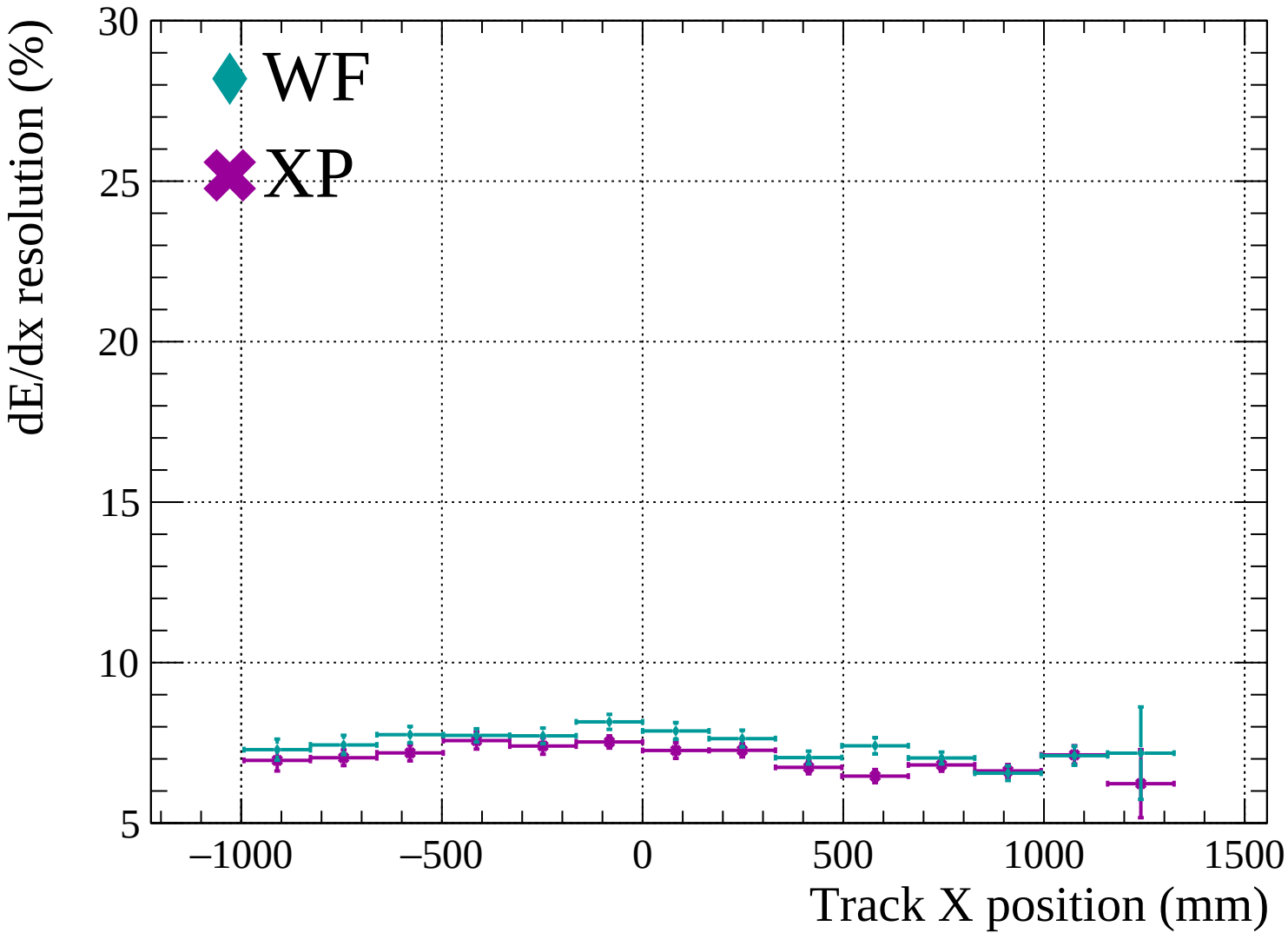


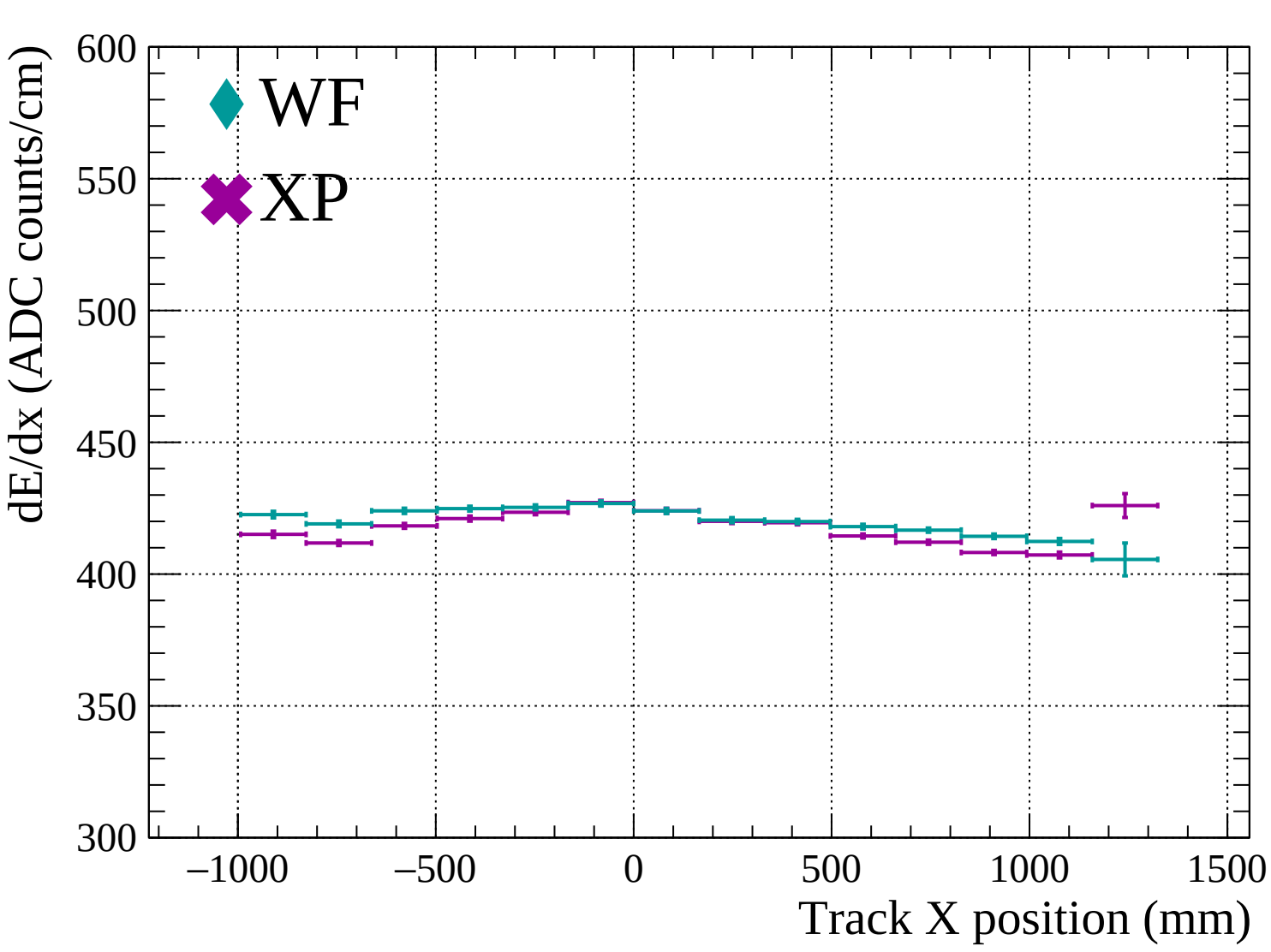


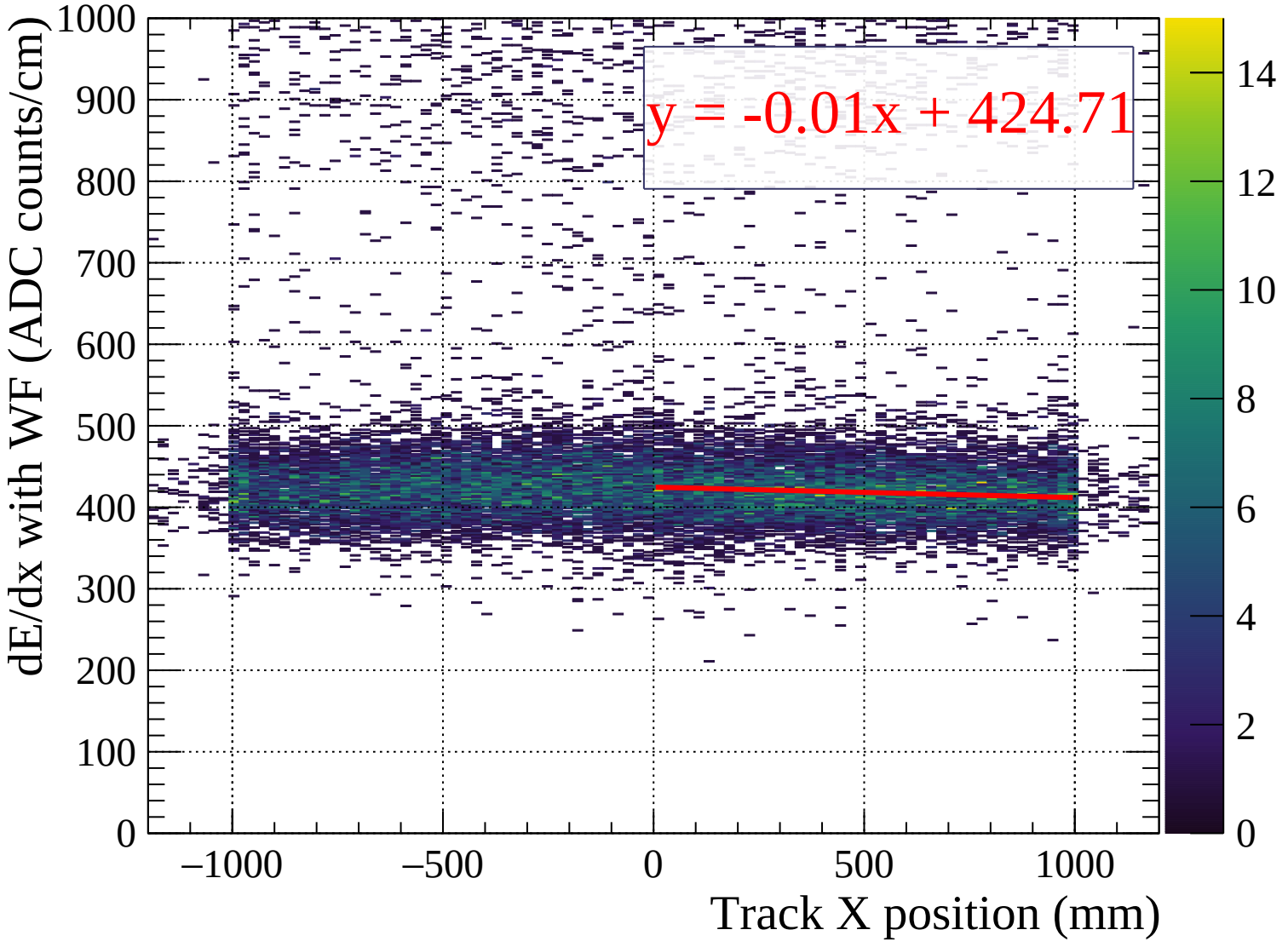


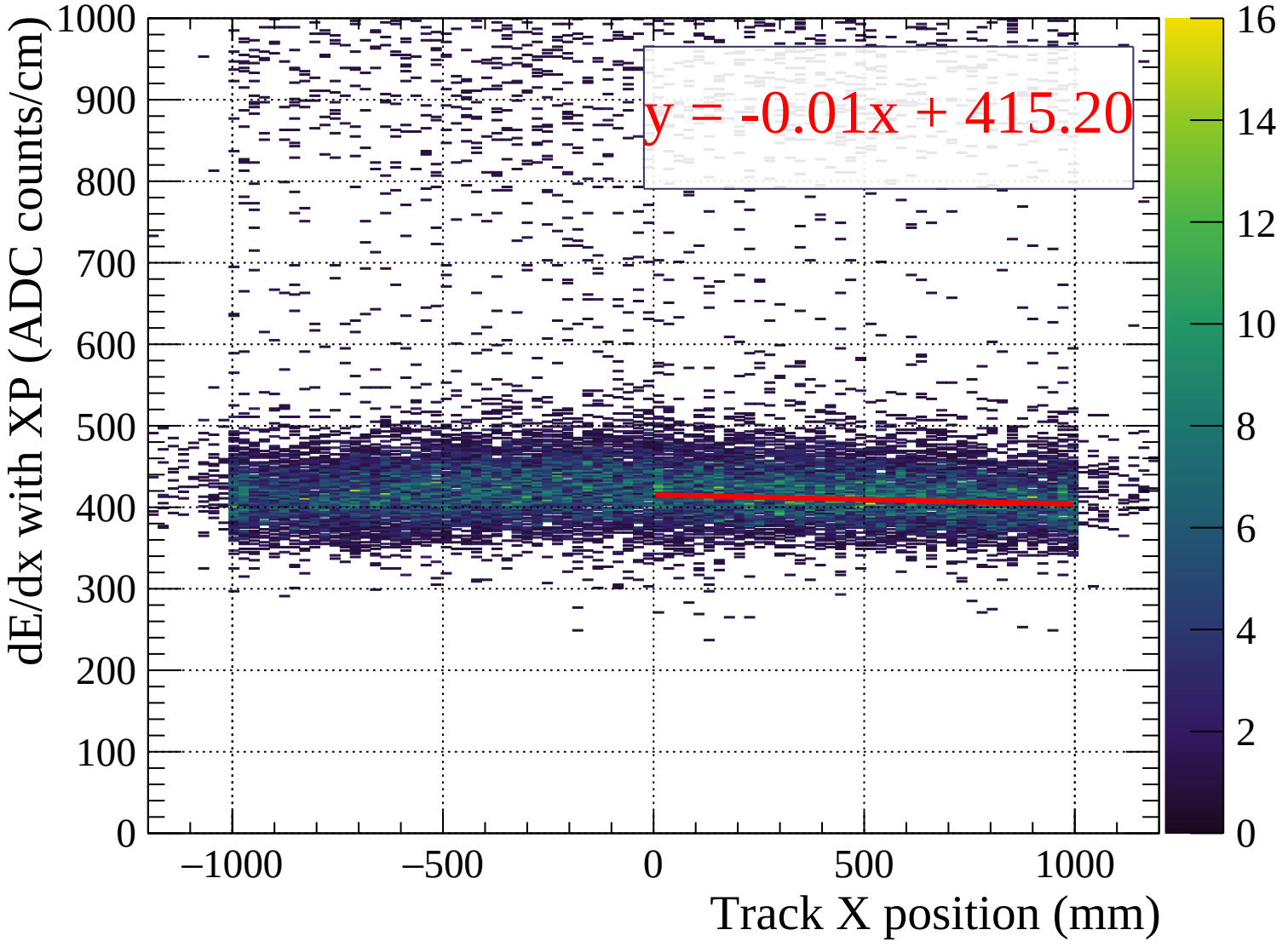


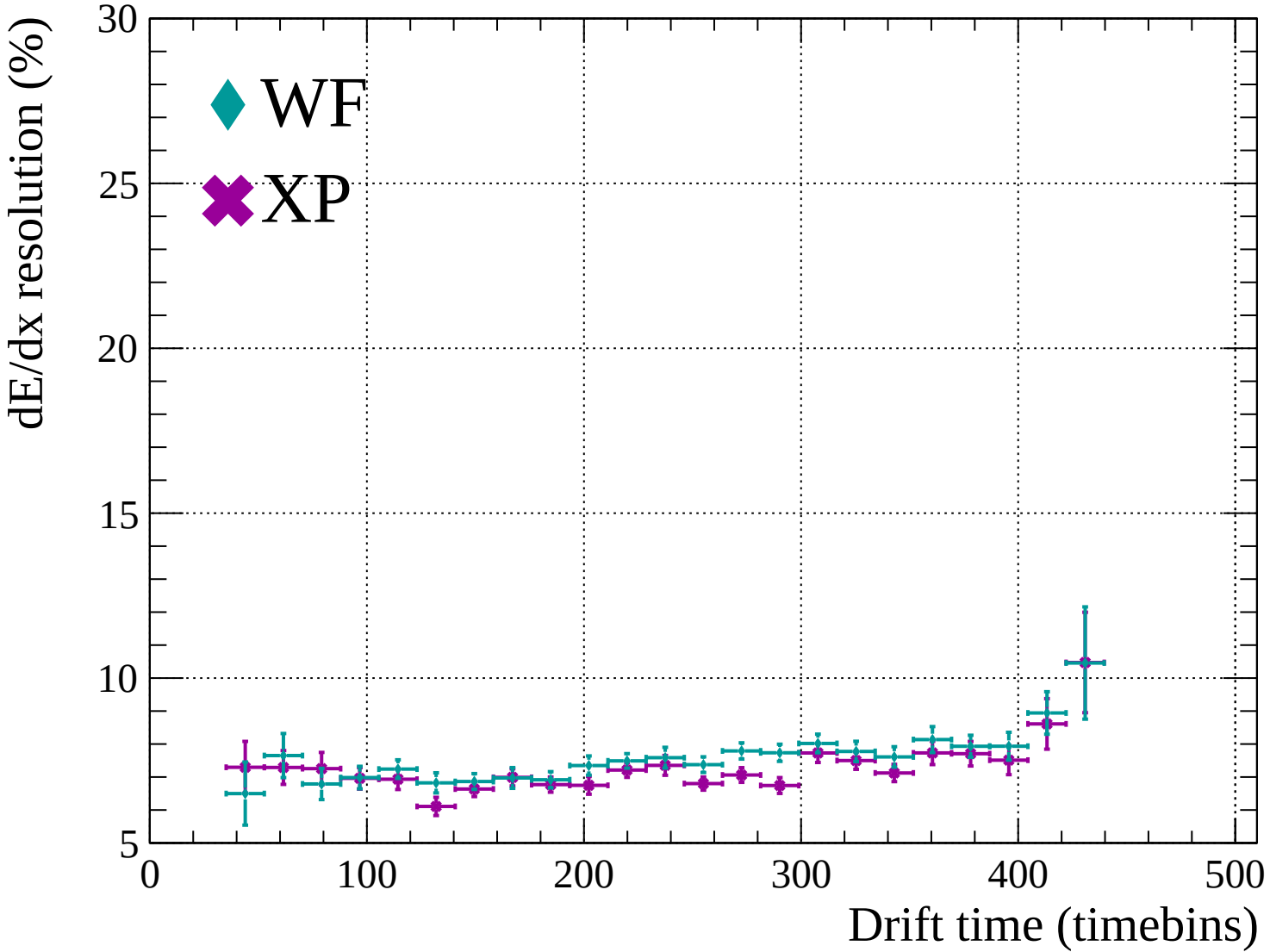


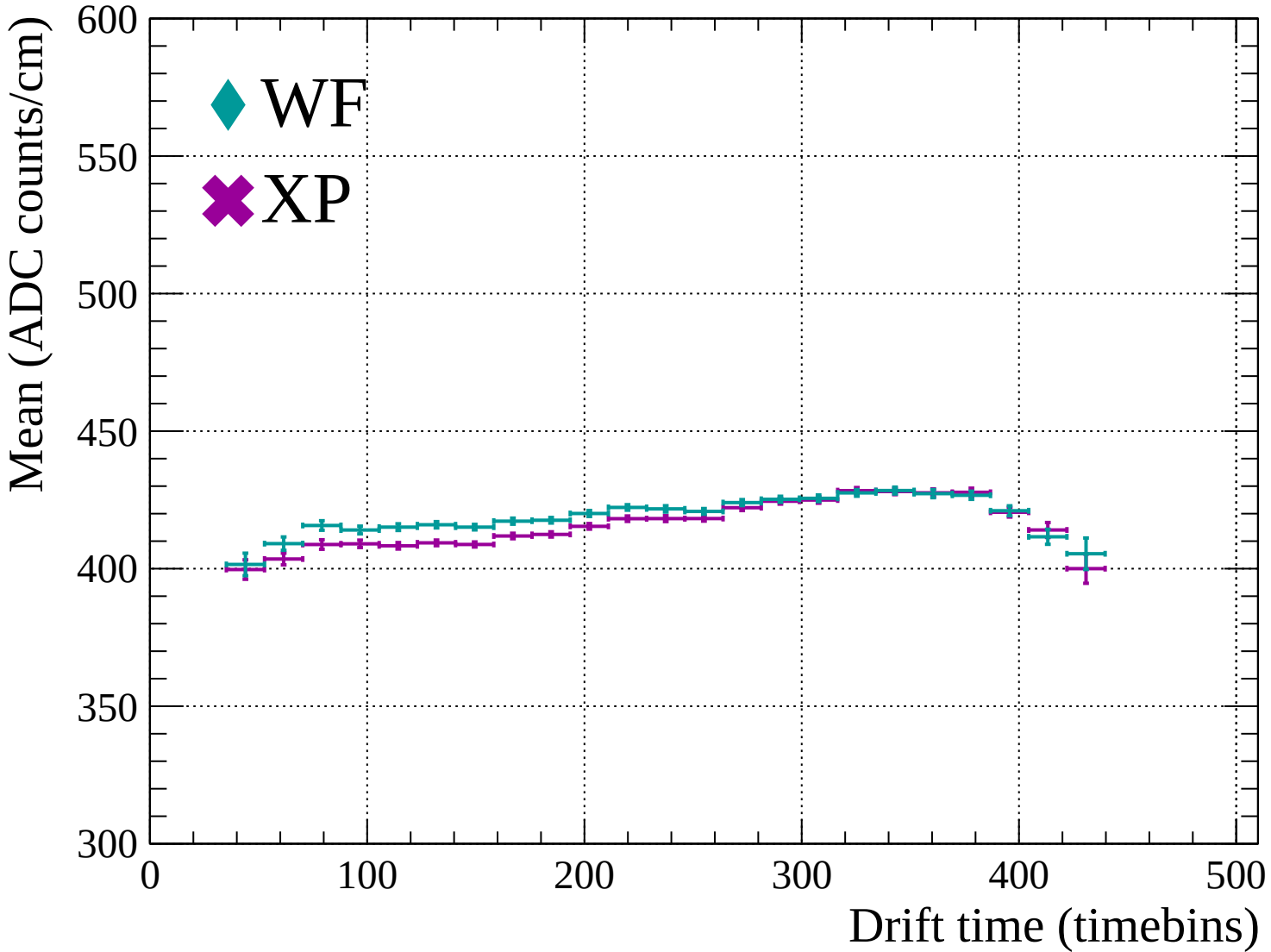


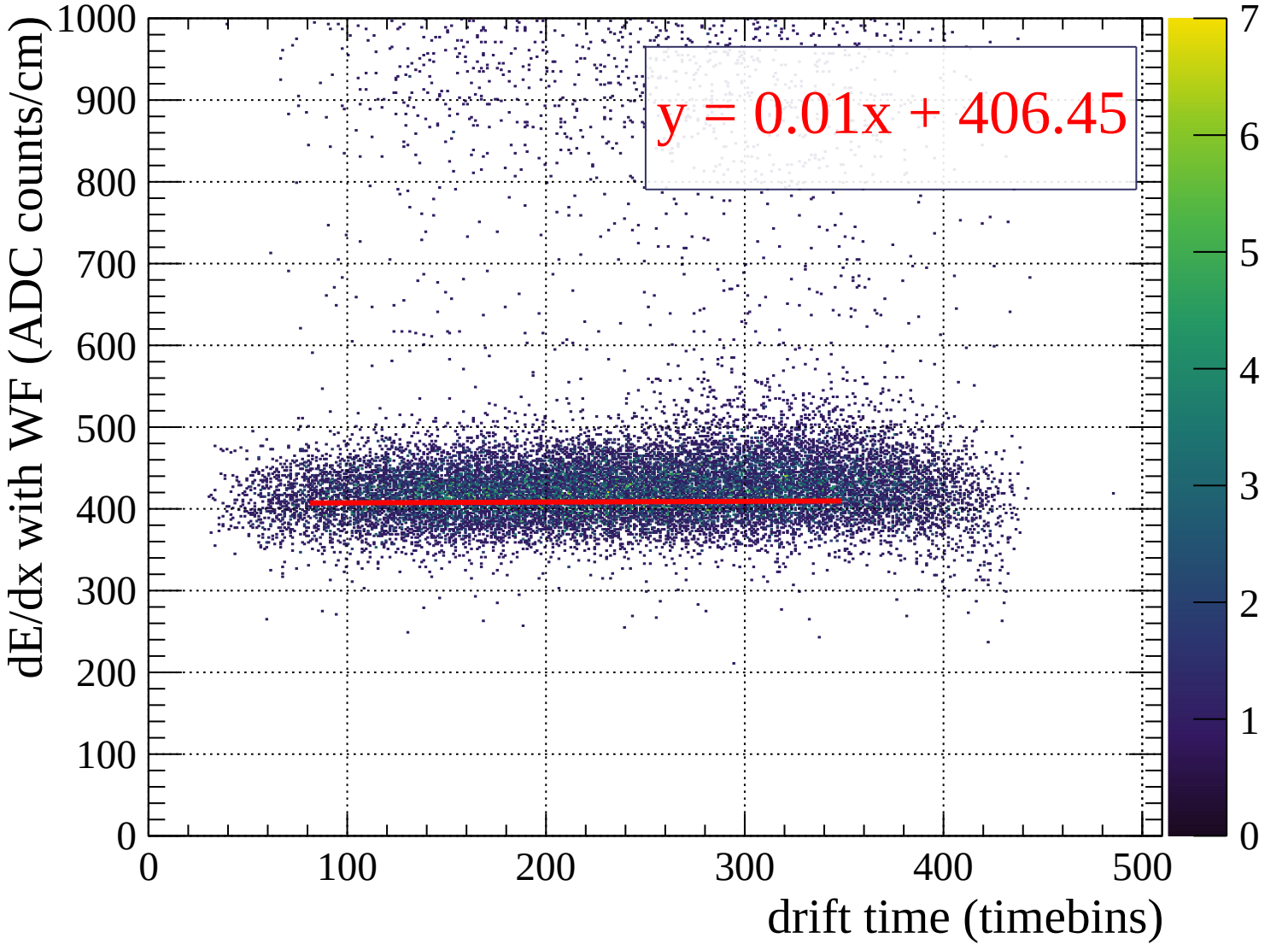


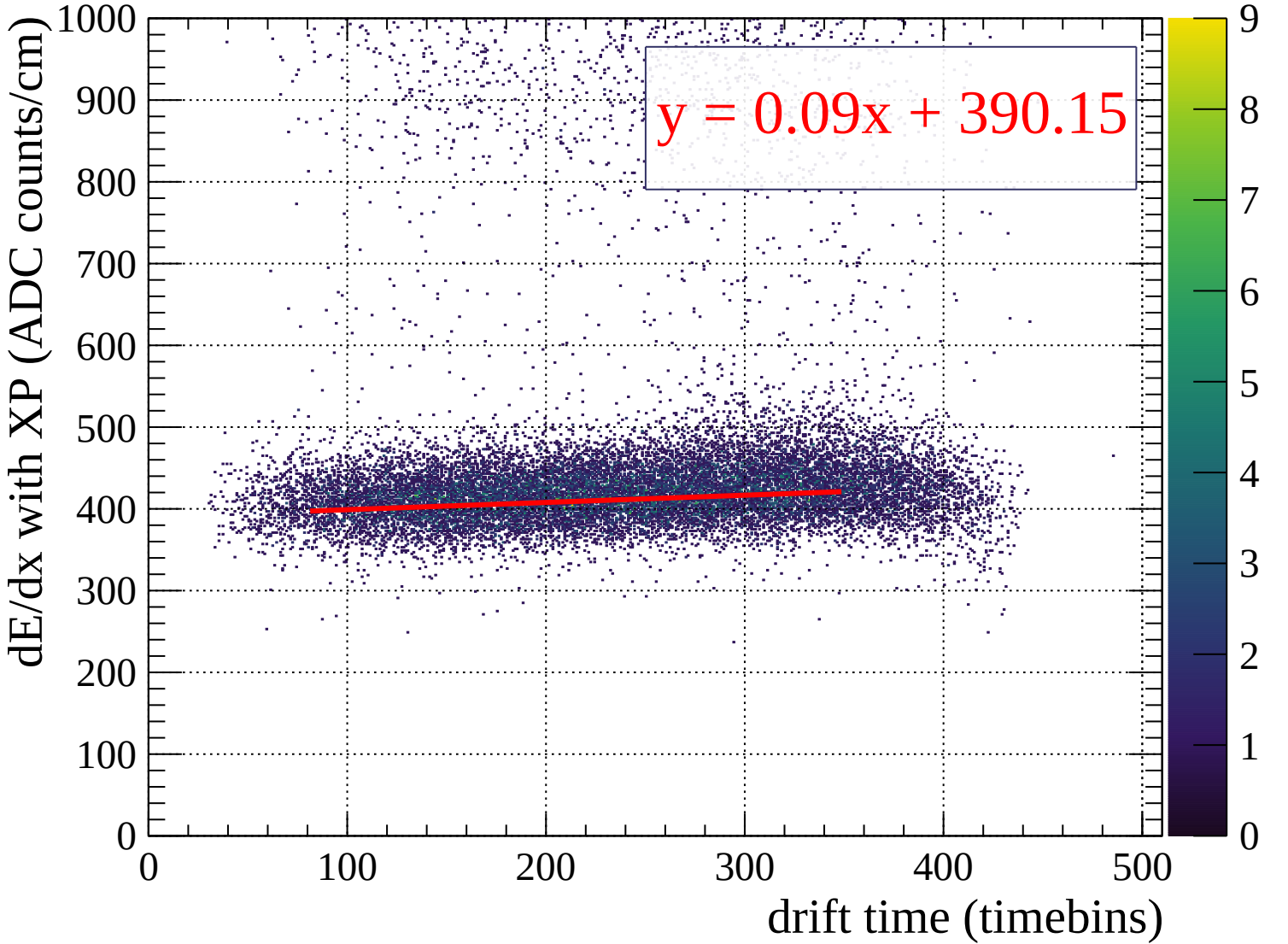


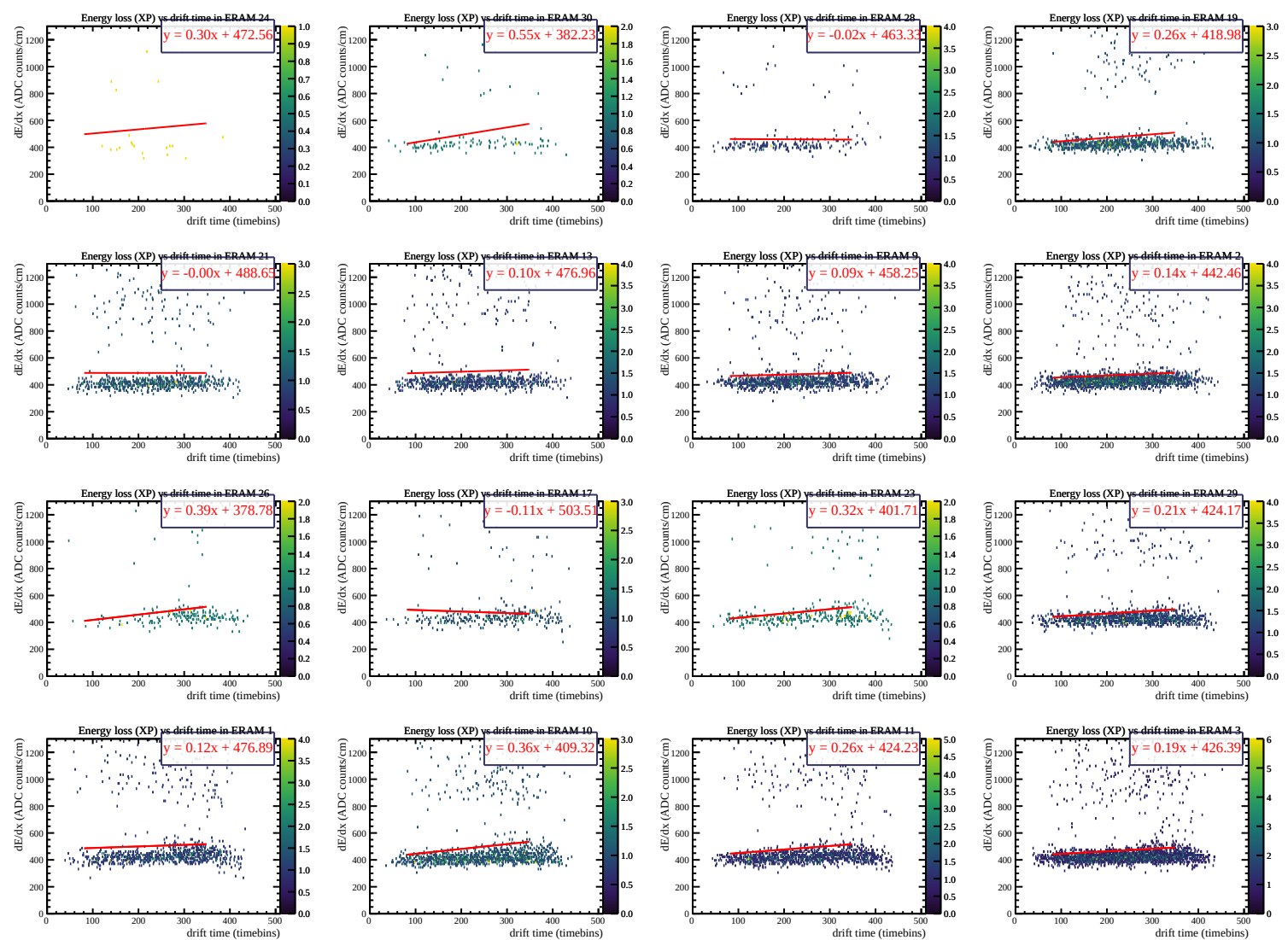


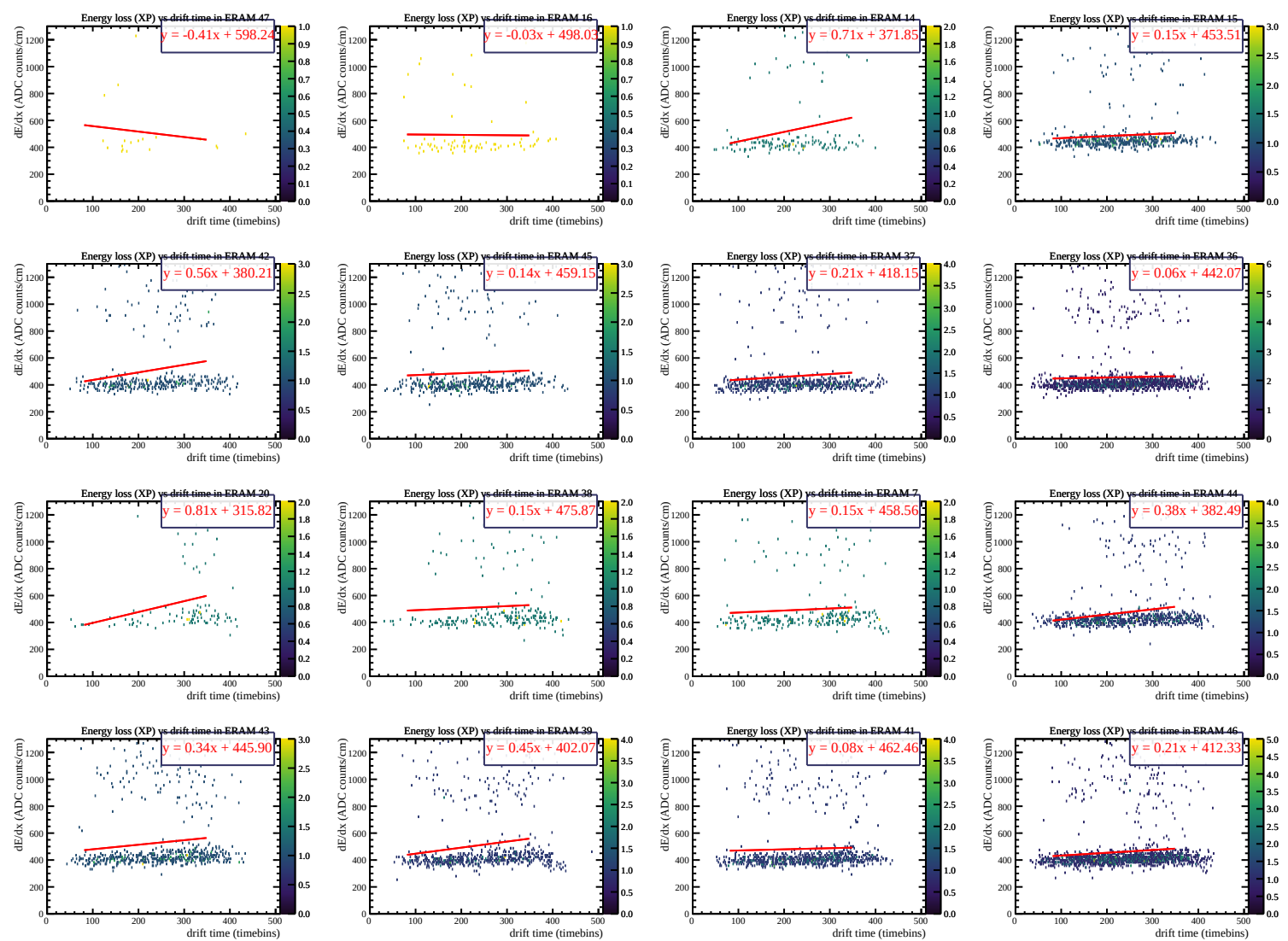


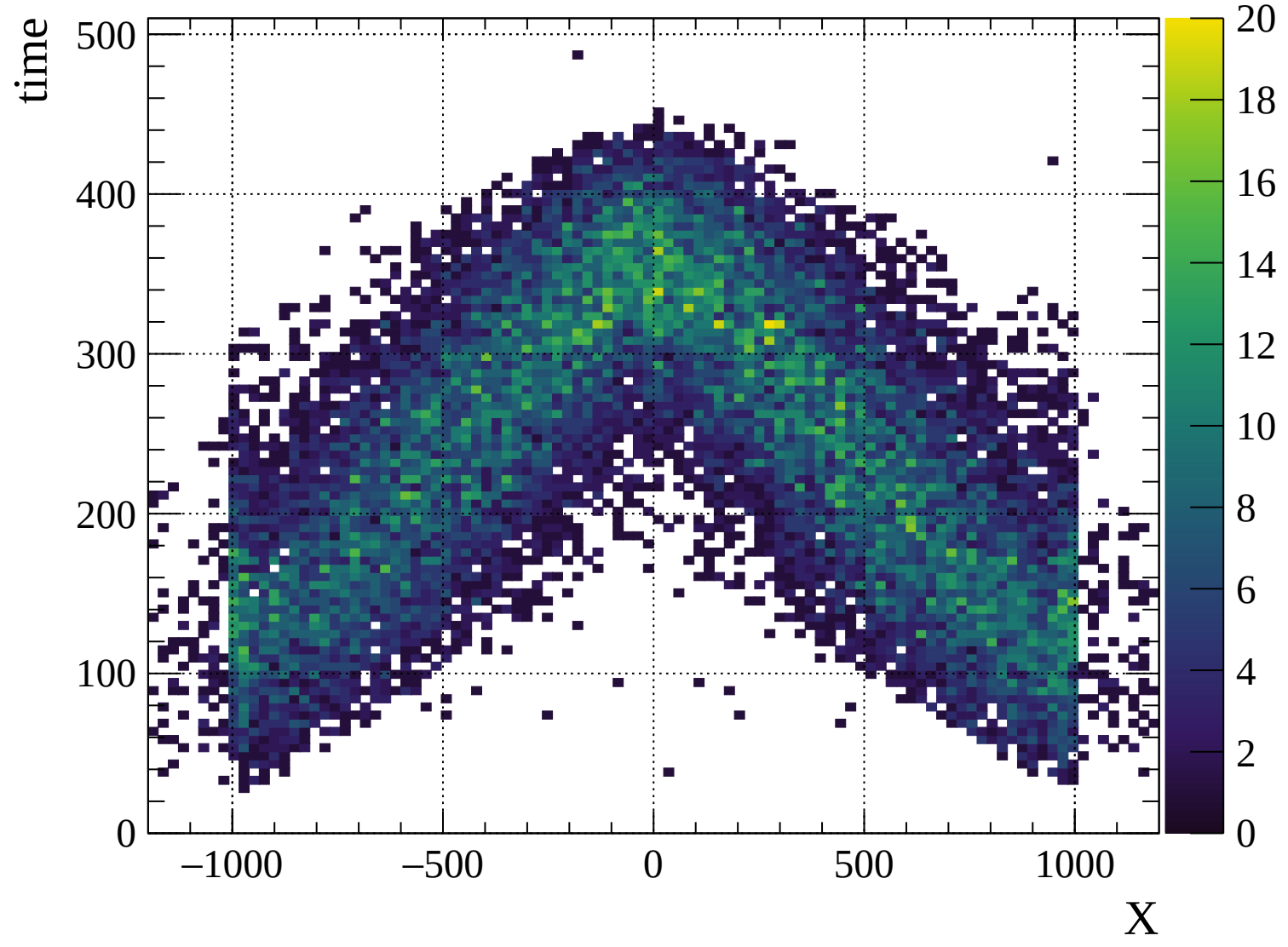


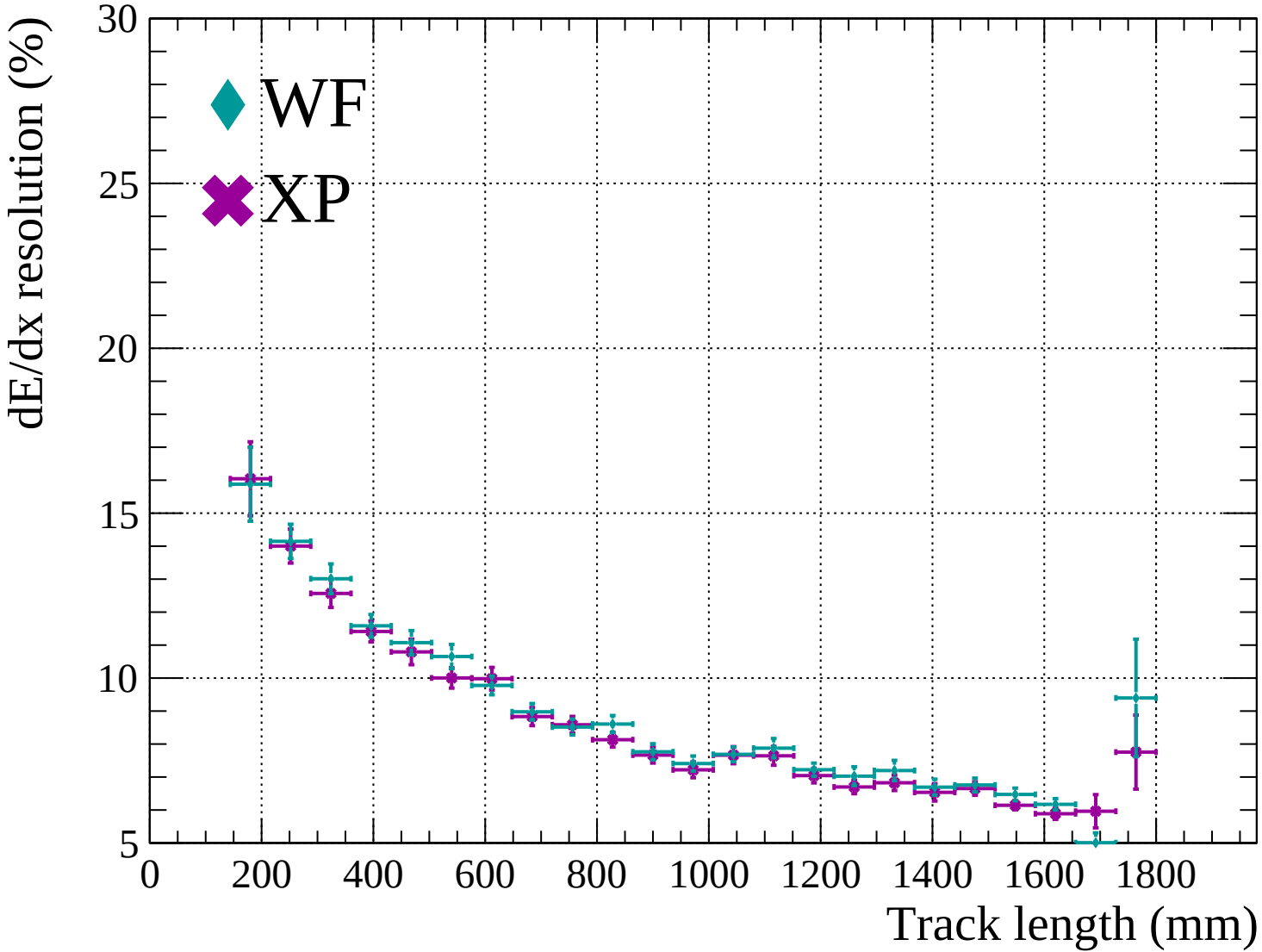


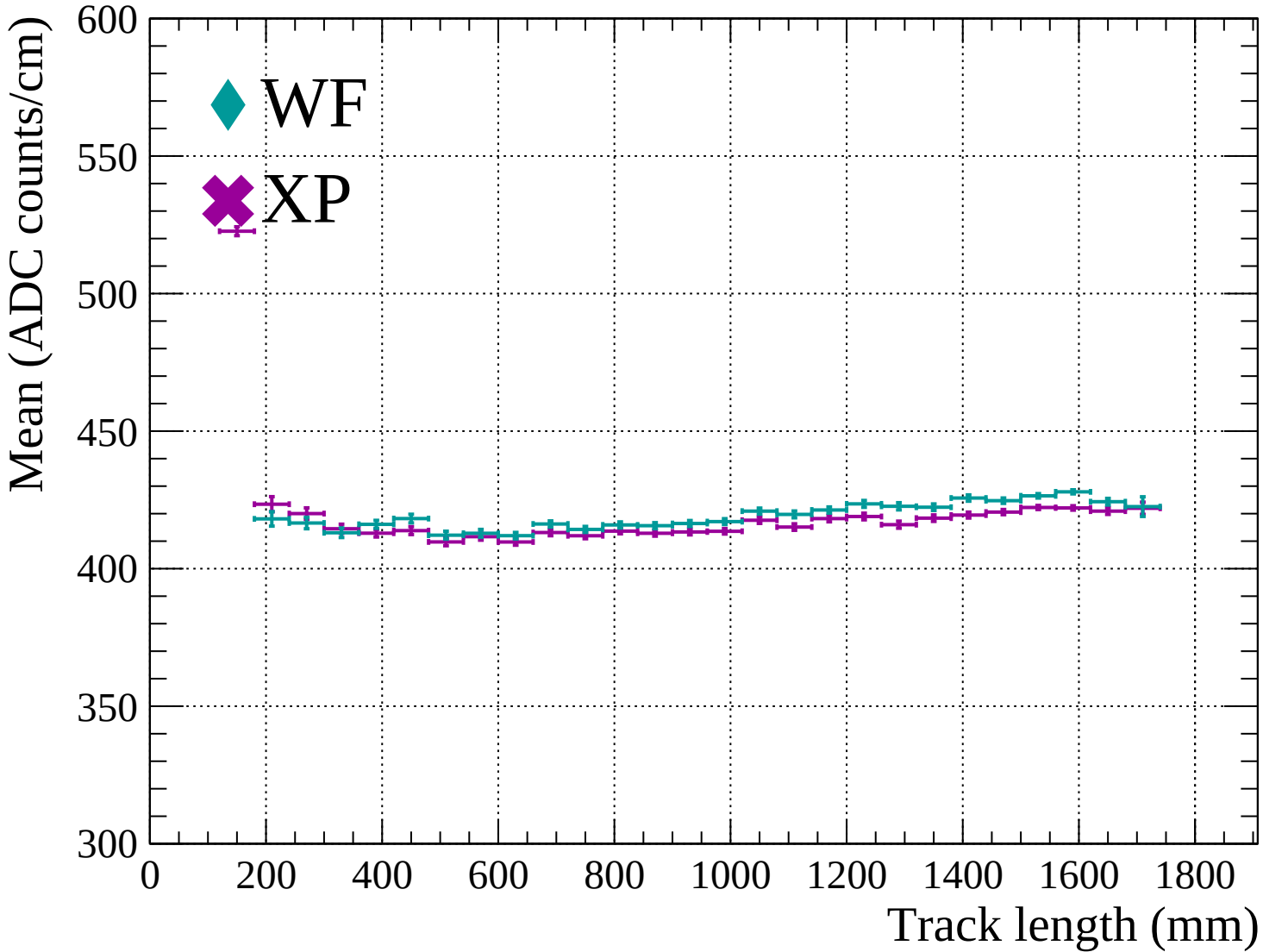


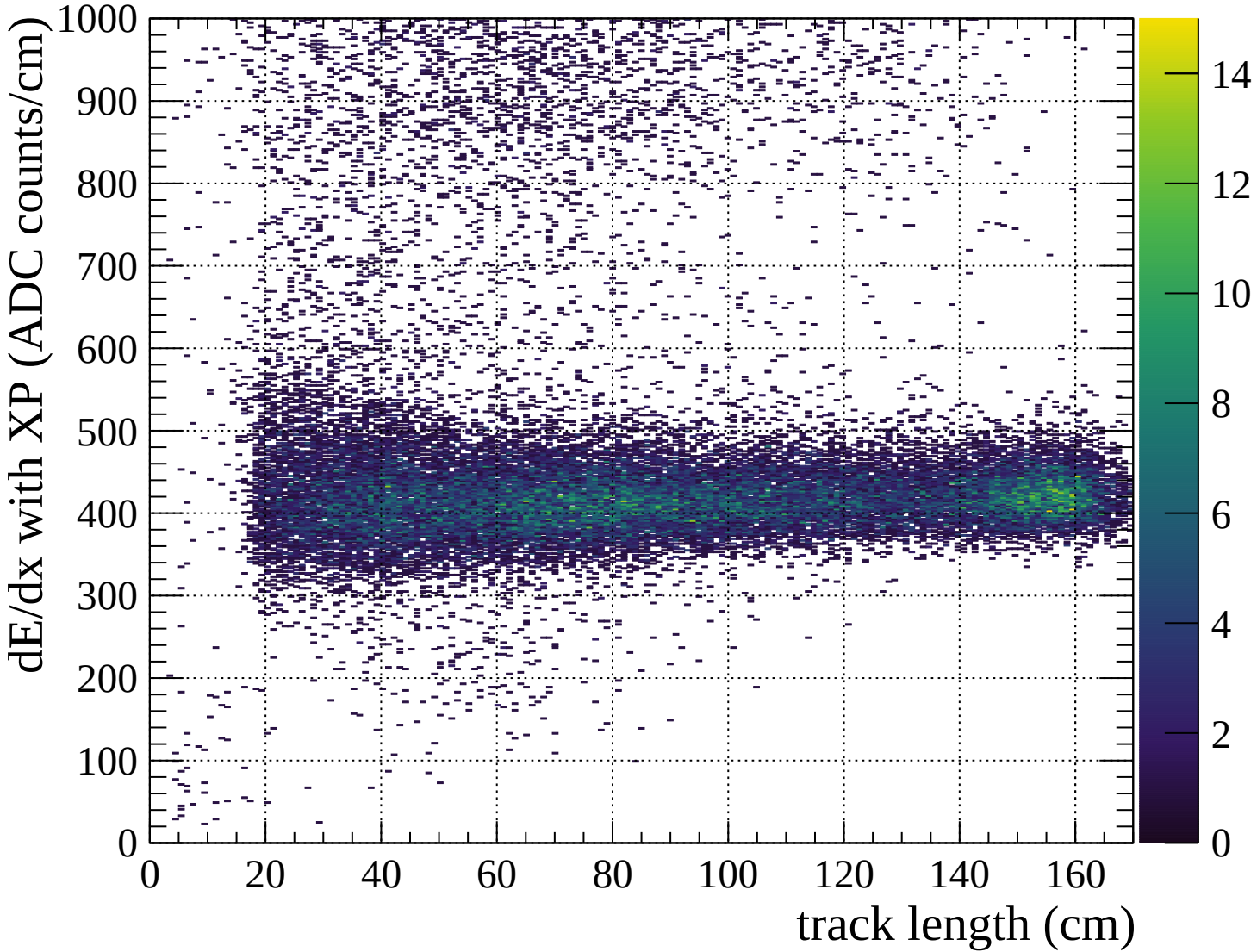


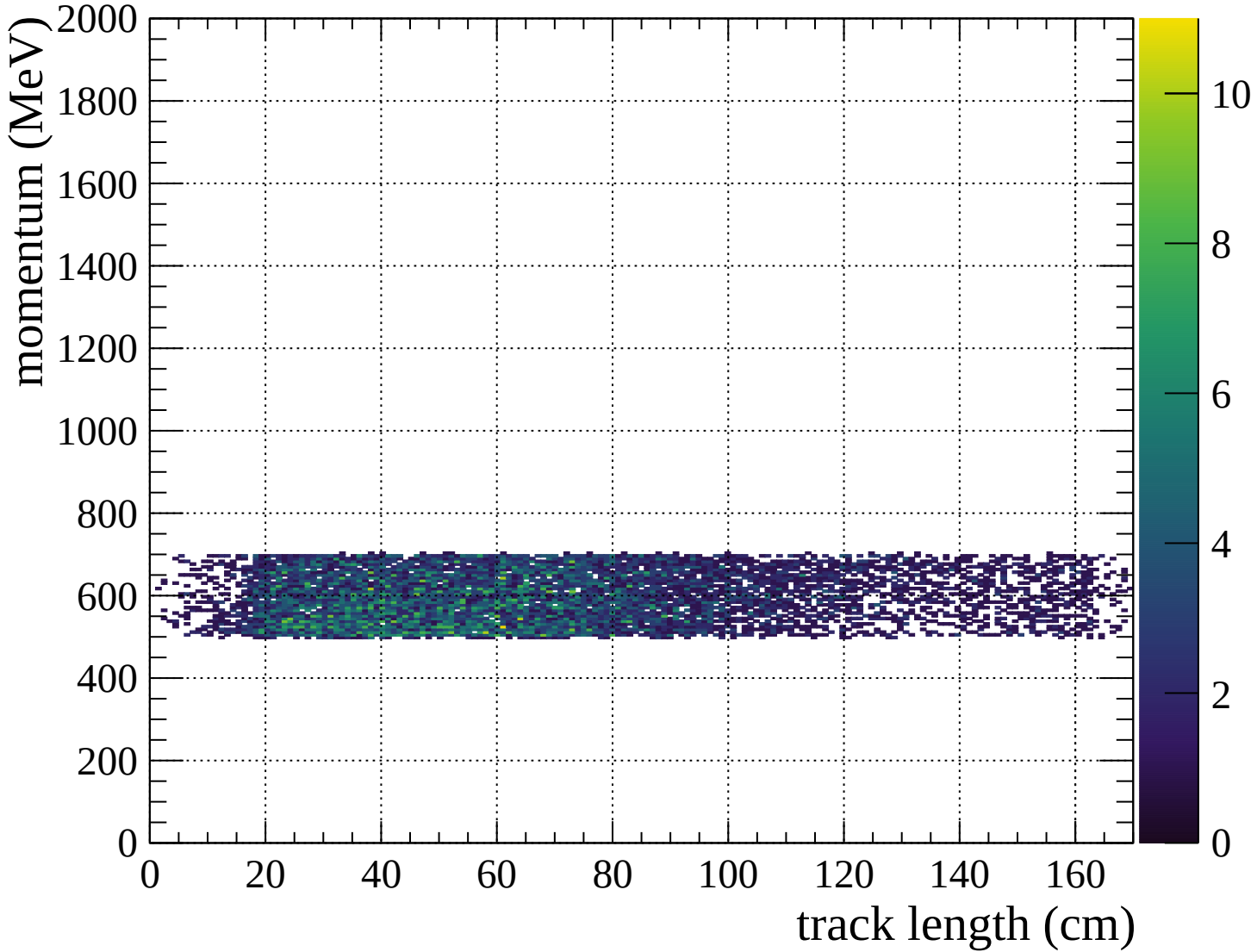


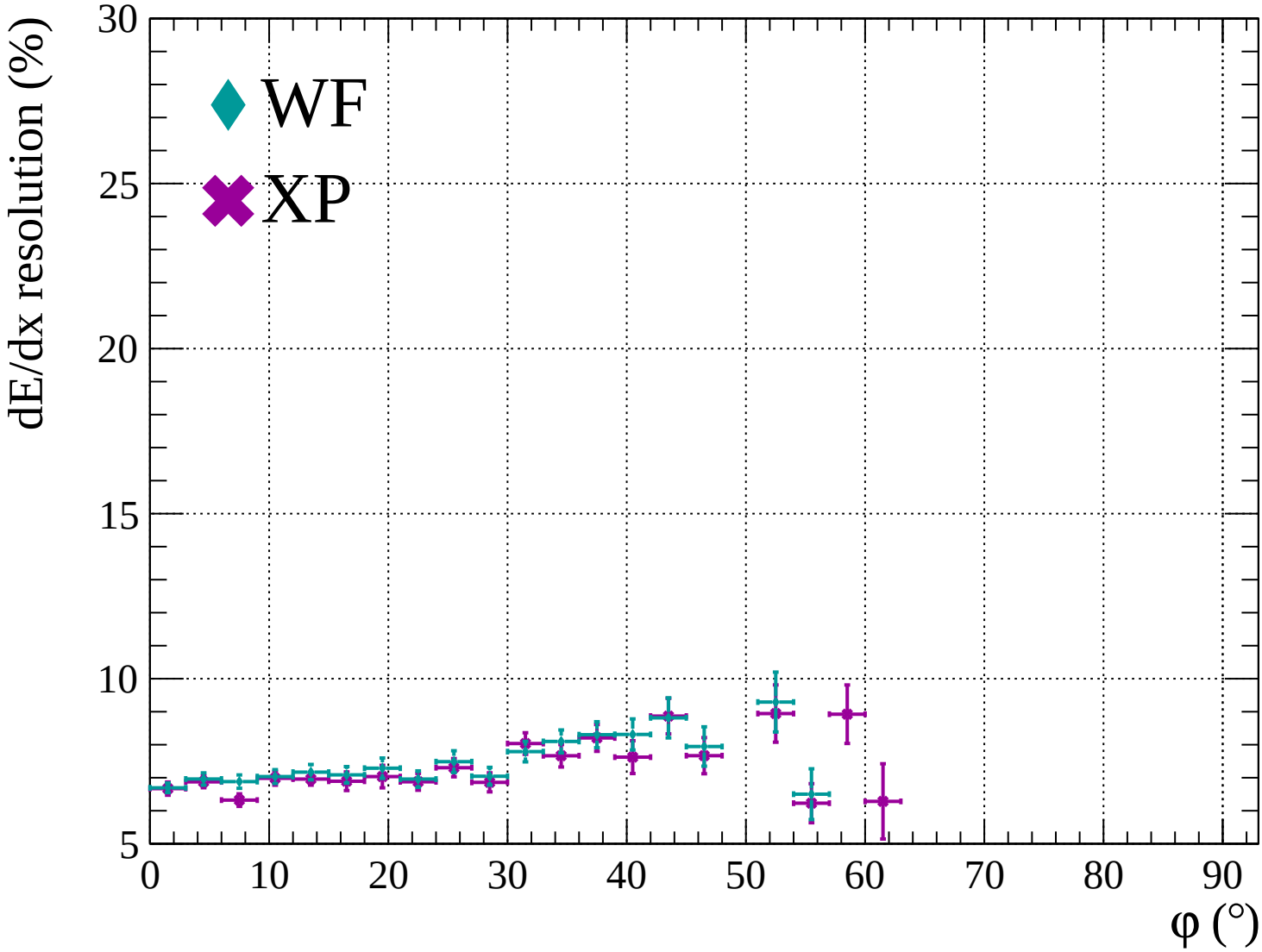


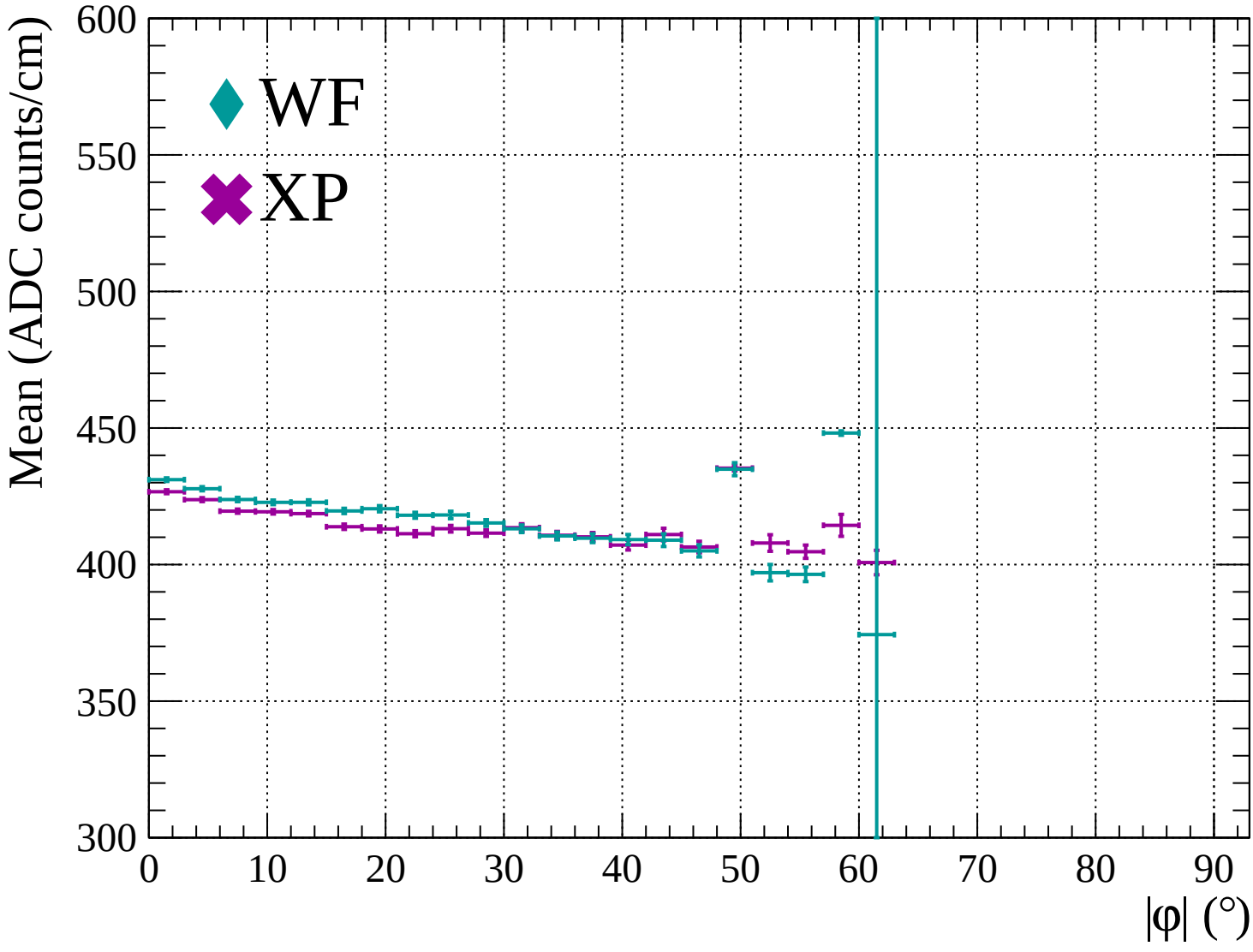


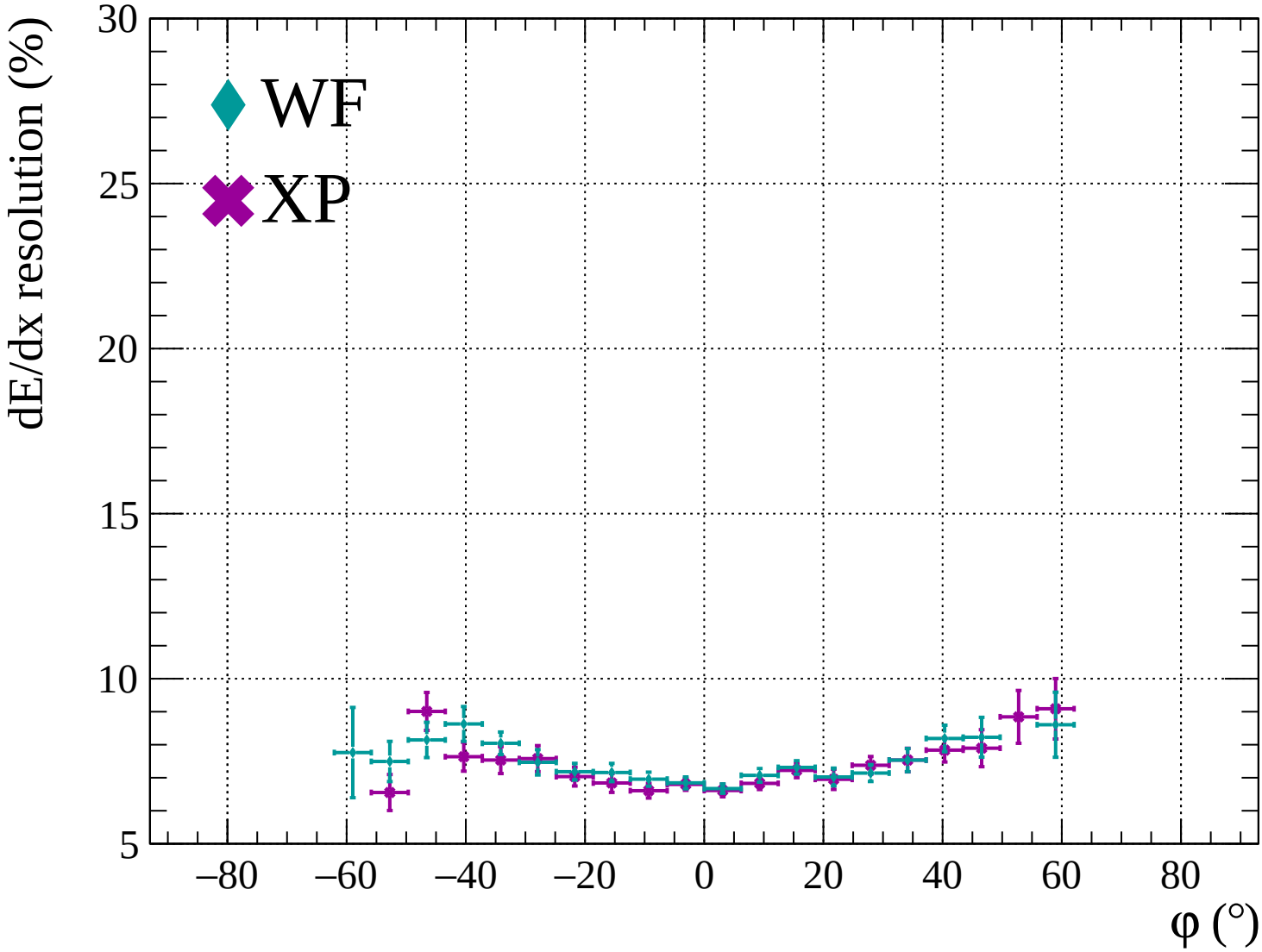


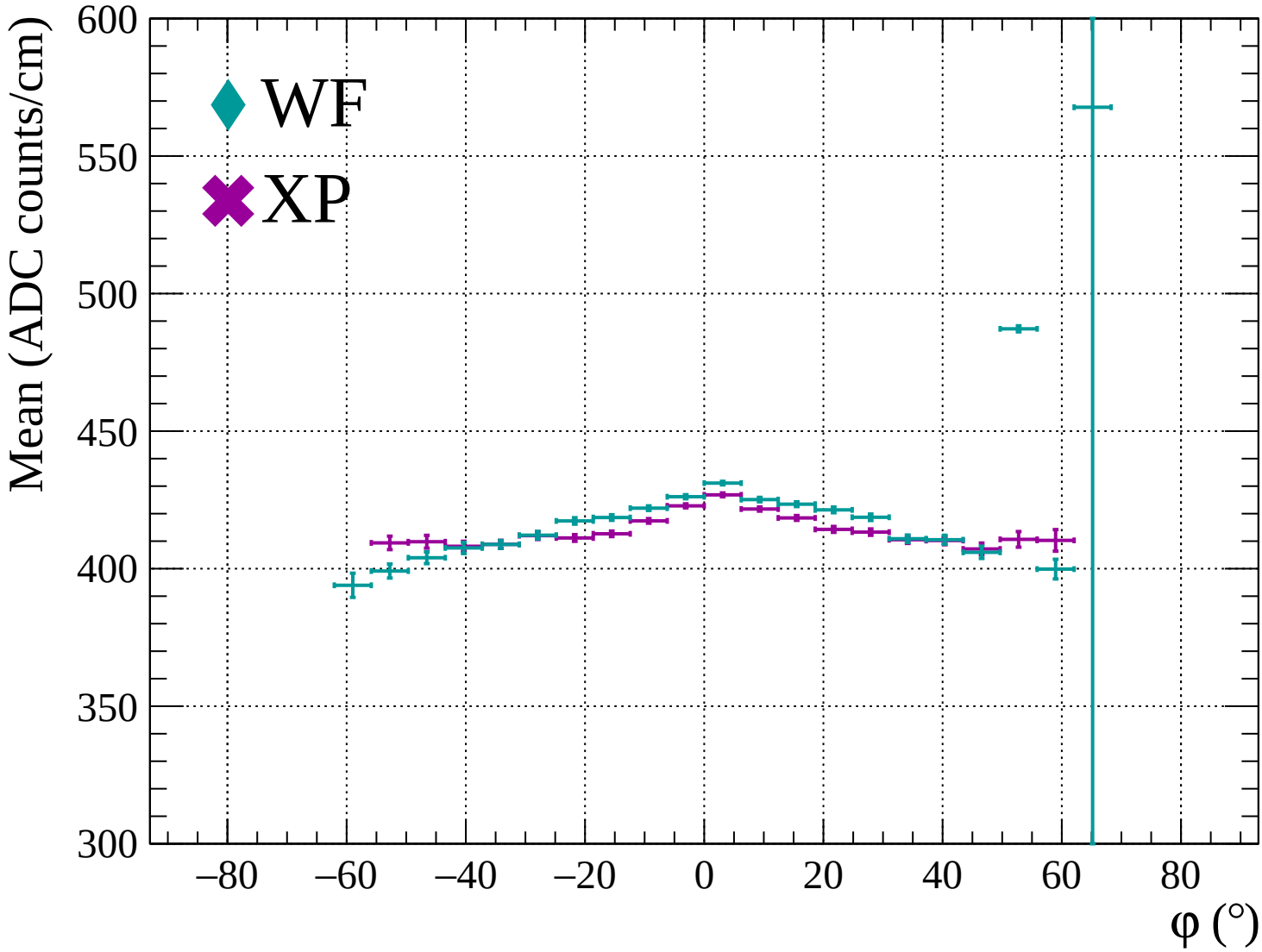


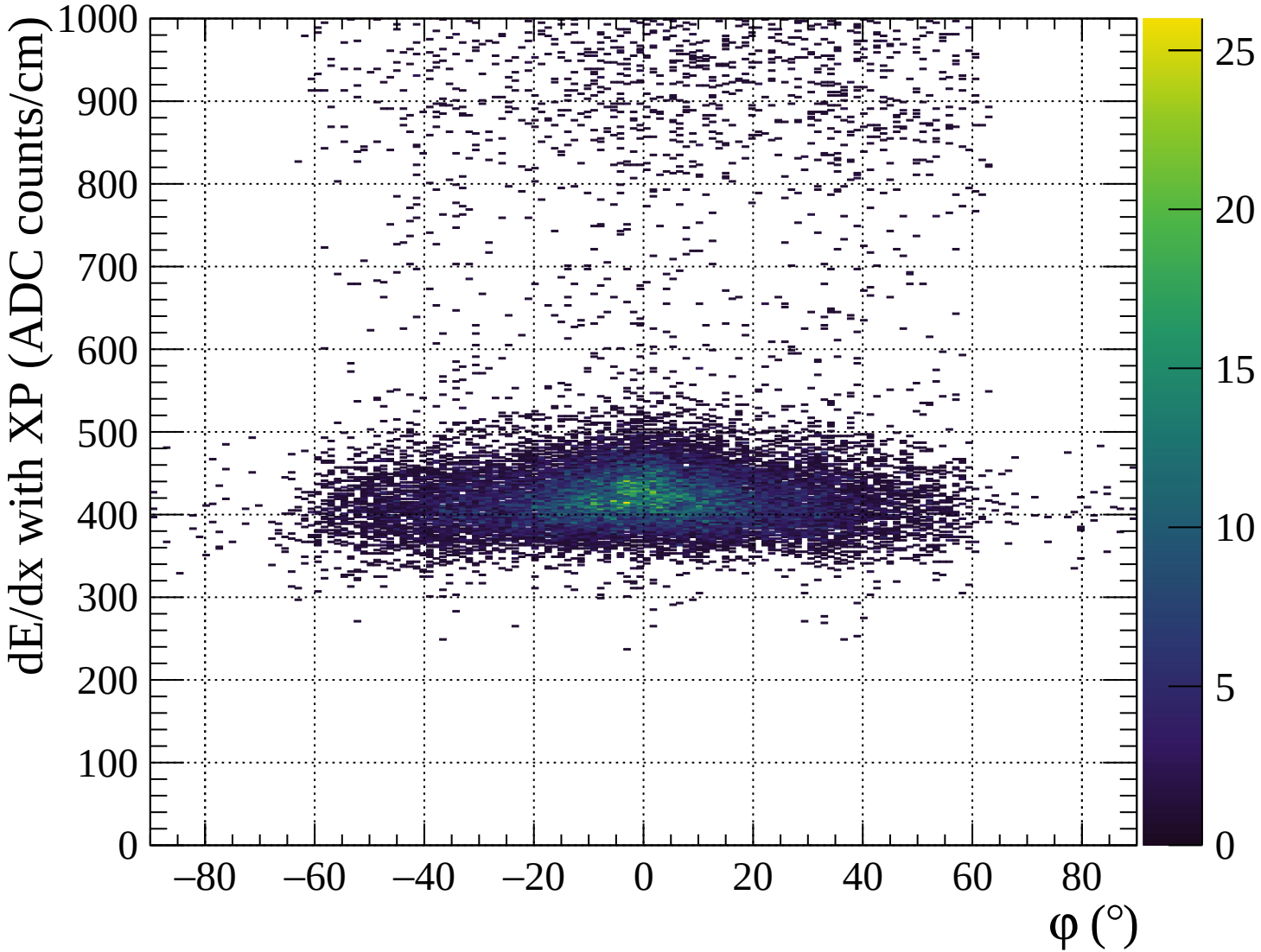


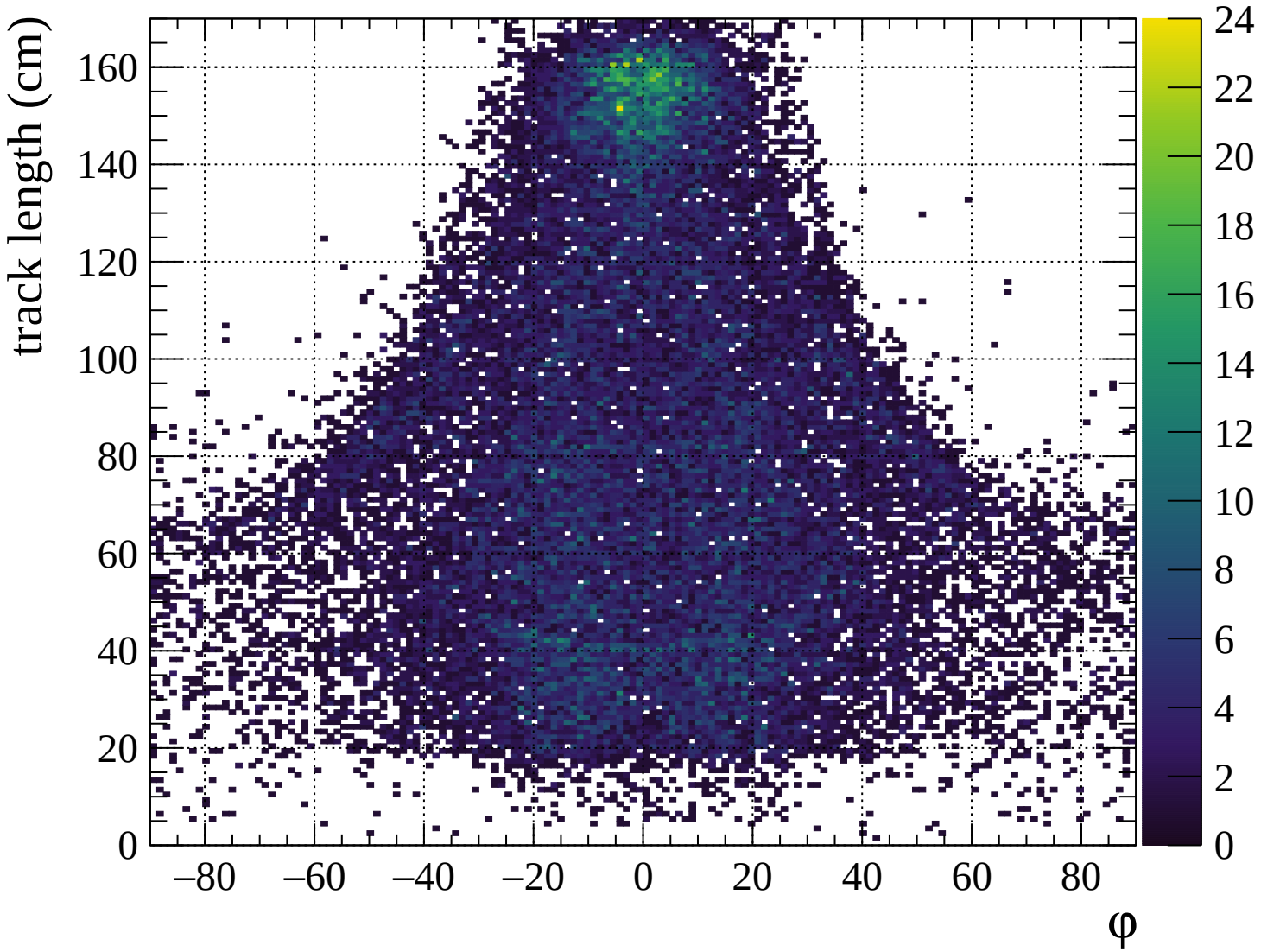


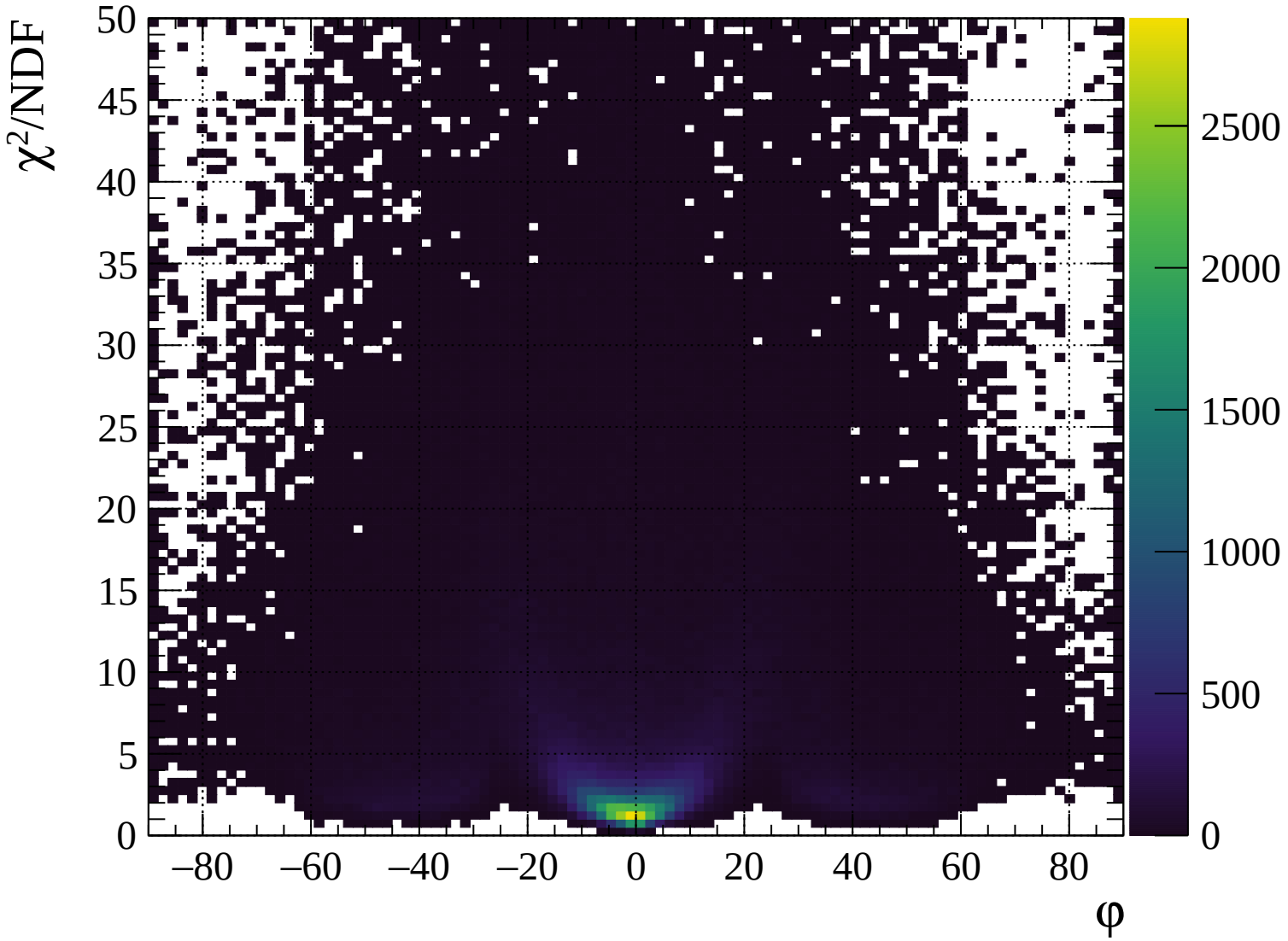


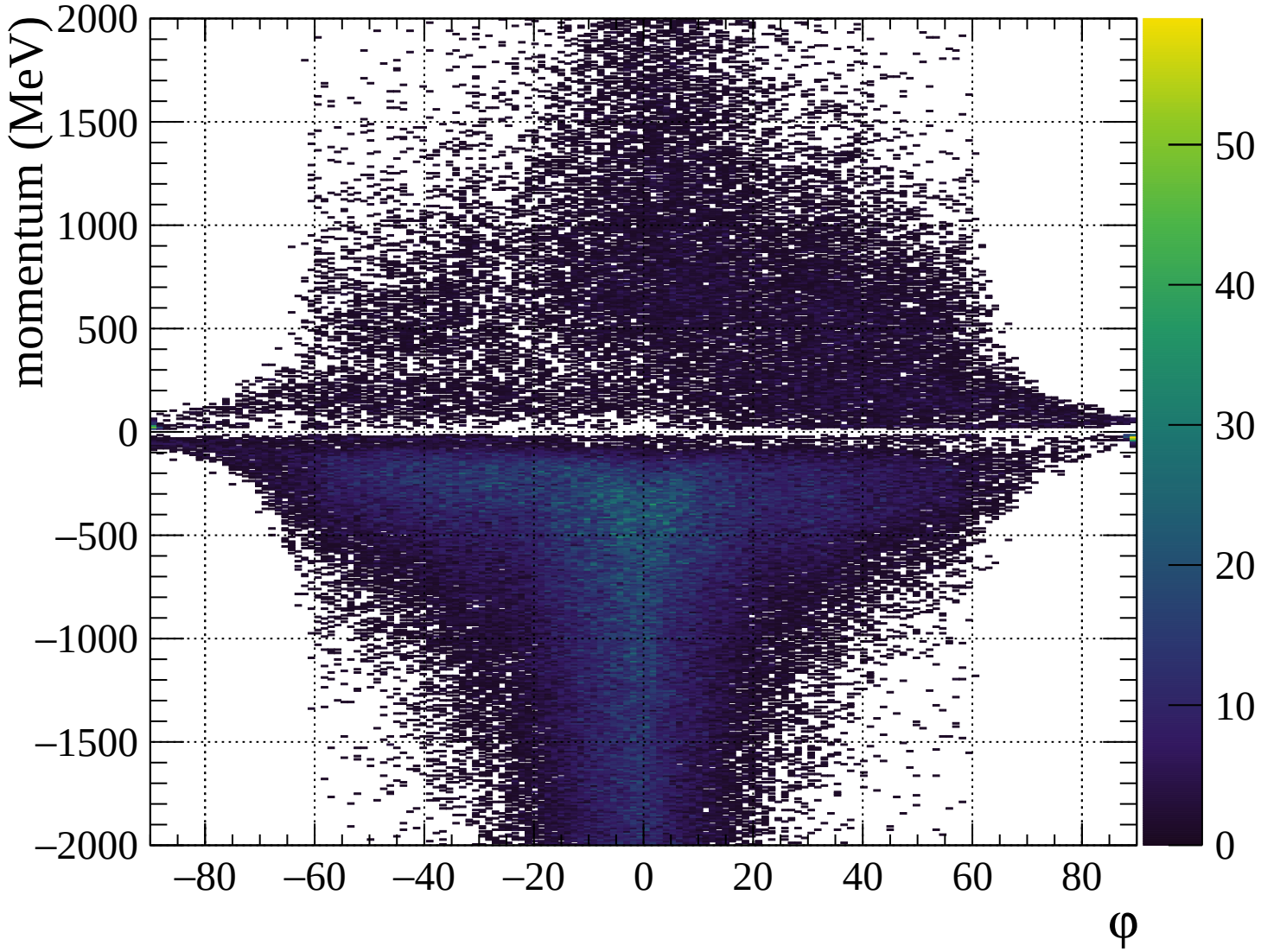


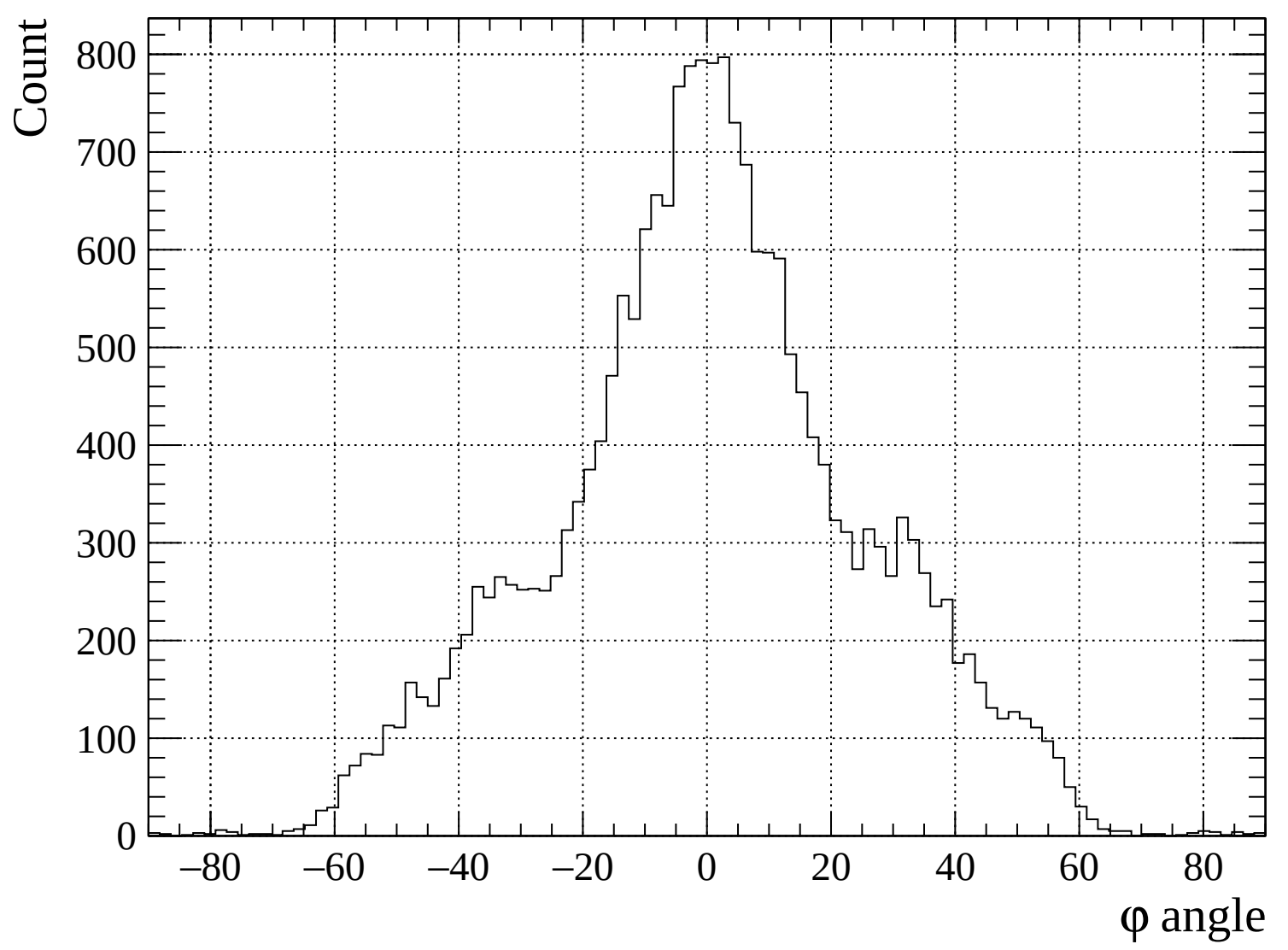


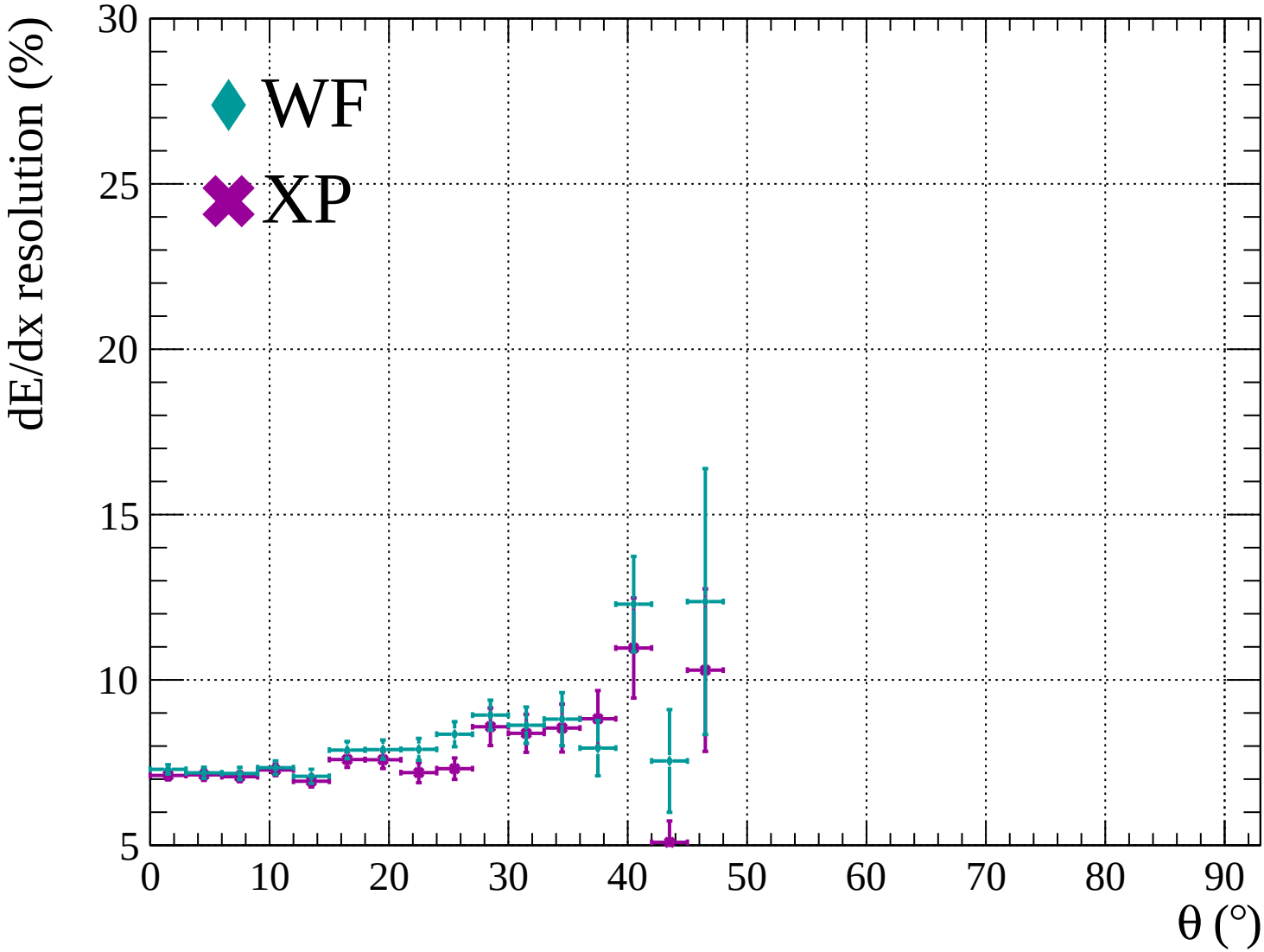


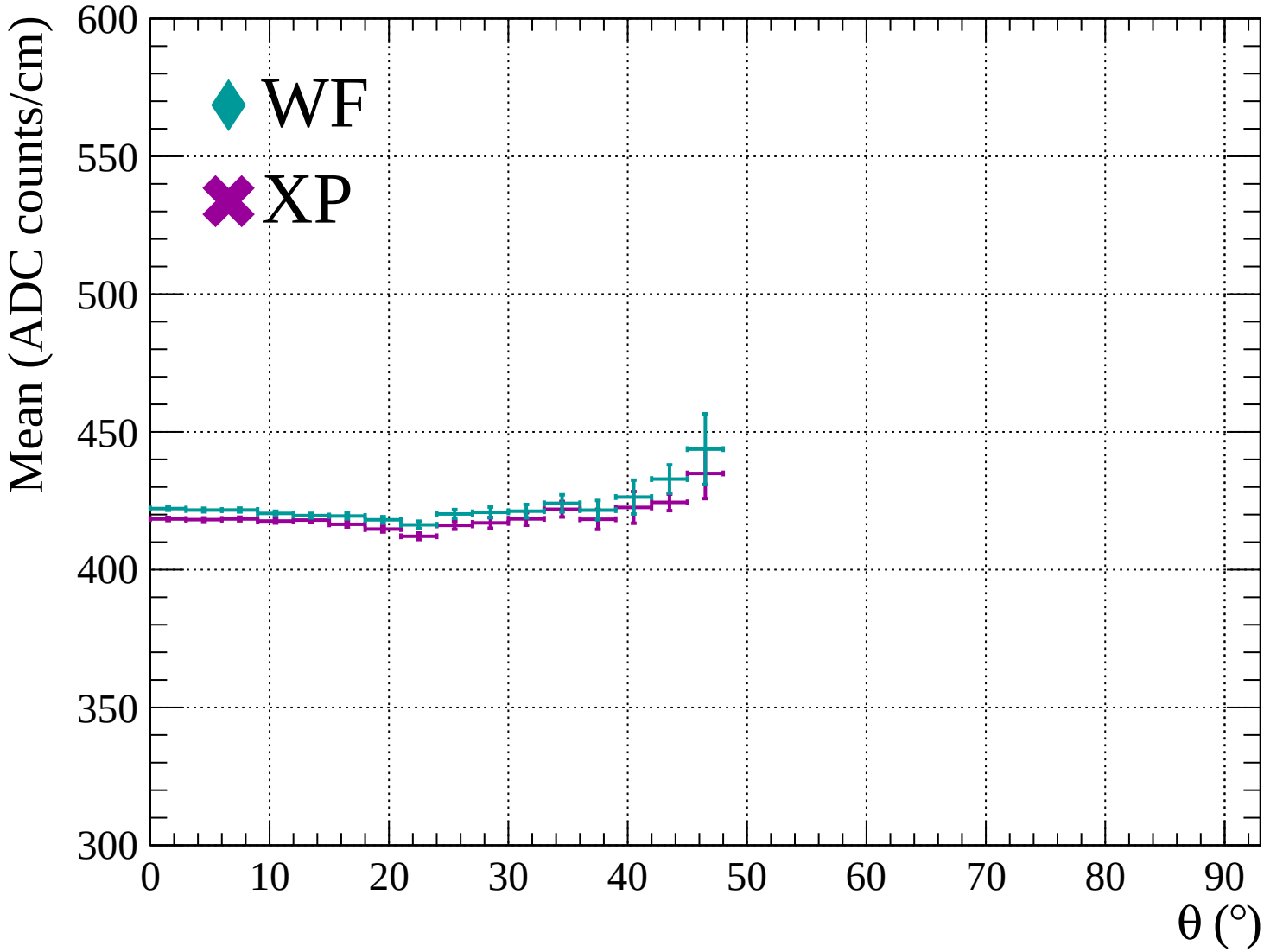


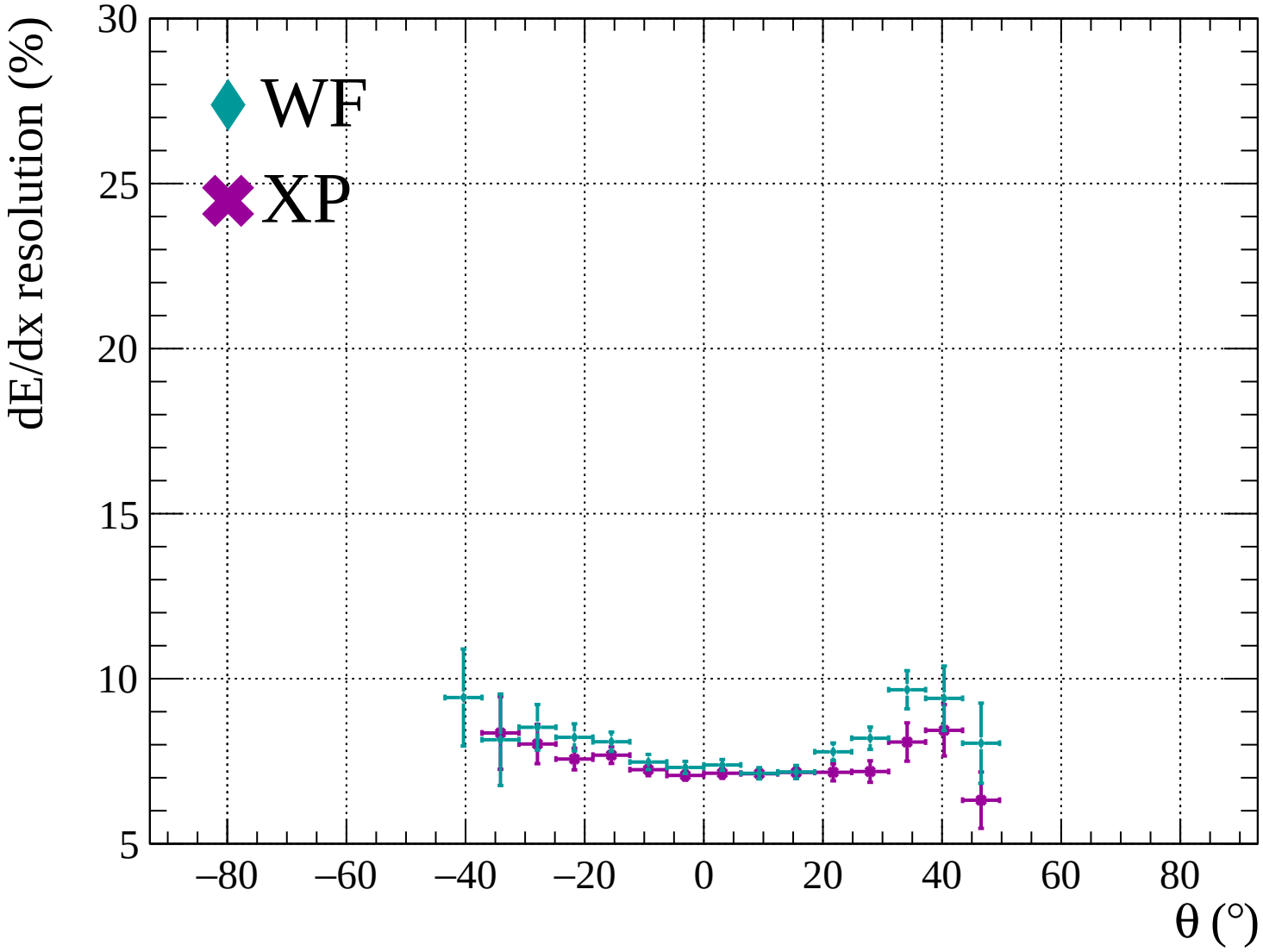


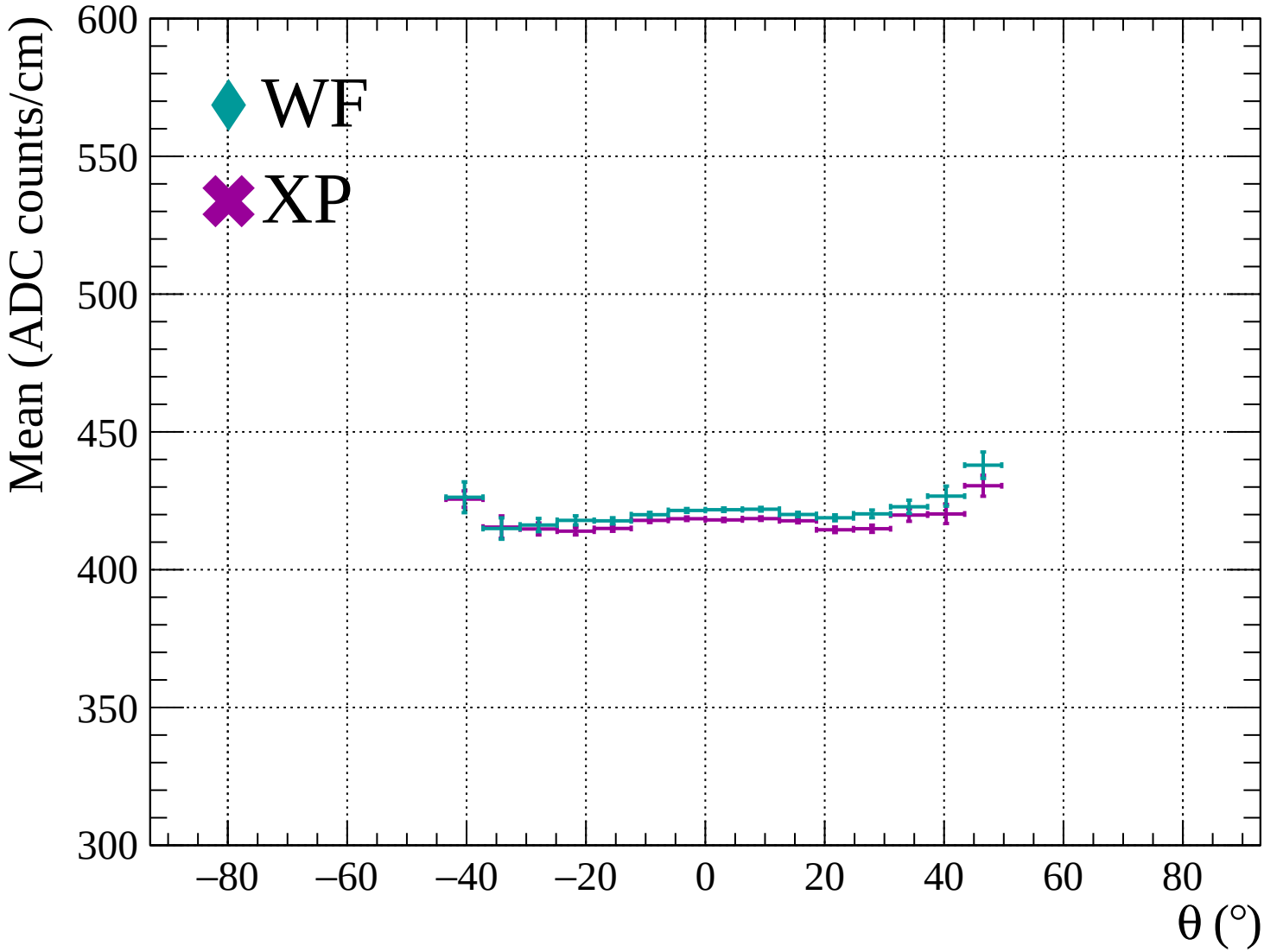


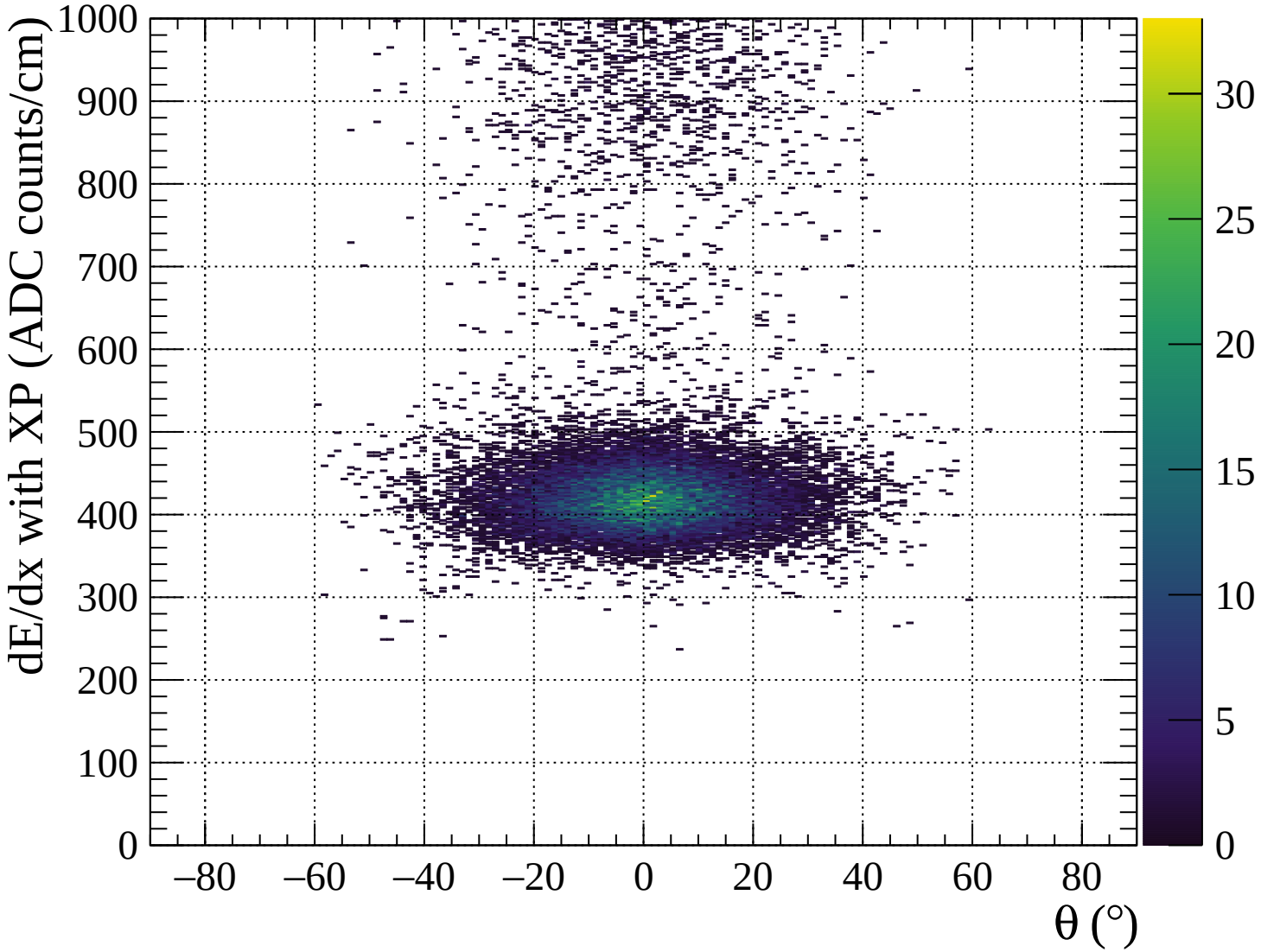


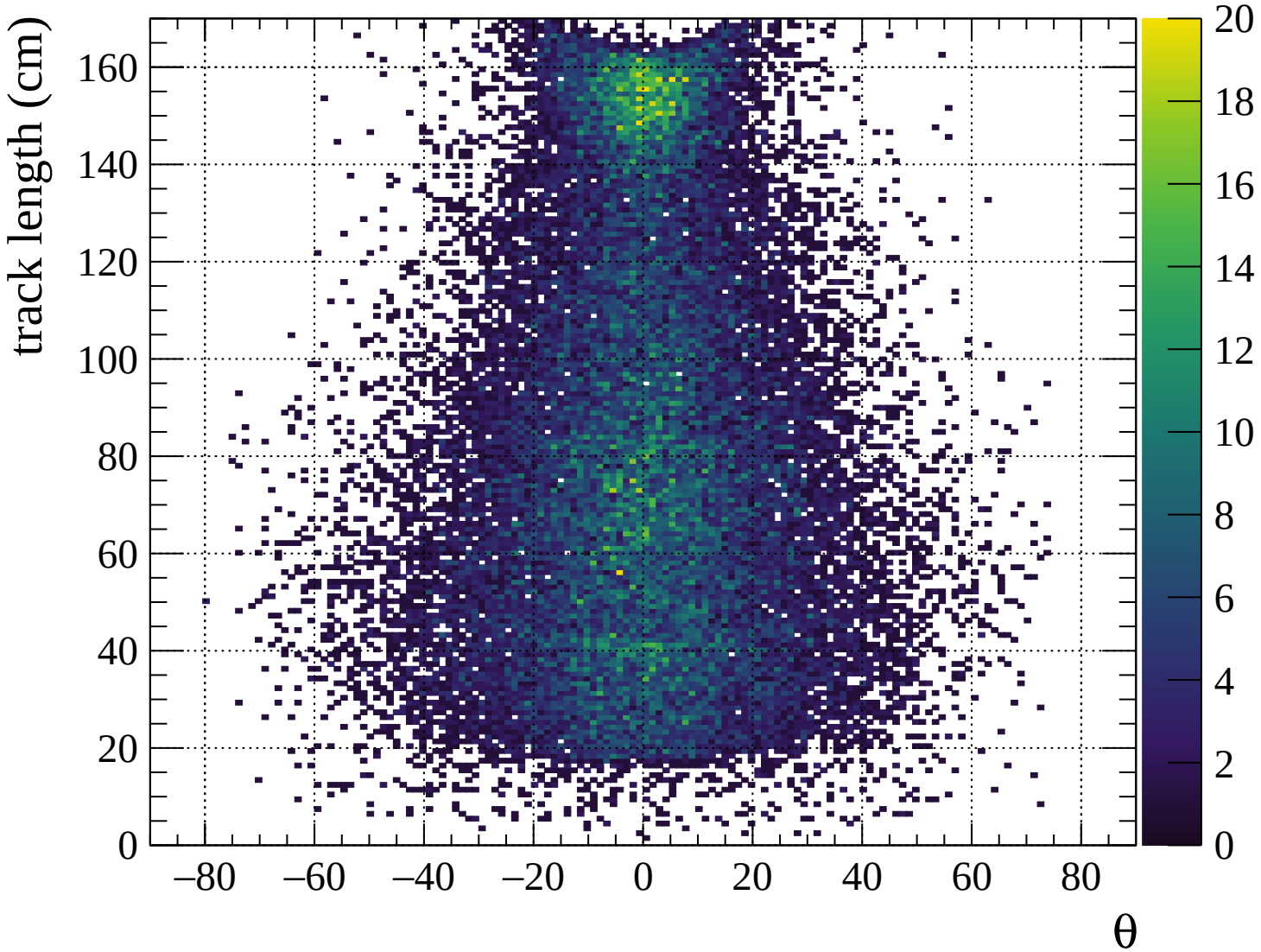


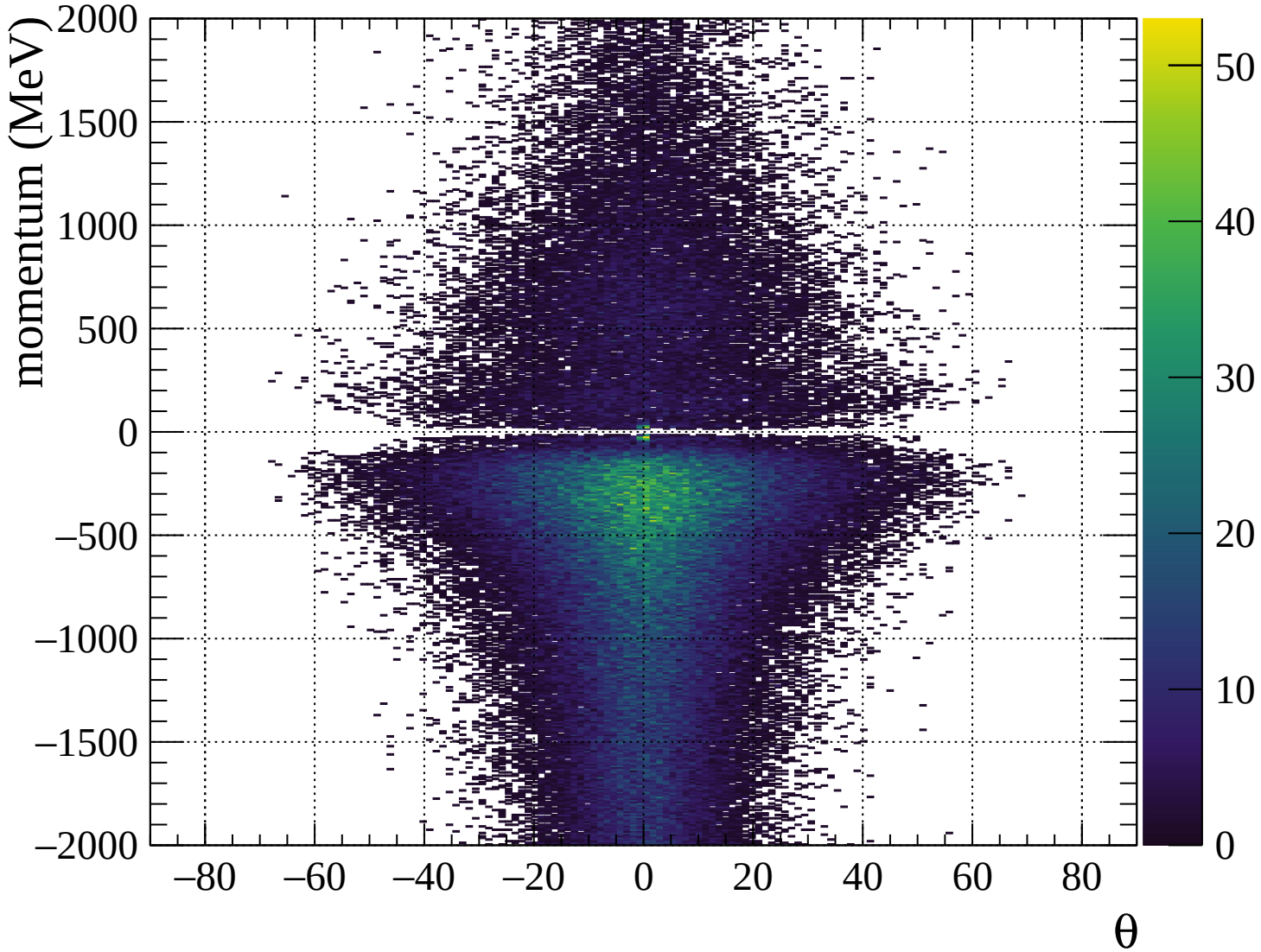


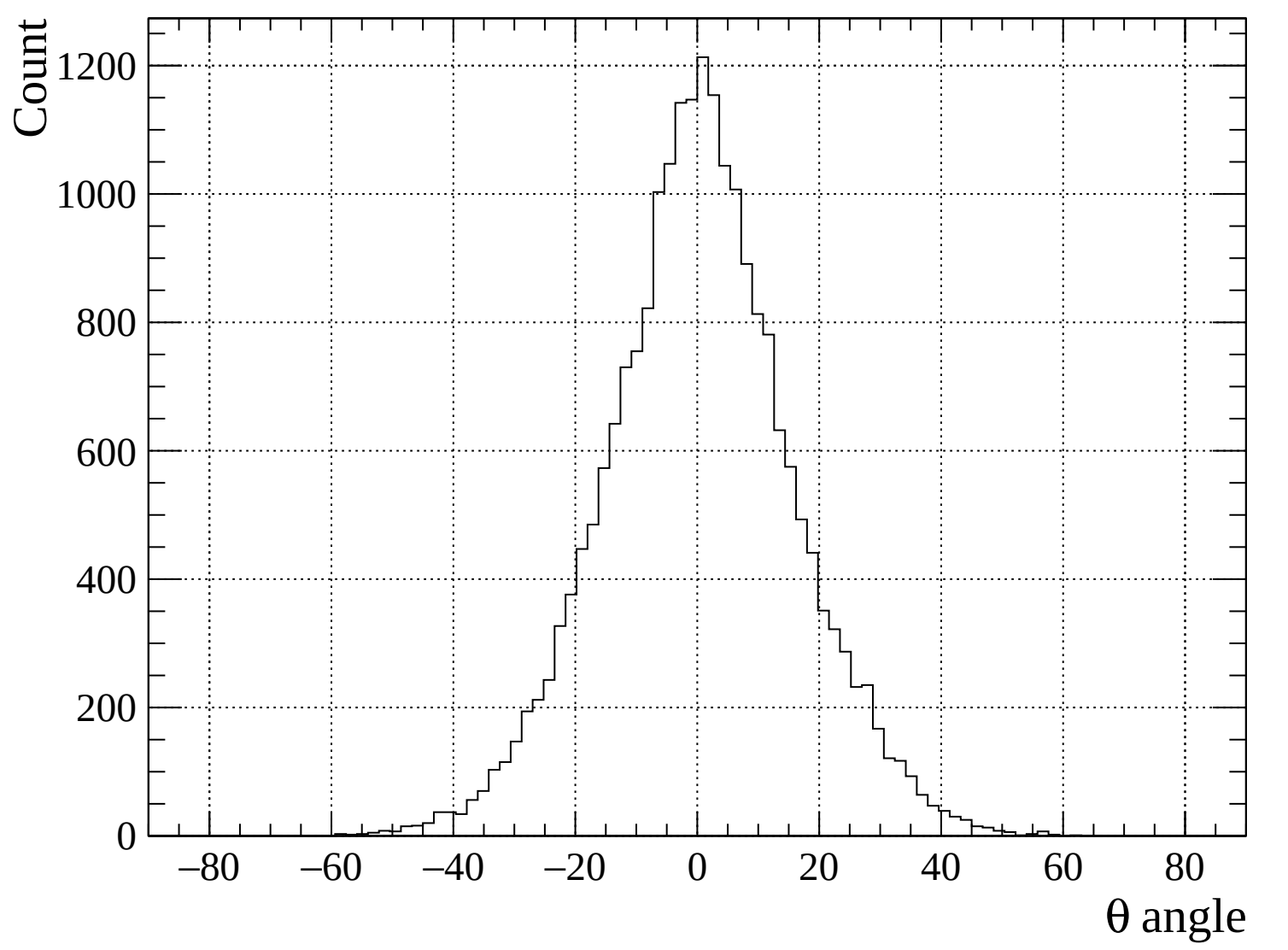


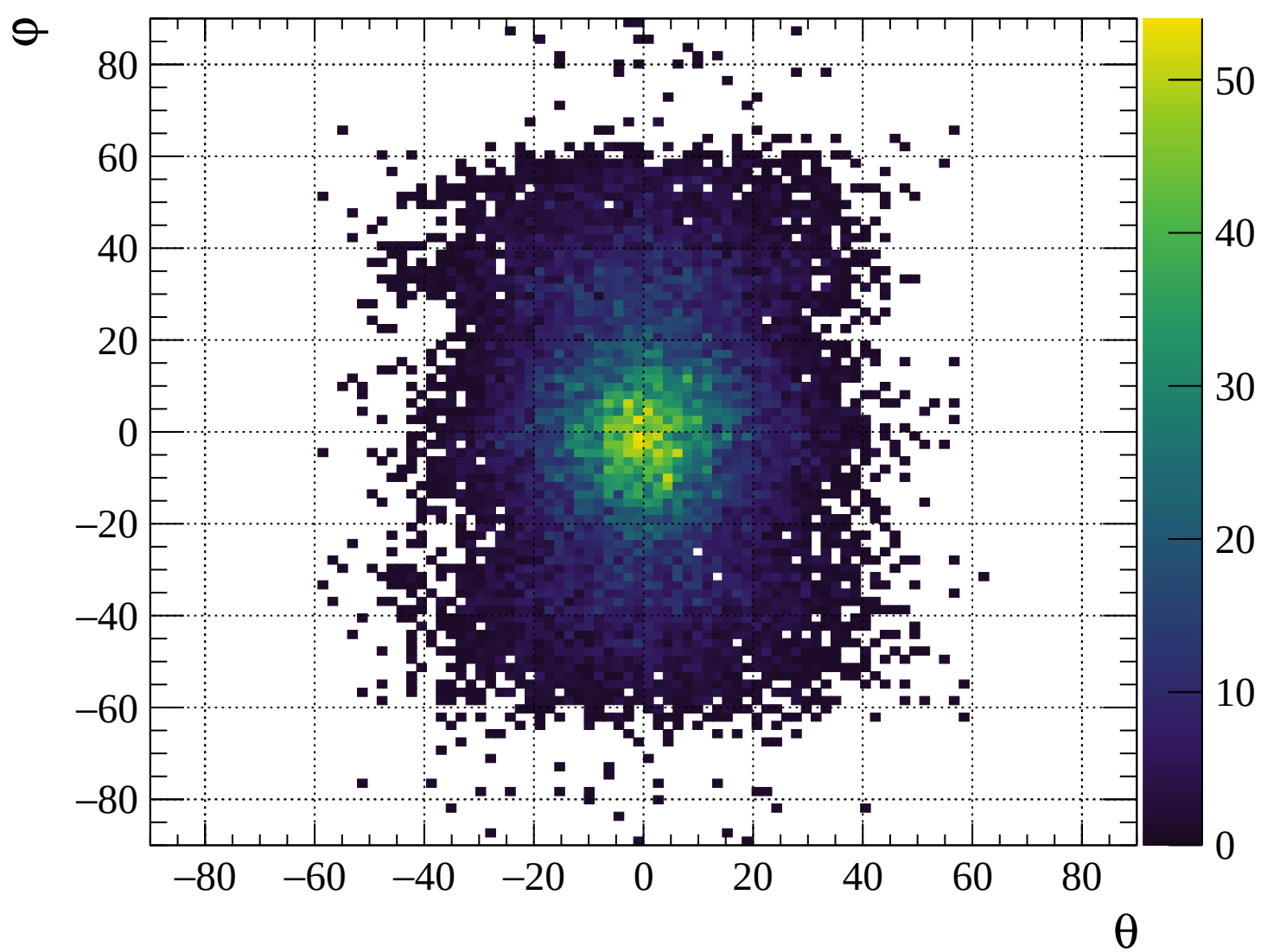


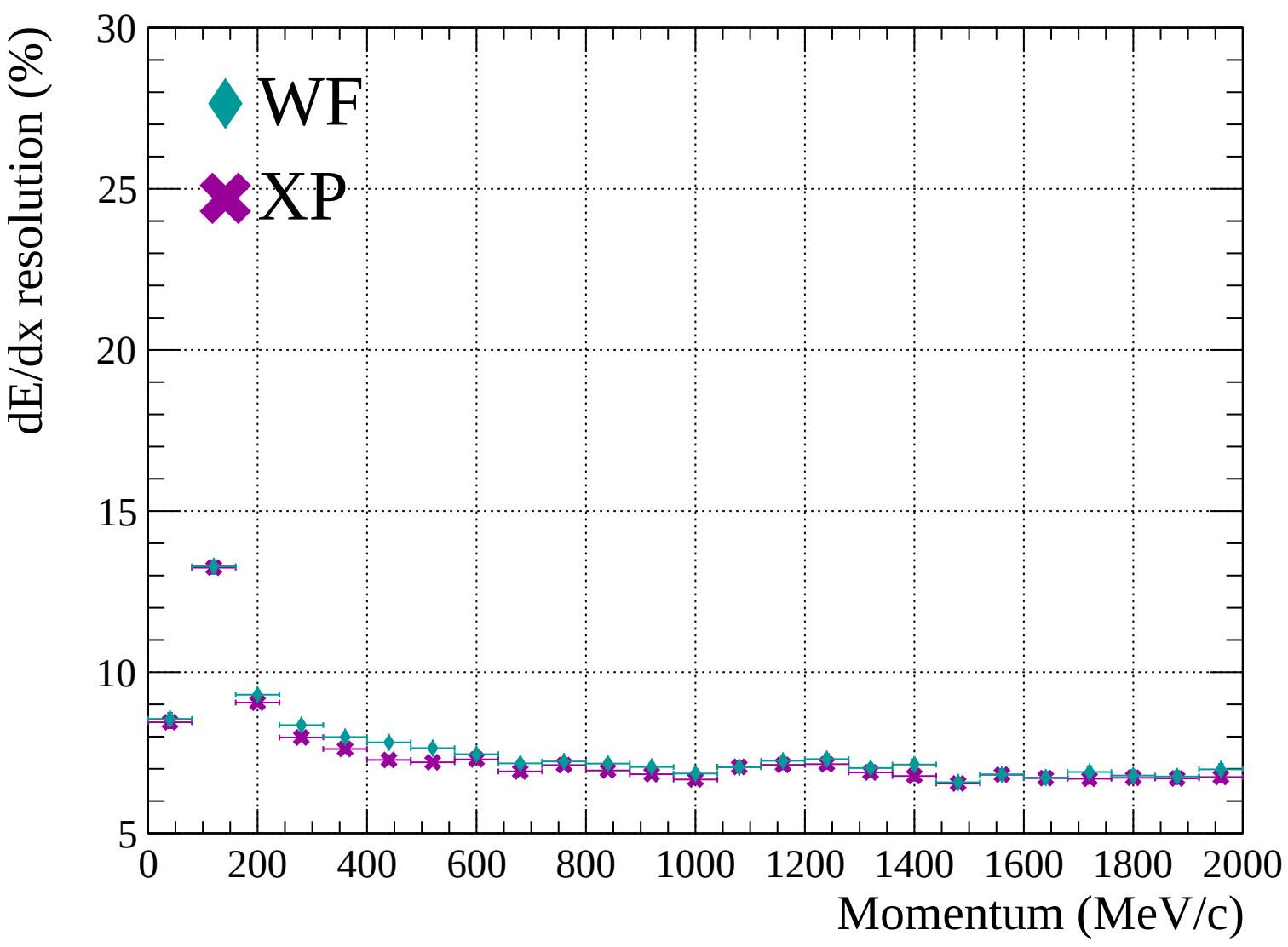


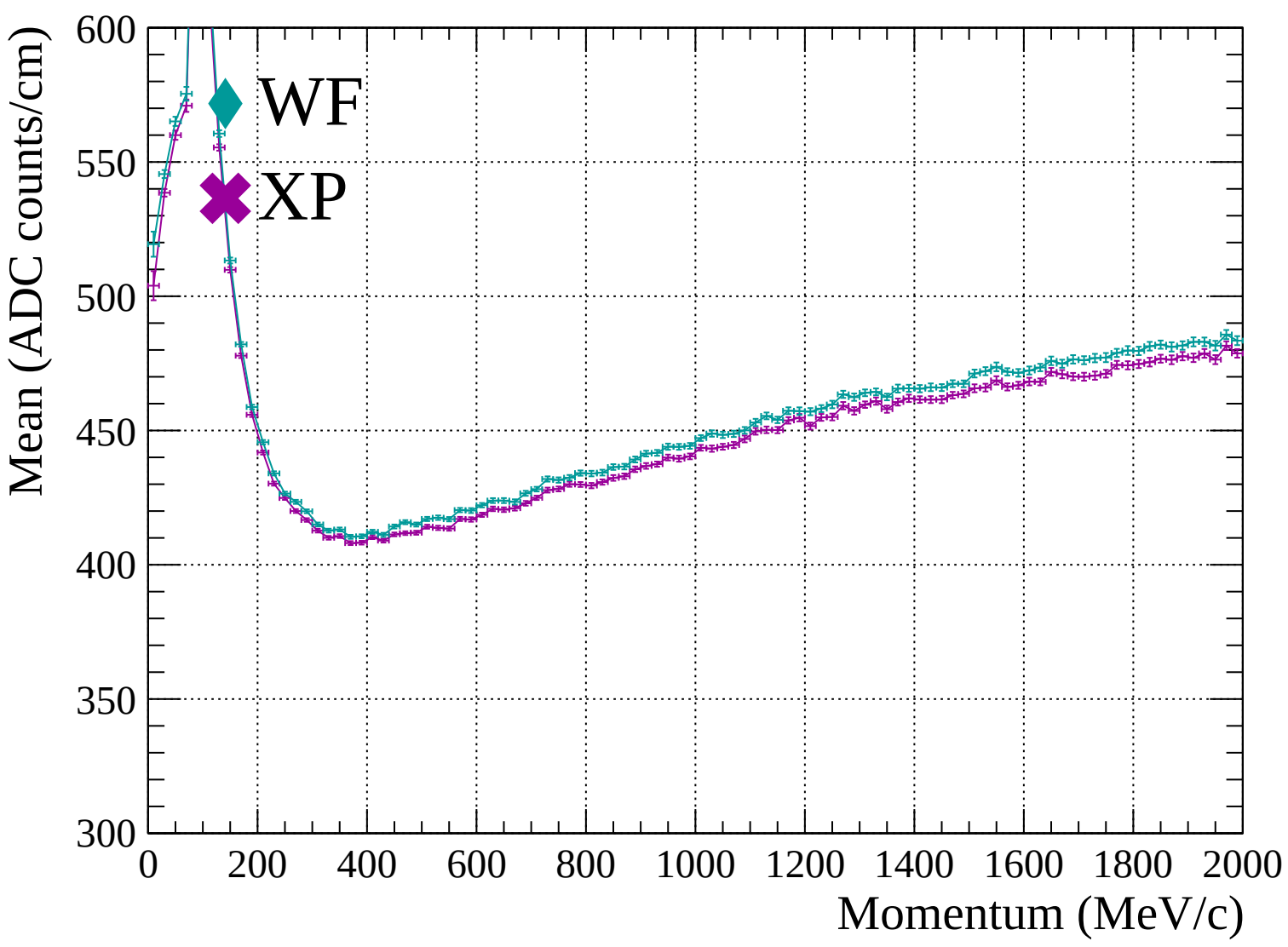


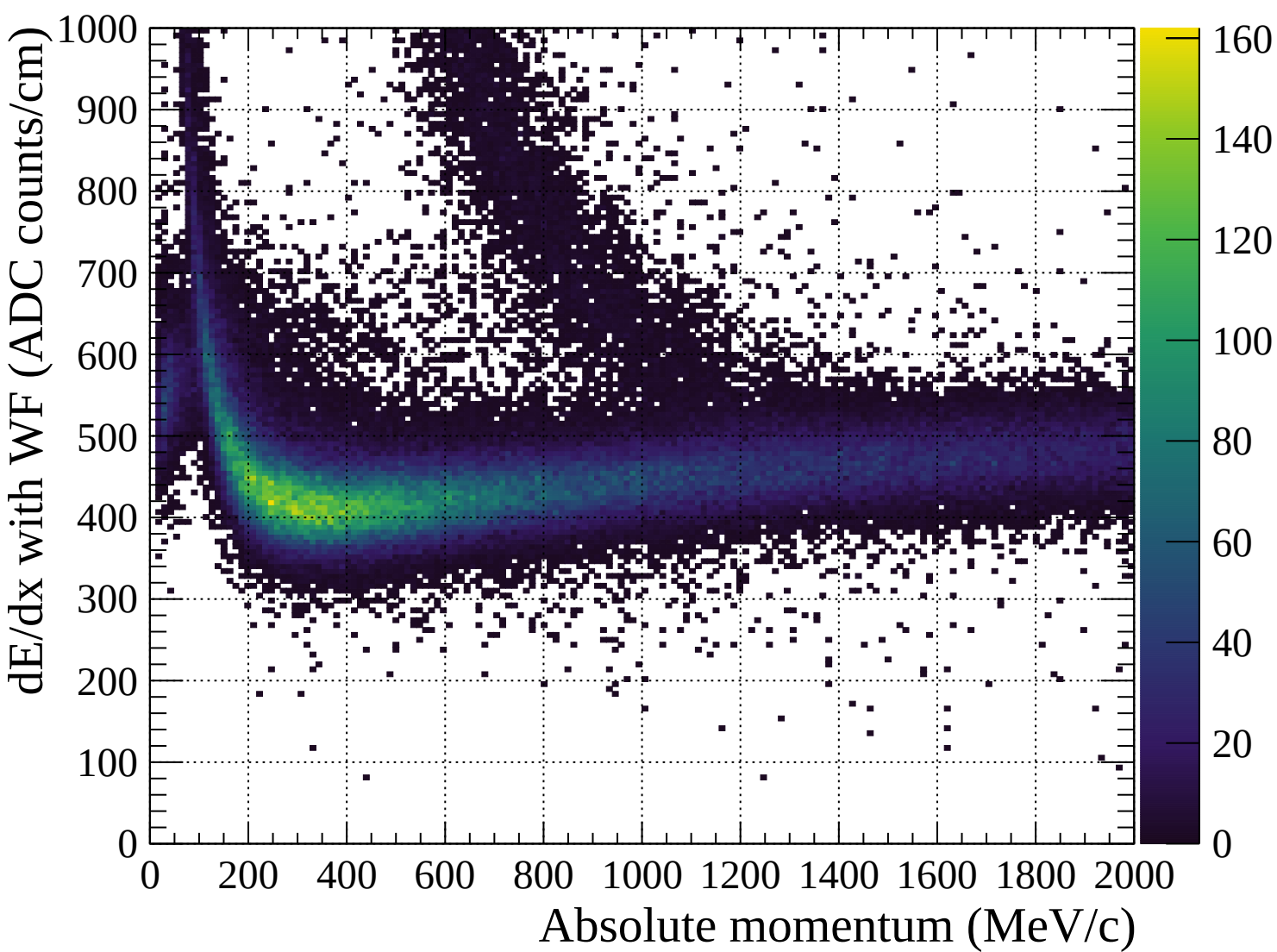


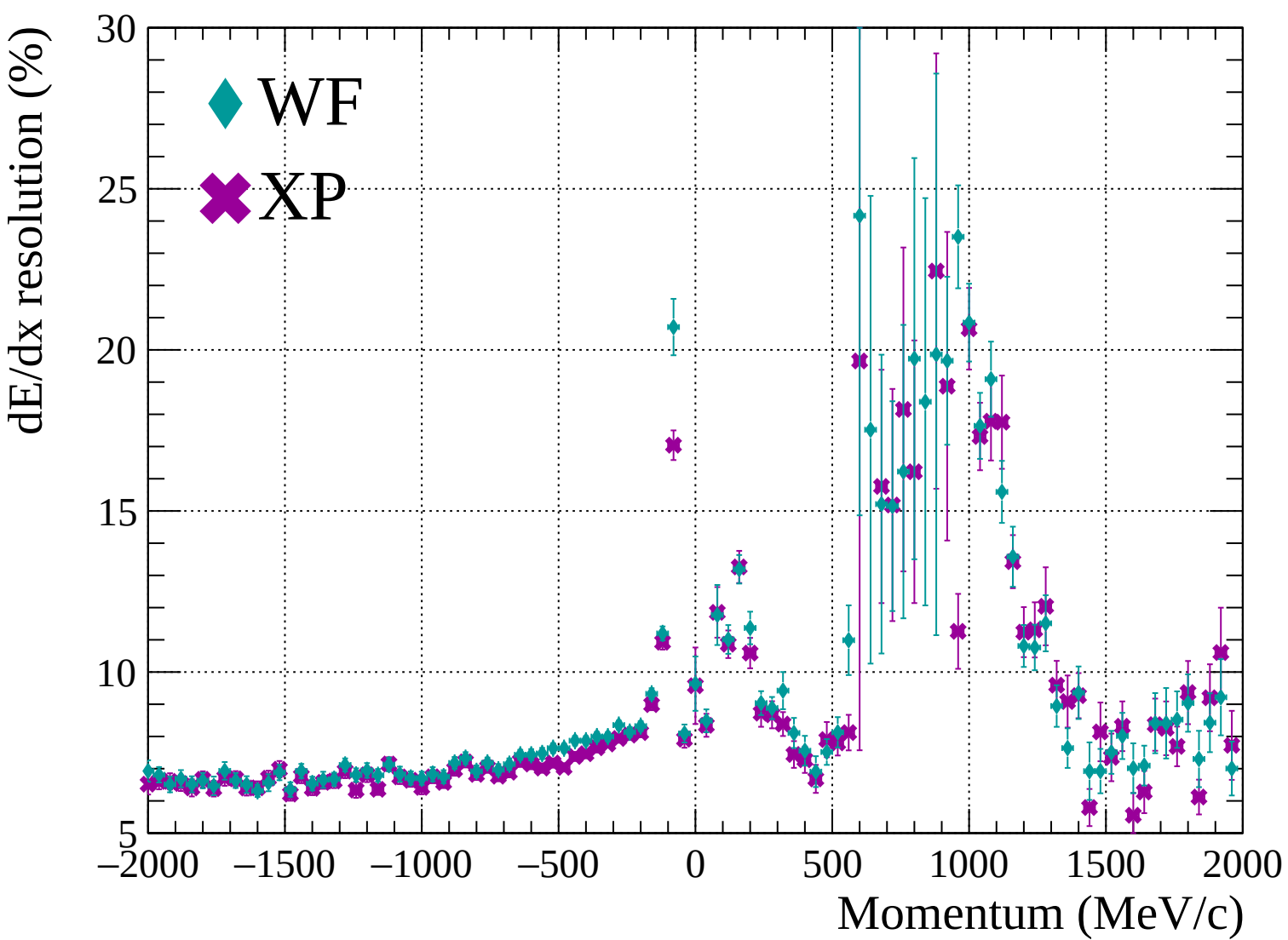


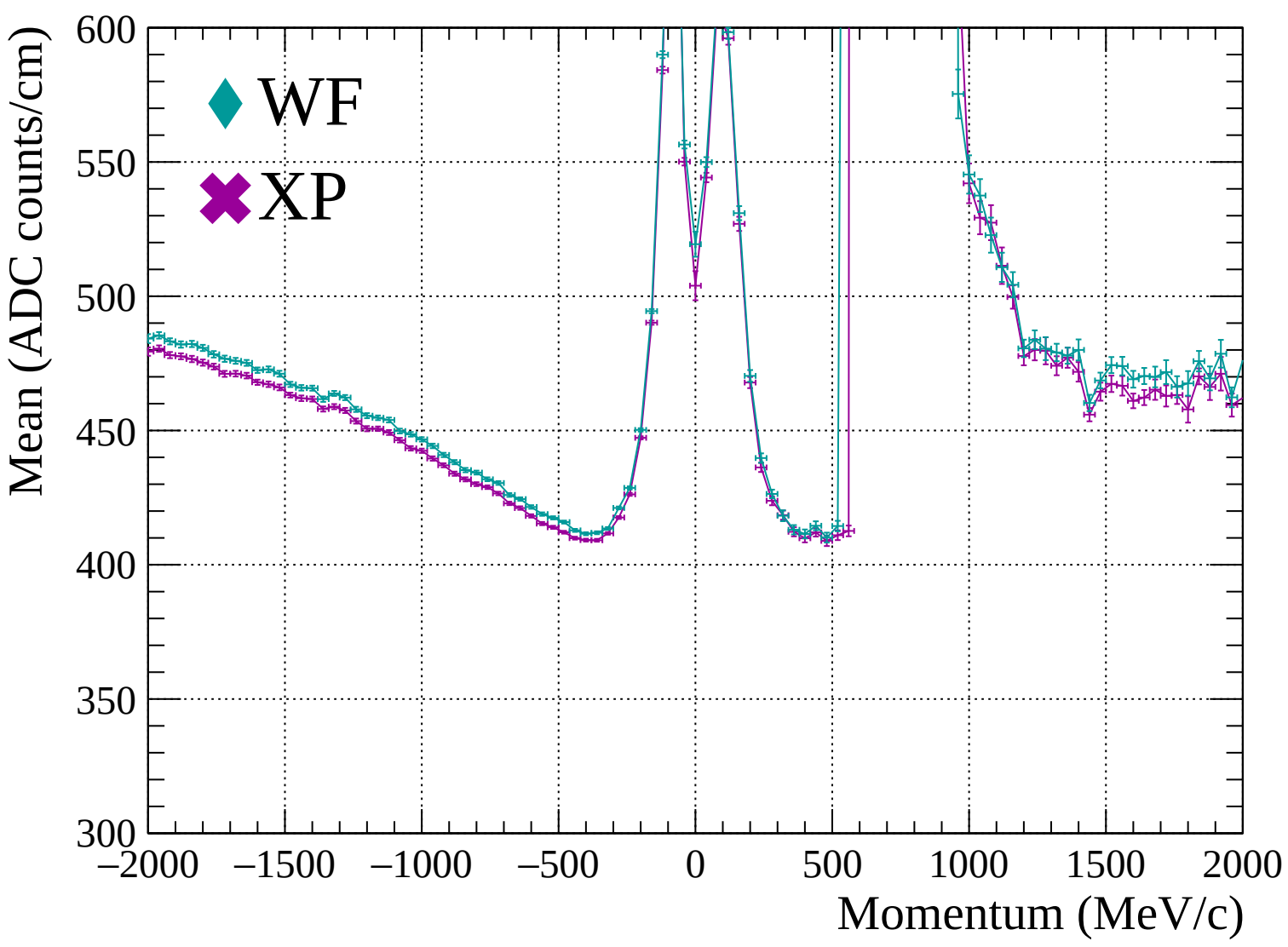


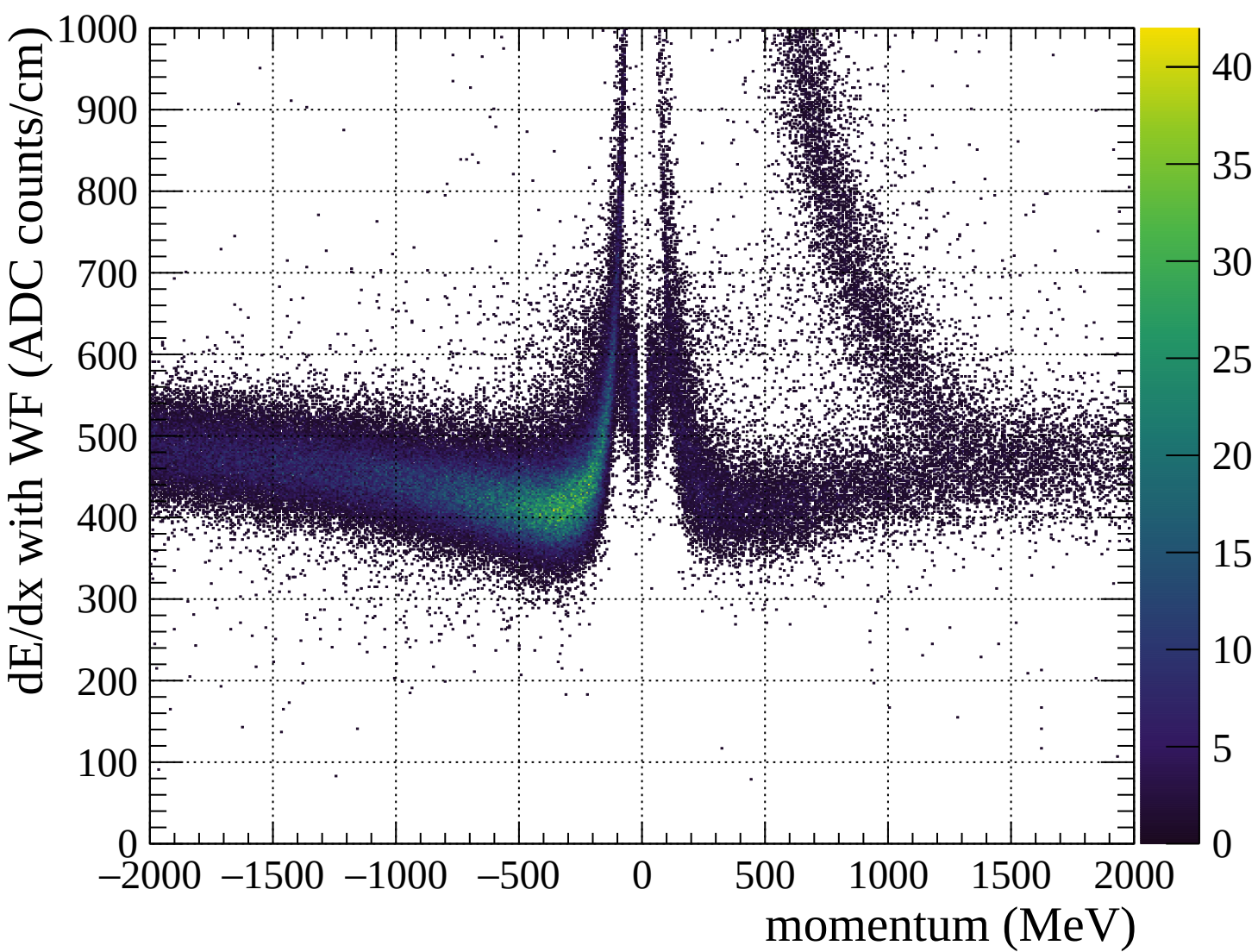


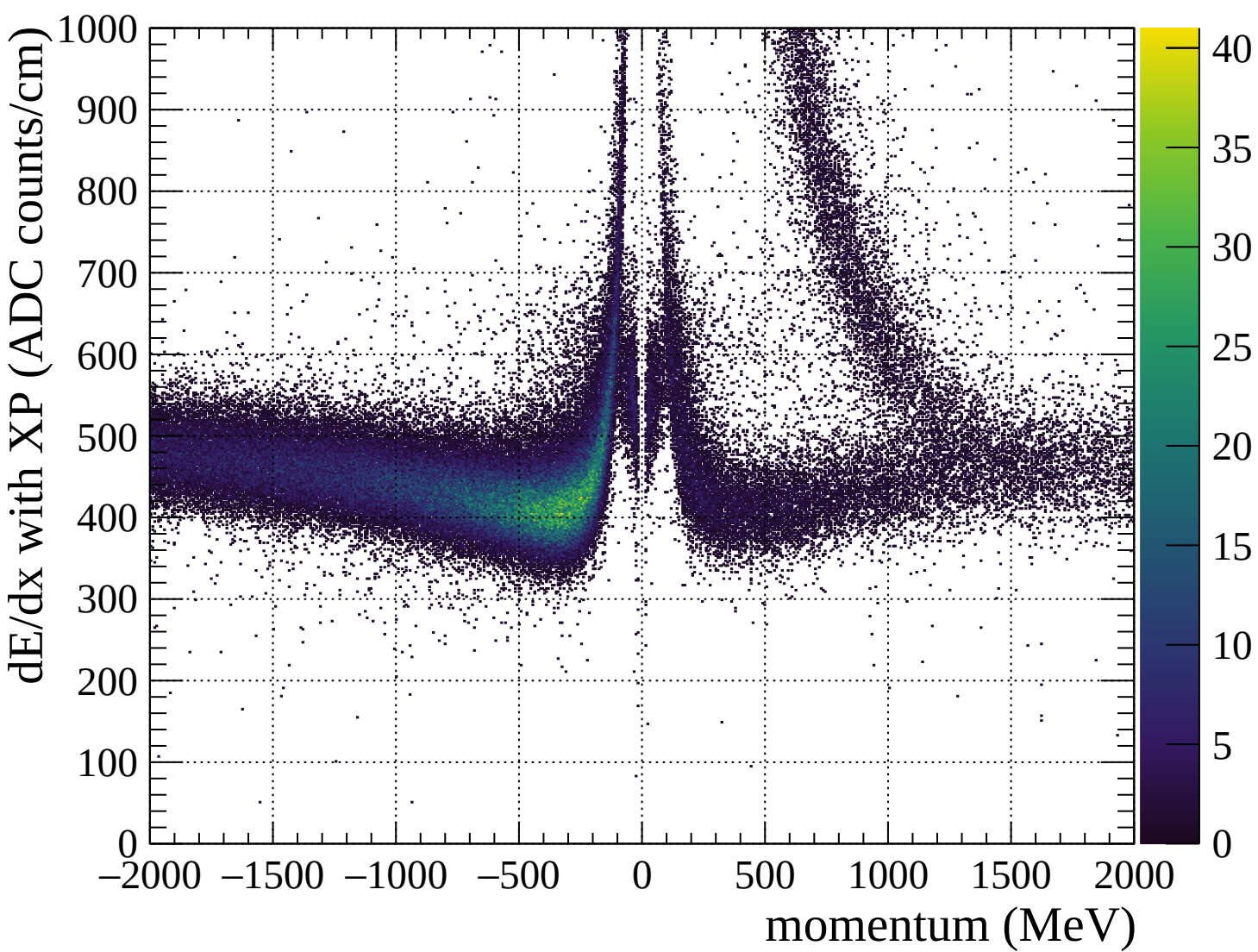


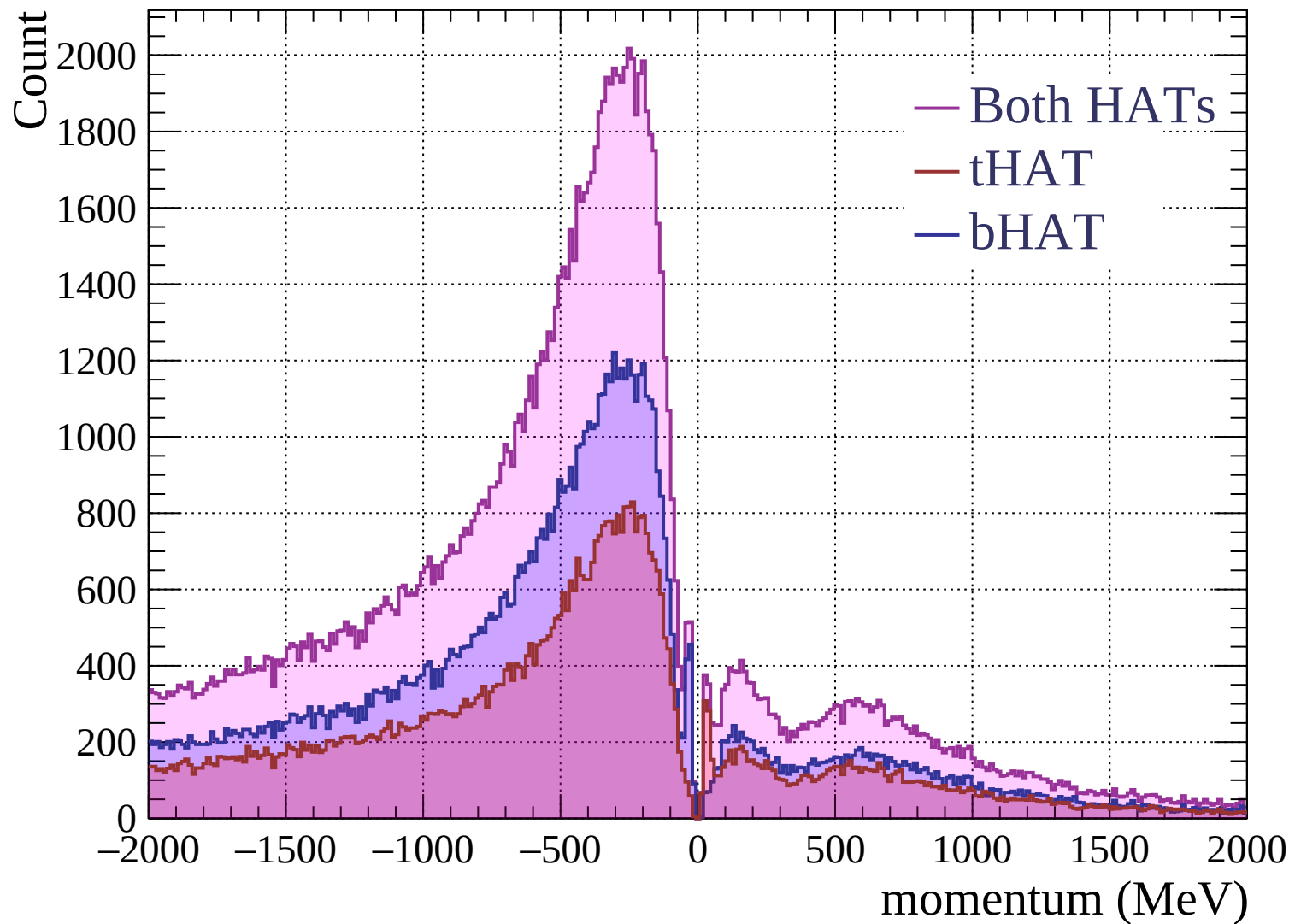


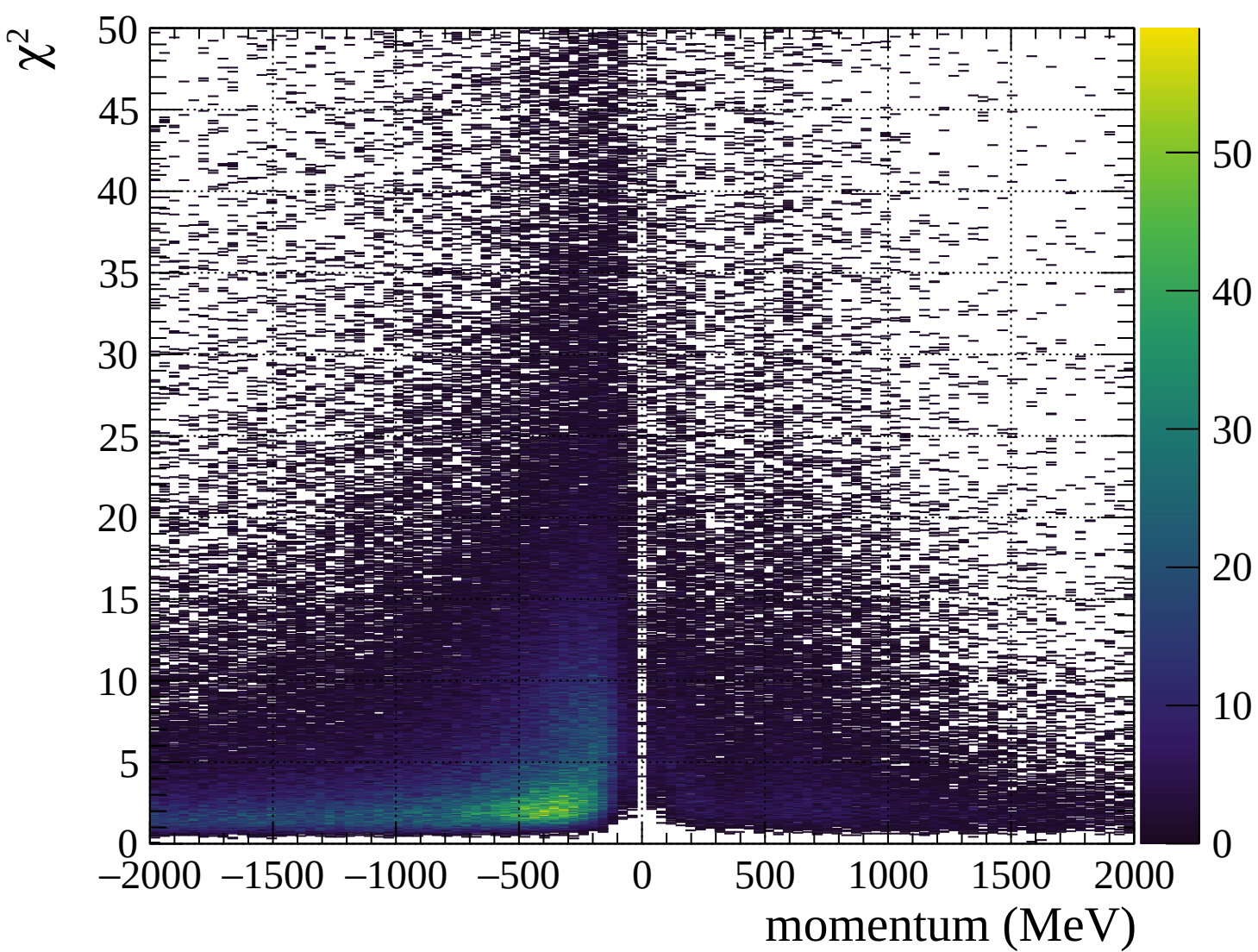




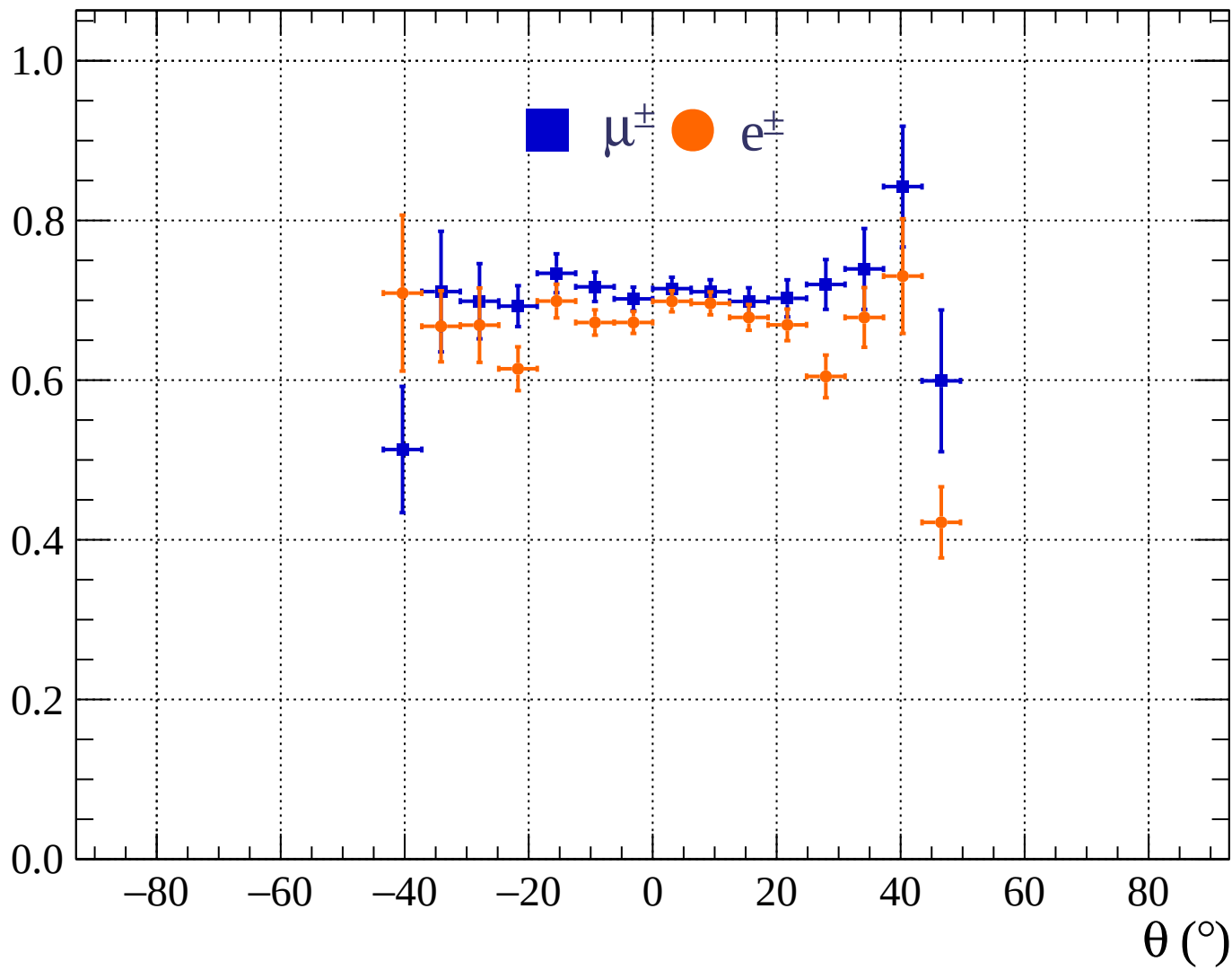


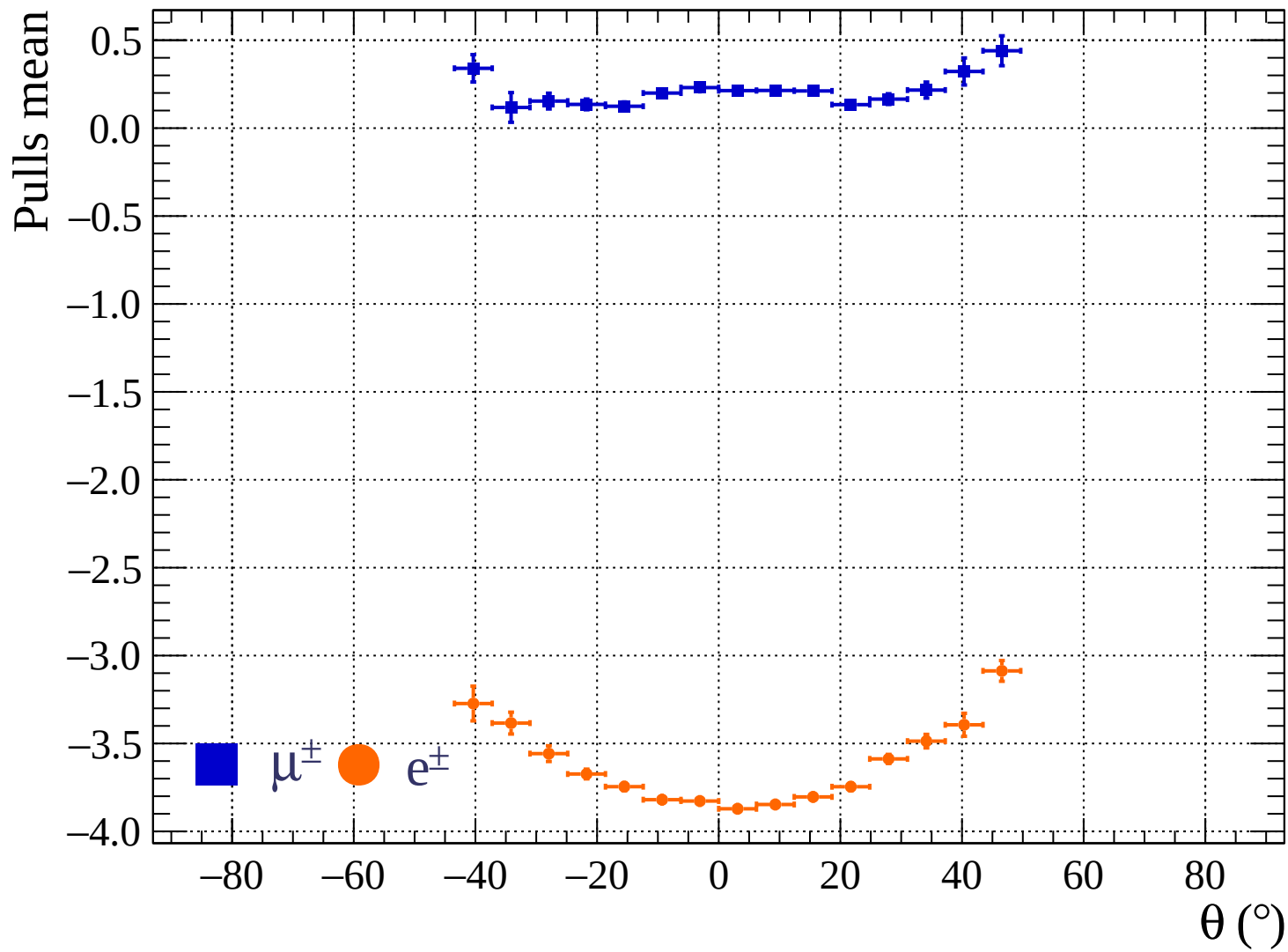




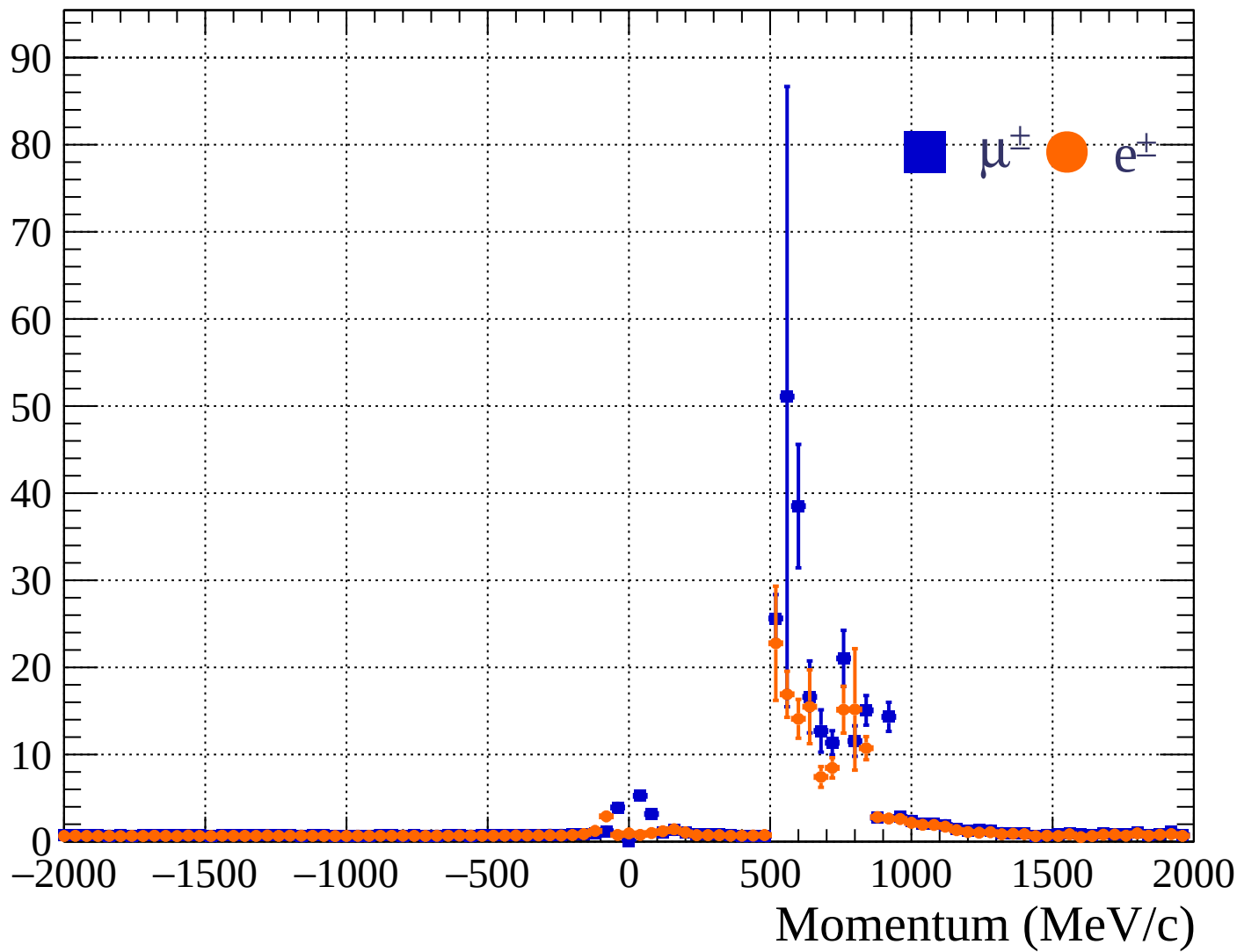


Pulls standard deviation

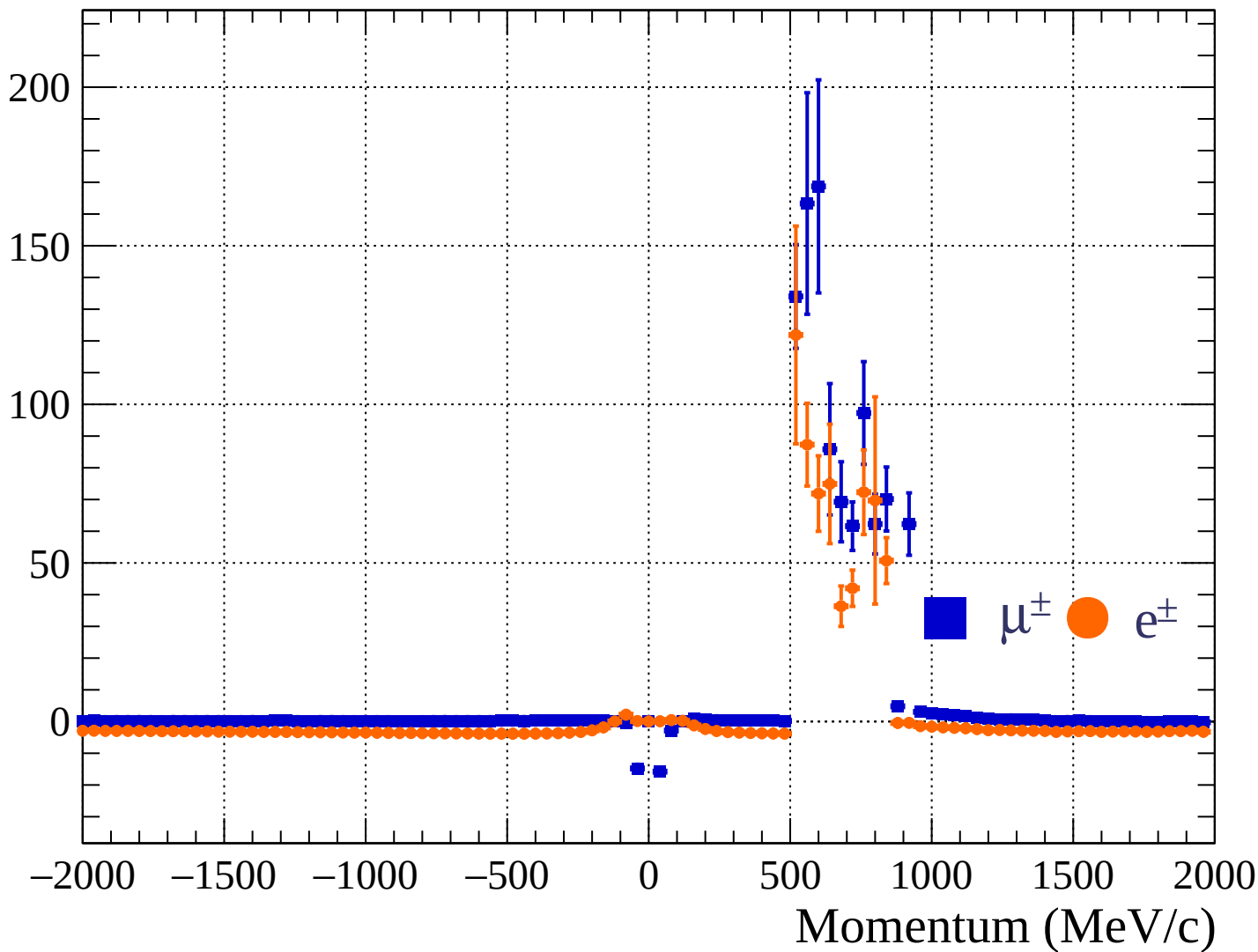


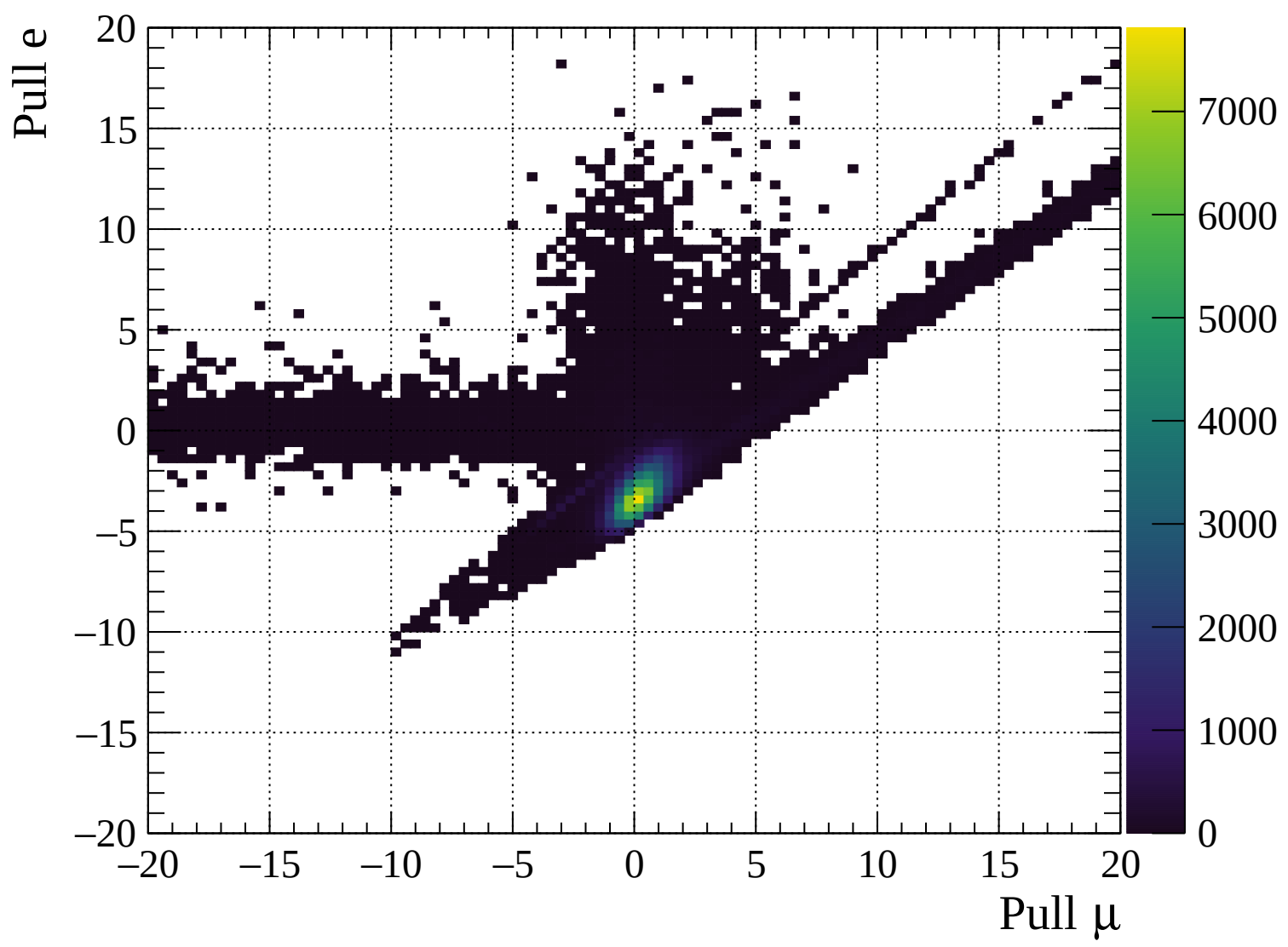


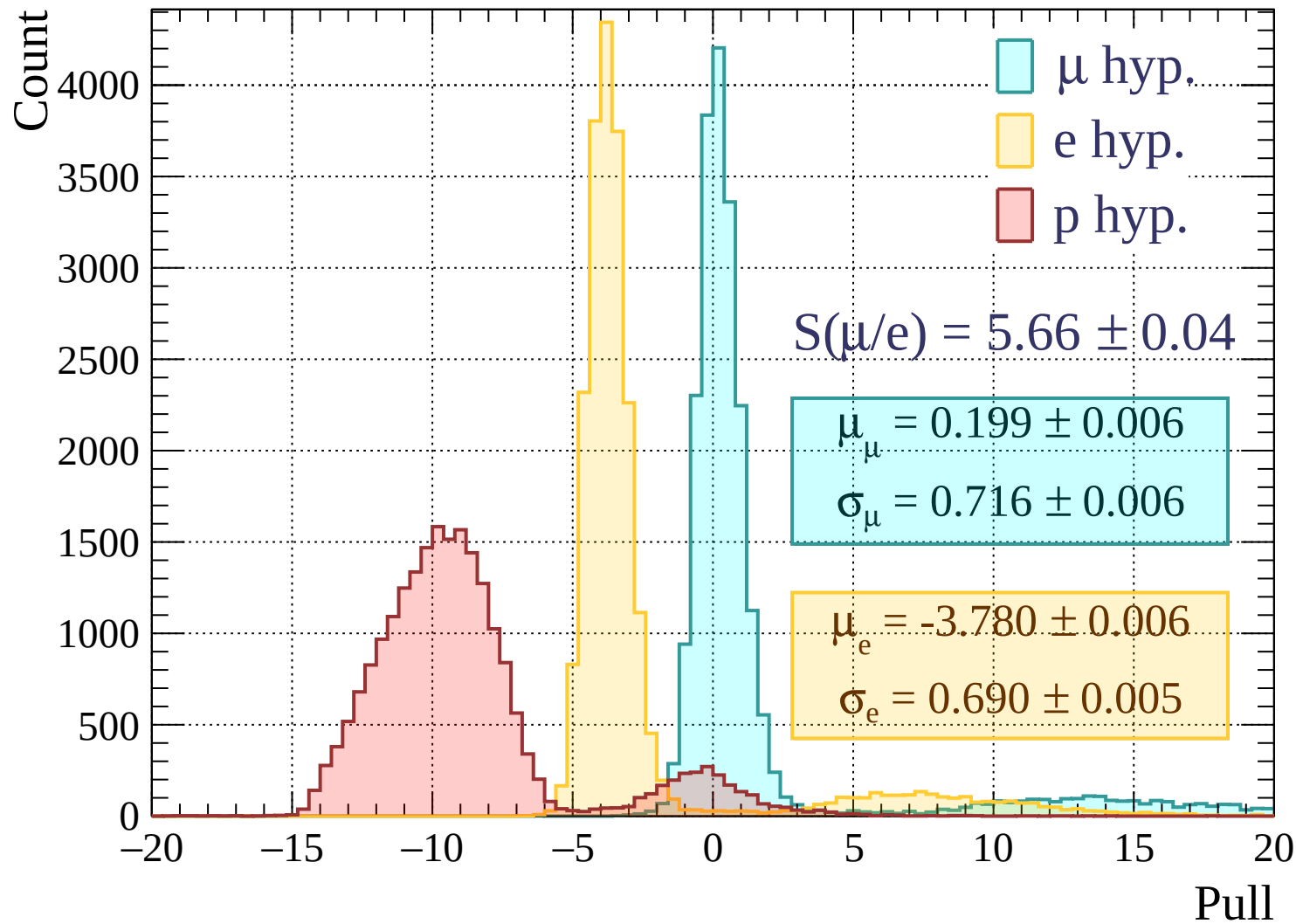
Pulls standard deviation

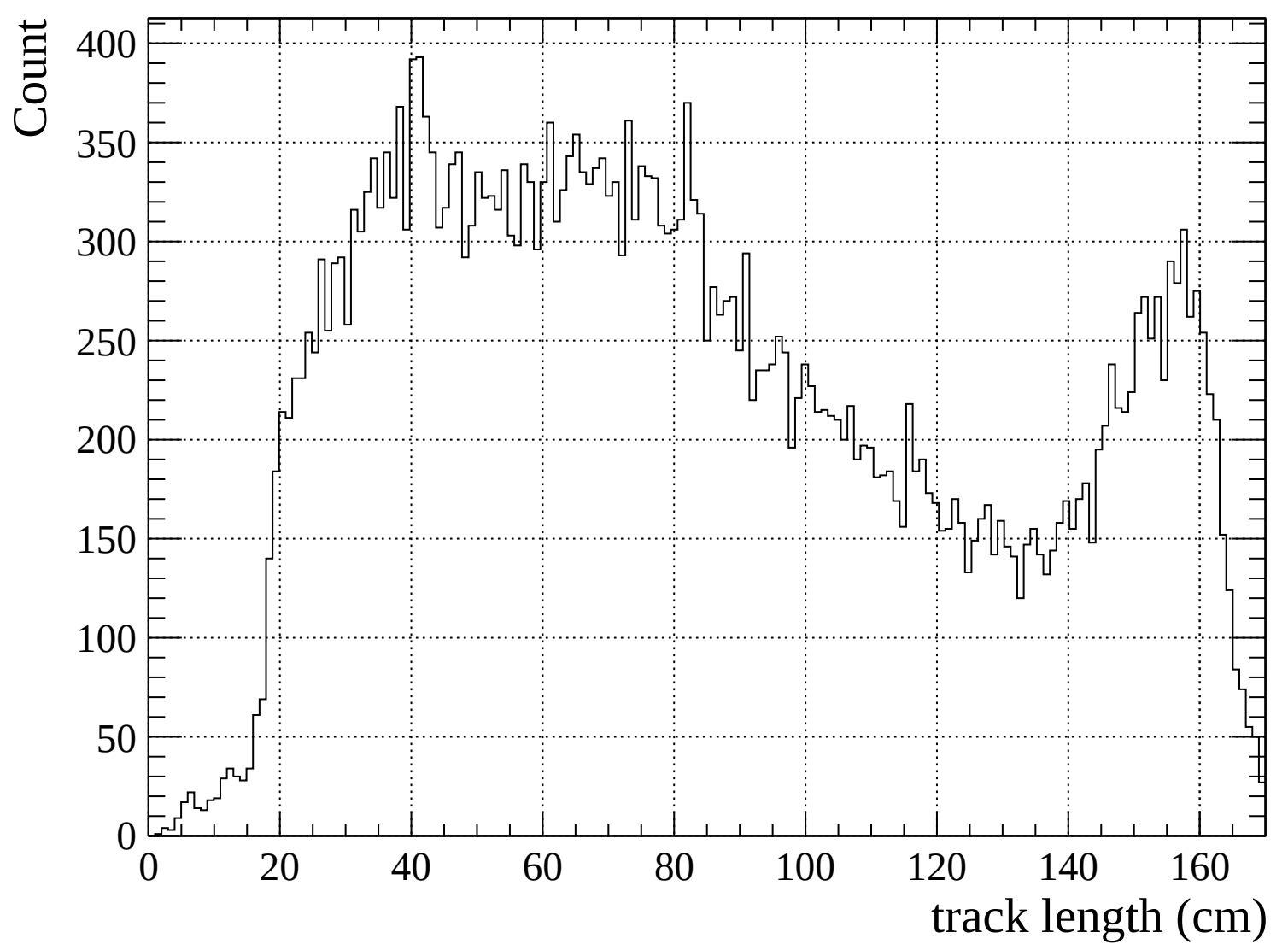


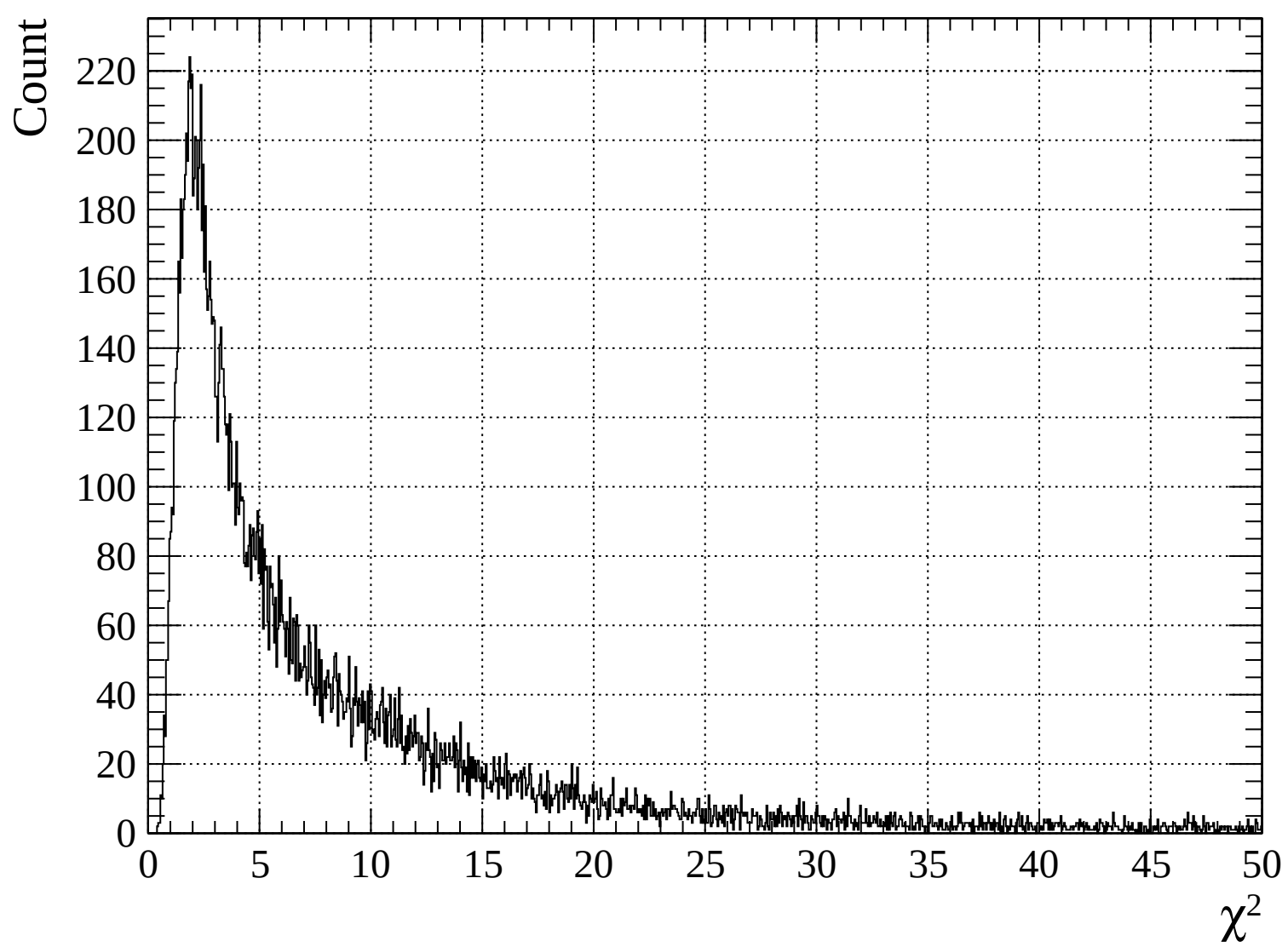
Pulls mean

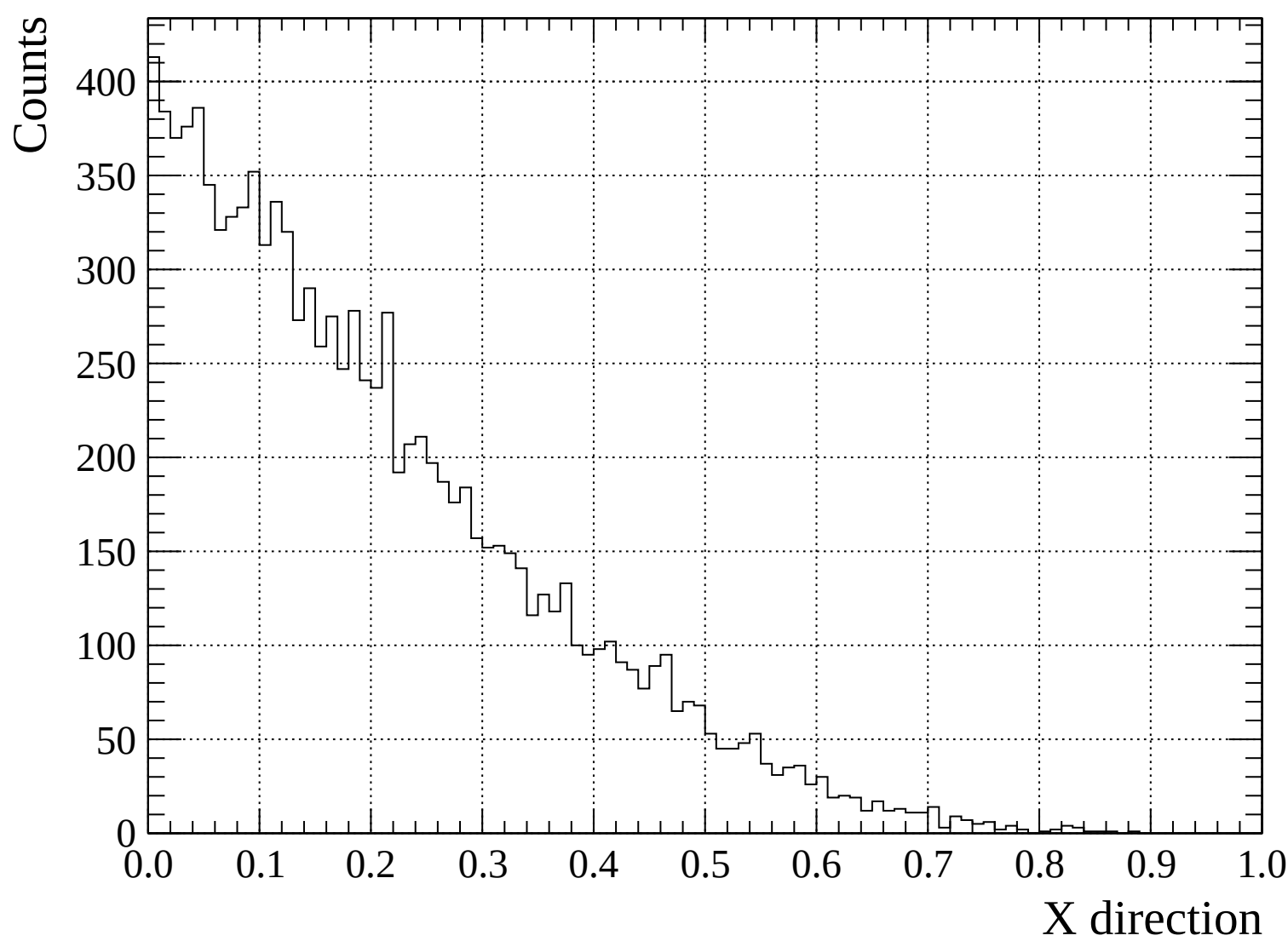


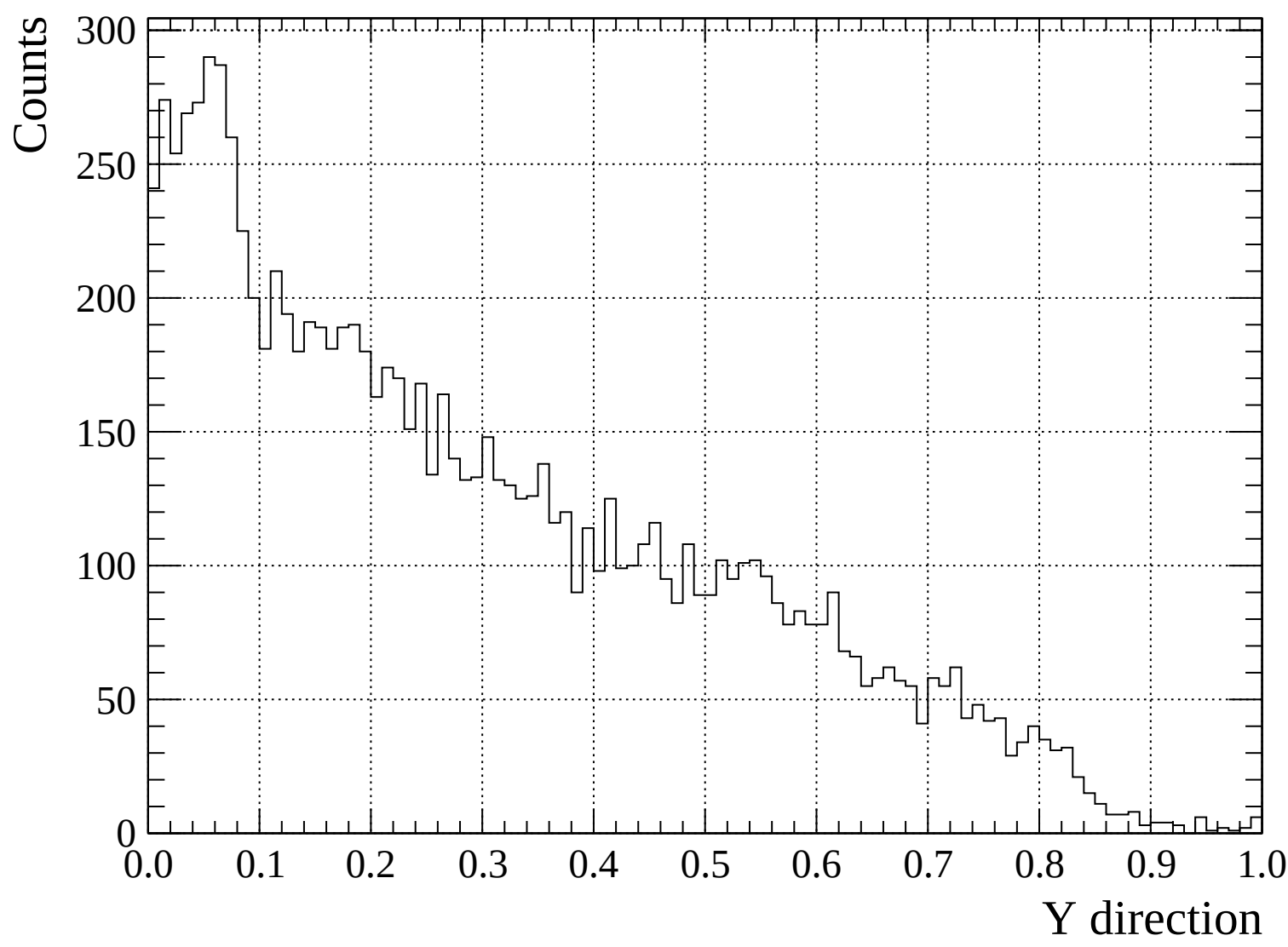


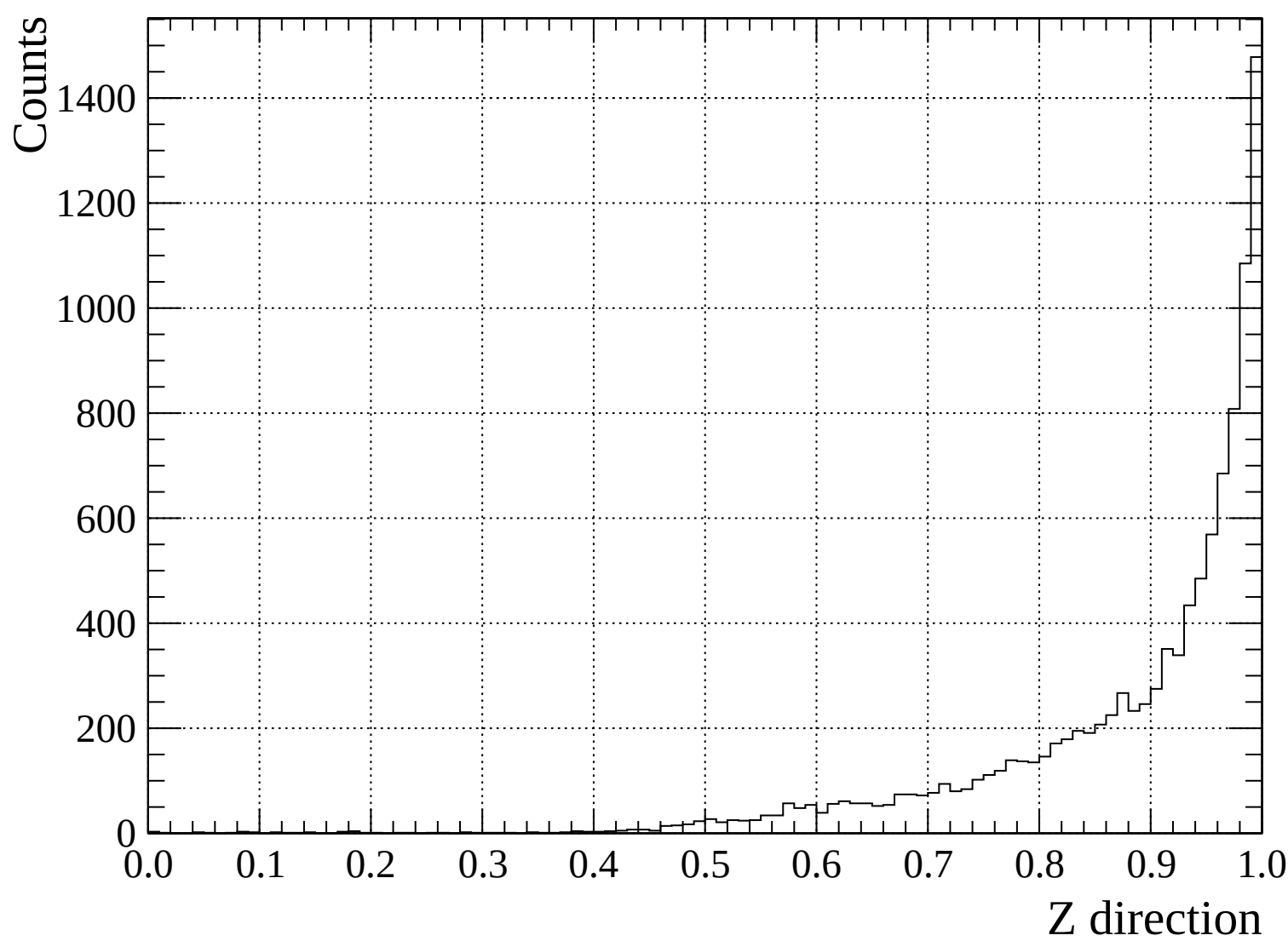


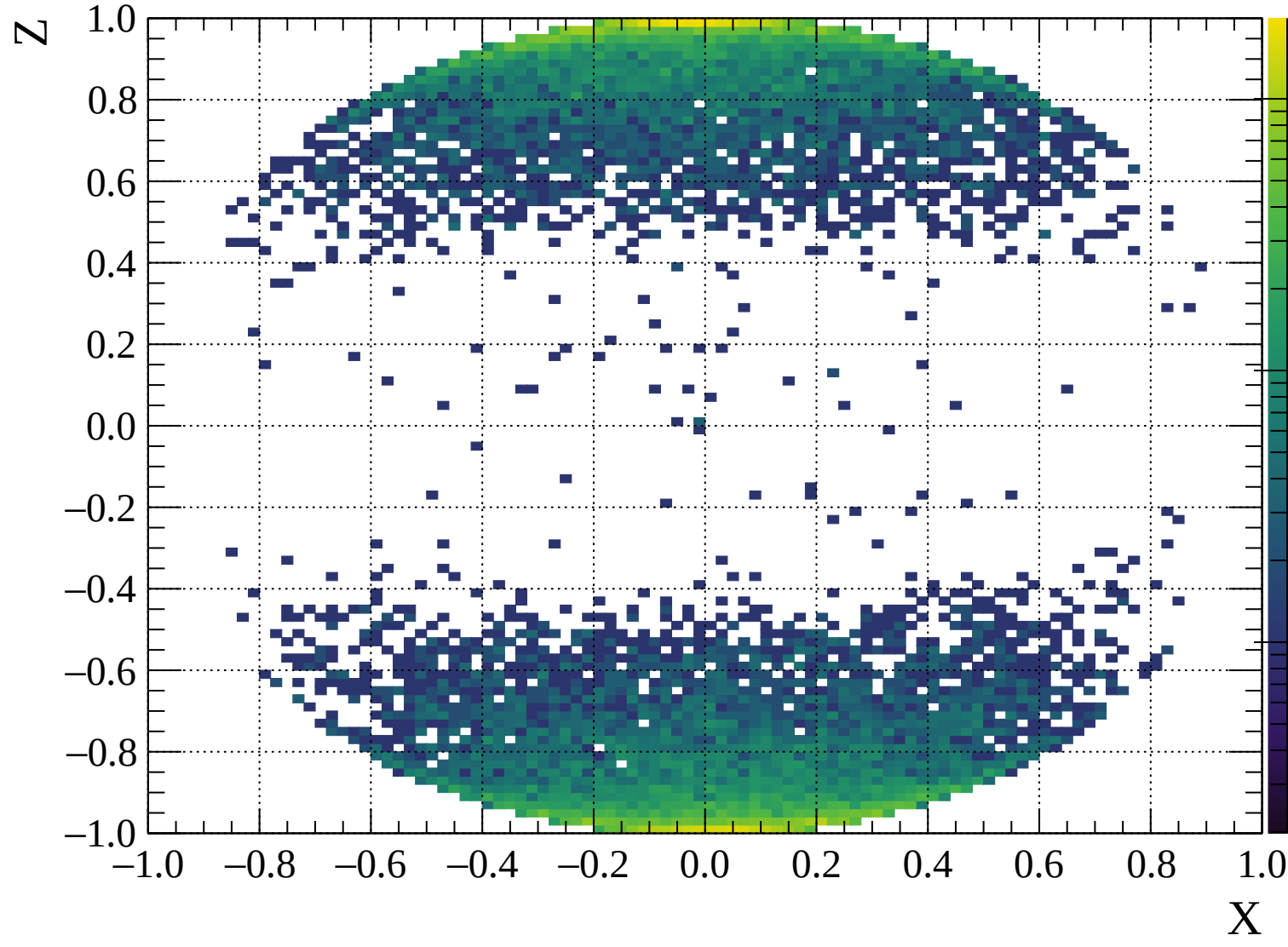


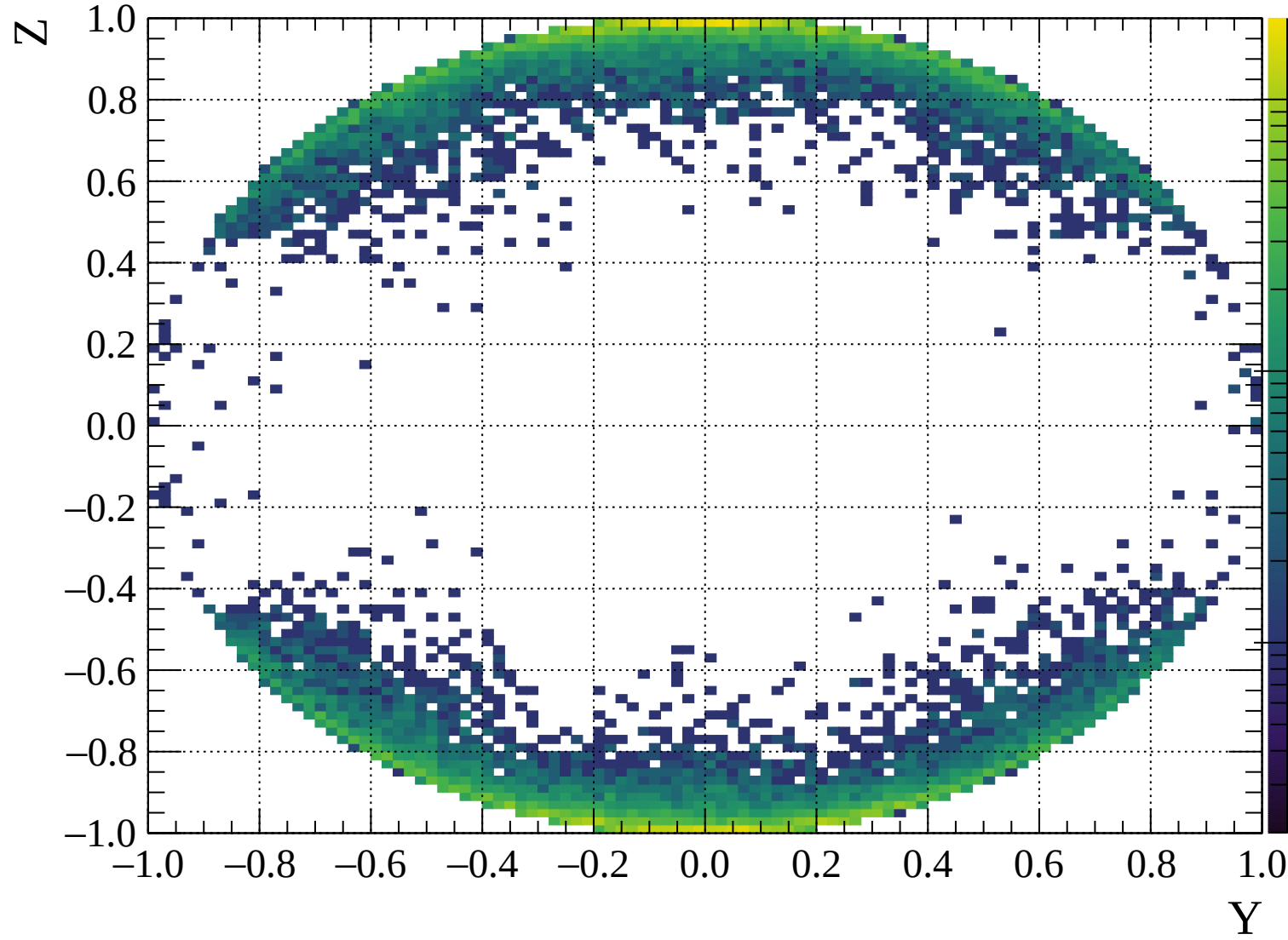


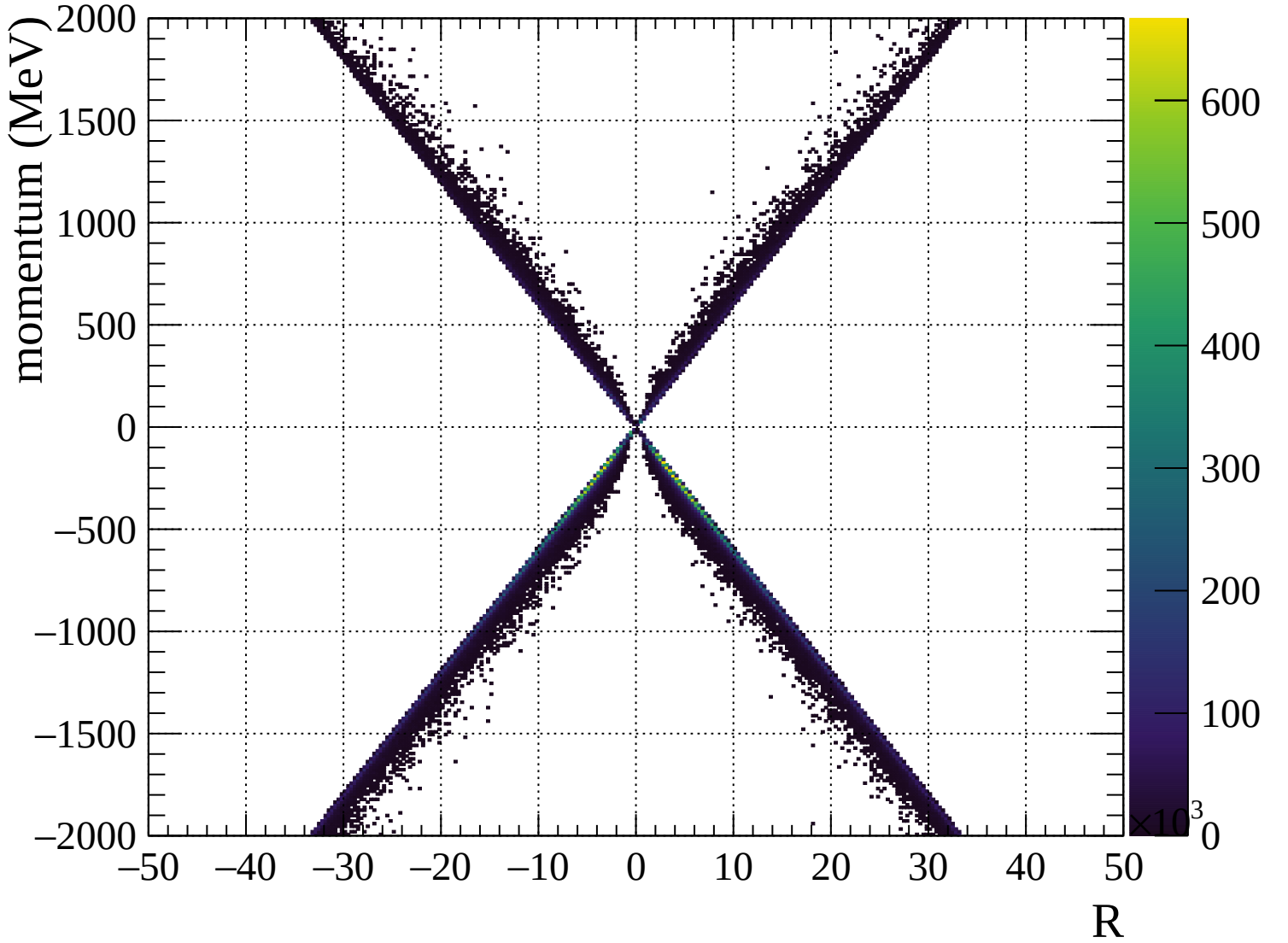


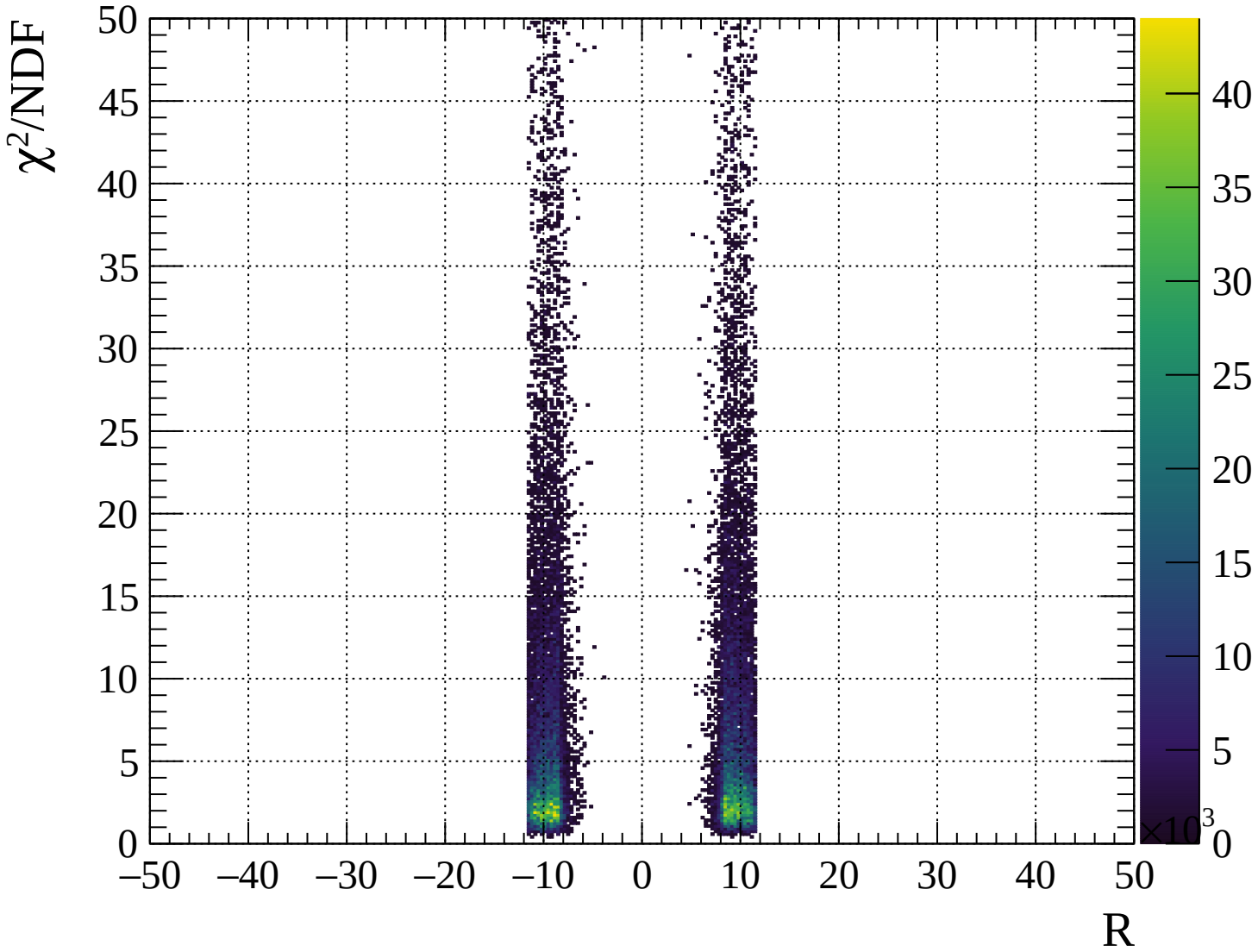




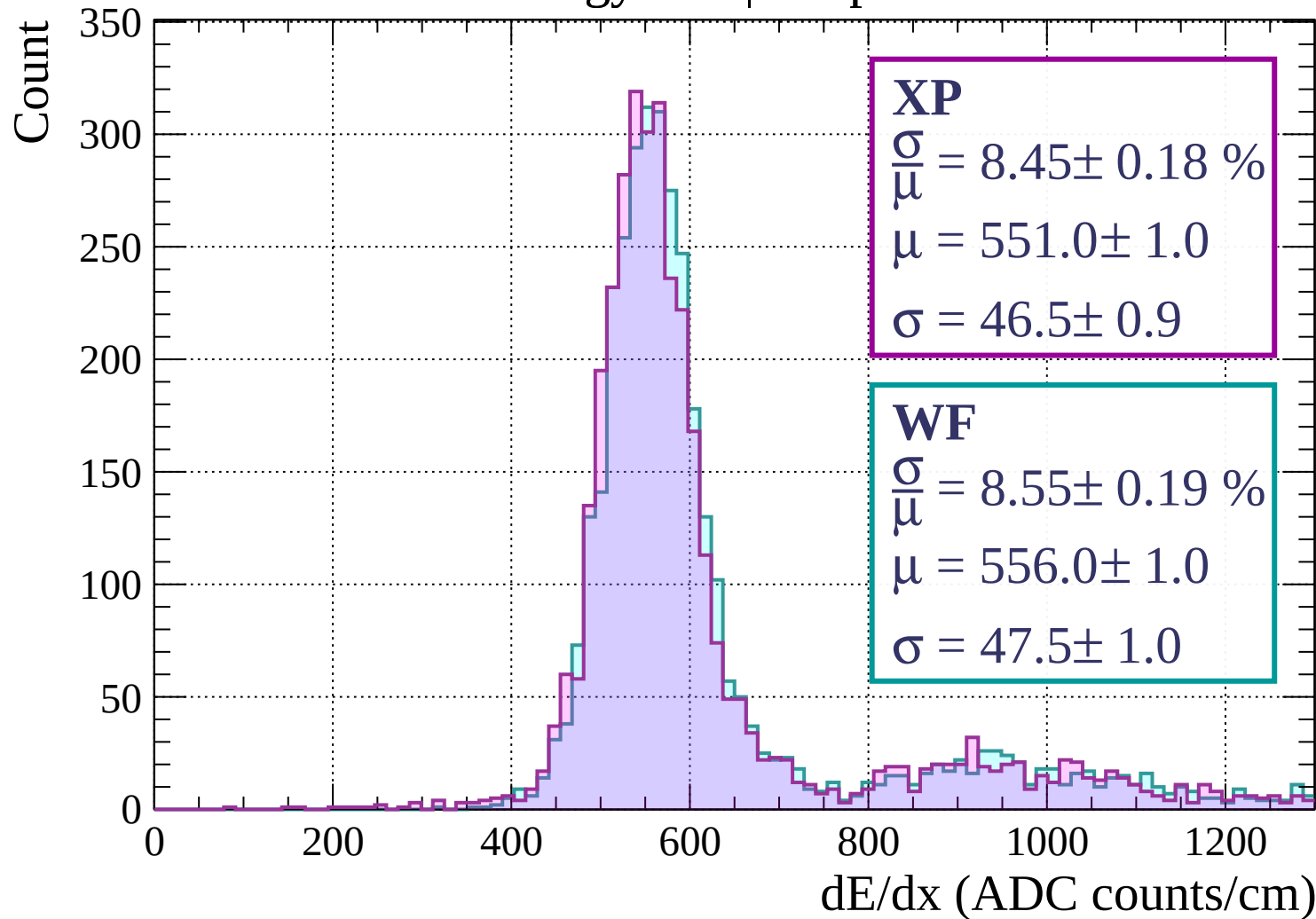




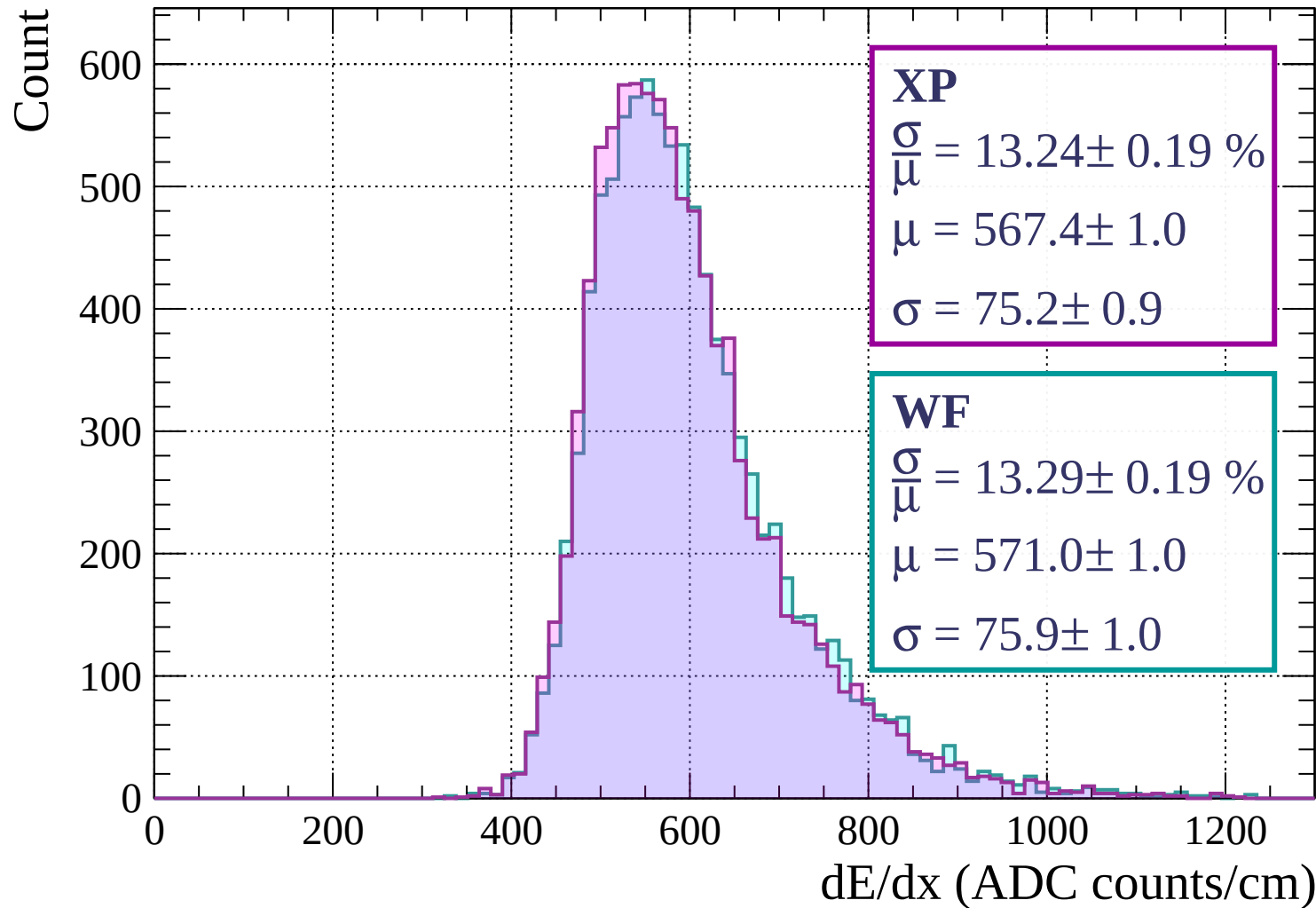




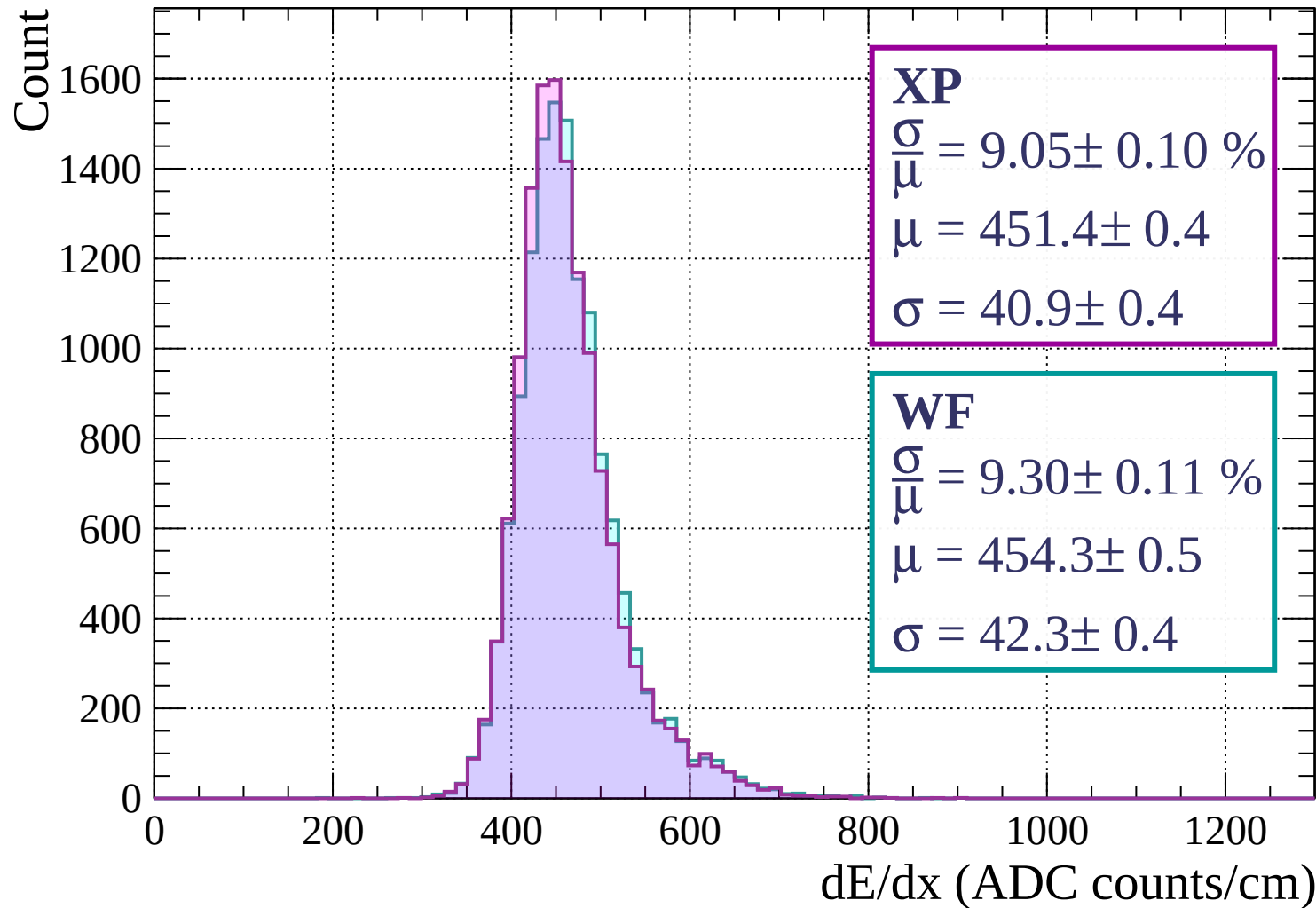
Energy loss | $0 < p < 80$



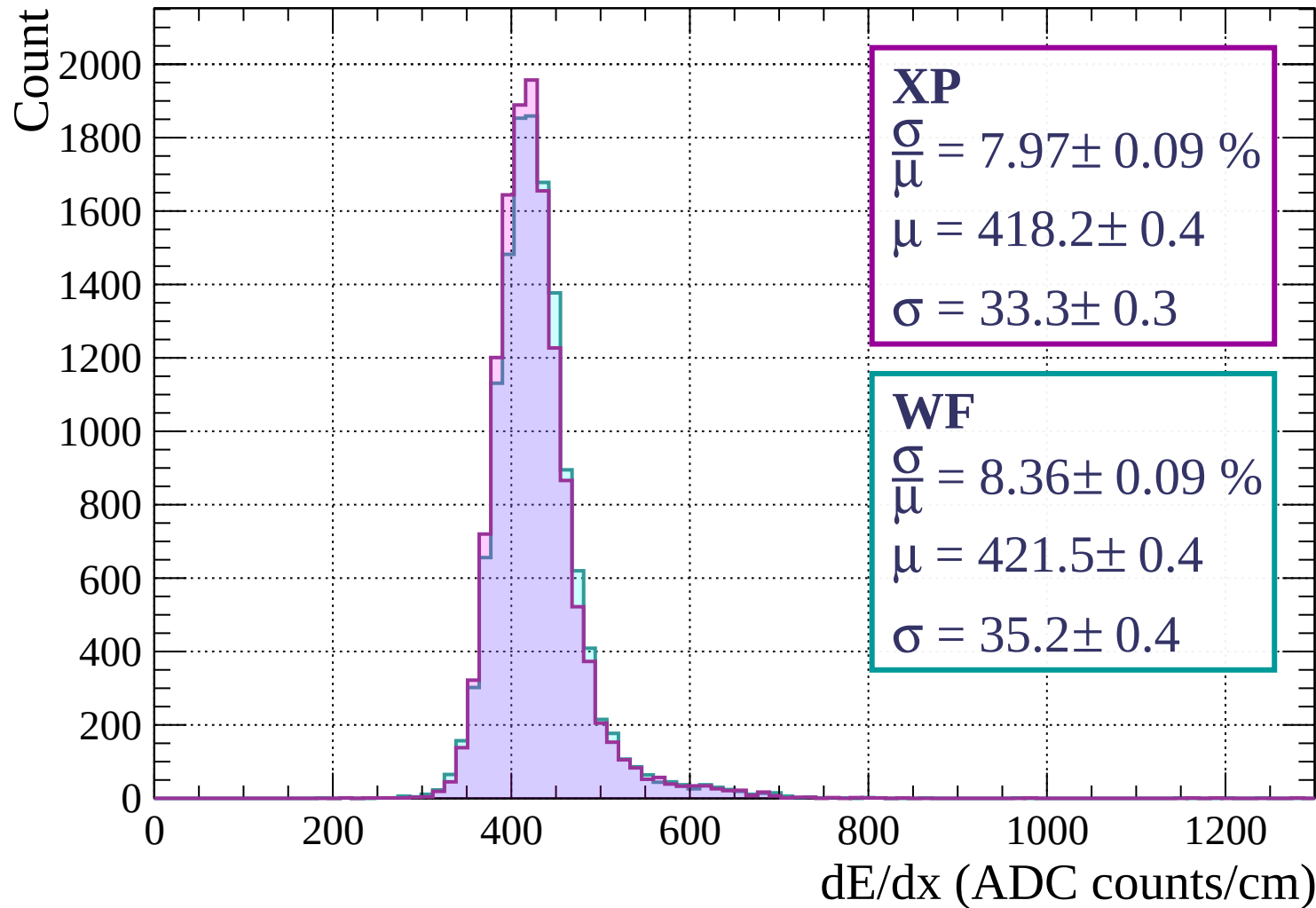
Energy loss | $80 < p < 160$



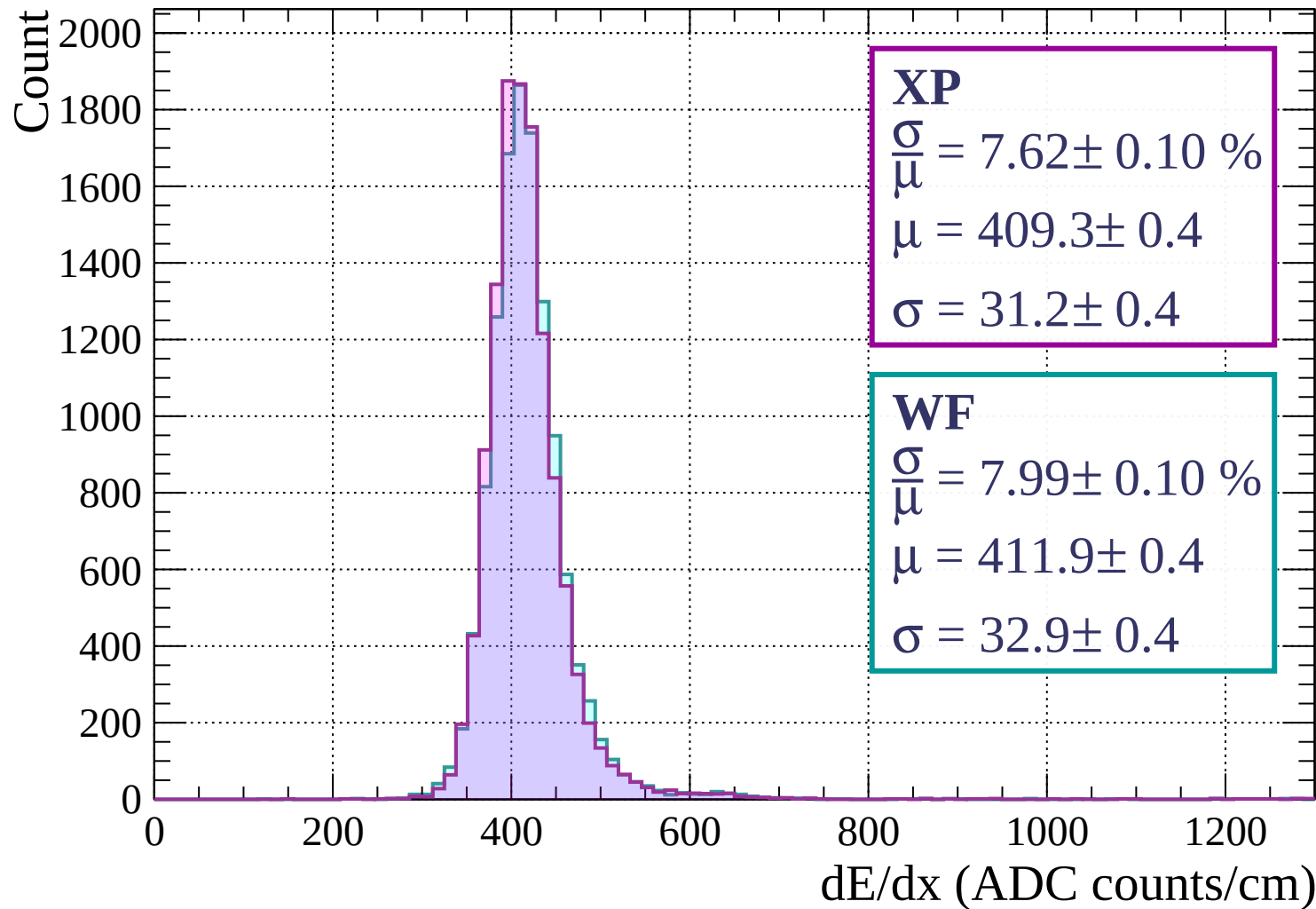
Energy loss | $160 < p < 240$



Energy loss | $240 < p < 320$



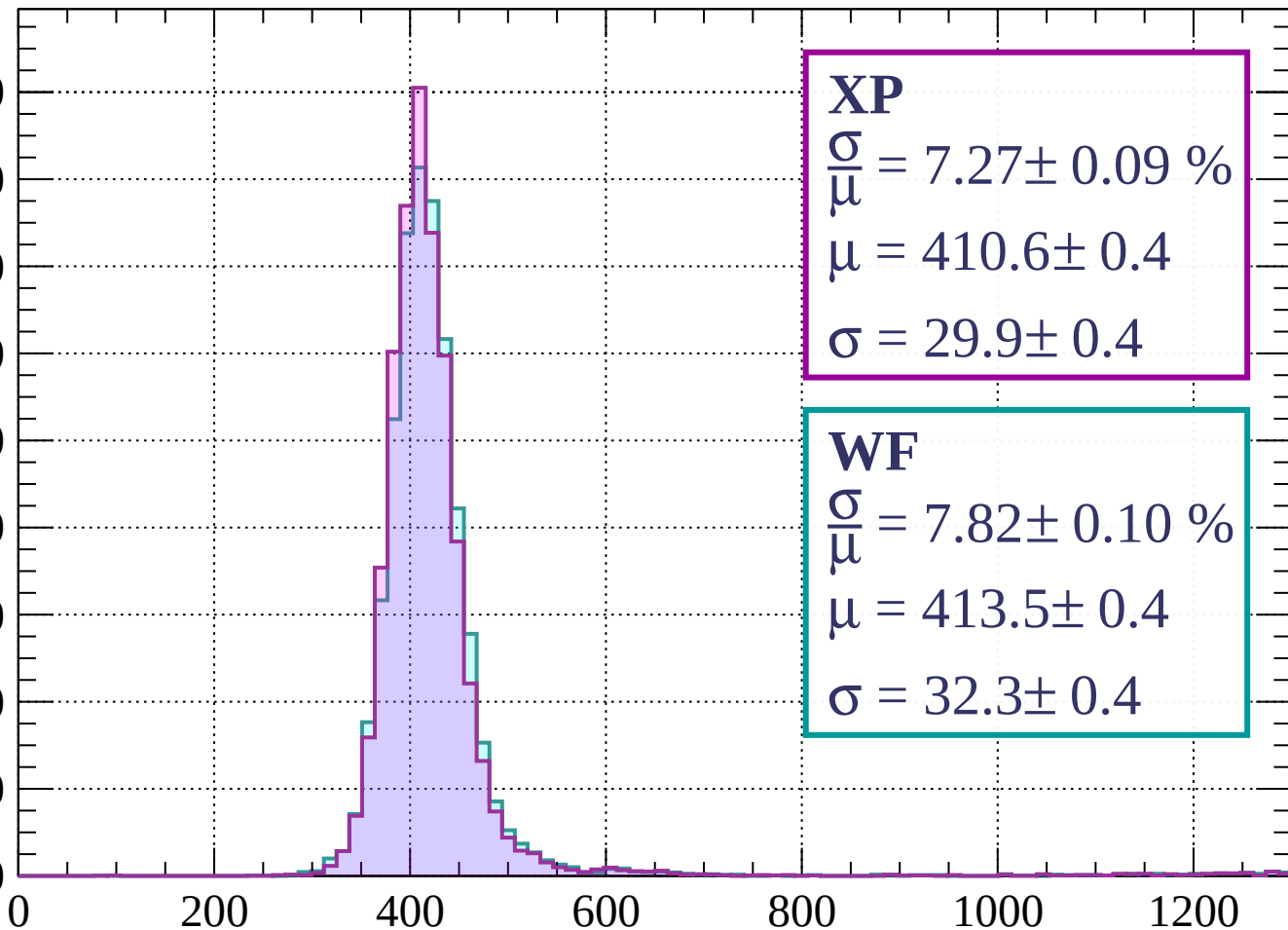
Energy loss | $320 < p < 400$



Energy loss | $400 < p < 480$

Count

1800
1600
1400
1200
1000
800
600
400
200
0



XP

$$\frac{\sigma}{\mu} = 7.27 \pm 0.09 \%$$

$$\mu = 410.6 \pm 0.4$$

$$\sigma = 29.9 \pm 0.4$$

WF

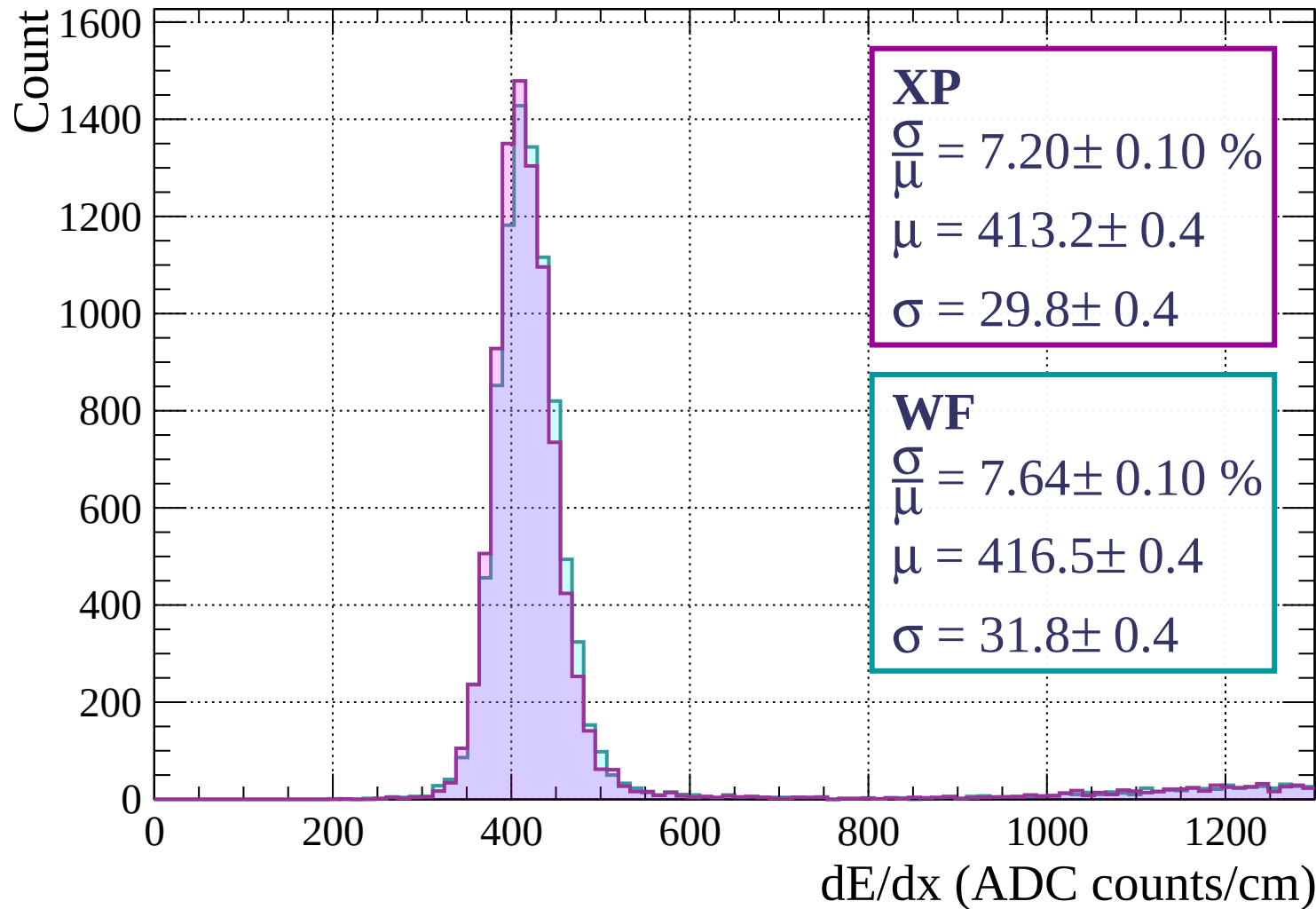
$$\frac{\sigma}{\mu} = 7.82 \pm 0.10 \%$$

$$\mu = 413.5 \pm 0.4$$

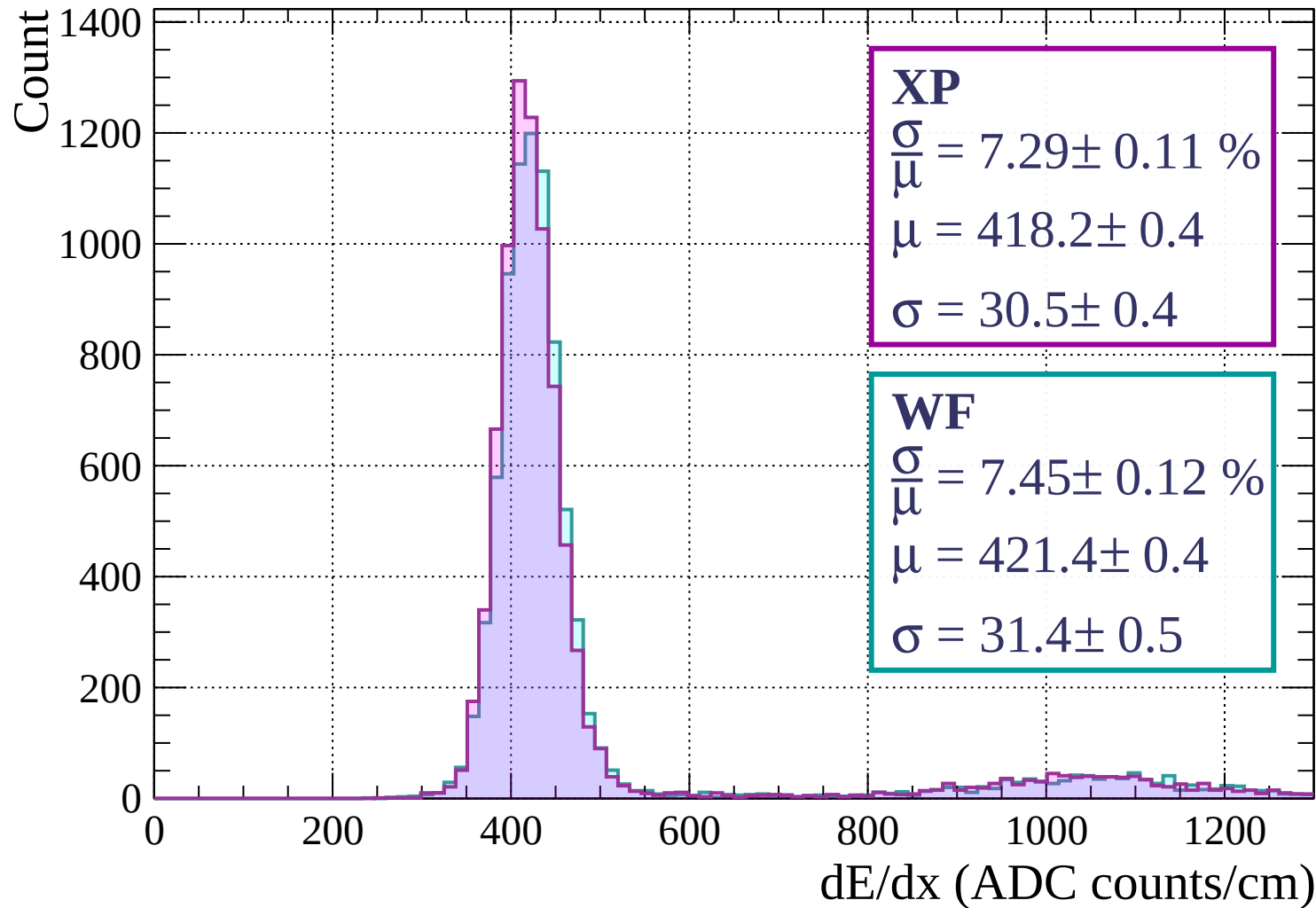
$$\sigma = 32.3 \pm 0.4$$

dE/dx (ADC counts/cm)

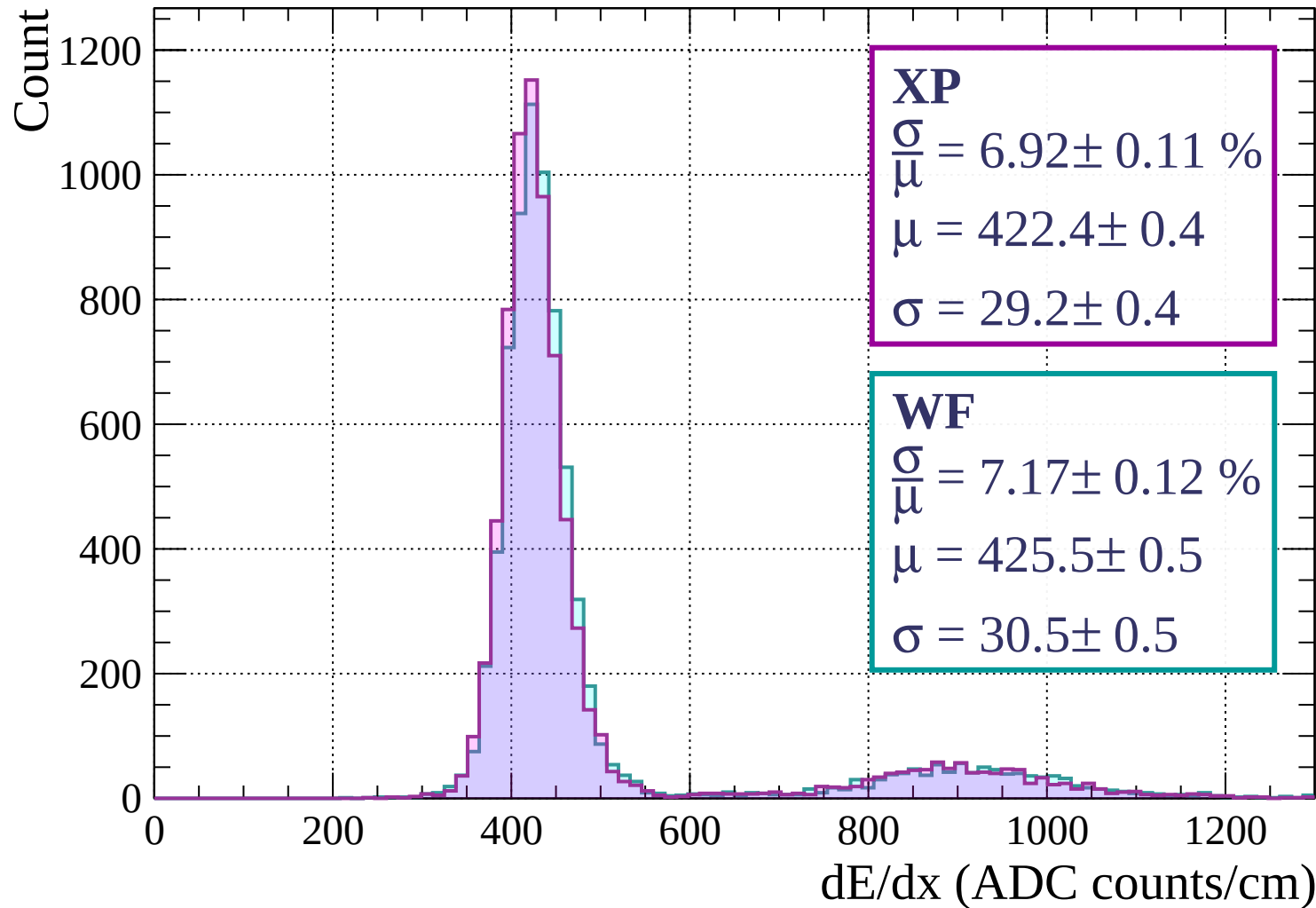
Energy loss | $480 < p < 560$



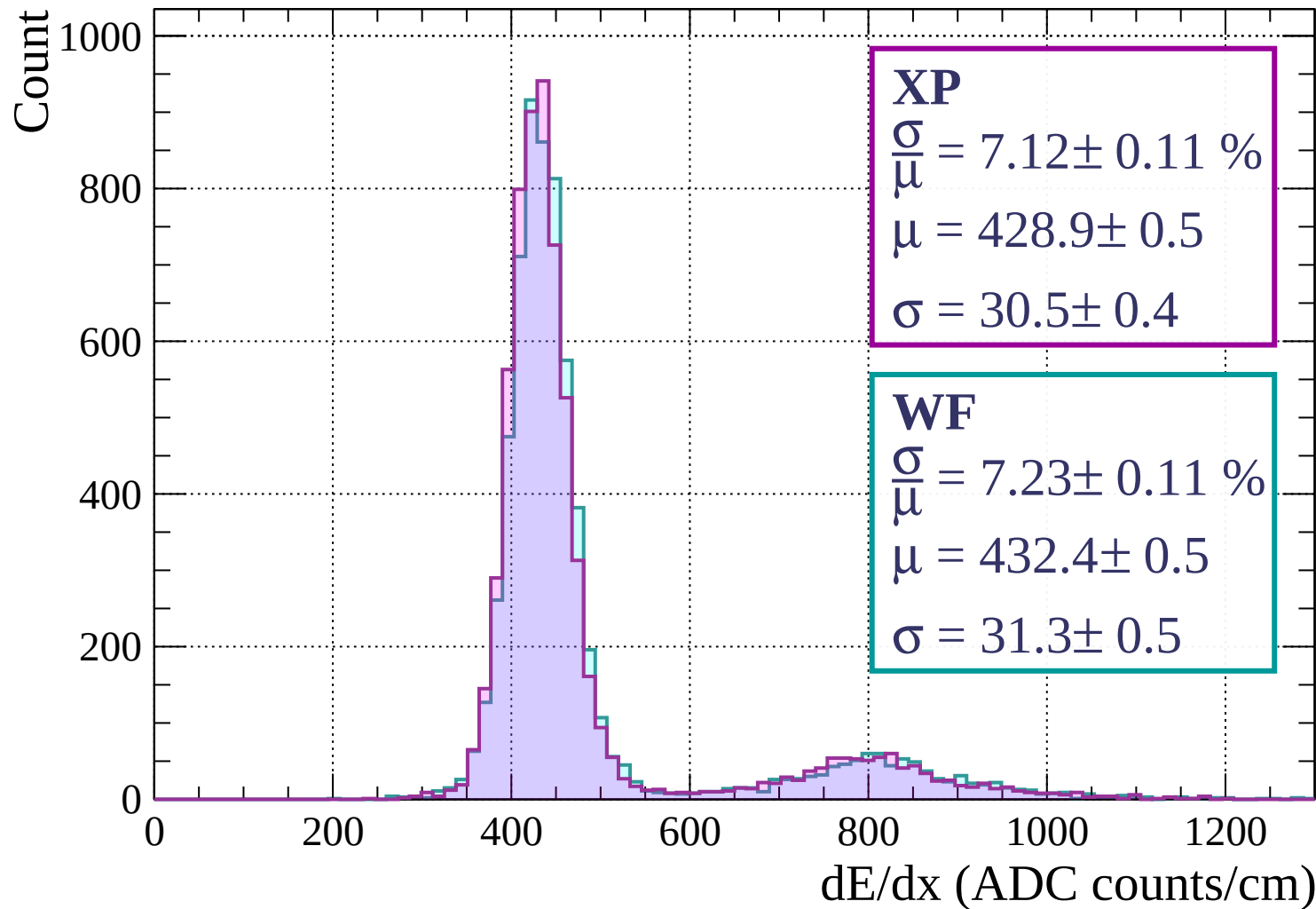
Energy loss | $560 < p < 640$



Energy loss | $640 < p < 720$

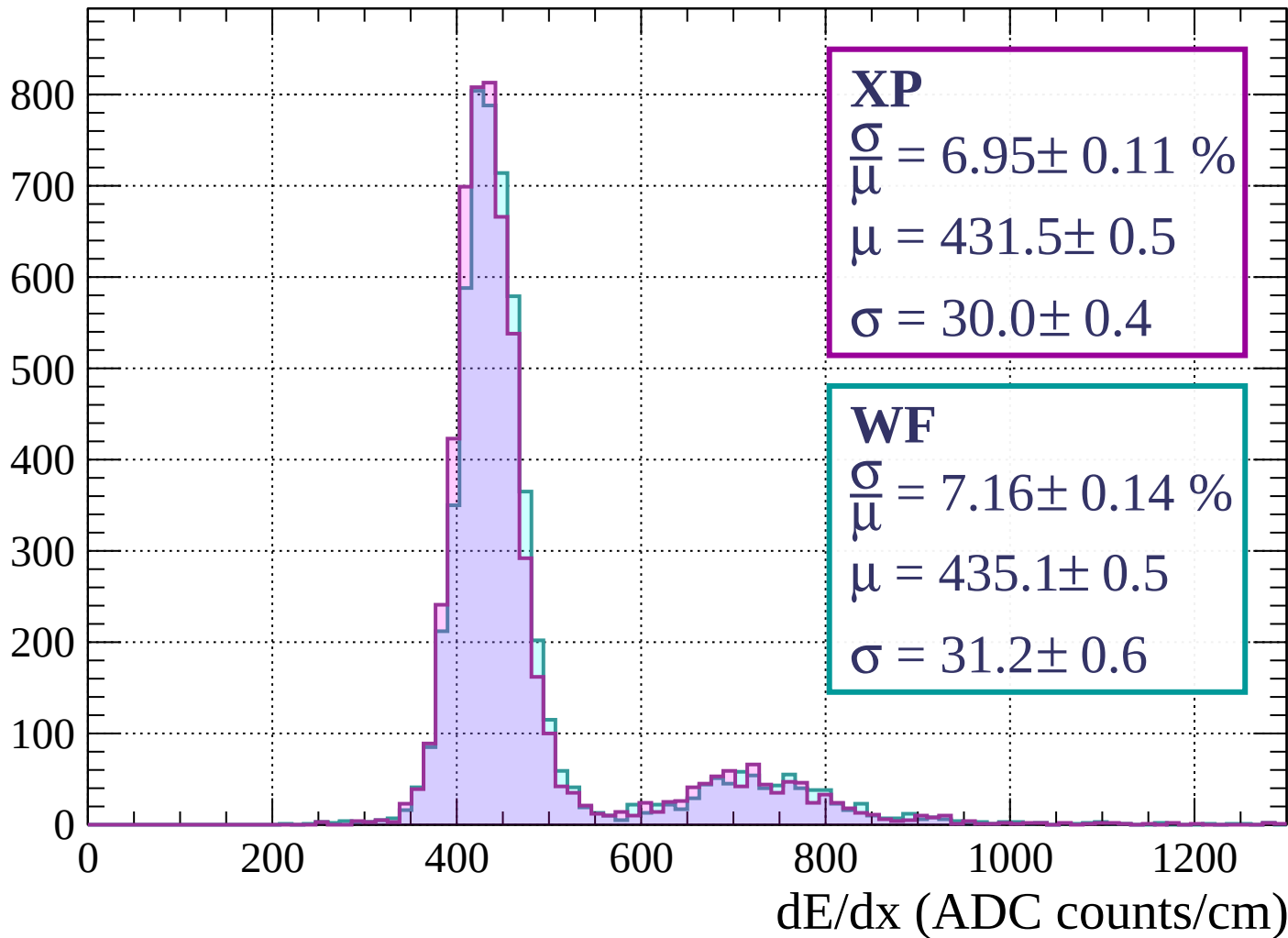


Energy loss | $720 < p < 800$



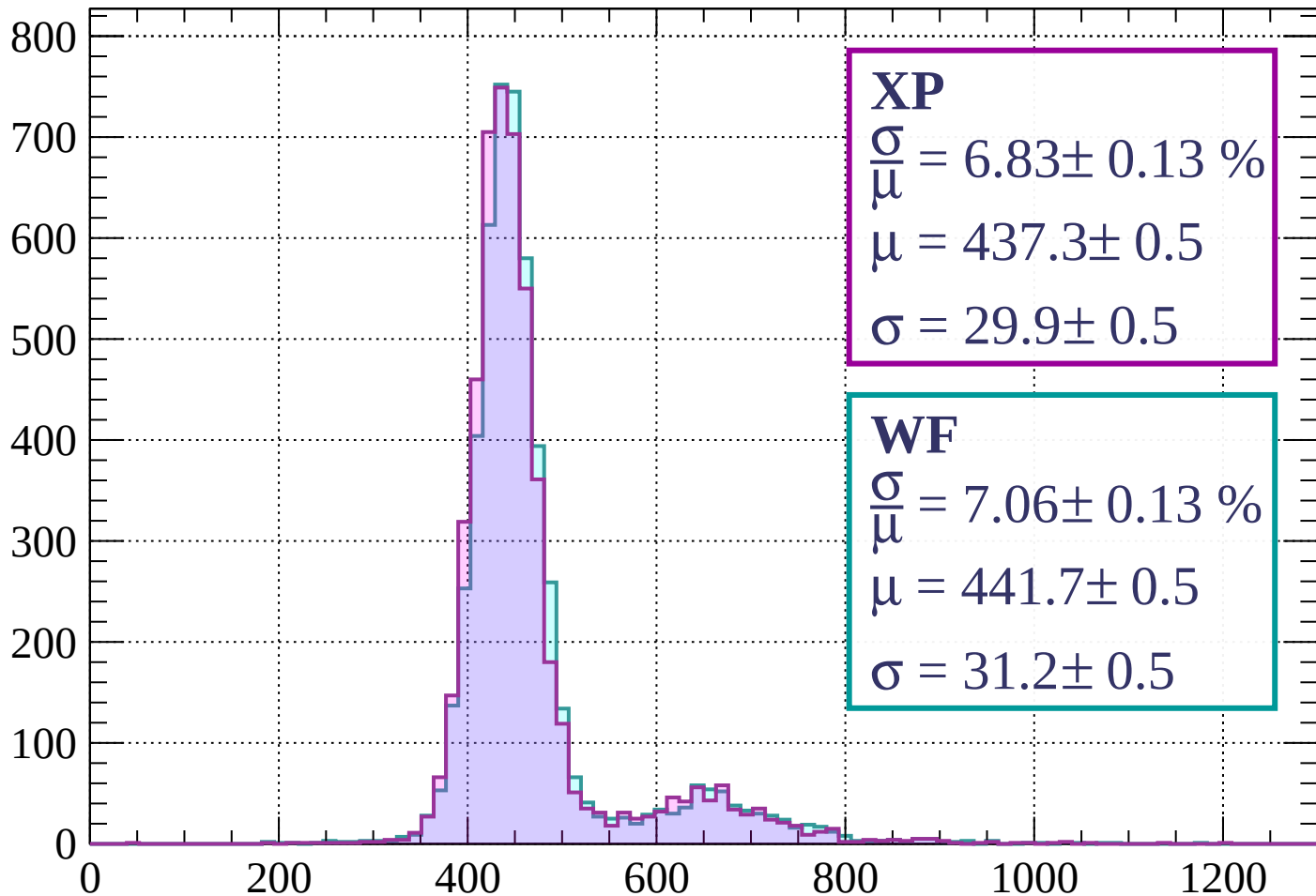
Energy loss | $800 < p < 880$

Count



Energy loss | $880 < p < 960$

Count



XP

$$\frac{\sigma}{\mu} = 6.83 \pm 0.13 \%$$

$$\mu = 437.3 \pm 0.5$$

$$\sigma = 29.9 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 7.06 \pm 0.13 \%$$

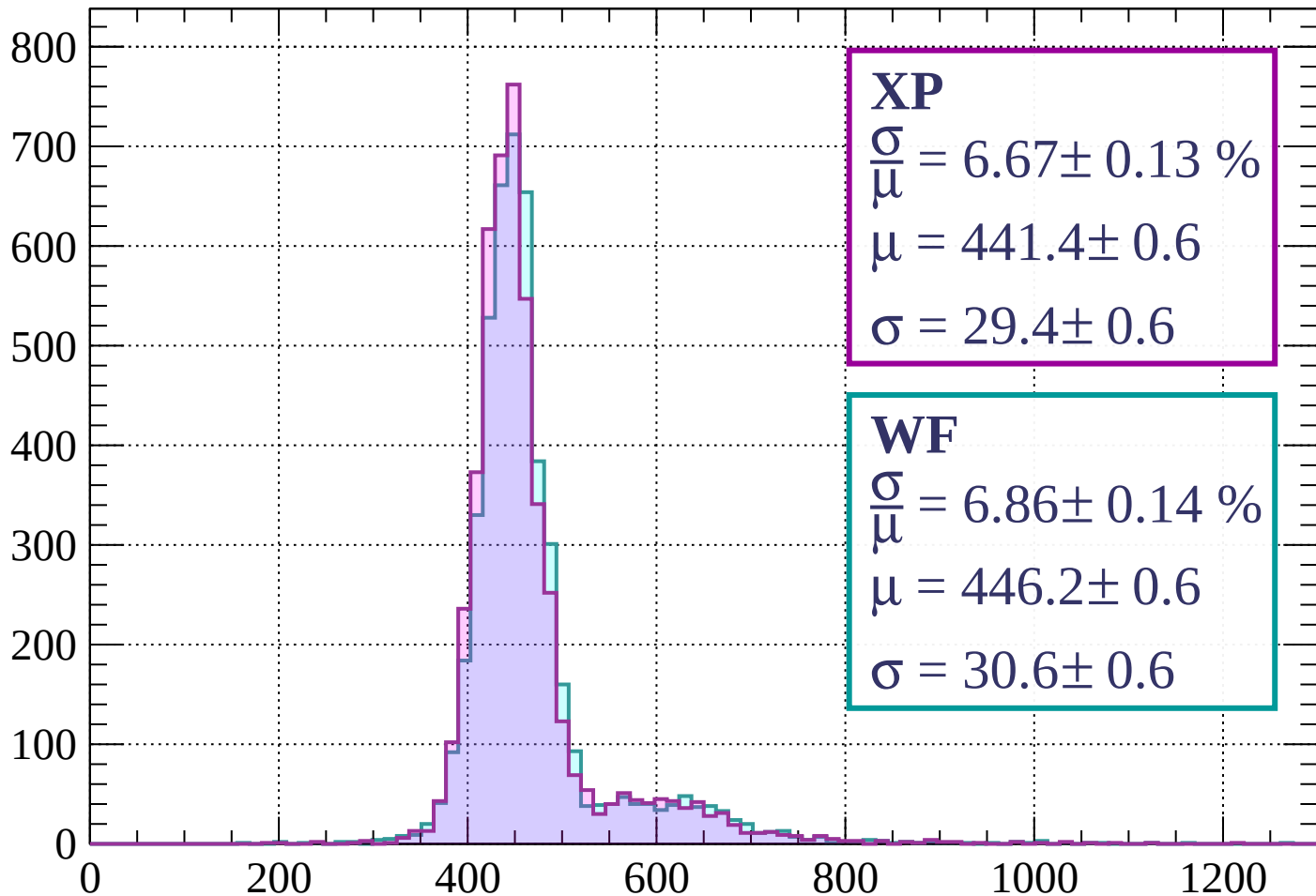
$$\mu = 441.7 \pm 0.5$$

$$\sigma = 31.2 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $960 < p < 1040$

Count



XP

$$\frac{\sigma}{\mu} = 6.67 \pm 0.13 \%$$

$$\mu = 441.4 \pm 0.6$$

$$\sigma = 29.4 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 6.86 \pm 0.14 \%$$

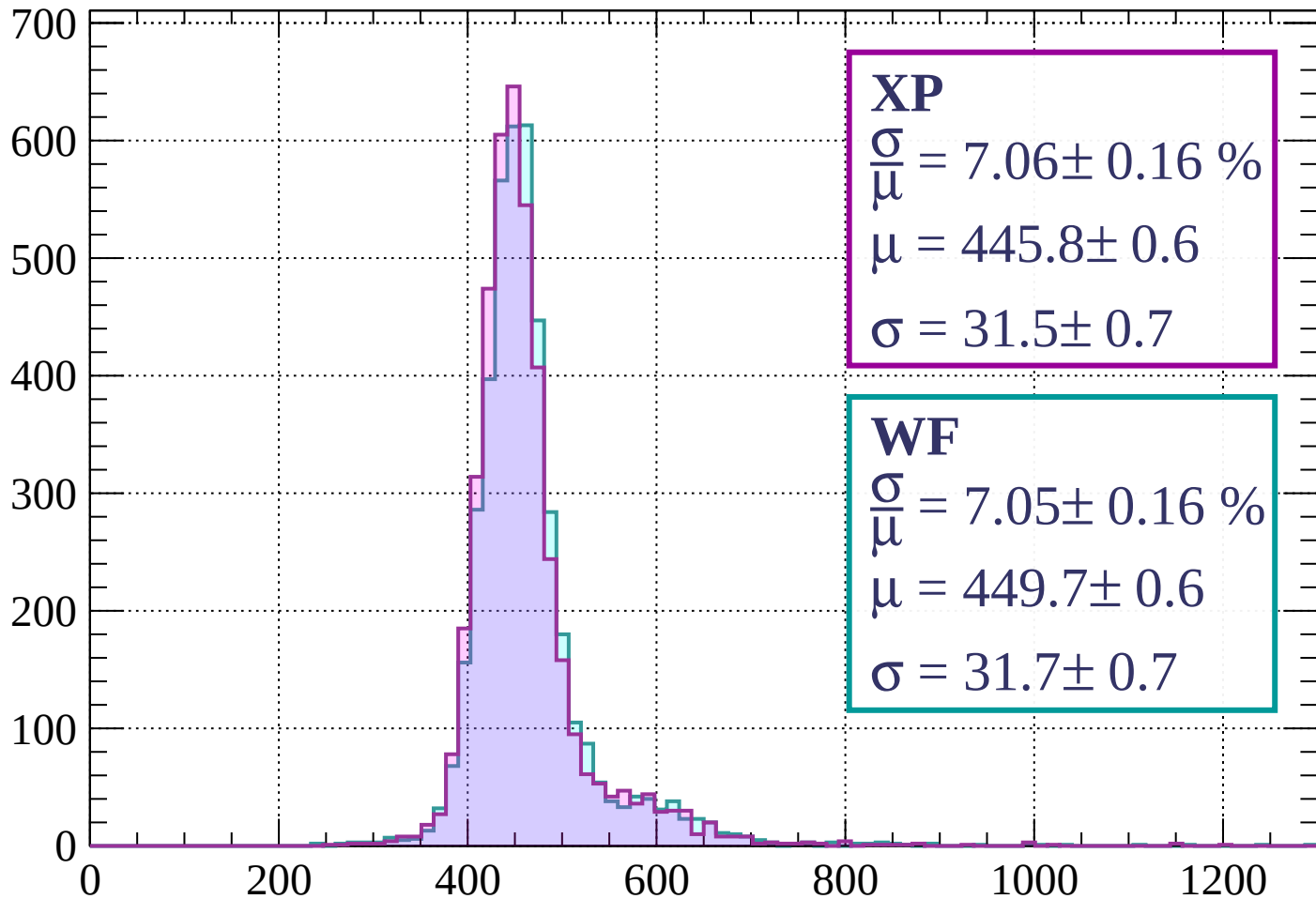
$$\mu = 446.2 \pm 0.6$$

$$\sigma = 30.6 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $1040 < p < 1120$

Count



XP

$$\frac{\sigma}{\mu} = 7.06 \pm 0.16 \%$$

$$\mu = 445.8 \pm 0.6$$

$$\sigma = 31.5 \pm 0.7$$

WF

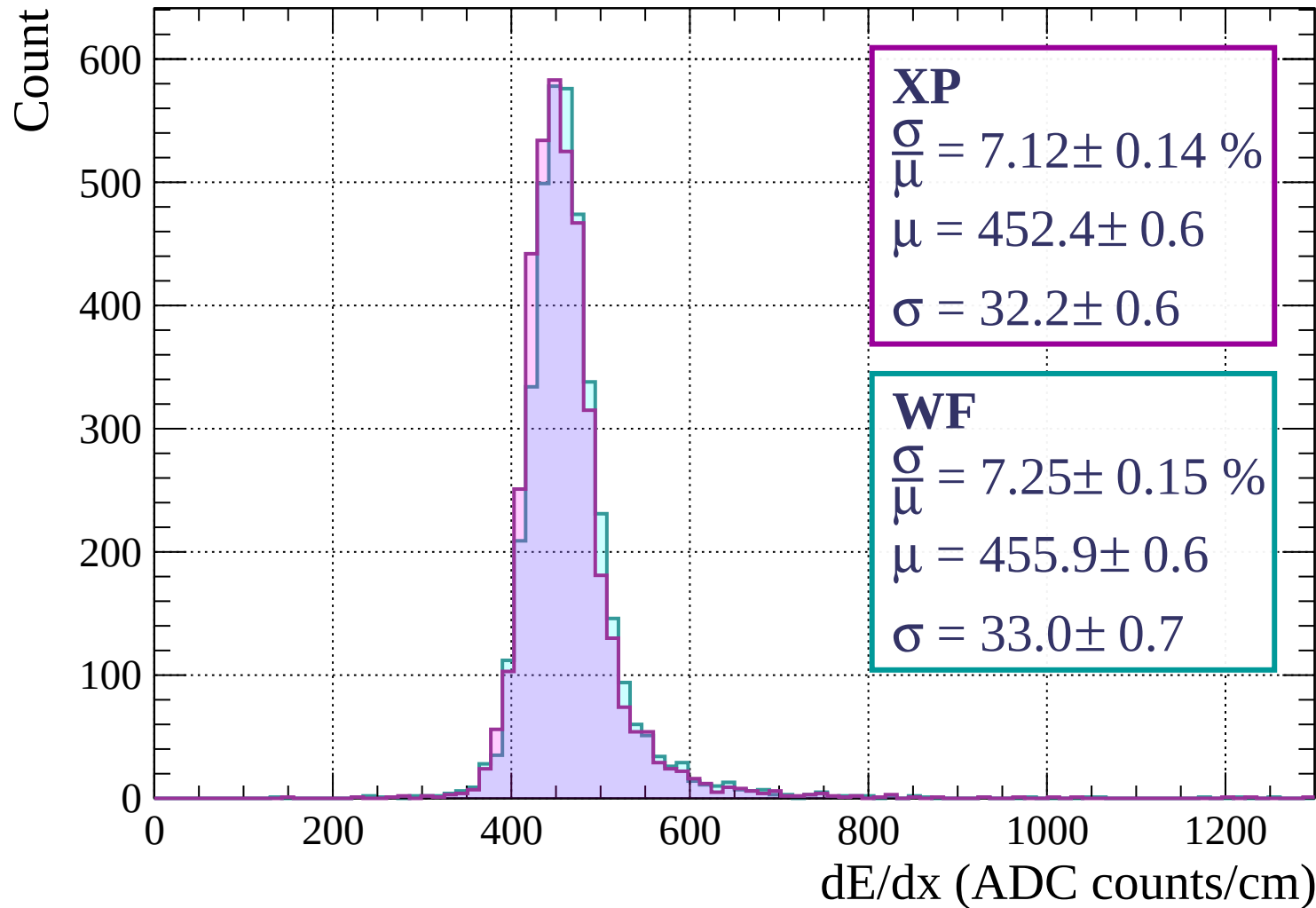
$$\frac{\sigma}{\mu} = 7.05 \pm 0.16 \%$$

$$\mu = 449.7 \pm 0.6$$

$$\sigma = 31.7 \pm 0.7$$

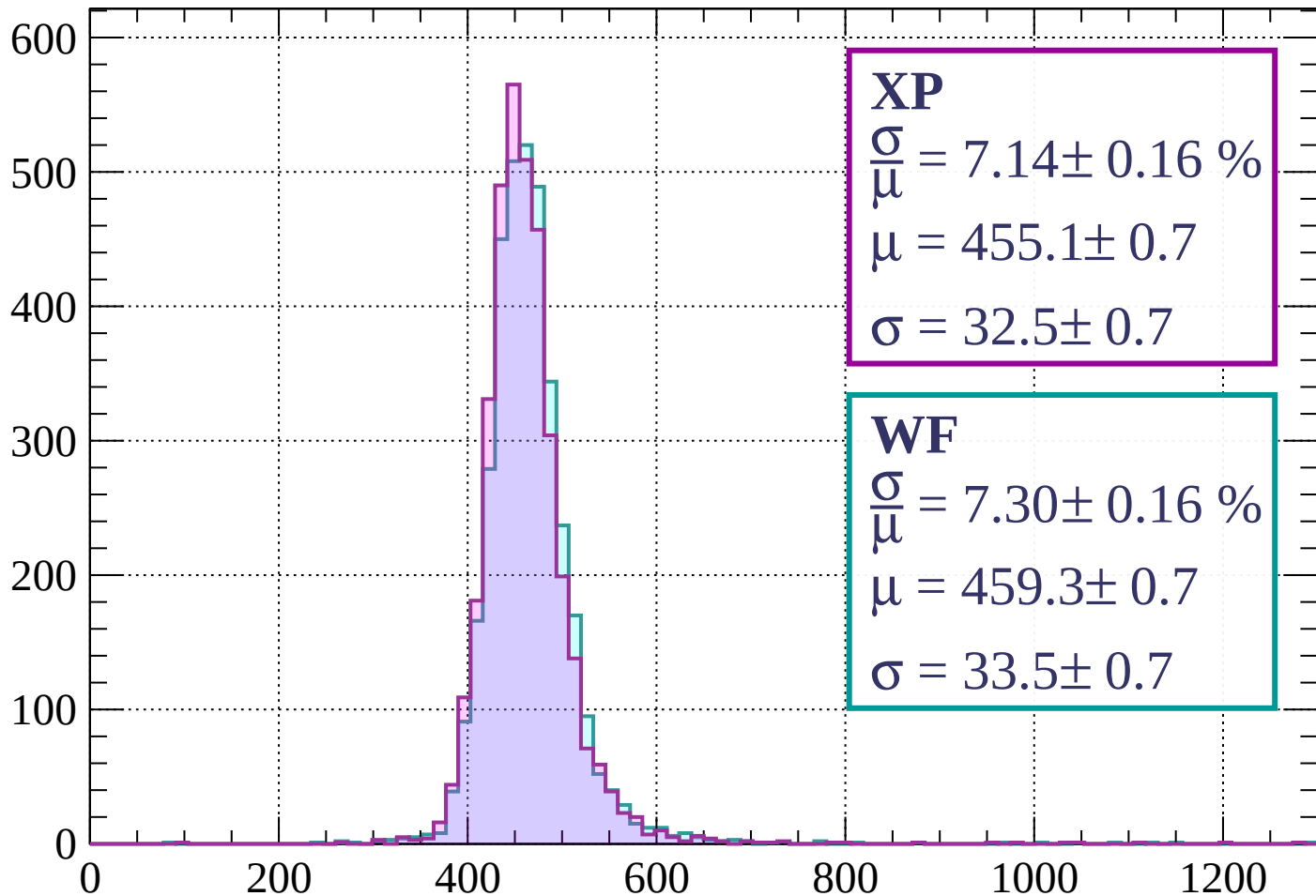
dE/dx (ADC counts/cm)

Energy loss | $1120 < p < 1200$



Energy loss | $1200 < p < 1280$

Count



XP

$$\frac{\sigma}{\mu} = 7.14 \pm 0.16 \%$$

$$\mu = 455.1 \pm 0.7$$

$$\sigma = 32.5 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 7.30 \pm 0.16 \%$$

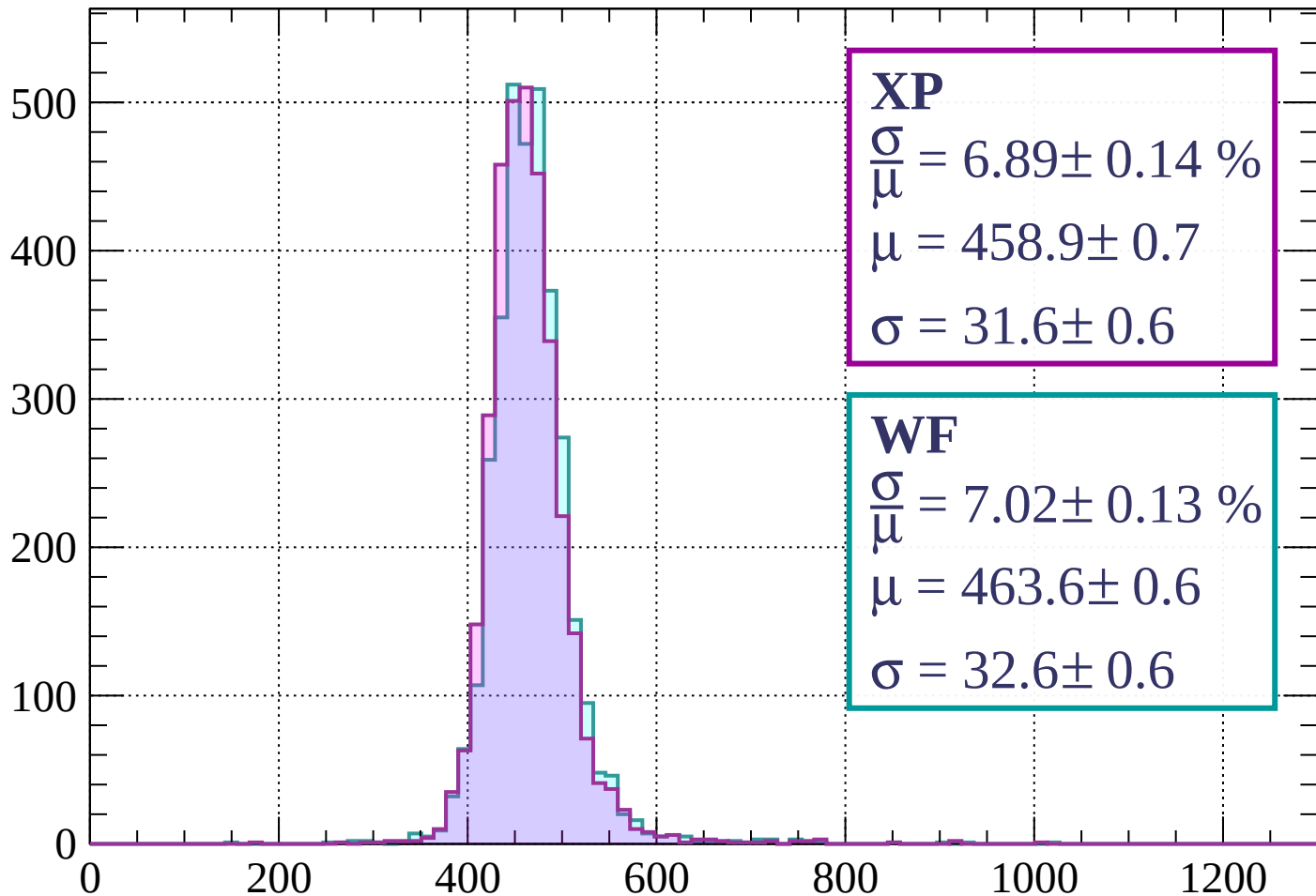
$$\mu = 459.3 \pm 0.7$$

$$\sigma = 33.5 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1280 < p < 1360$

Count



XP

$$\frac{\sigma}{\mu} = 6.89 \pm 0.14 \%$$

$$\mu = 458.9 \pm 0.7$$

$$\sigma = 31.6 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 7.02 \pm 0.13 \%$$

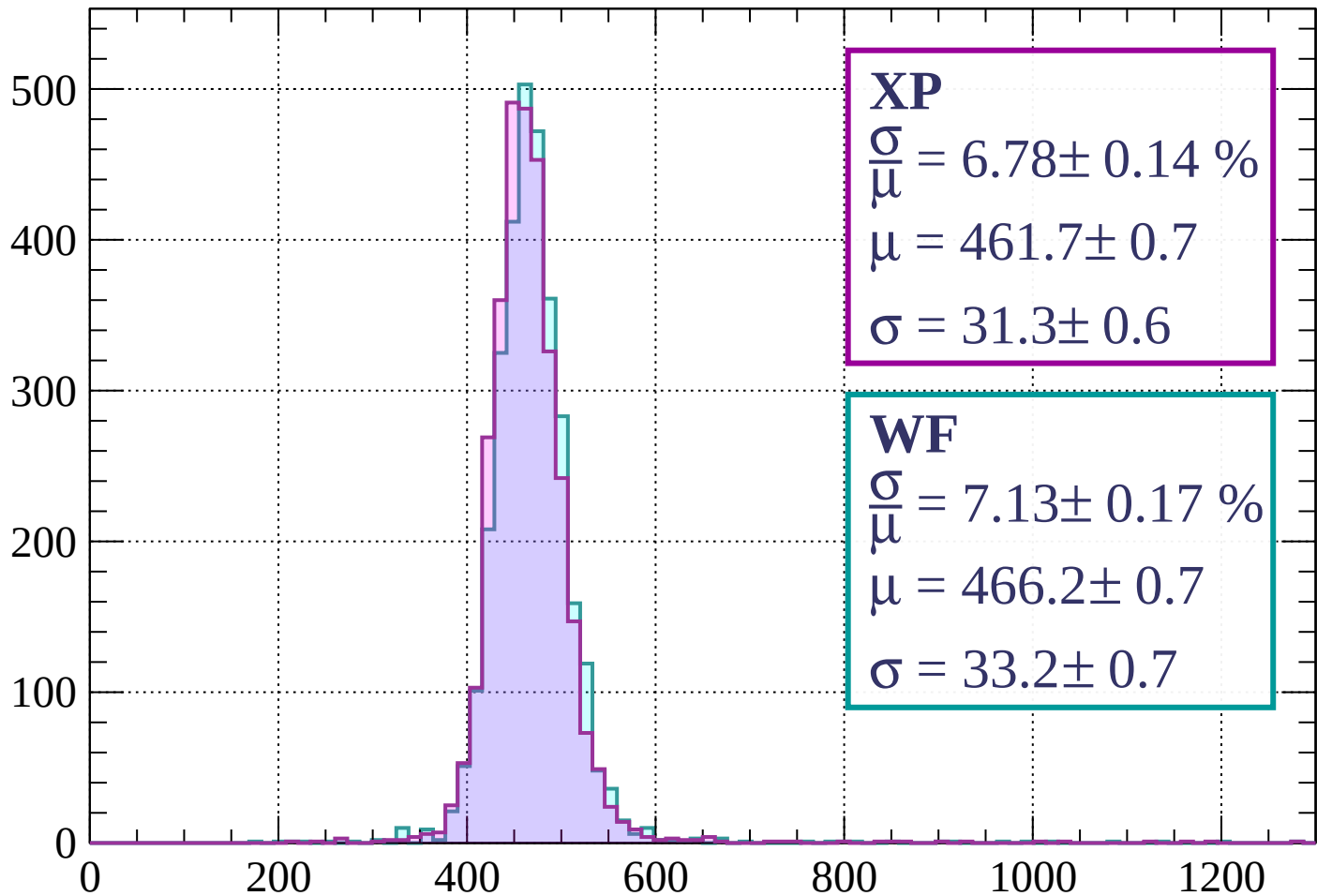
$$\mu = 463.6 \pm 0.6$$

$$\sigma = 32.6 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $1360 < p < 1440$

Count



XP

$$\frac{\sigma}{\mu} = 6.78 \pm 0.14 \%$$

$$\mu = 461.7 \pm 0.7$$

$$\sigma = 31.3 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 7.13 \pm 0.17 \%$$

$$\mu = 466.2 \pm 0.7$$

$$\sigma = 33.2 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1440 < p < 1520$

Count

500

400

300

200

100

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 6.55 \pm 0.16 \%$$

$$\mu = 463.3 \pm 0.7$$

$$\sigma = 30.3 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 6.58 \pm 0.14 \%$$

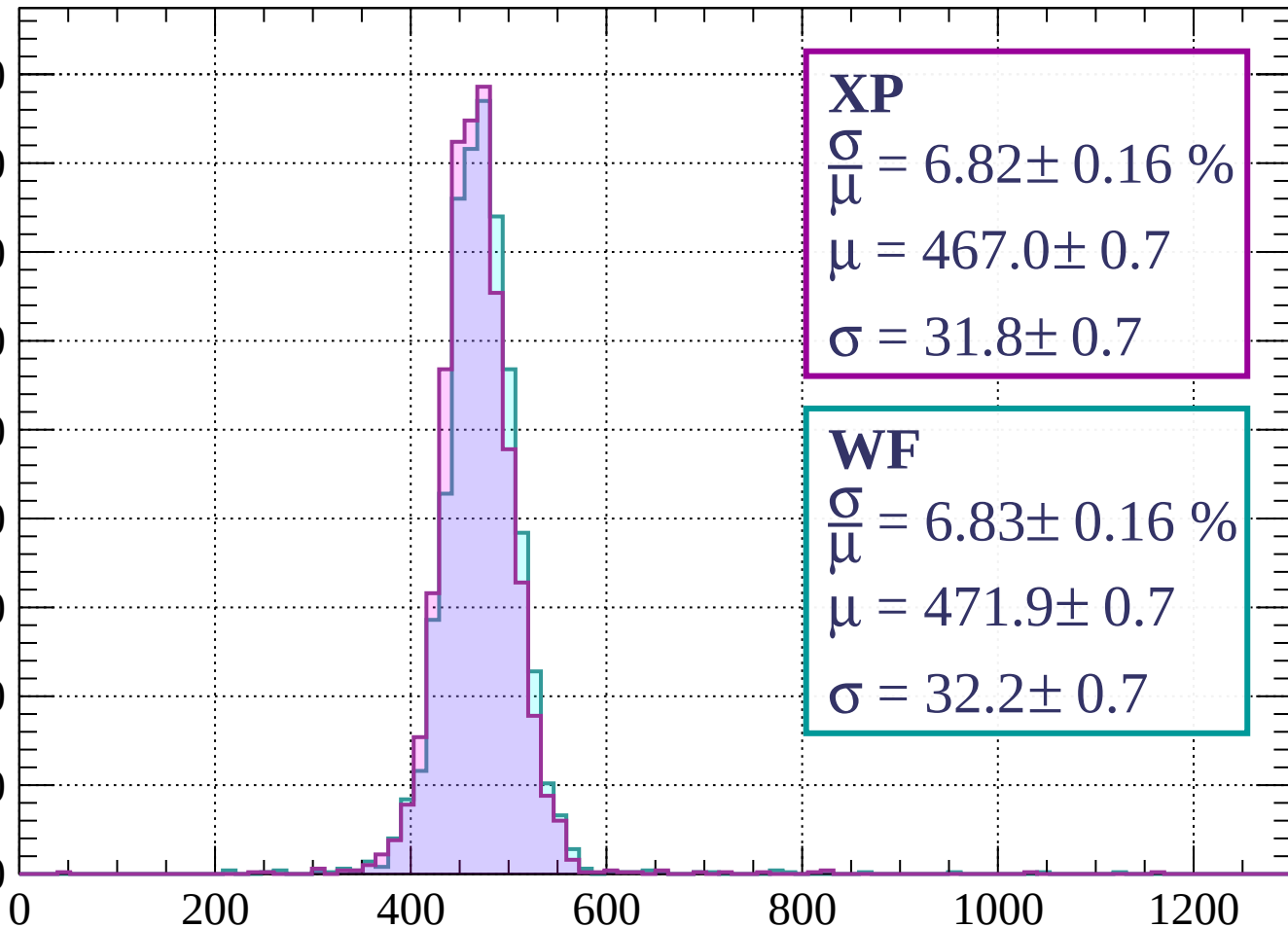
$$\mu = 467.7 \pm 0.6$$

$$\sigma = 30.8 \pm 0.6$$

Energy loss | $1520 < p < 1600$

Count

450
400
350
300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 6.82 \pm 0.16 \%$$

$$\mu = 467.0 \pm 0.7$$

$$\sigma = 31.8 \pm 0.7$$

WF

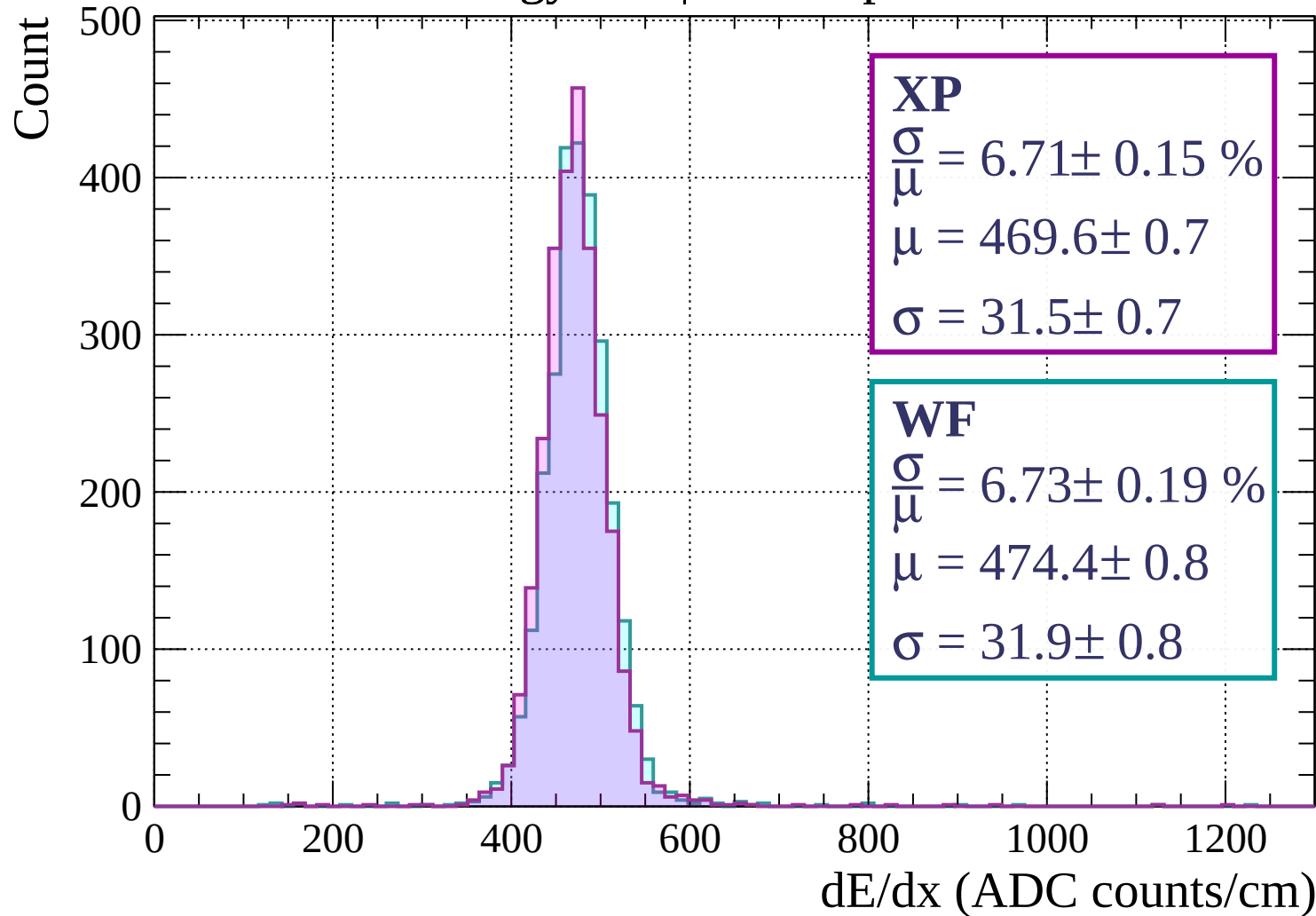
$$\frac{\sigma}{\mu} = 6.83 \pm 0.16 \%$$

$$\mu = 471.9 \pm 0.7$$

$$\sigma = 32.2 \pm 0.7$$

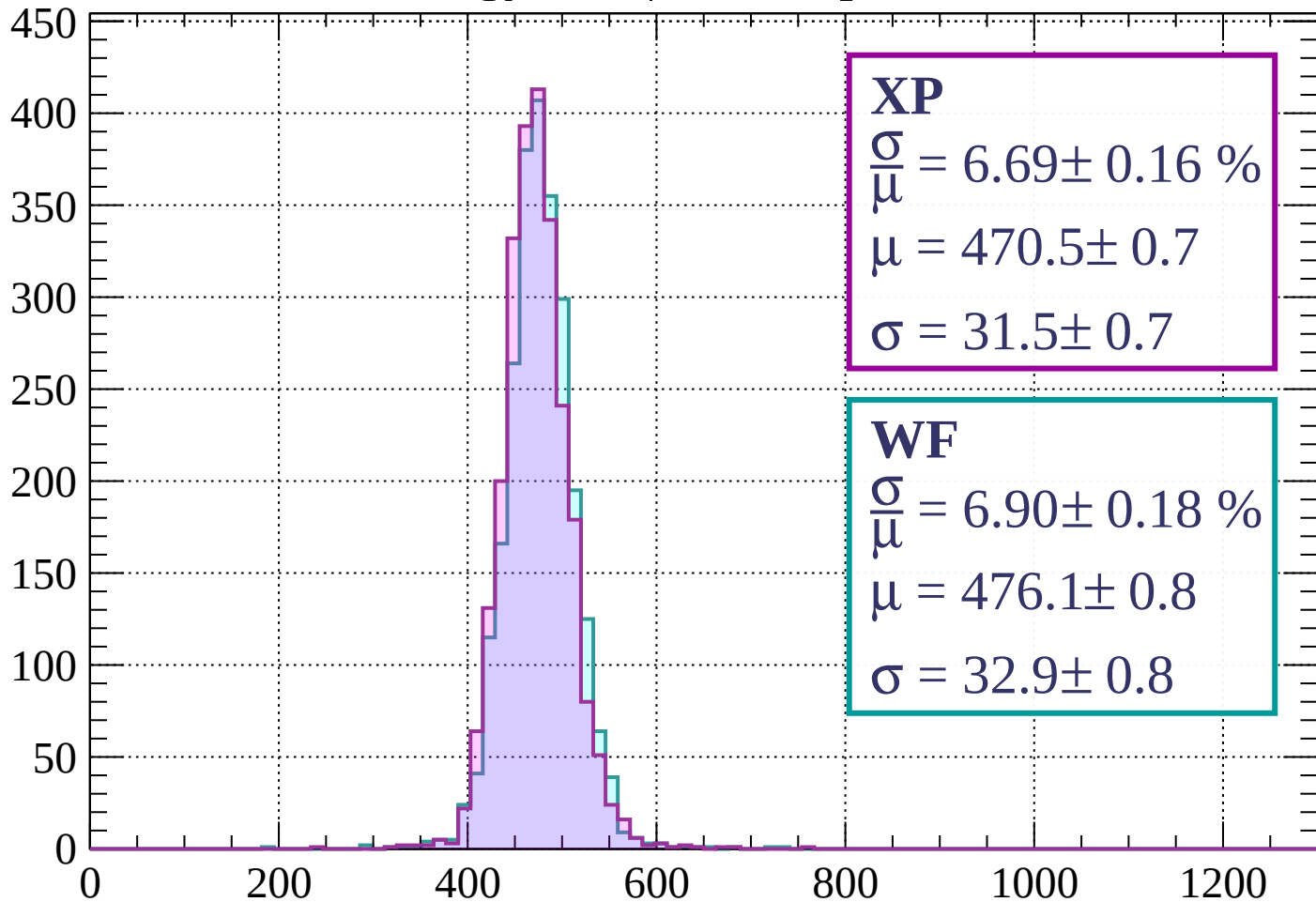
dE/dx (ADC counts/cm)

Energy loss | $1600 < p < 1680$



Energy loss | $1680 < p < 1760$

Count



XP

$$\frac{\sigma}{\mu} = 6.69 \pm 0.16 \%$$

$$\mu = 470.5 \pm 0.7$$

$$\sigma = 31.5 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 6.90 \pm 0.18 \%$$

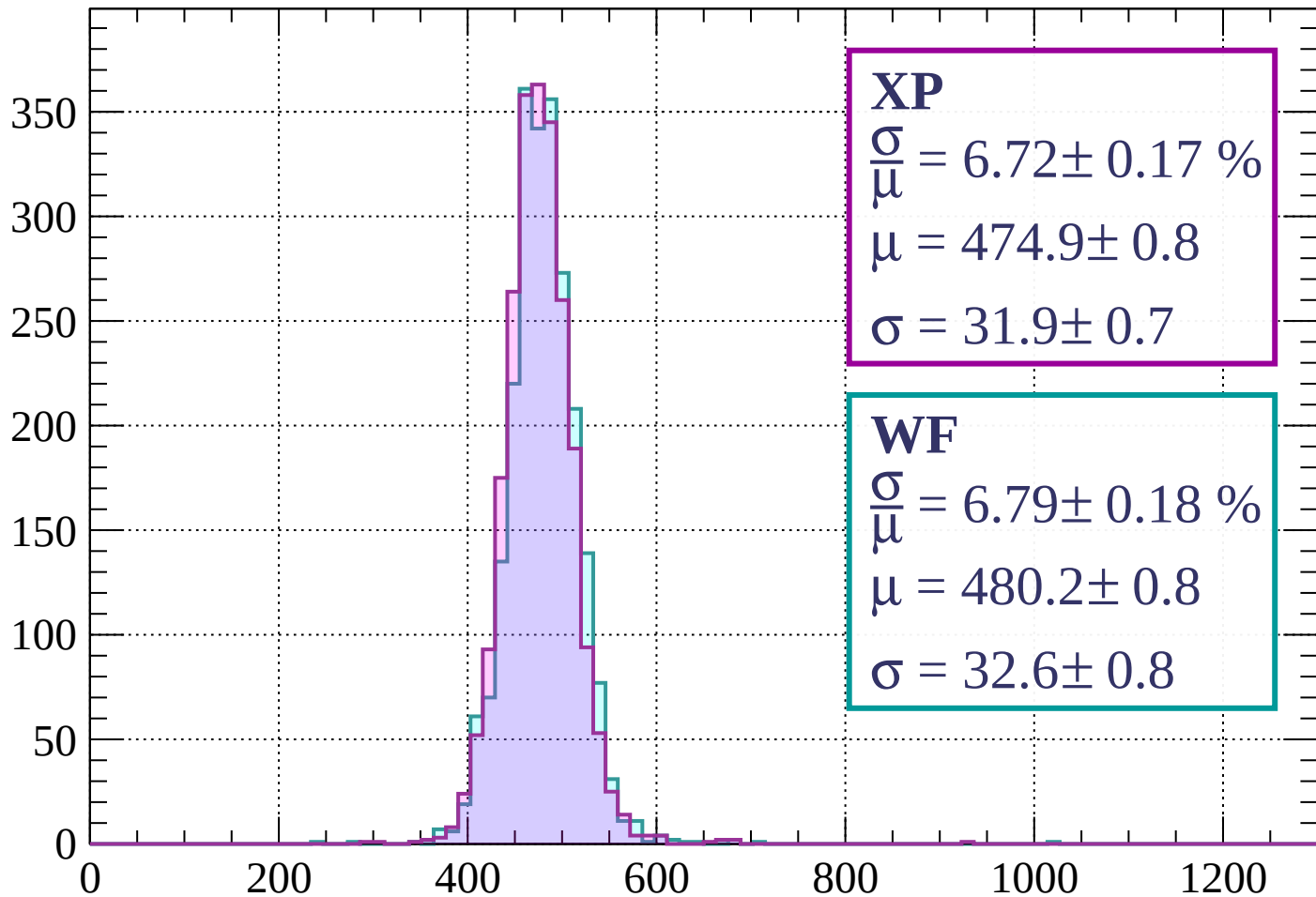
$$\mu = 476.1 \pm 0.8$$

$$\sigma = 32.9 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1760 < p < 1840$

Count



XP

$$\frac{\sigma}{\mu} = 6.72 \pm 0.17 \%$$

$$\mu = 474.9 \pm 0.8$$

$$\sigma = 31.9 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 6.79 \pm 0.18 \%$$

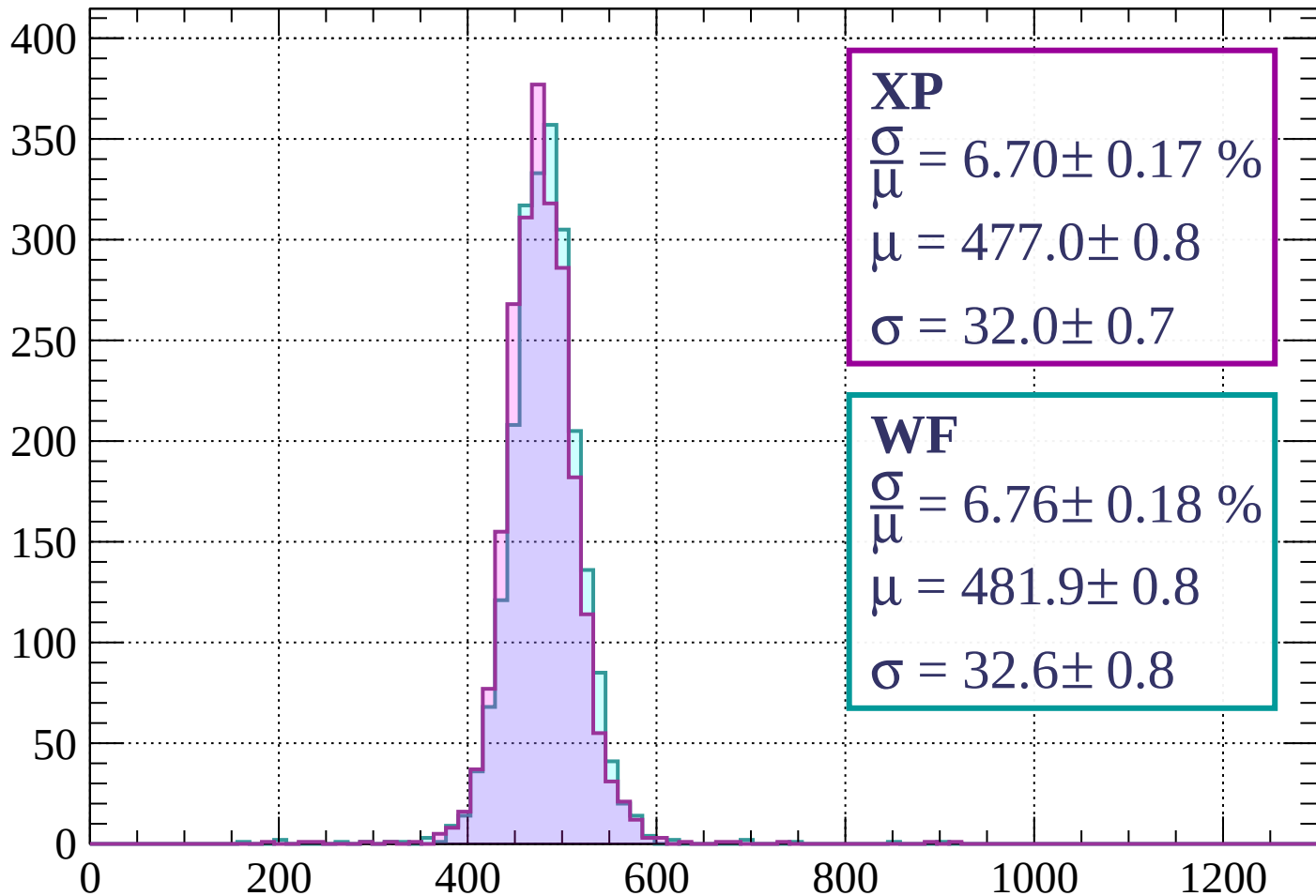
$$\mu = 480.2 \pm 0.8$$

$$\sigma = 32.6 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1840 < p < 1920$

Count



XP

$$\frac{\sigma}{\mu} = 6.70 \pm 0.17 \%$$

$$\mu = 477.0 \pm 0.8$$

$$\sigma = 32.0 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 6.76 \pm 0.18 \%$$

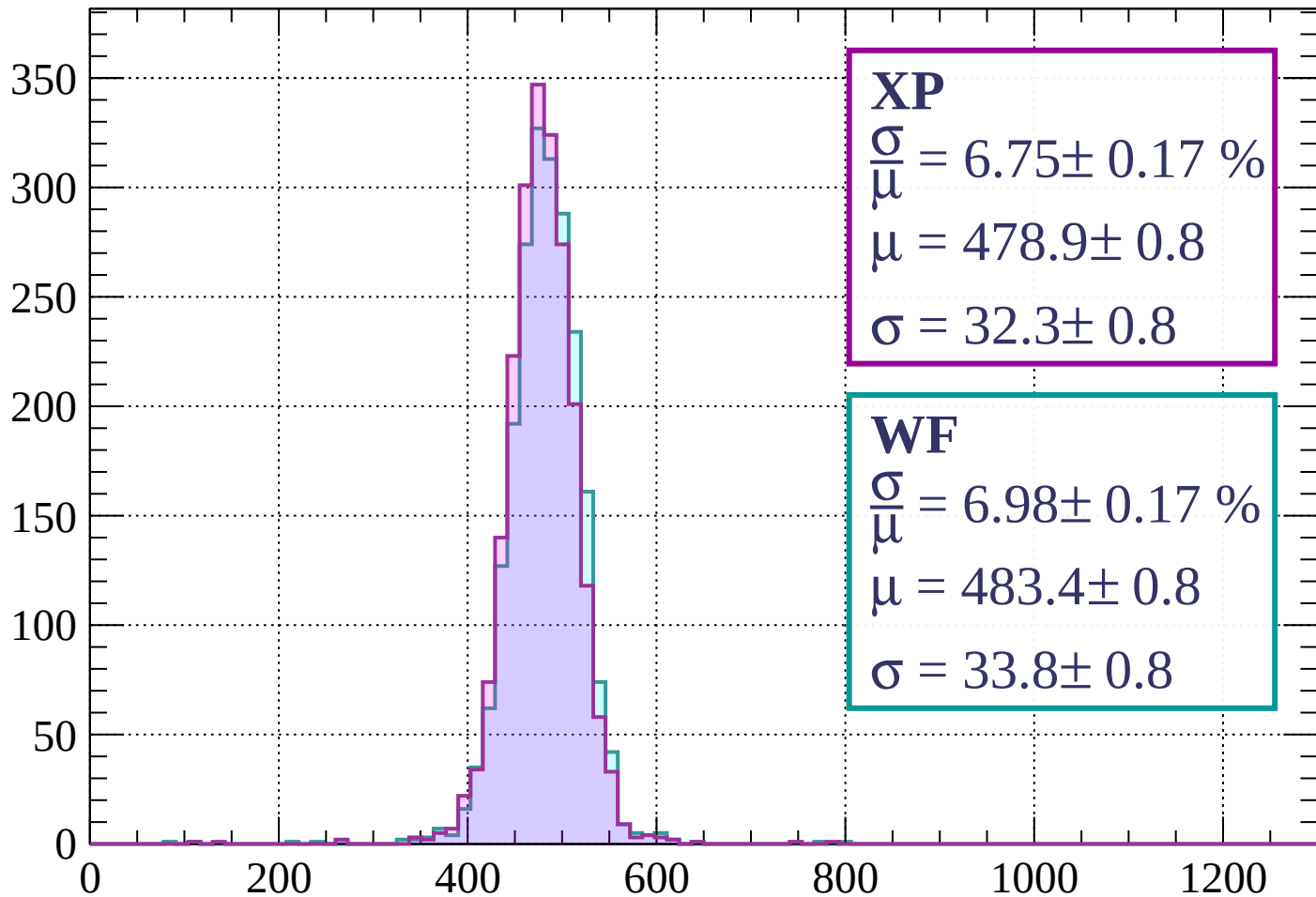
$$\mu = 481.9 \pm 0.8$$

$$\sigma = 32.6 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1920 < p < 2000$

Count



XP

$$\frac{\sigma}{\mu} = 6.75 \pm 0.17 \%$$

$$\mu = 478.9 \pm 0.8$$

$$\sigma = 32.3 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 6.98 \pm 0.17 \%$$

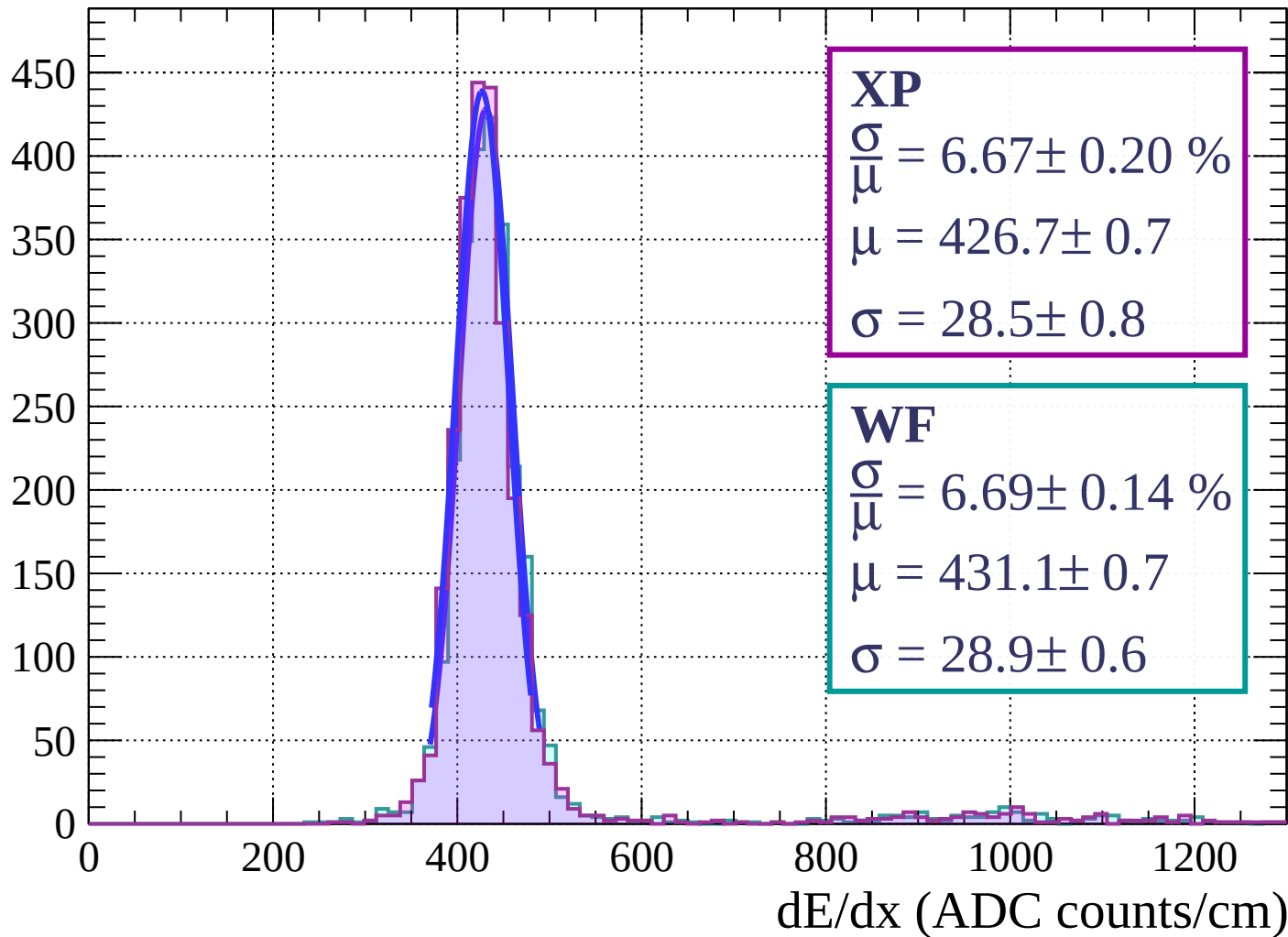
$$\mu = 483.4 \pm 0.8$$

$$\sigma = 33.8 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $0 < \phi < 3$

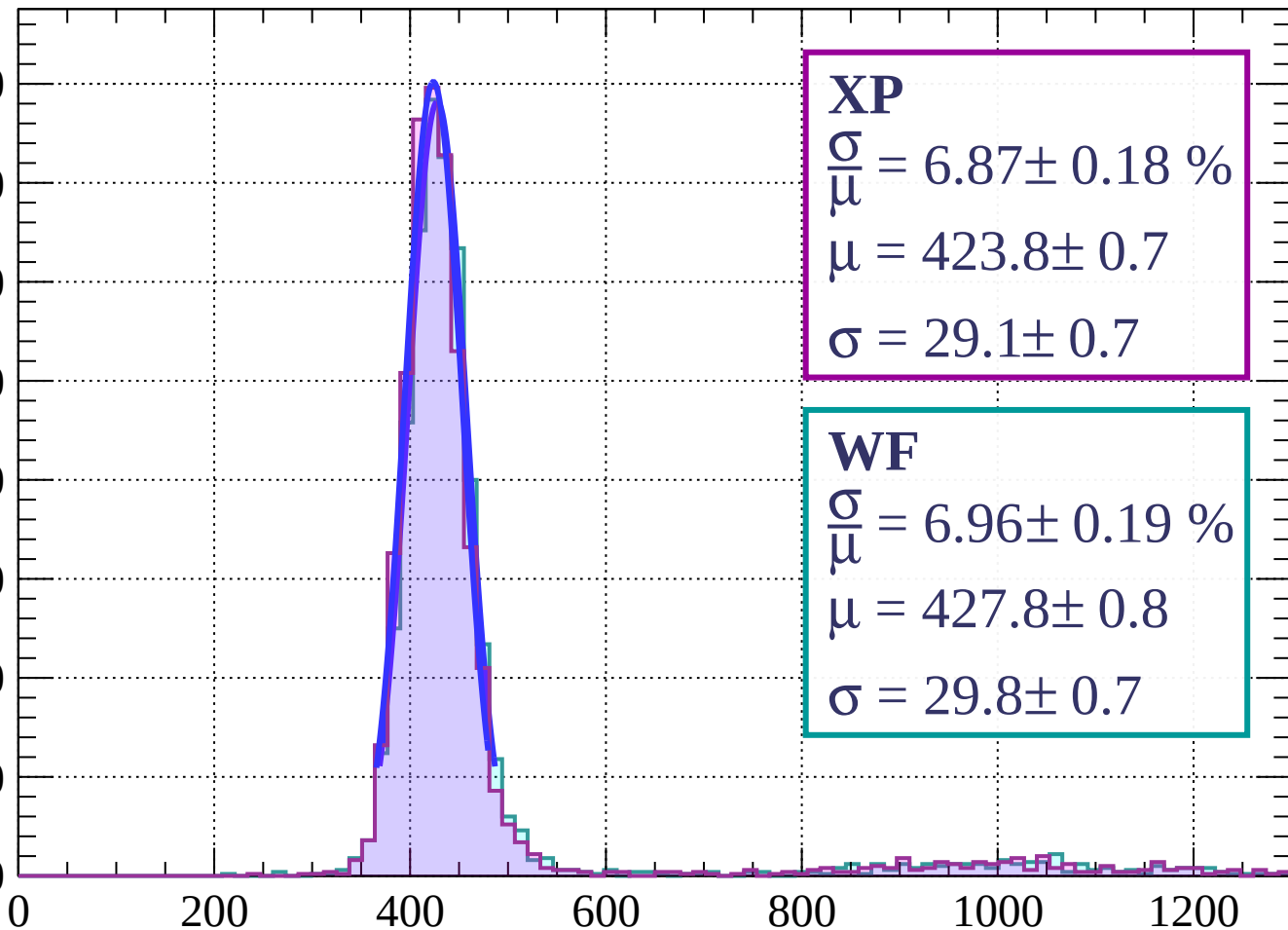
Count



Energy loss | $3 < \phi < 6$

Count

400
350
300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 6.87 \pm 0.18 \%$$

$$\mu = 423.8 \pm 0.7$$

$$\sigma = 29.1 \pm 0.7$$

WF

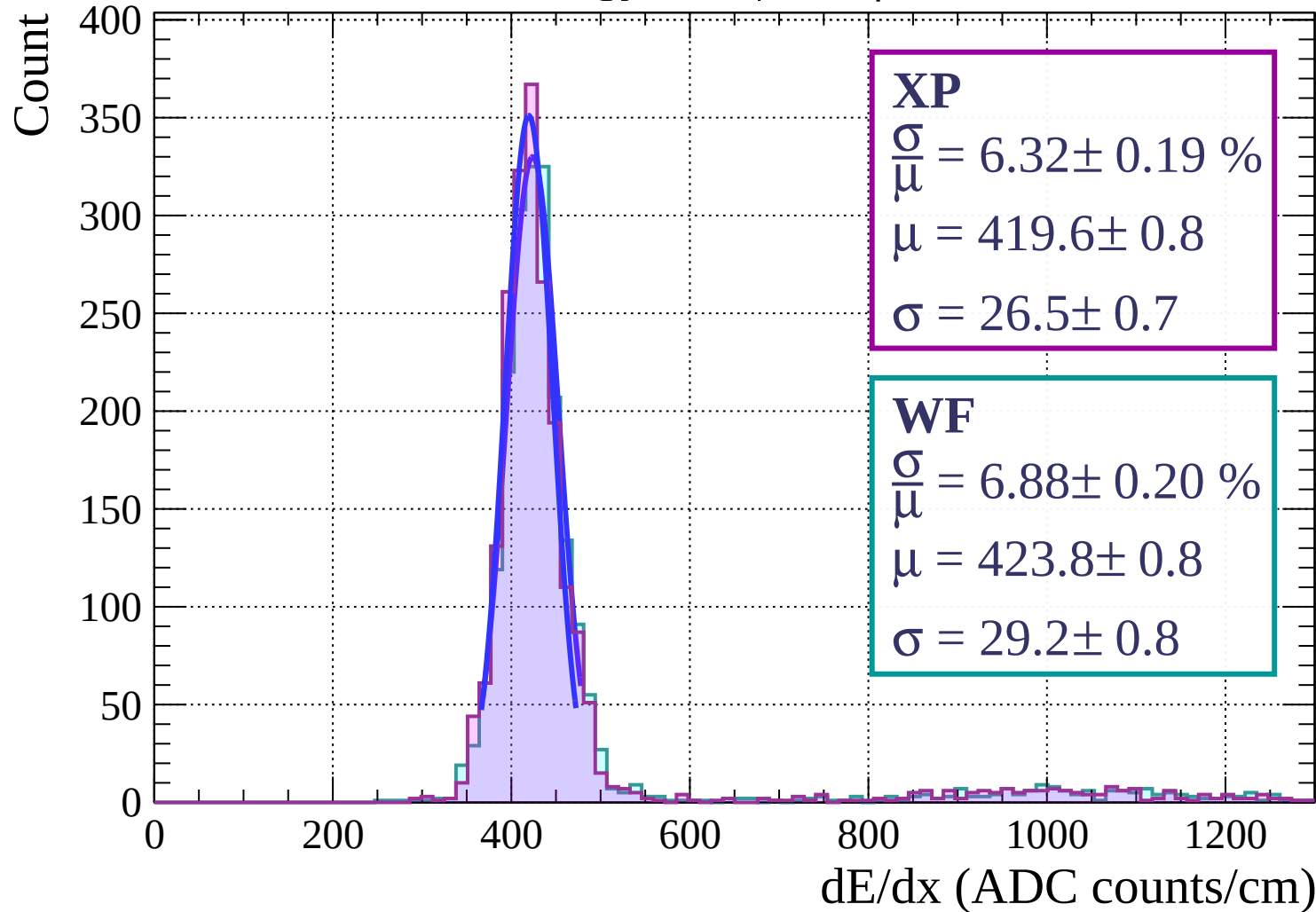
$$\frac{\sigma}{\mu} = 6.96 \pm 0.19 \%$$

$$\mu = 427.8 \pm 0.8$$

$$\sigma = 29.8 \pm 0.7$$

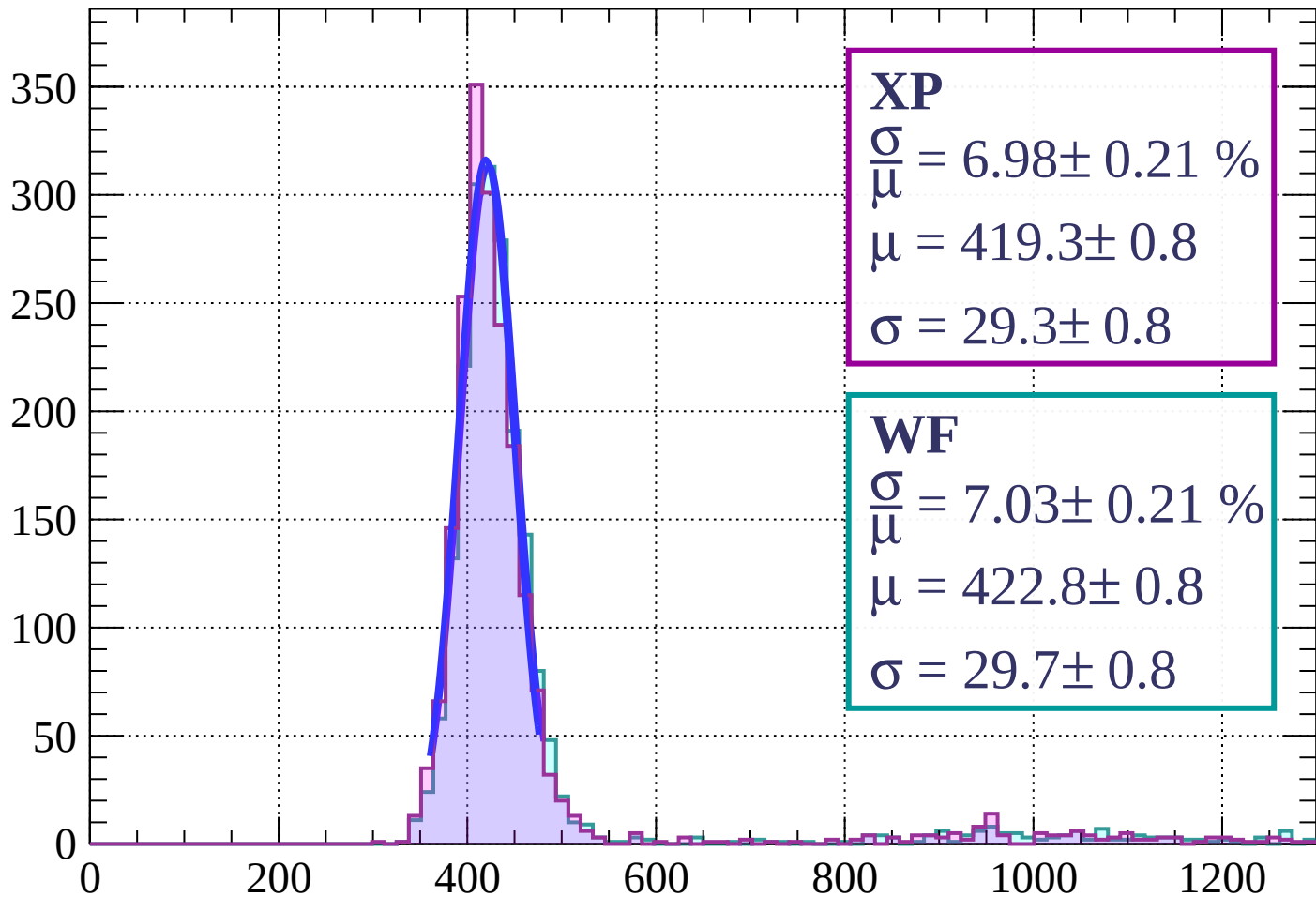
dE/dx (ADC counts/cm)

Energy loss | $6 < \phi < 9$



Energy loss | $9 < \phi < 12$

Count



XP

$$\frac{\sigma}{\mu} = 6.98 \pm 0.21 \%$$

$$\mu = 419.3 \pm 0.8$$

$$\sigma = 29.3 \pm 0.8$$

WF

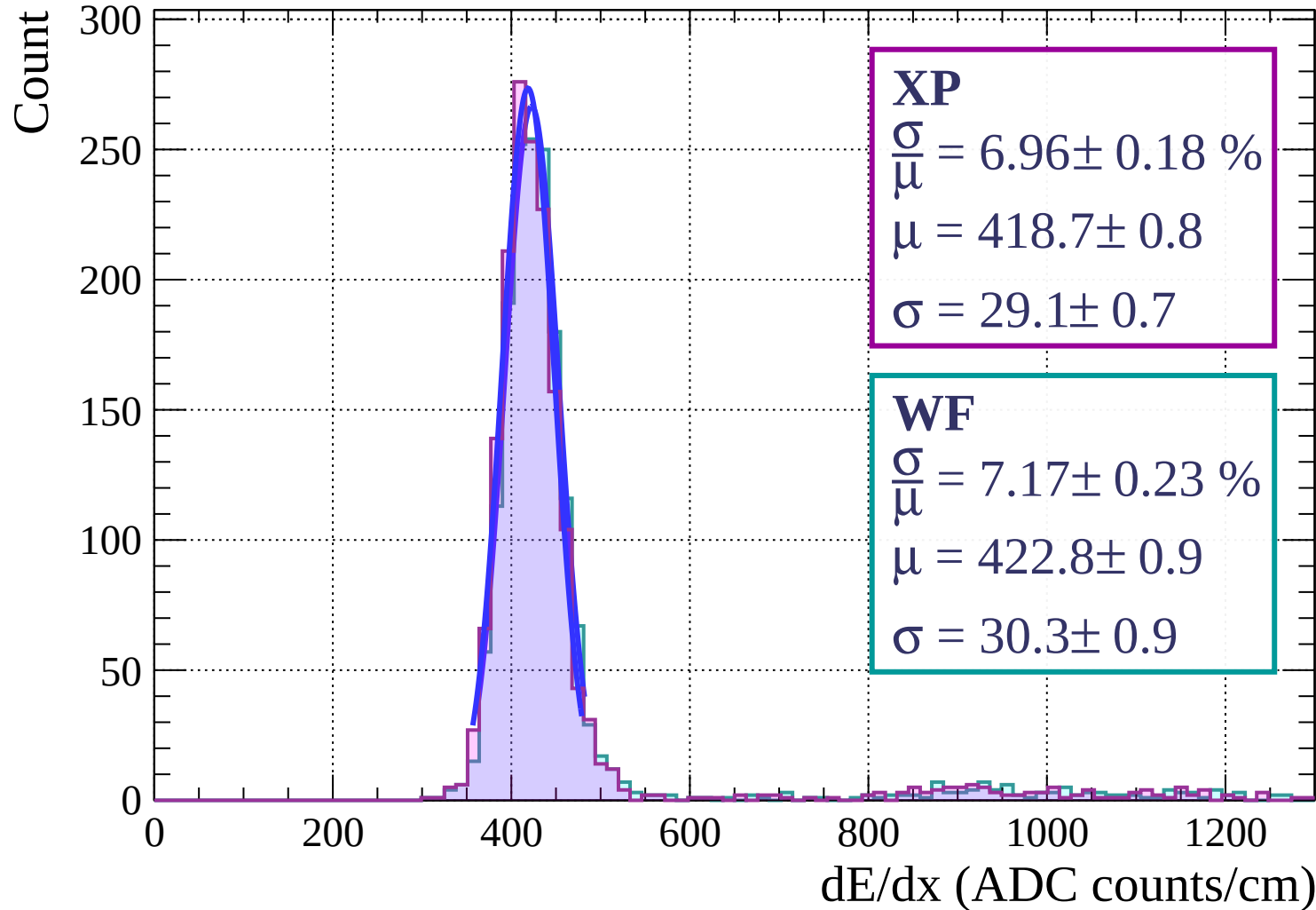
$$\frac{\sigma}{\mu} = 7.03 \pm 0.21 \%$$

$$\mu = 422.8 \pm 0.8$$

$$\sigma = 29.7 \pm 0.8$$

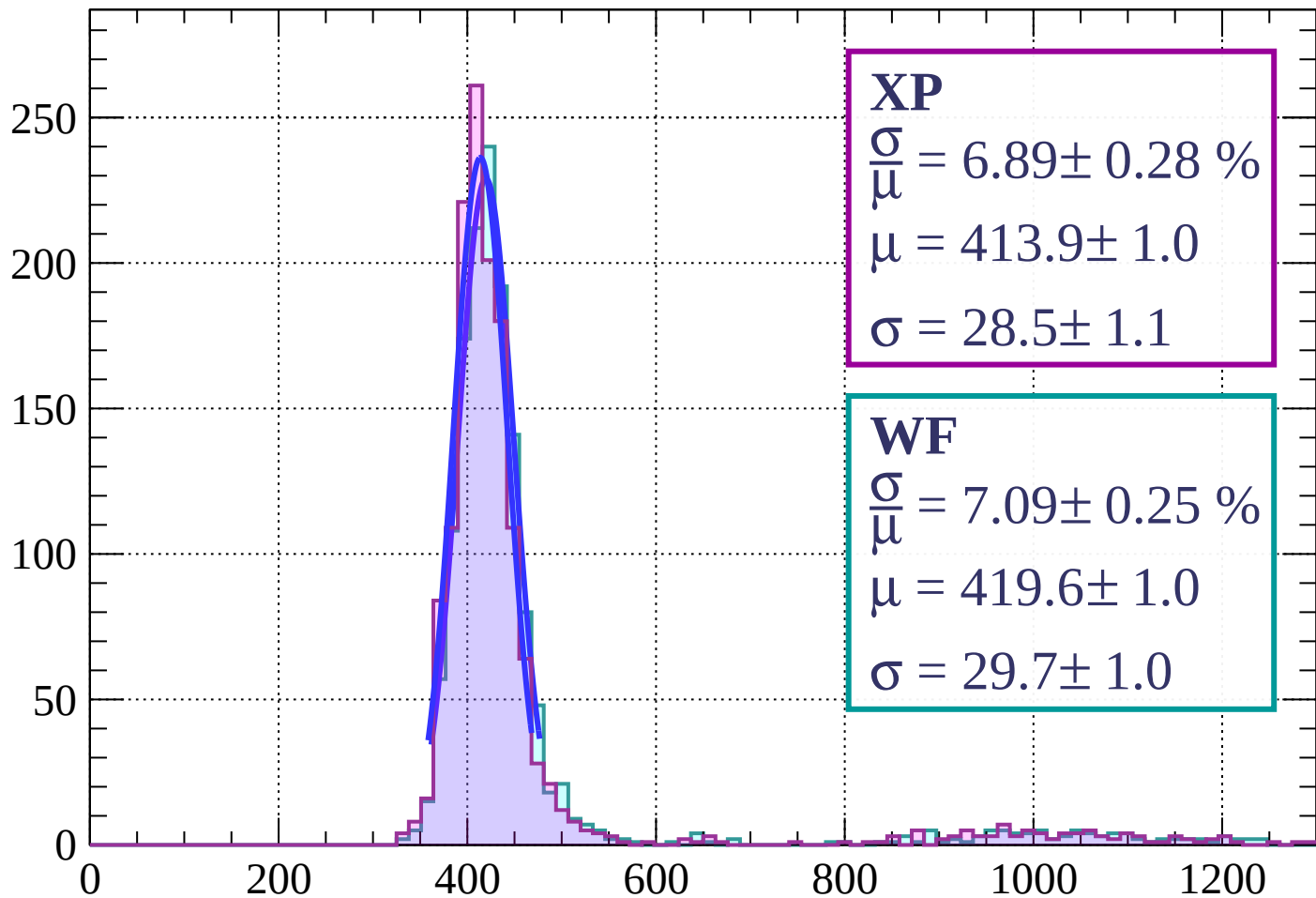
dE/dx (ADC counts/cm)

Energy loss | $12 < \phi < 15$



Energy loss | $15 < \phi < 18$

Count



XP

$$\frac{\sigma}{\mu} = 6.89 \pm 0.28 \%$$

$$\mu = 413.9 \pm 1.0$$

$$\sigma = 28.5 \pm 1.1$$

WF

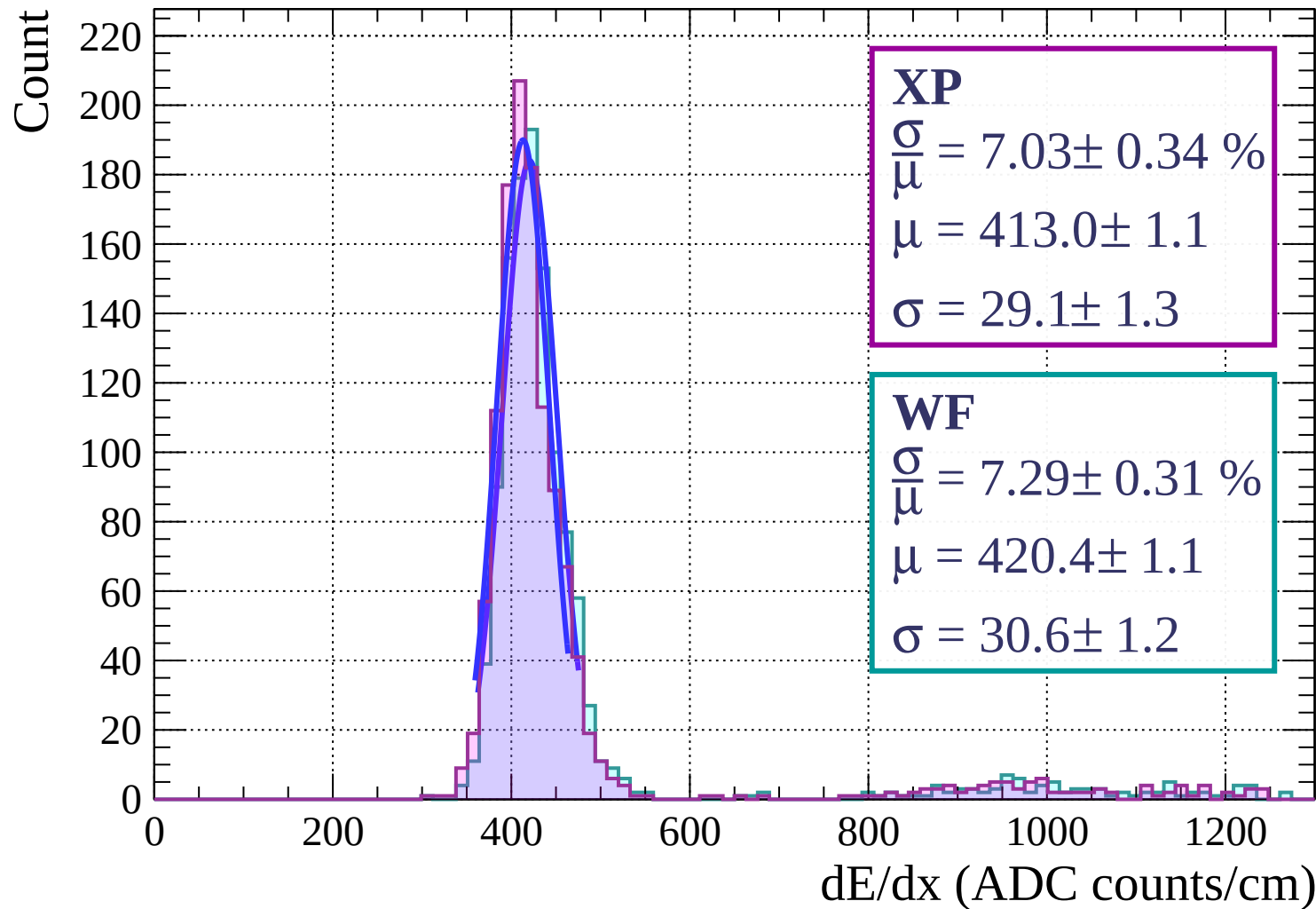
$$\frac{\sigma}{\mu} = 7.09 \pm 0.25 \%$$

$$\mu = 419.6 \pm 1.0$$

$$\sigma = 29.7 \pm 1.0$$

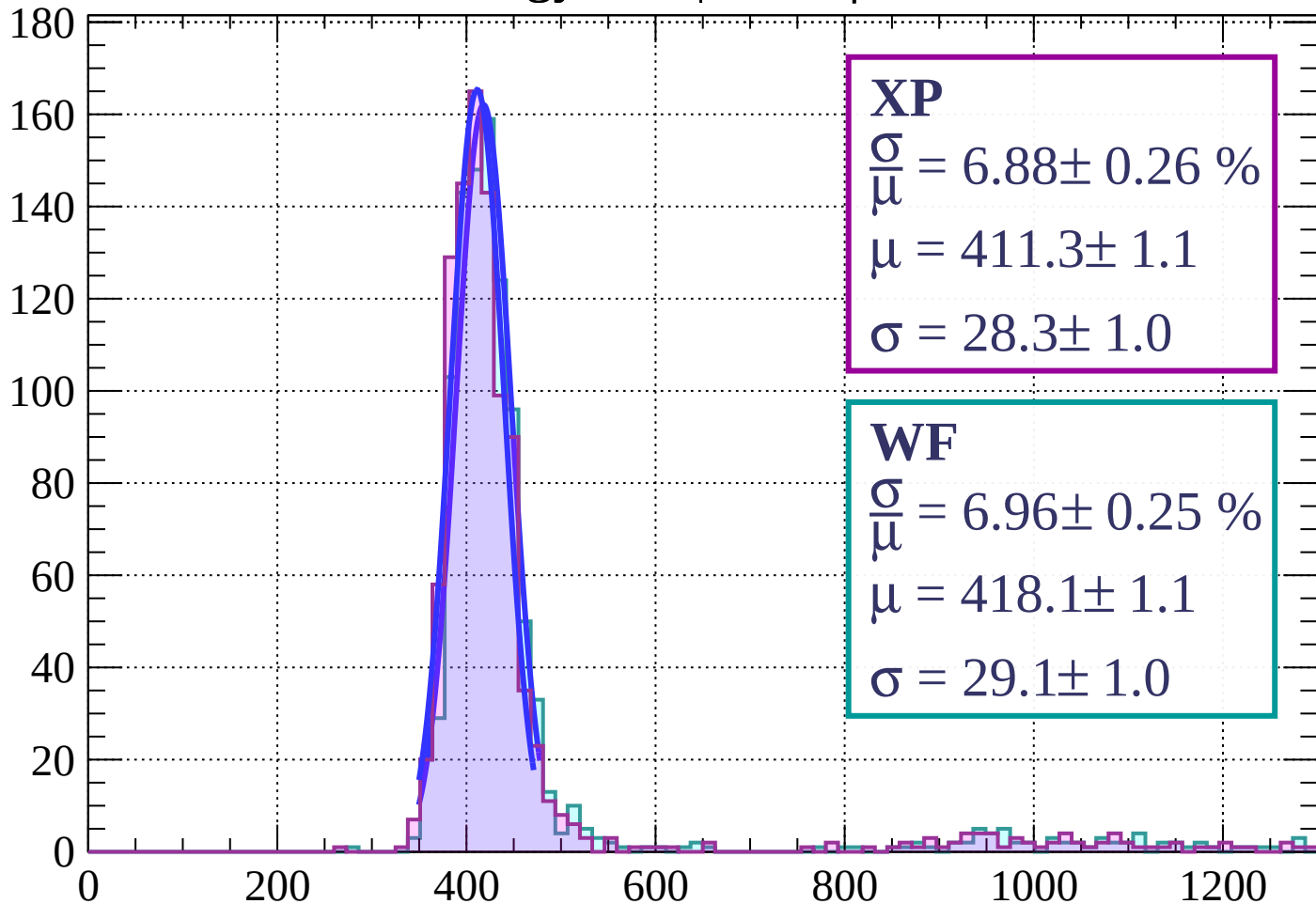
dE/dx (ADC counts/cm)

Energy loss | $18 < \phi < 21$



Energy loss | $21 < \phi < 24$

Count



XP

$$\frac{\sigma}{\mu} = 6.88 \pm 0.26 \%$$

$$\mu = 411.3 \pm 1.1$$

$$\sigma = 28.3 \pm 1.0$$

WF

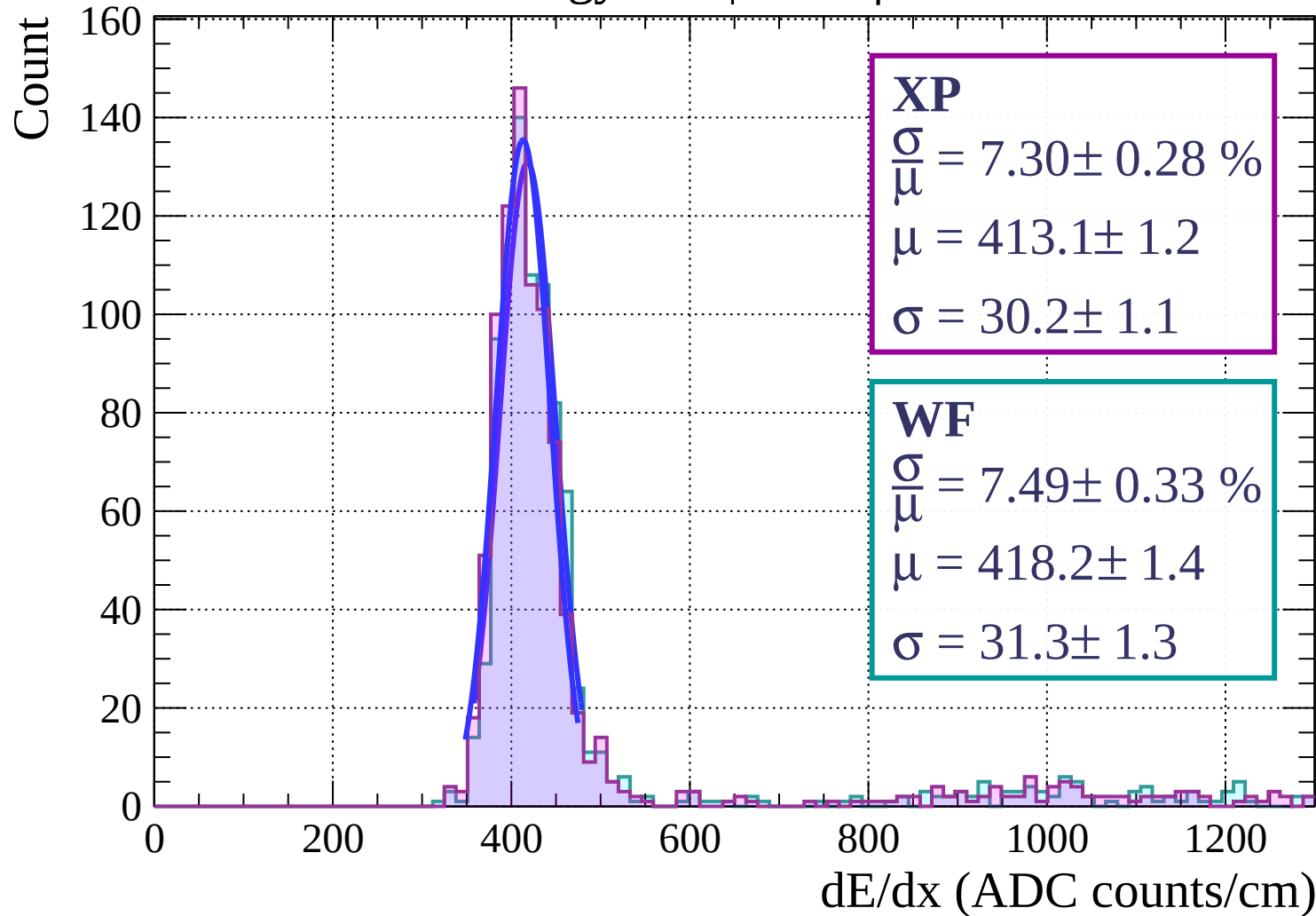
$$\frac{\sigma}{\mu} = 6.96 \pm 0.25 \%$$

$$\mu = 418.1 \pm 1.1$$

$$\sigma = 29.1 \pm 1.0$$

dE/dx (ADC counts/cm)

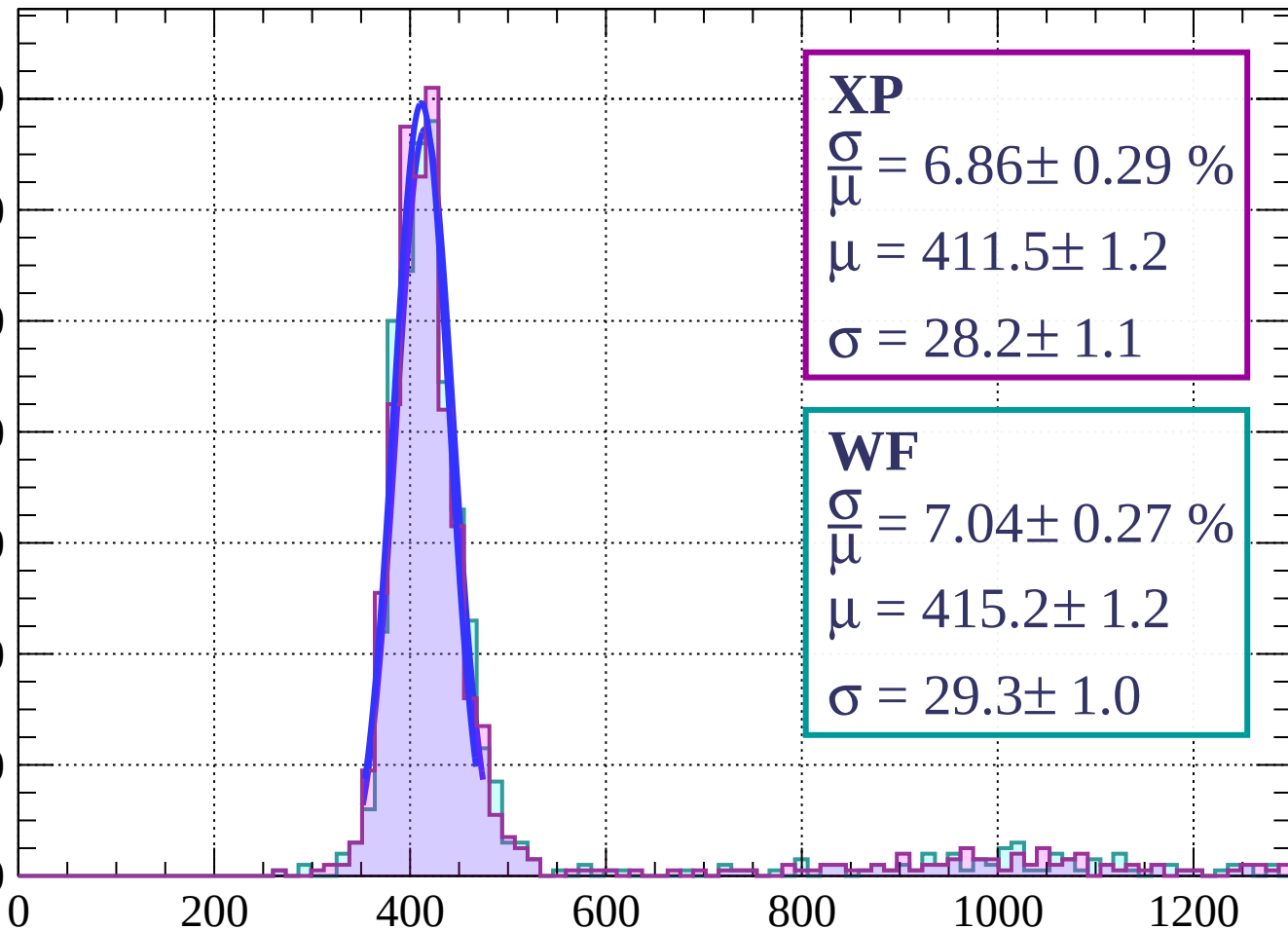
Energy loss | $24 < \phi < 27$



Energy loss | $27 < \phi < 30$

Count

140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 6.86 \pm 0.29 \%$$

$$\mu = 411.5 \pm 1.2$$

$$\sigma = 28.2 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 7.04 \pm 0.27 \%$$

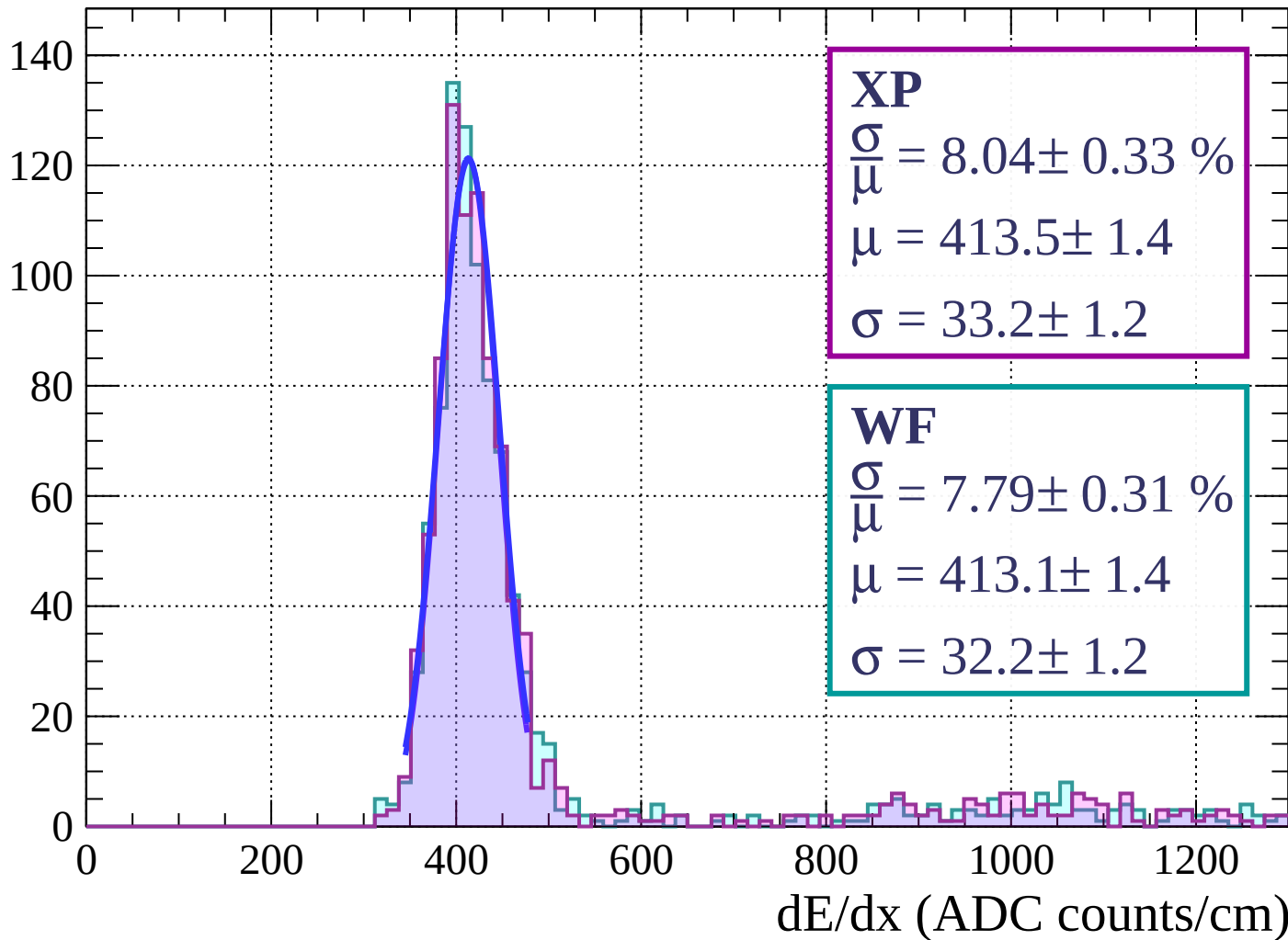
$$\mu = 415.2 \pm 1.2$$

$$\sigma = 29.3 \pm 1.0$$

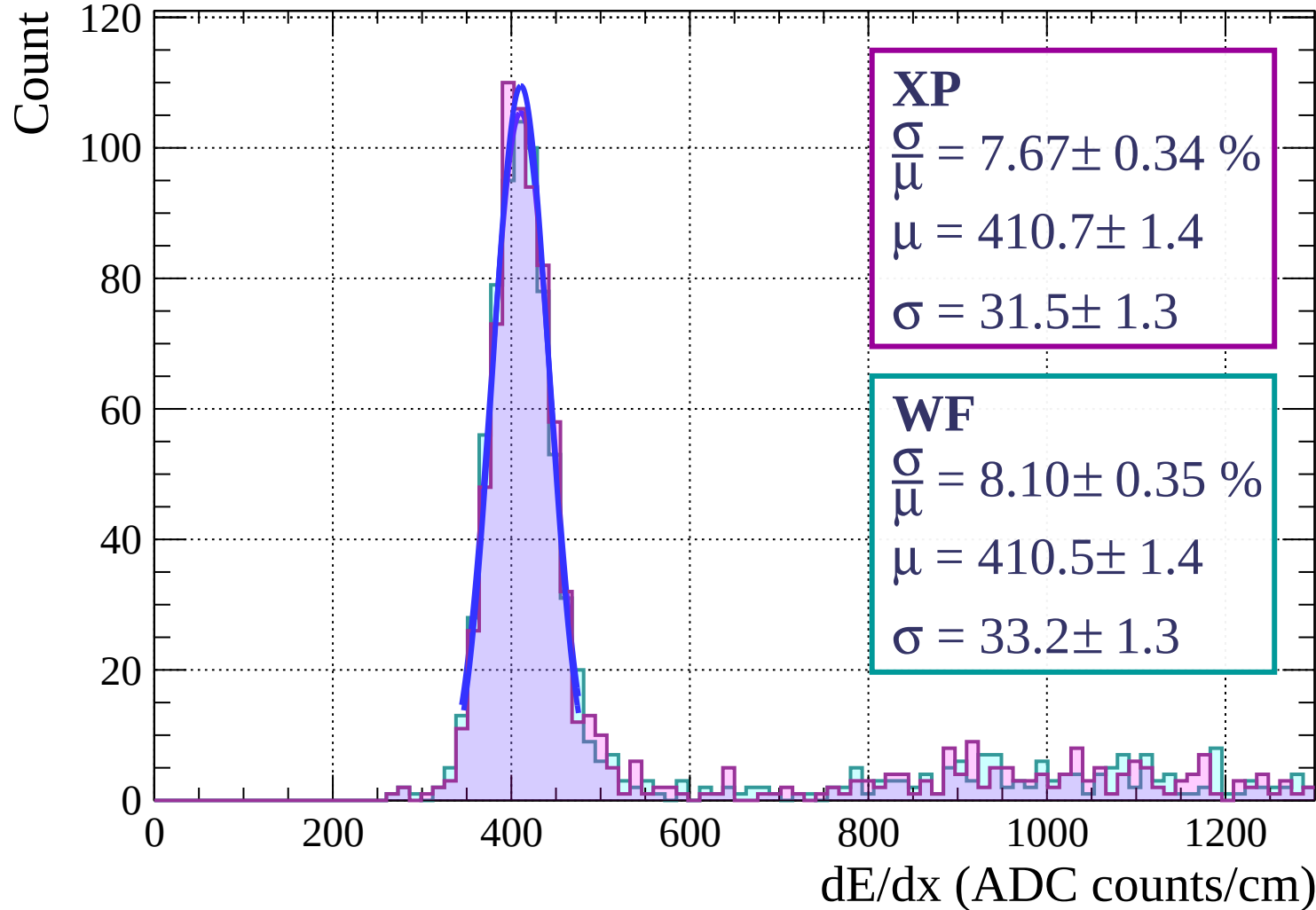
dE/dx (ADC counts/cm)

Energy loss | $30 < \phi < 33$

Count

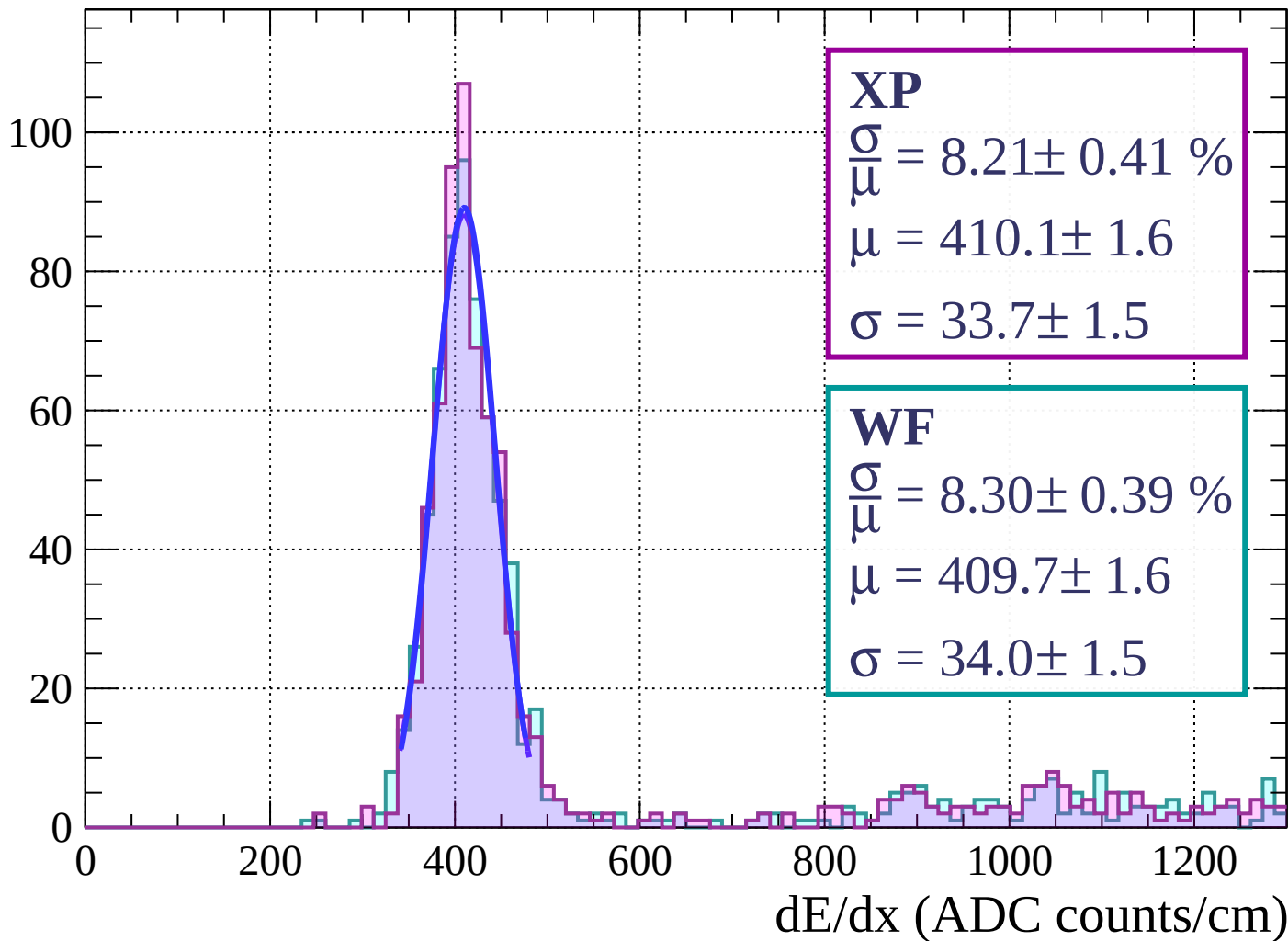


Energy loss | $33 < \phi < 36$



Energy loss | $36 < \phi < 39$

Count



Energy loss | $39 < \phi < 42$

Count

XP

$$\frac{\sigma}{\mu} = 7.62 \pm 0.50 \%$$

$$\mu = 407.1 \pm 1.8$$

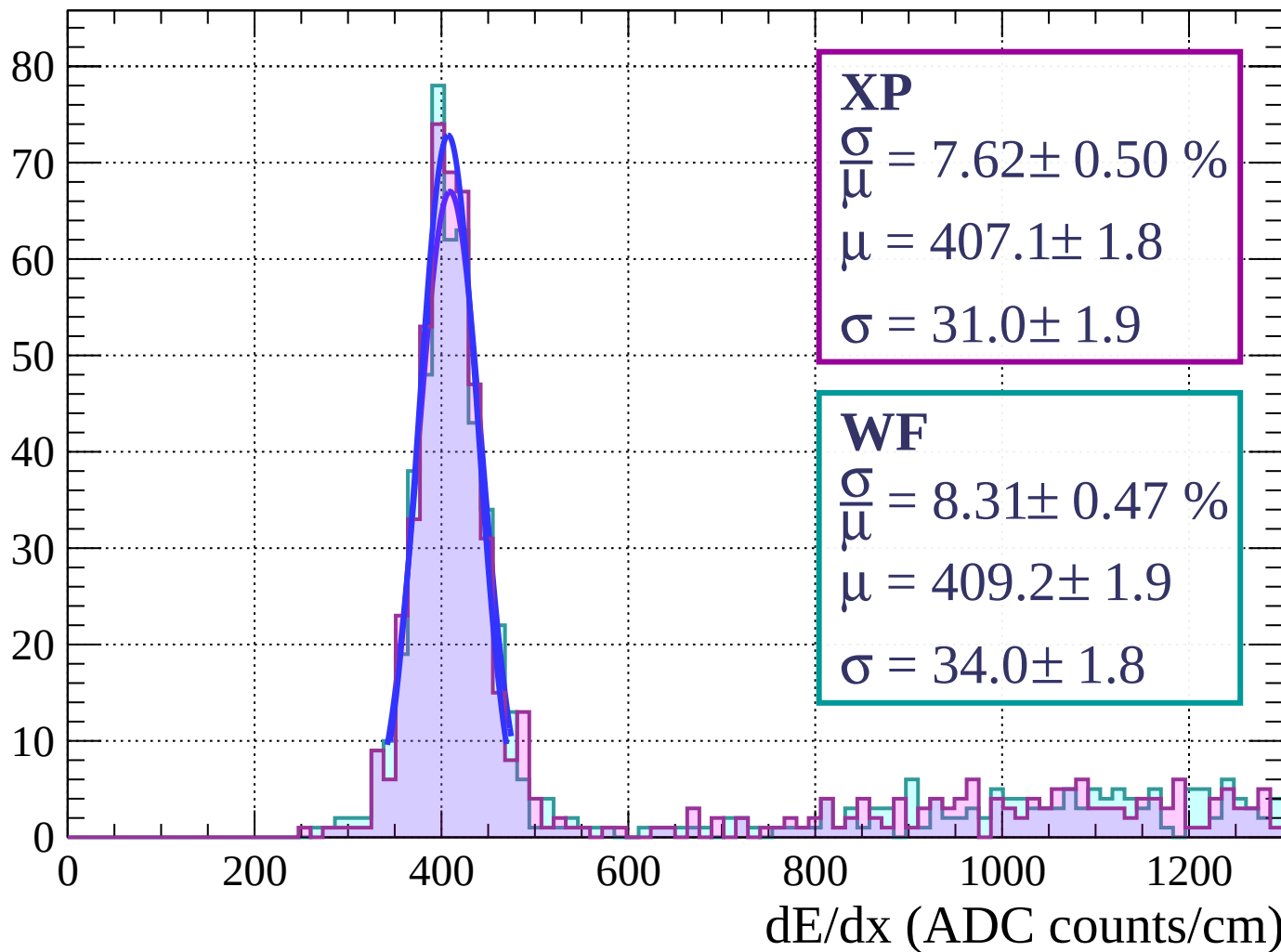
$$\sigma = 31.0 \pm 1.9$$

WF

$$\frac{\sigma}{\mu} = 8.31 \pm 0.47 \%$$

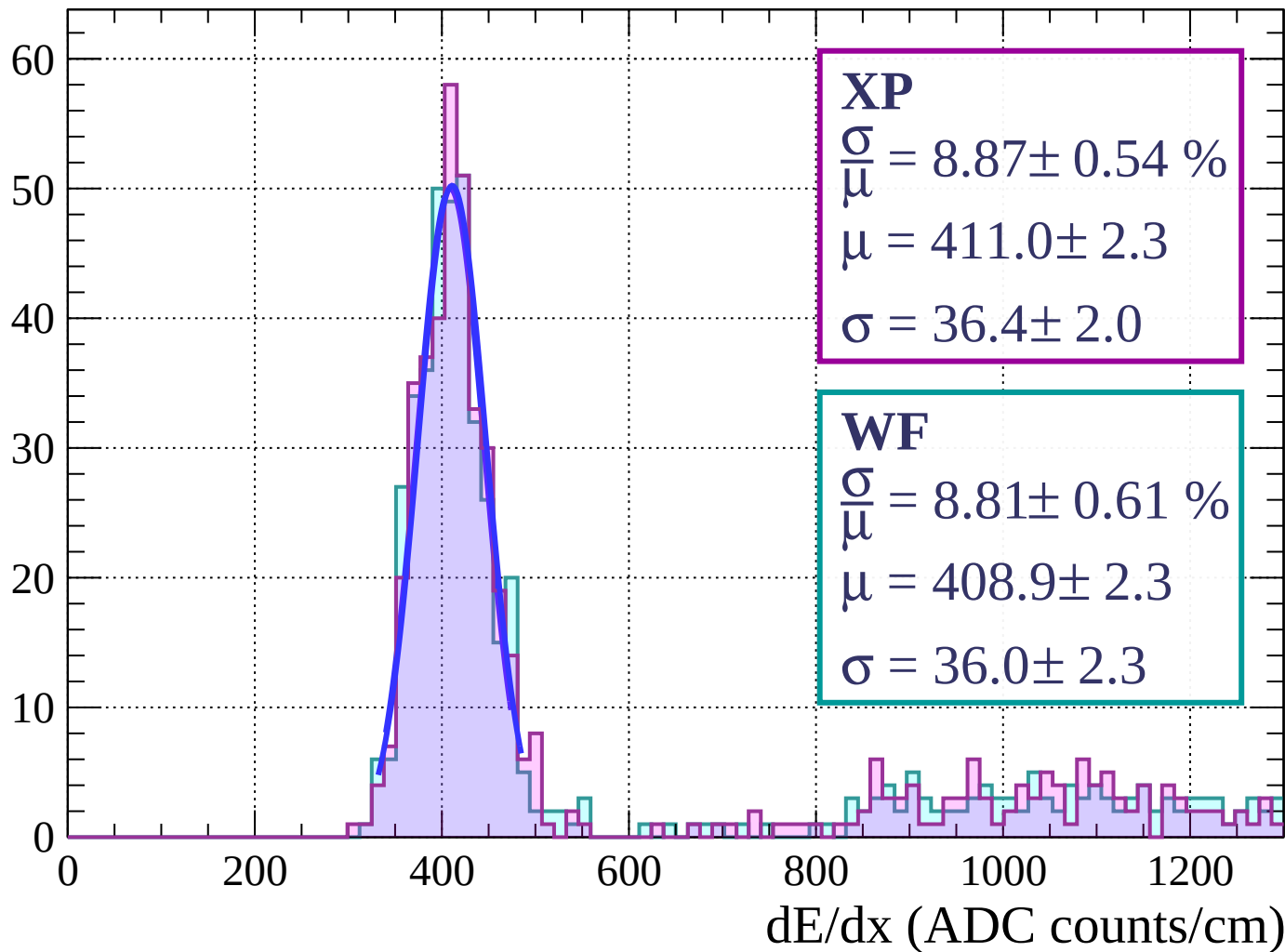
$$\mu = 409.2 \pm 1.9$$

$$\sigma = 34.0 \pm 1.8$$



Energy loss | $42 < \phi < 45$

Count



Energy loss | $45 < \phi < 48$

Count

XP

$$\frac{\sigma}{\mu} = 7.67 \pm 0.55 \%$$

$$\mu = 406.4 \pm 2.1$$

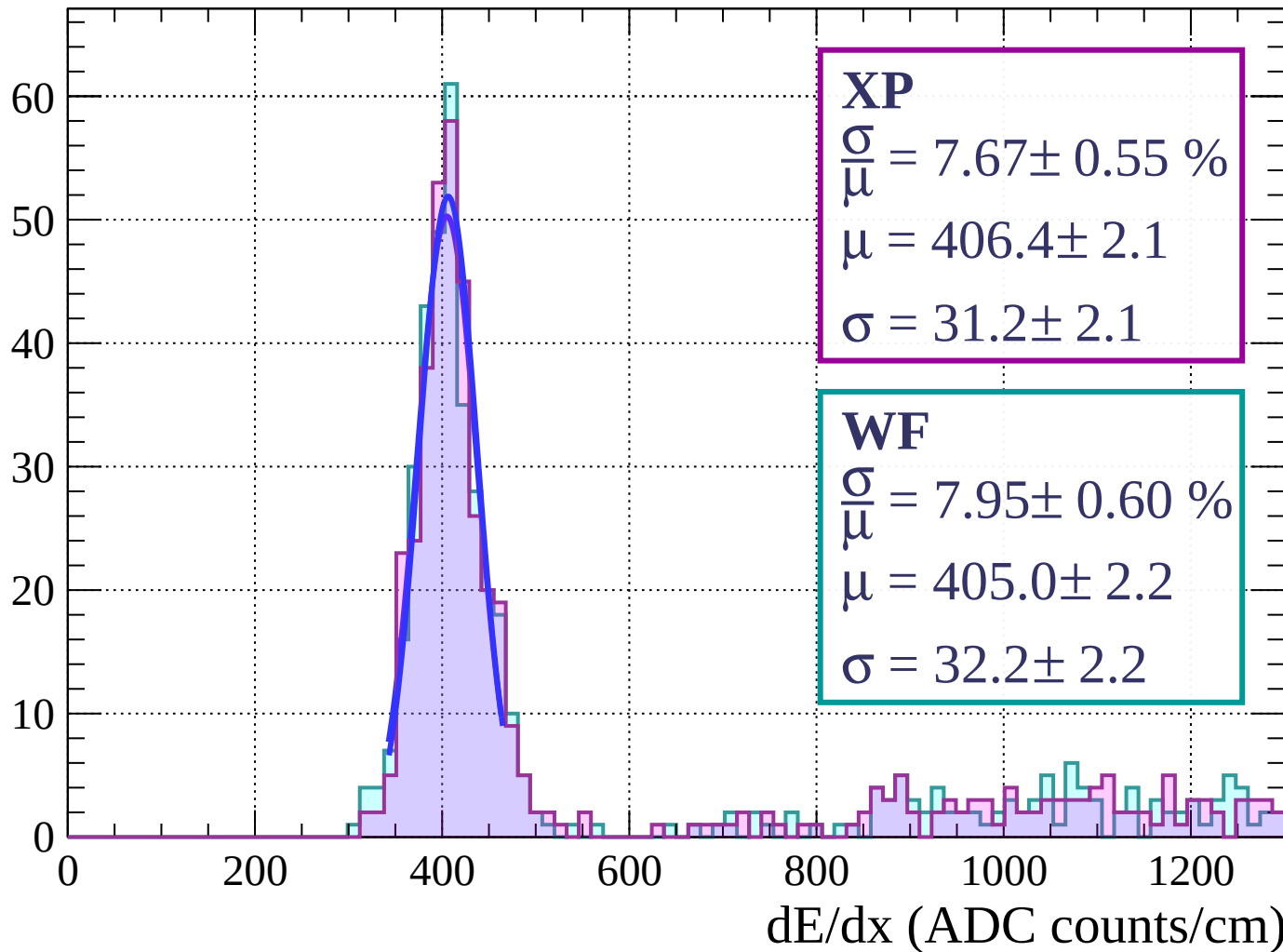
$$\sigma = 31.2 \pm 2.1$$

WF

$$\frac{\sigma}{\mu} = 7.95 \pm 0.60 \%$$

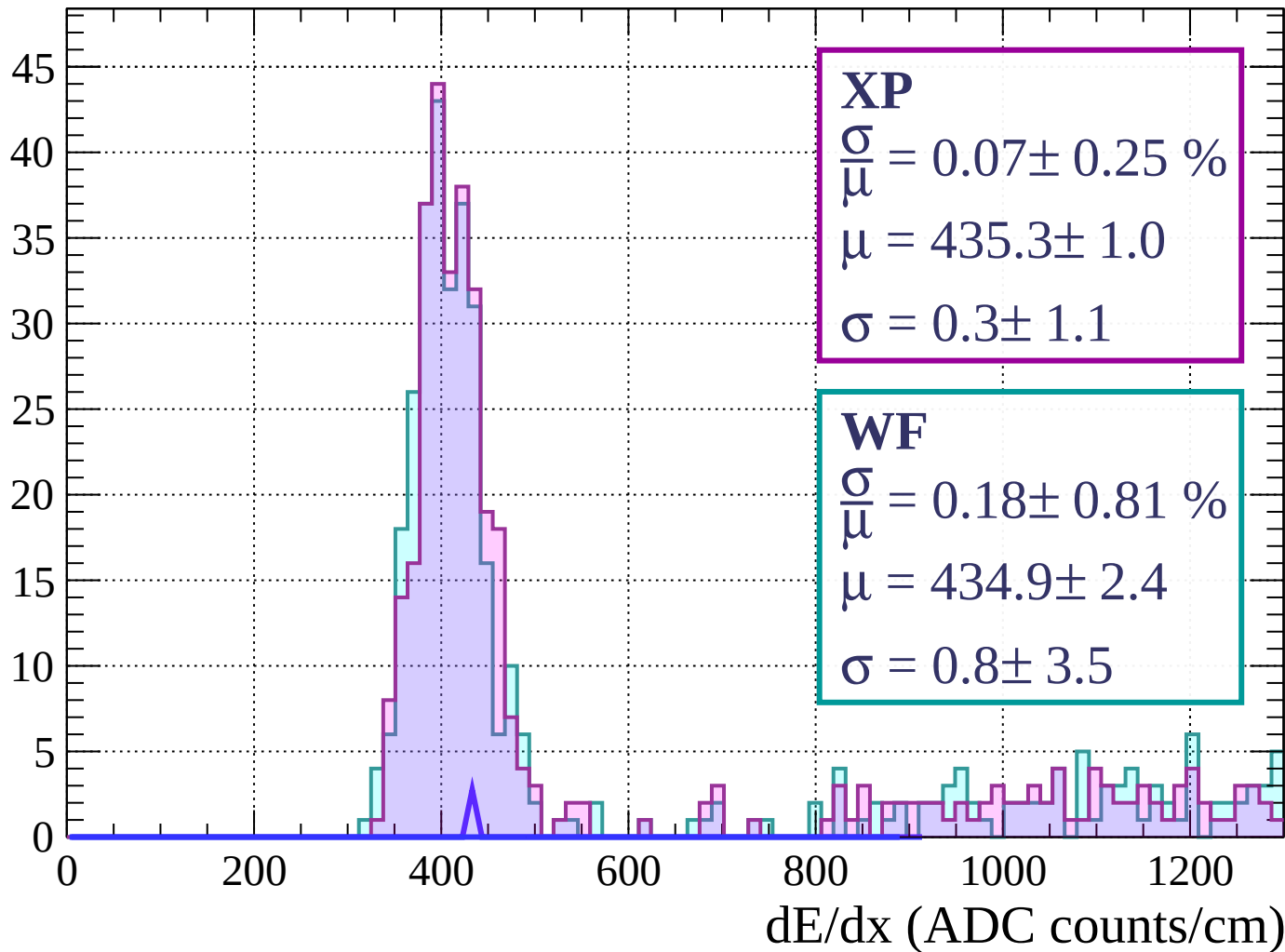
$$\mu = 405.0 \pm 2.2$$

$$\sigma = 32.2 \pm 2.2$$



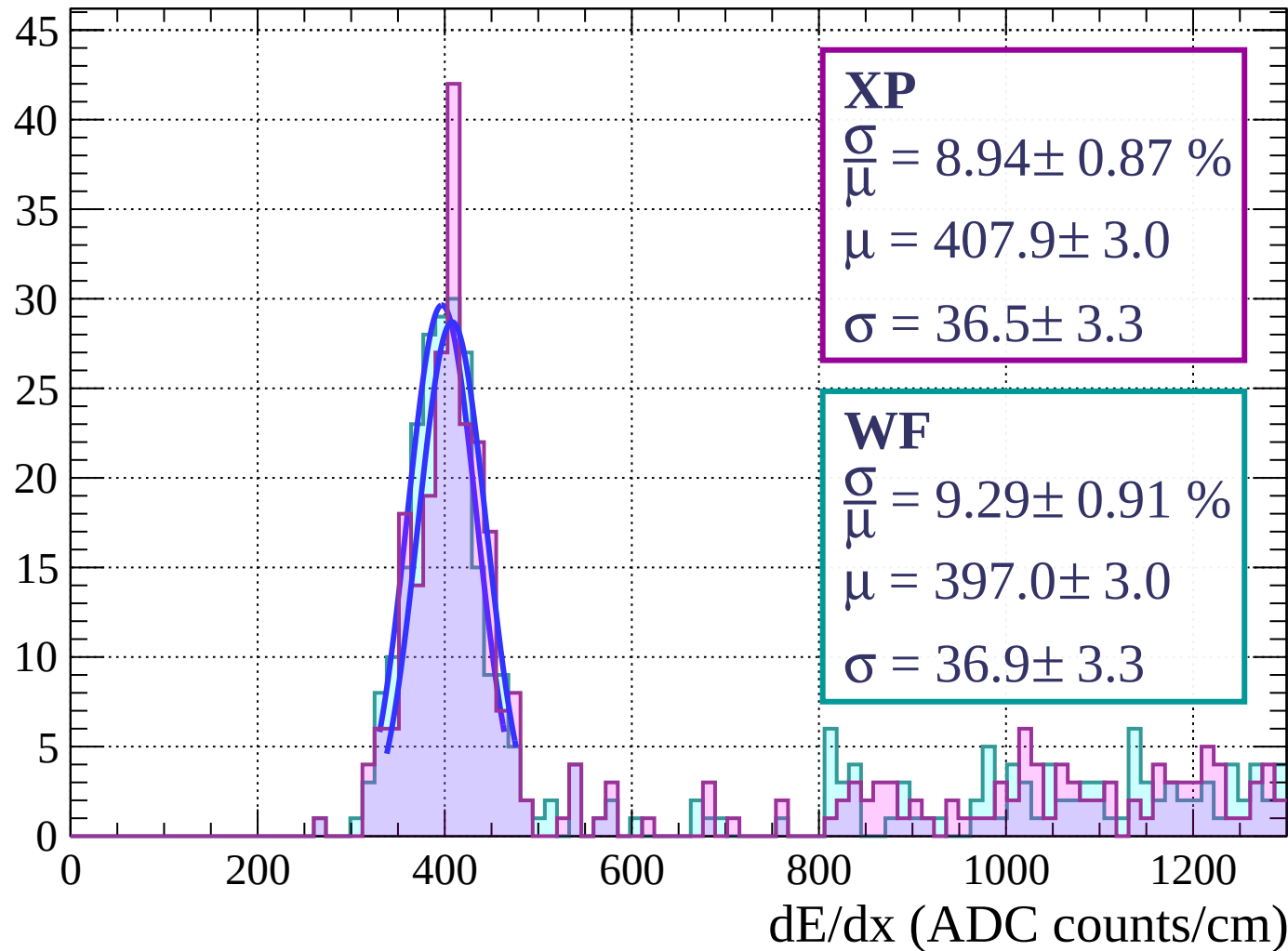
Energy loss | $48 < \phi < 51$

Count



Energy loss | $51 < \phi < 54$

Count



Energy loss | $54 < \phi < 57$

Count

XP

$$\frac{\sigma}{\mu} = 6.23 \pm 0.59 \%$$

$$\mu = 404.7 \pm 2.4$$

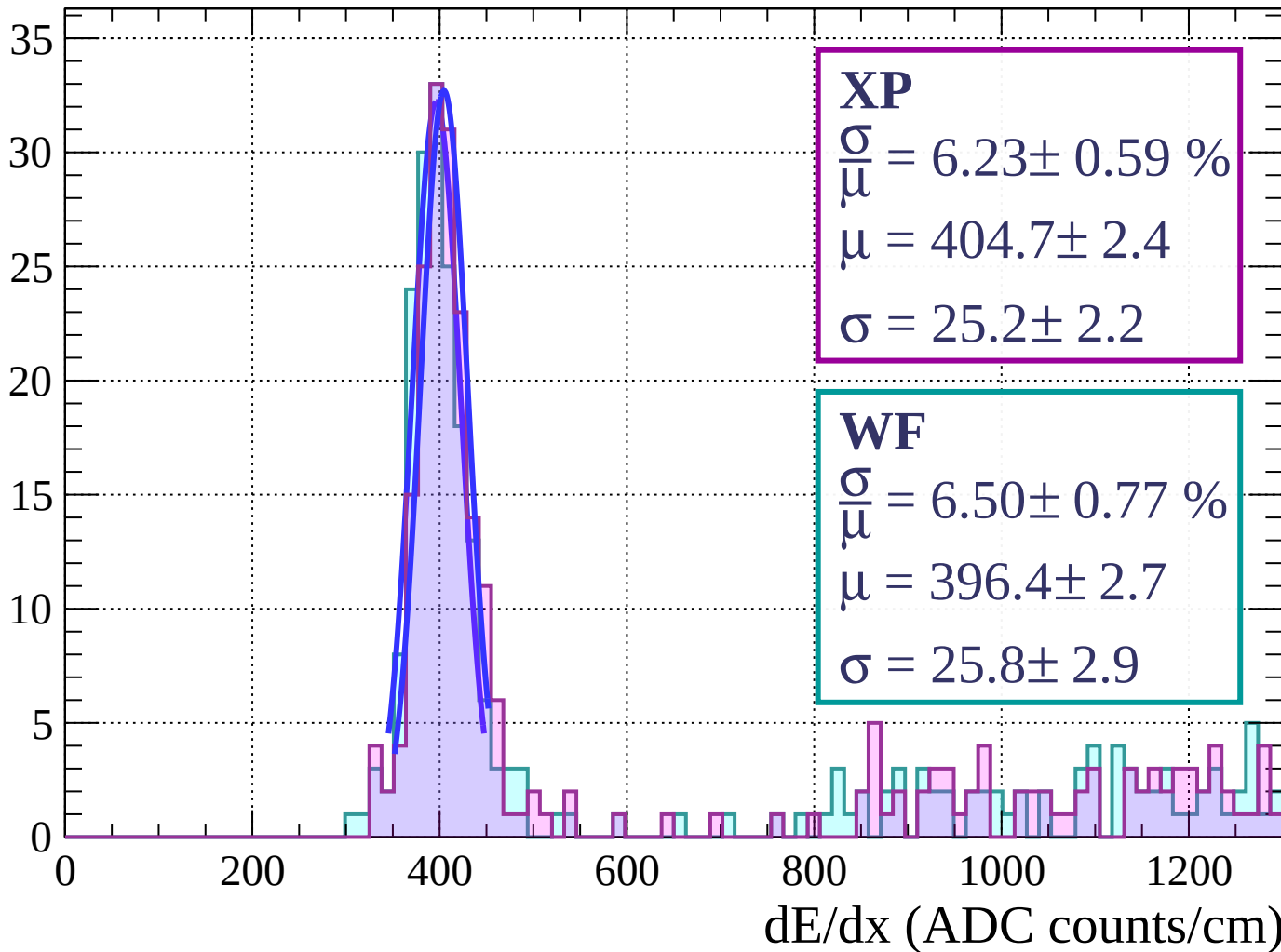
$$\sigma = 25.2 \pm 2.2$$

WF

$$\frac{\sigma}{\mu} = 6.50 \pm 0.77 \%$$

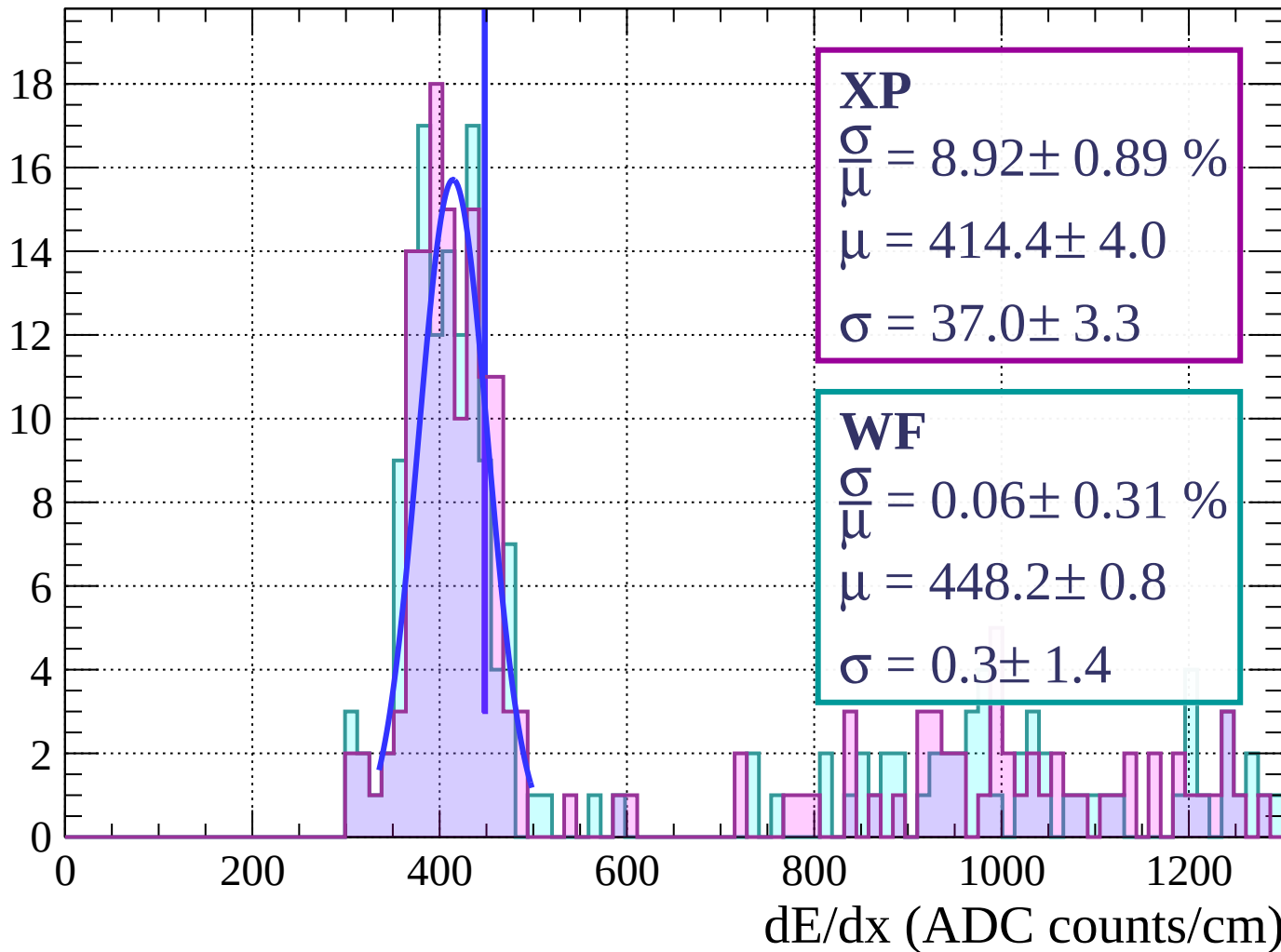
$$\mu = 396.4 \pm 2.7$$

$$\sigma = 25.8 \pm 2.9$$



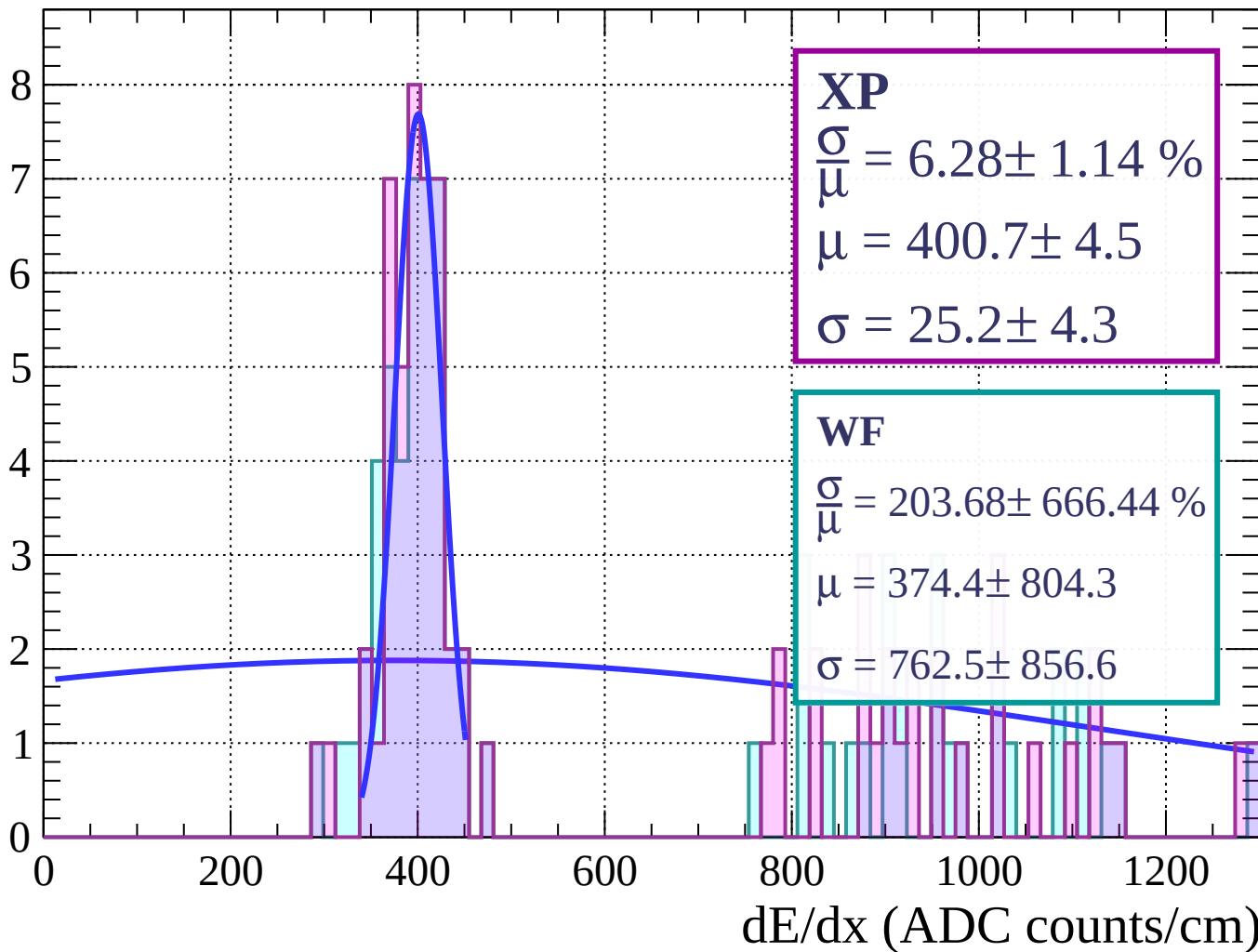
Energy loss | $57 < \phi < 60$

Count



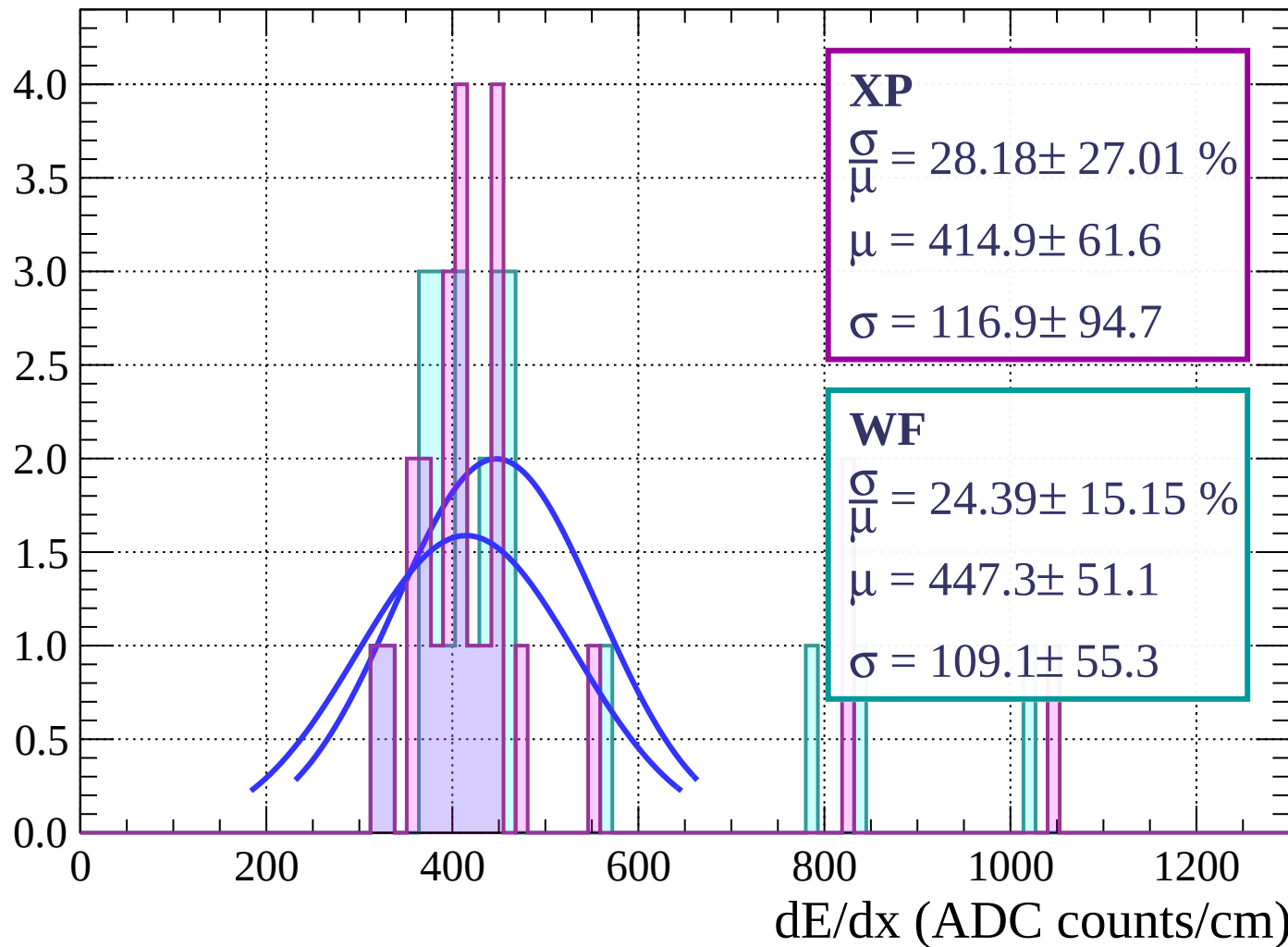
Energy loss | $60 < \phi < 63$

Count



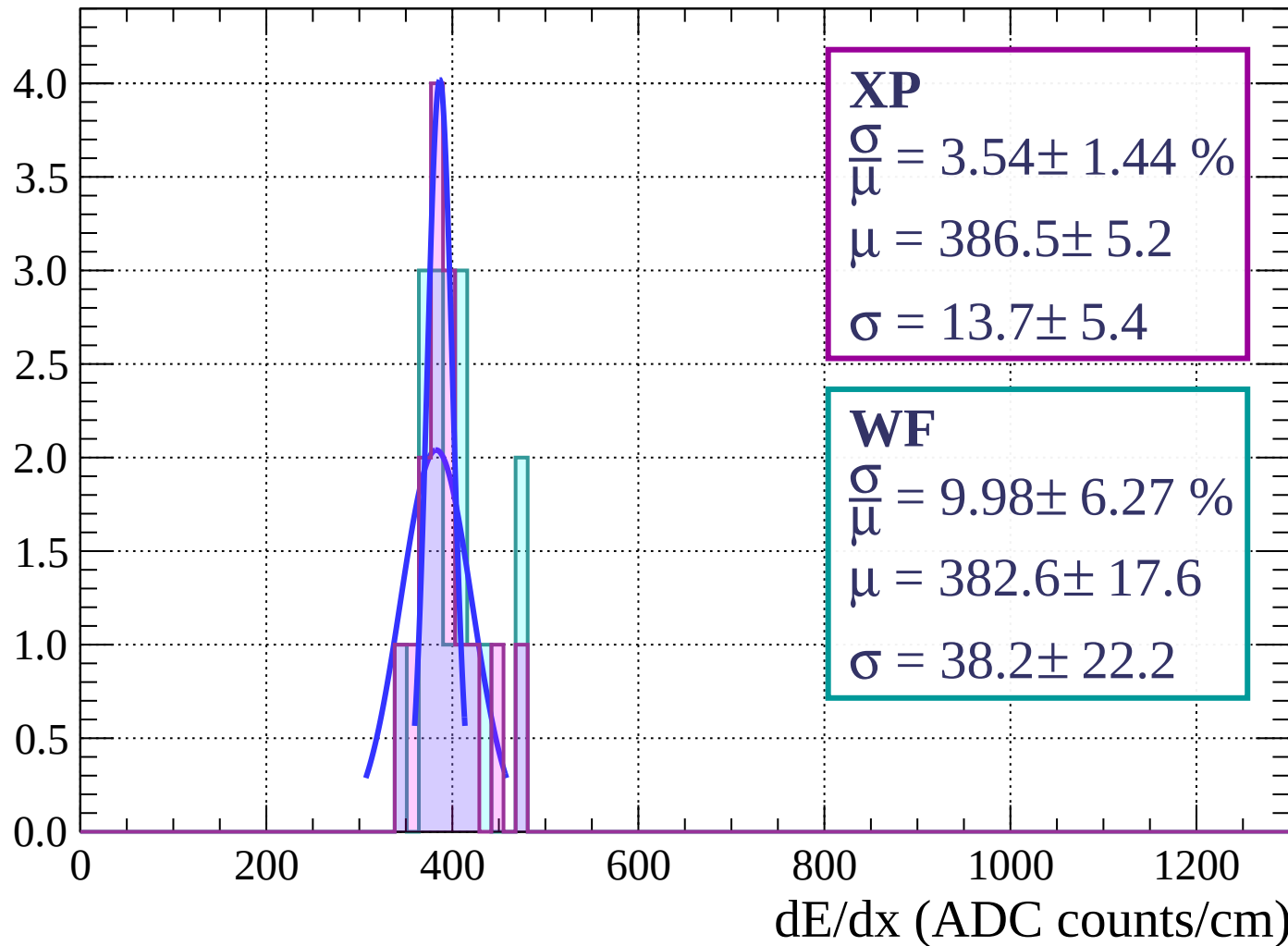
Energy loss | $63 < \phi < 66$

Count



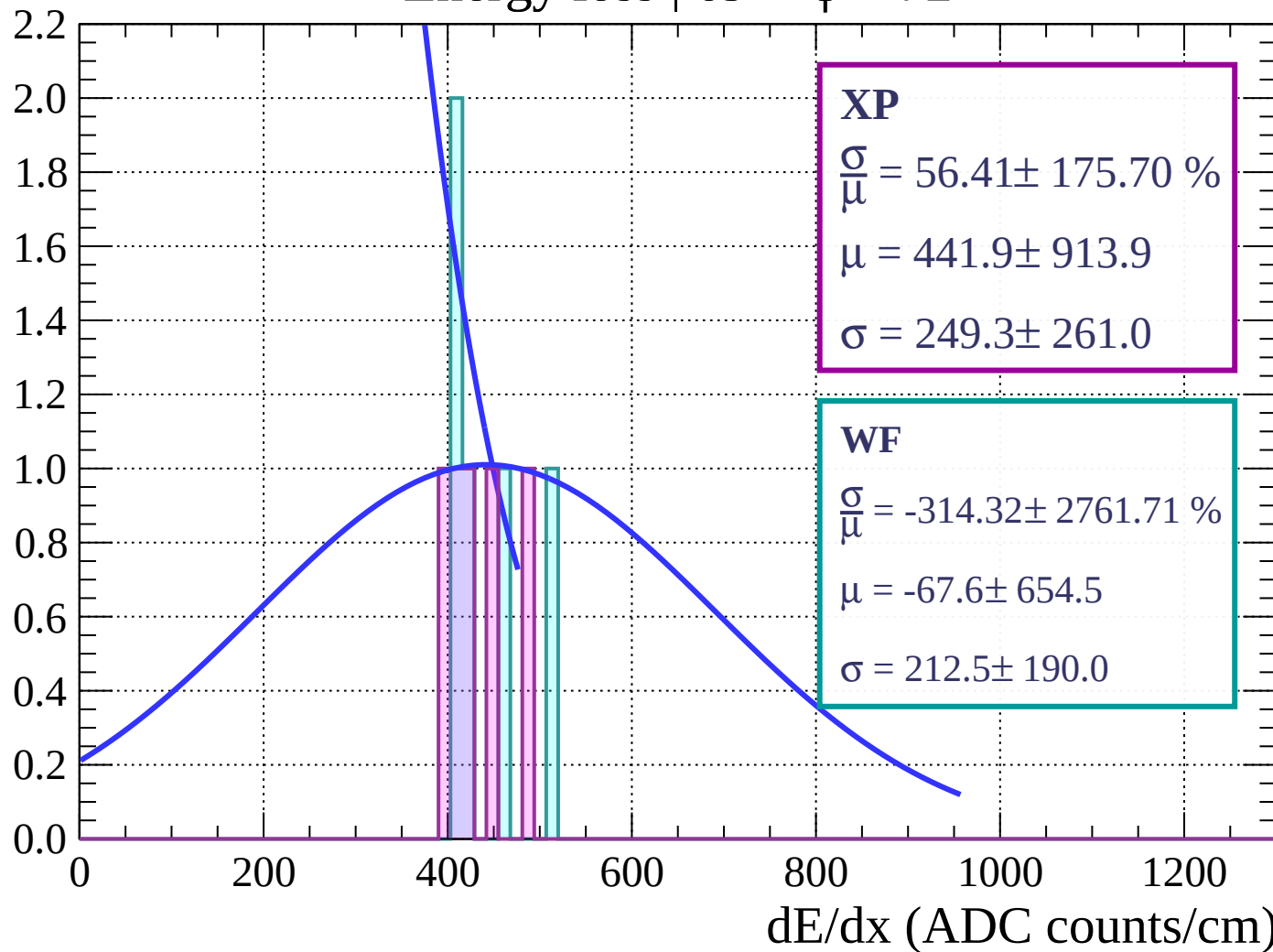
Energy loss | $66 < \phi < 69$

Count

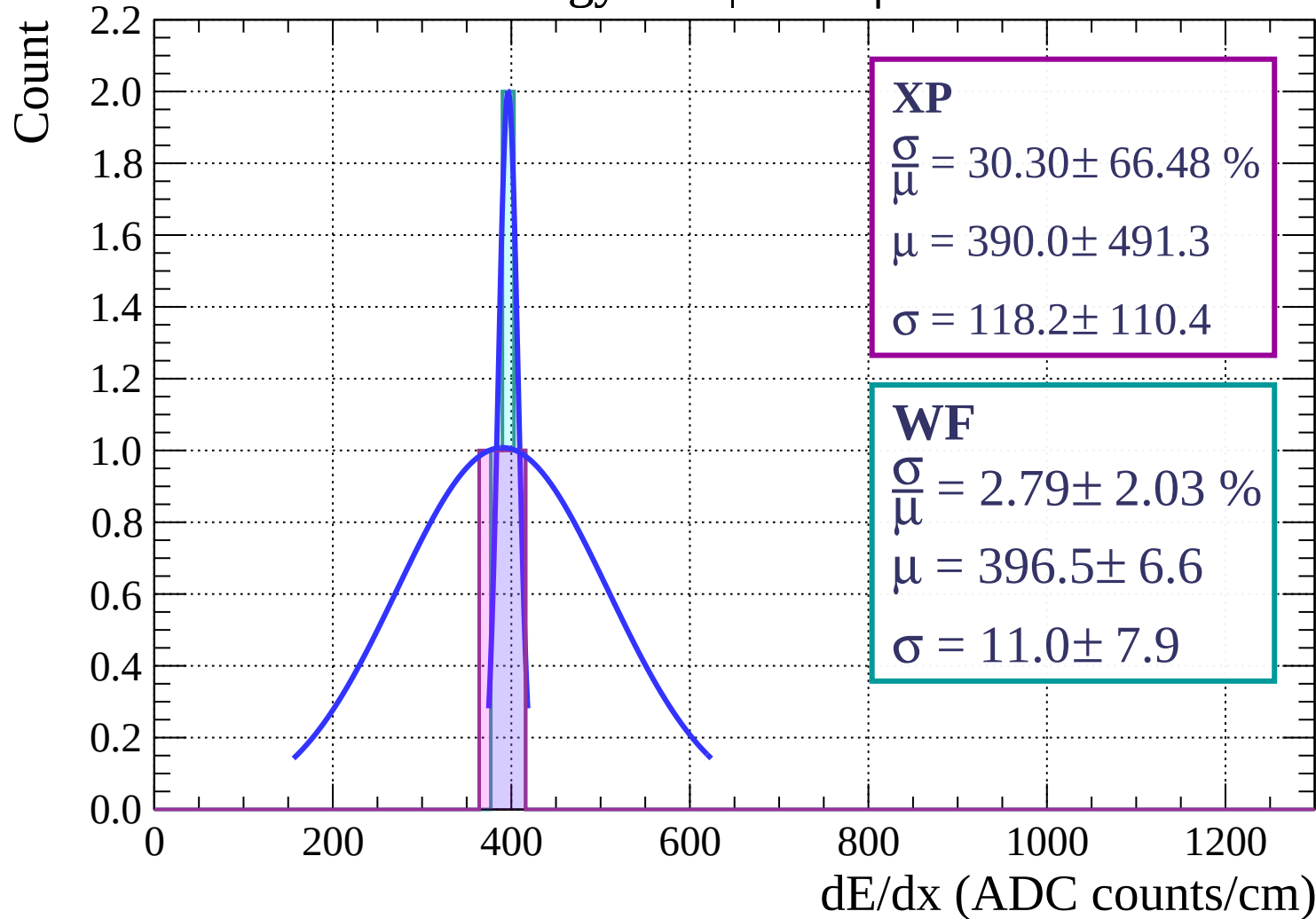


Energy loss | $69 < \phi < 72$

Count

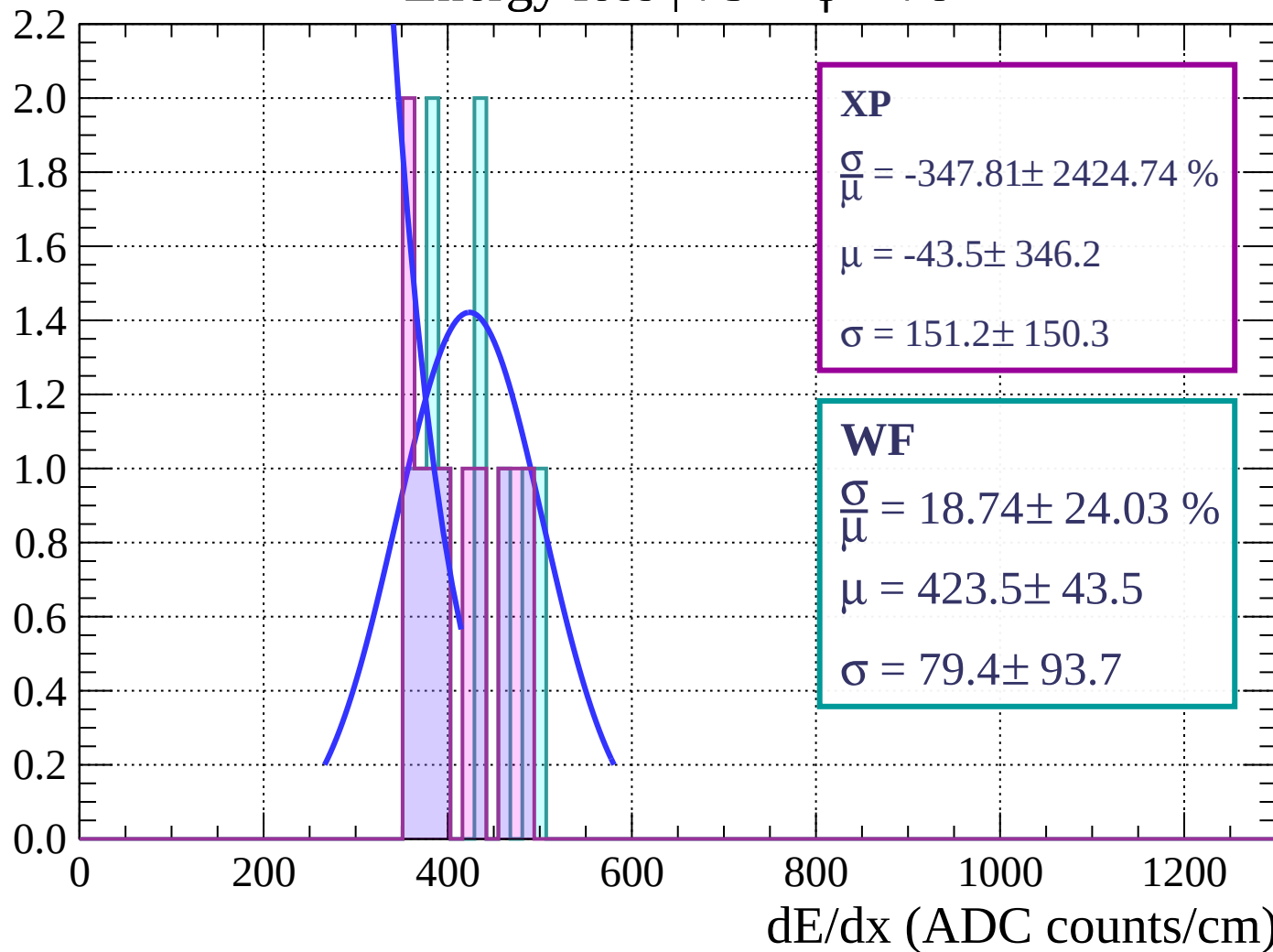


Energy loss | $72 < \phi < 75$



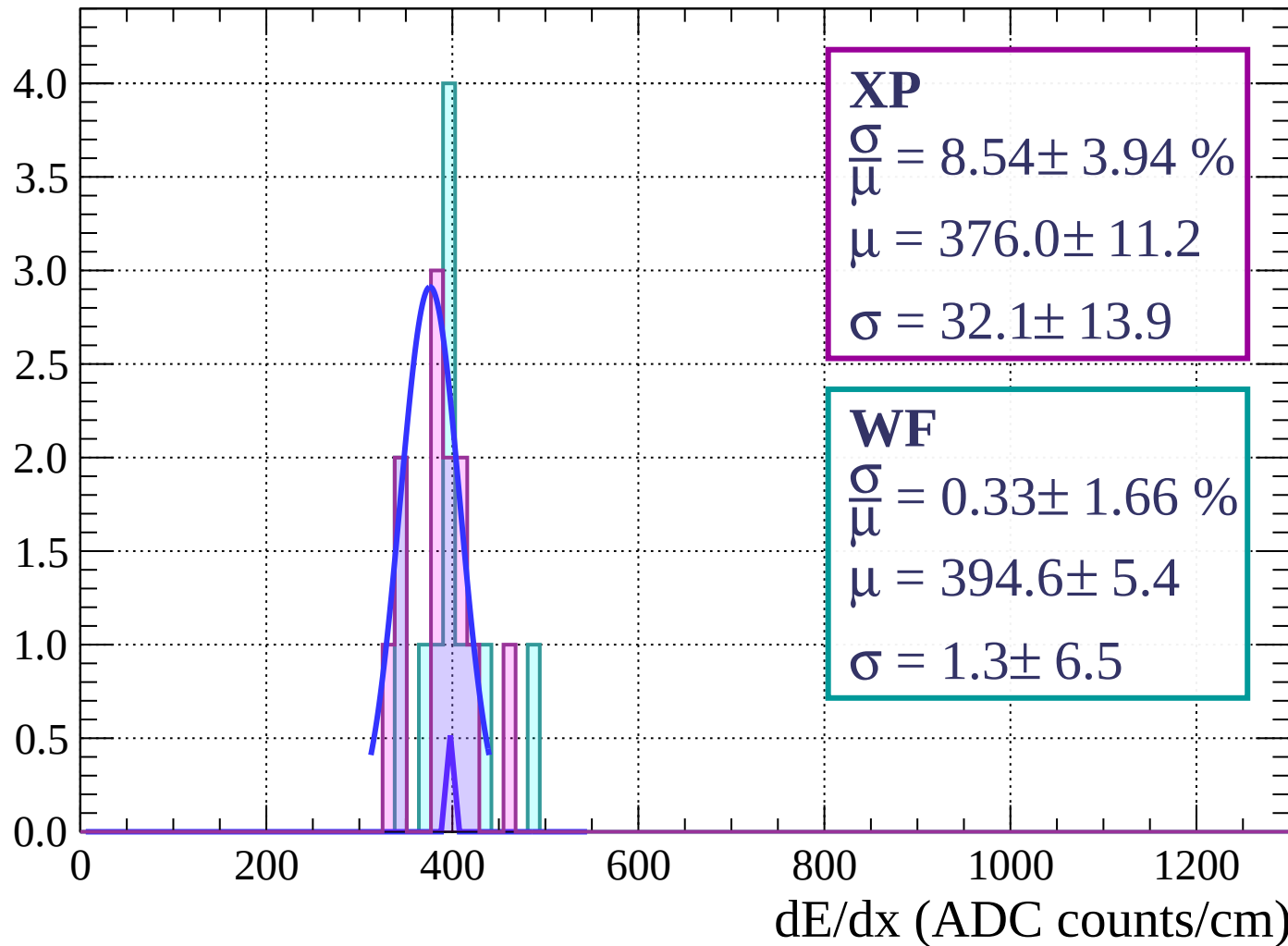
Energy loss | $75 < \phi < 78$

Count



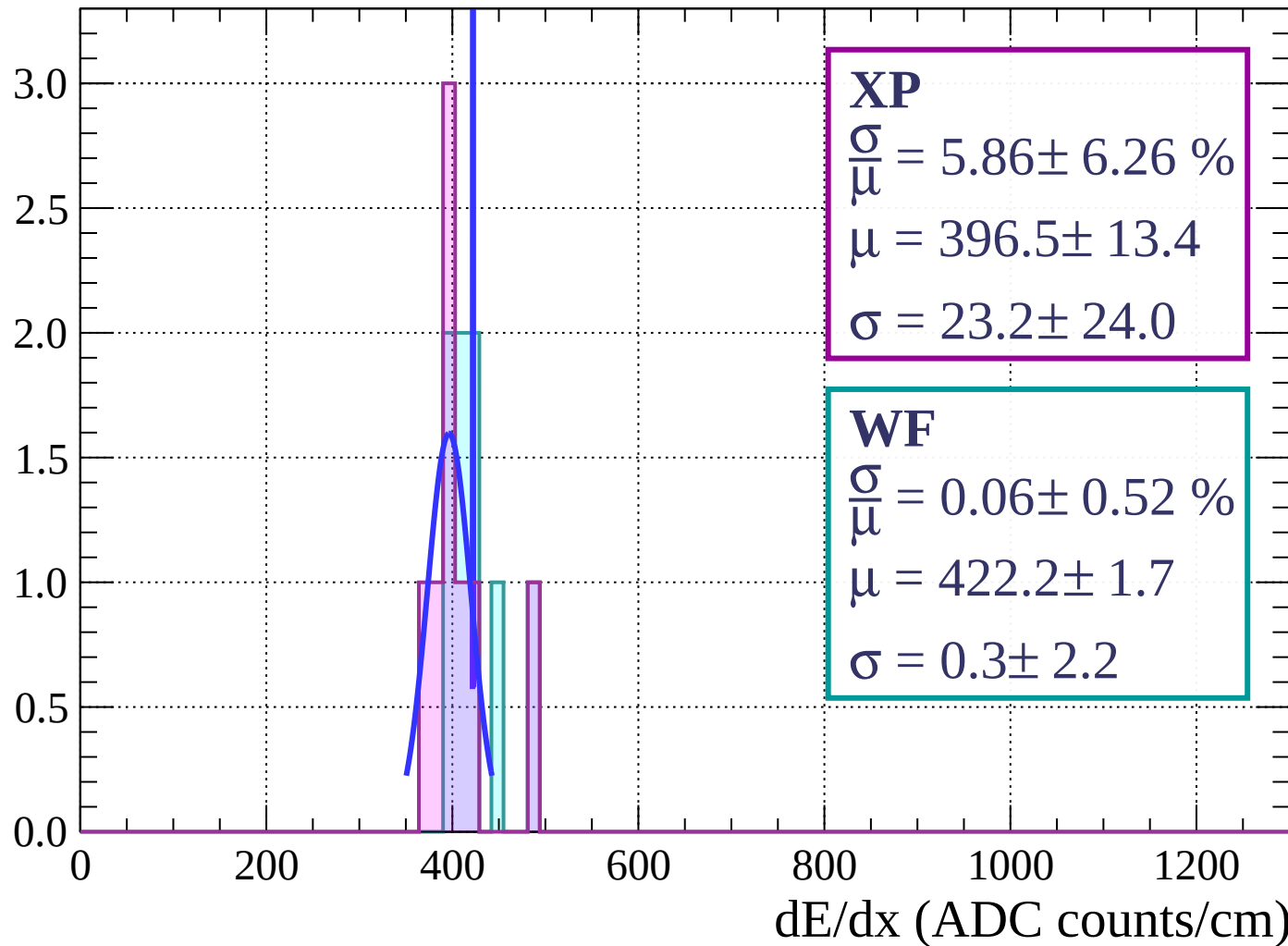
Energy loss | $78 < \phi < 81$

Count



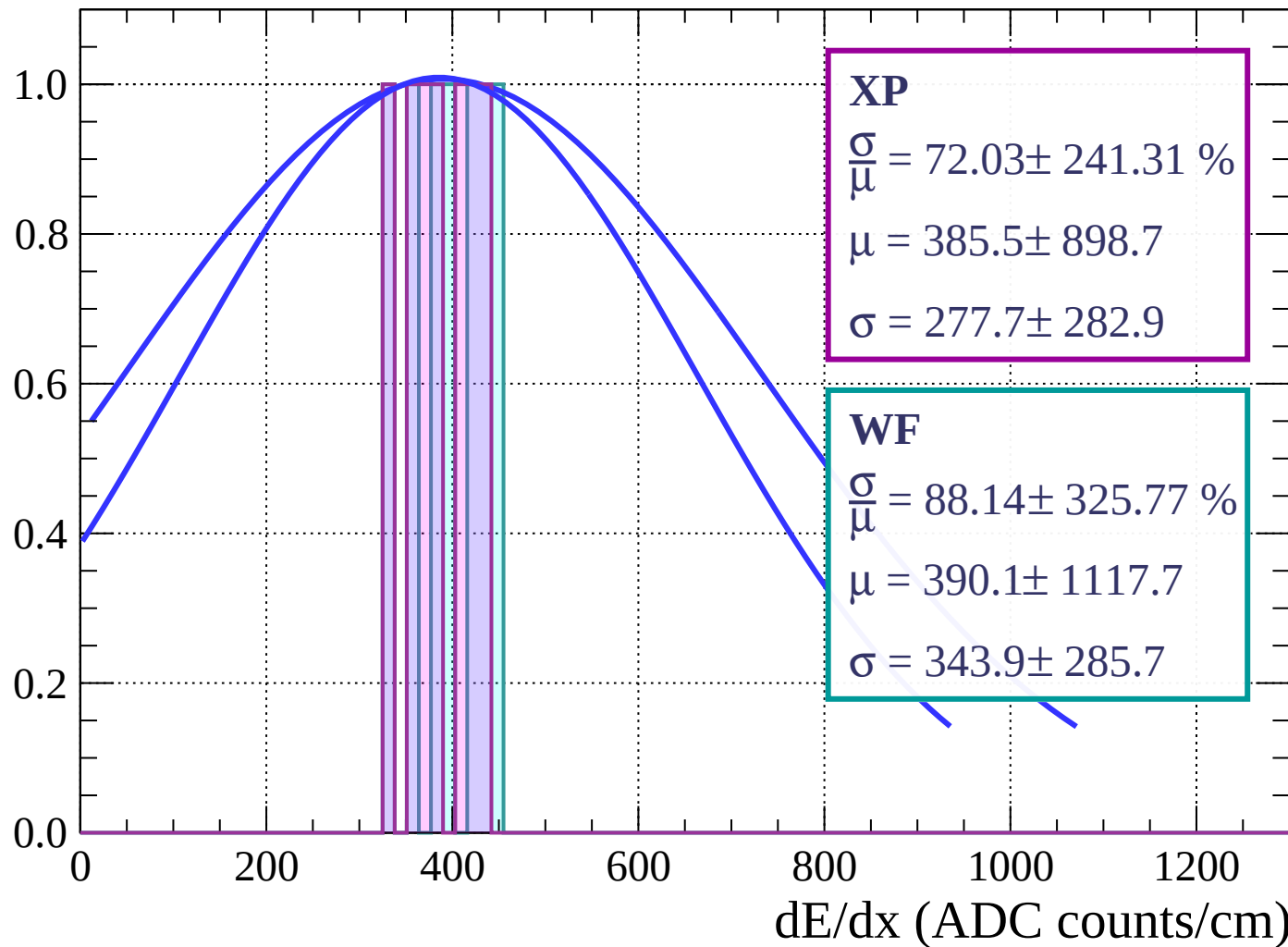
Energy loss | $81 < \phi < 84$

Count



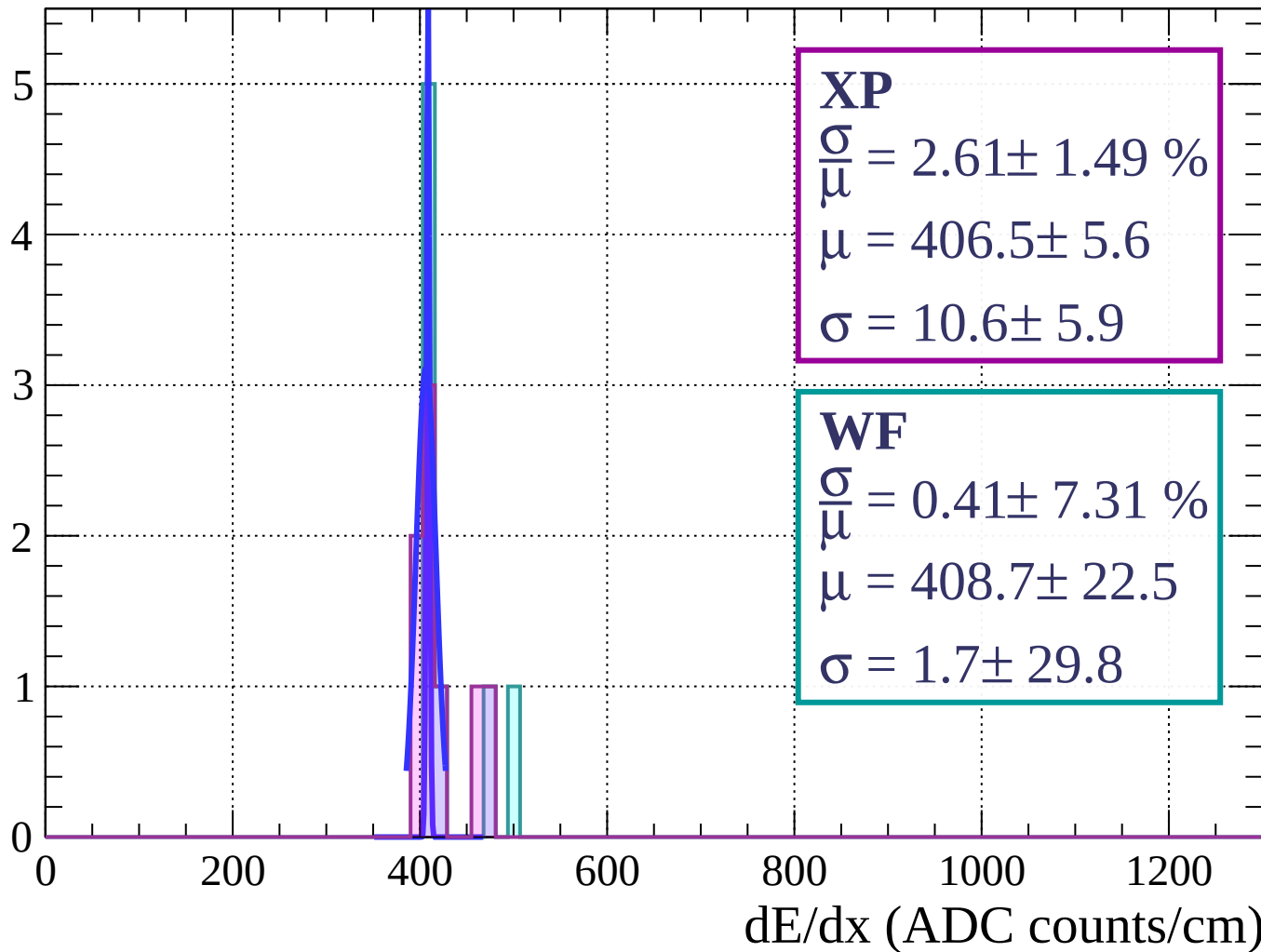
Energy loss | $84 < \phi < 87$

Count



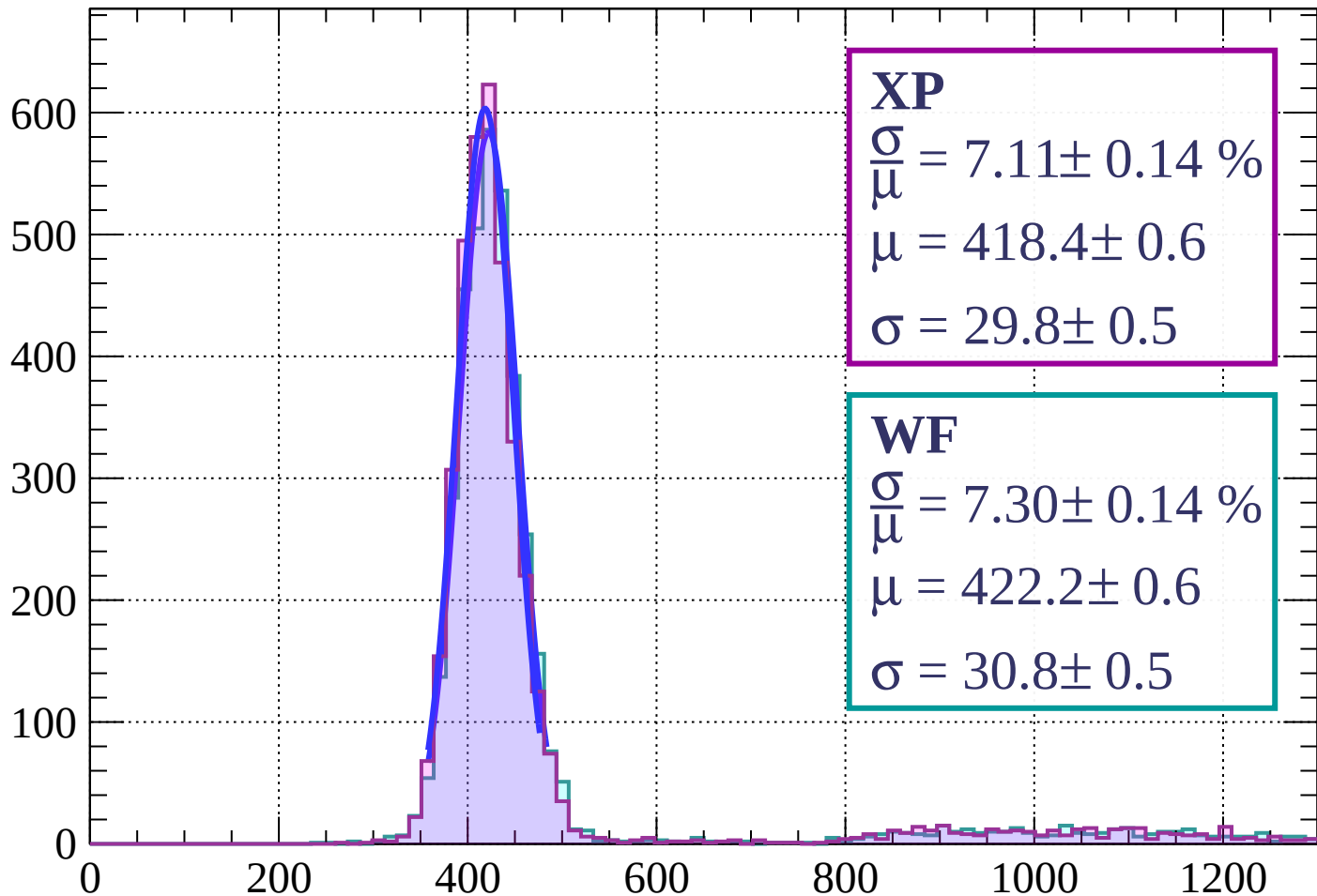
Energy loss | $87 < \phi < 90$

Count



Energy loss | $0 < \theta < 3$

Count



XP

$$\frac{\sigma}{\mu} = 7.11 \pm 0.14 \%$$

$$\mu = 418.4 \pm 0.6$$

$$\sigma = 29.8 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 7.30 \pm 0.14 \%$$

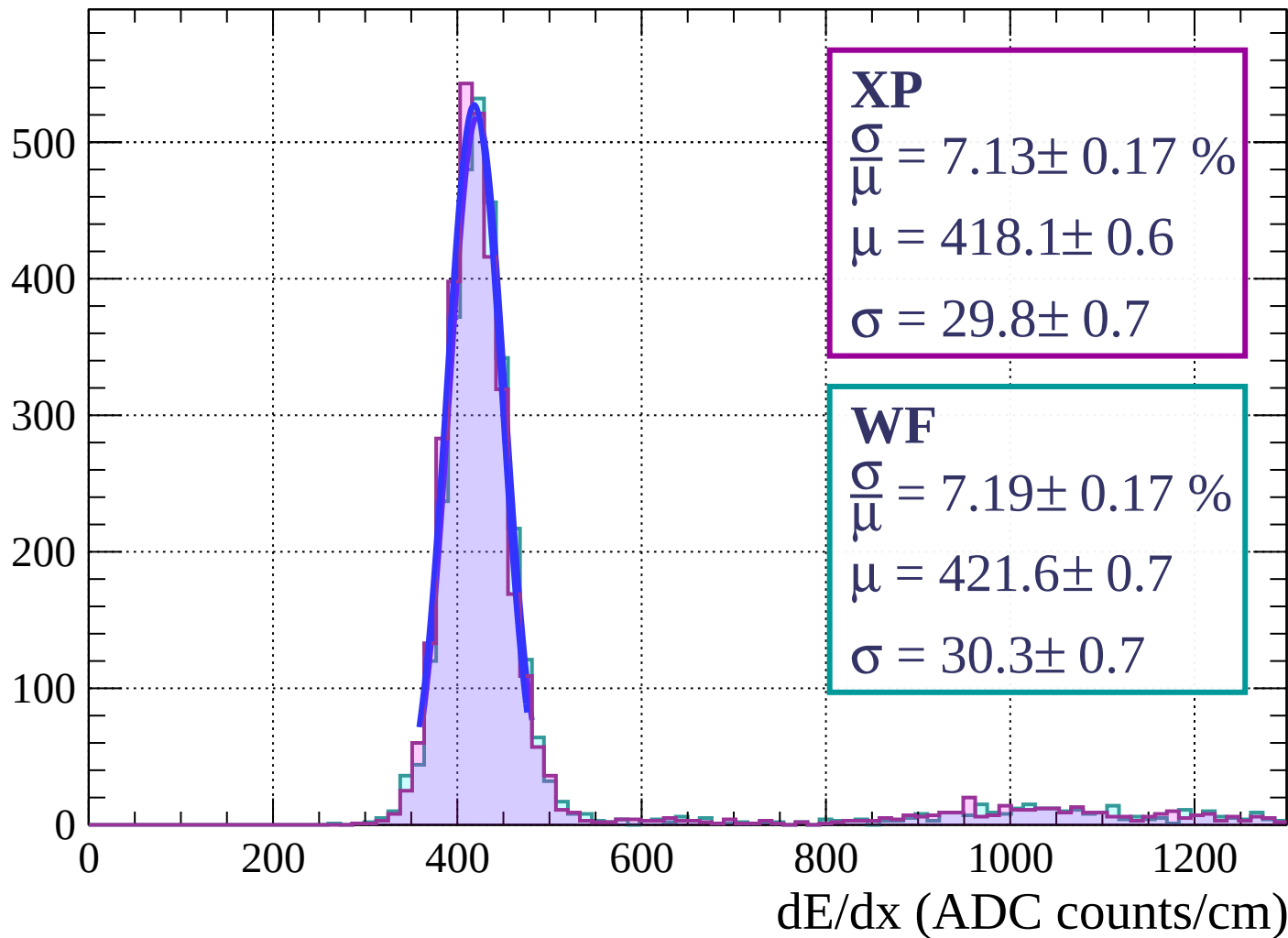
$$\mu = 422.2 \pm 0.6$$

$$\sigma = 30.8 \pm 0.5$$

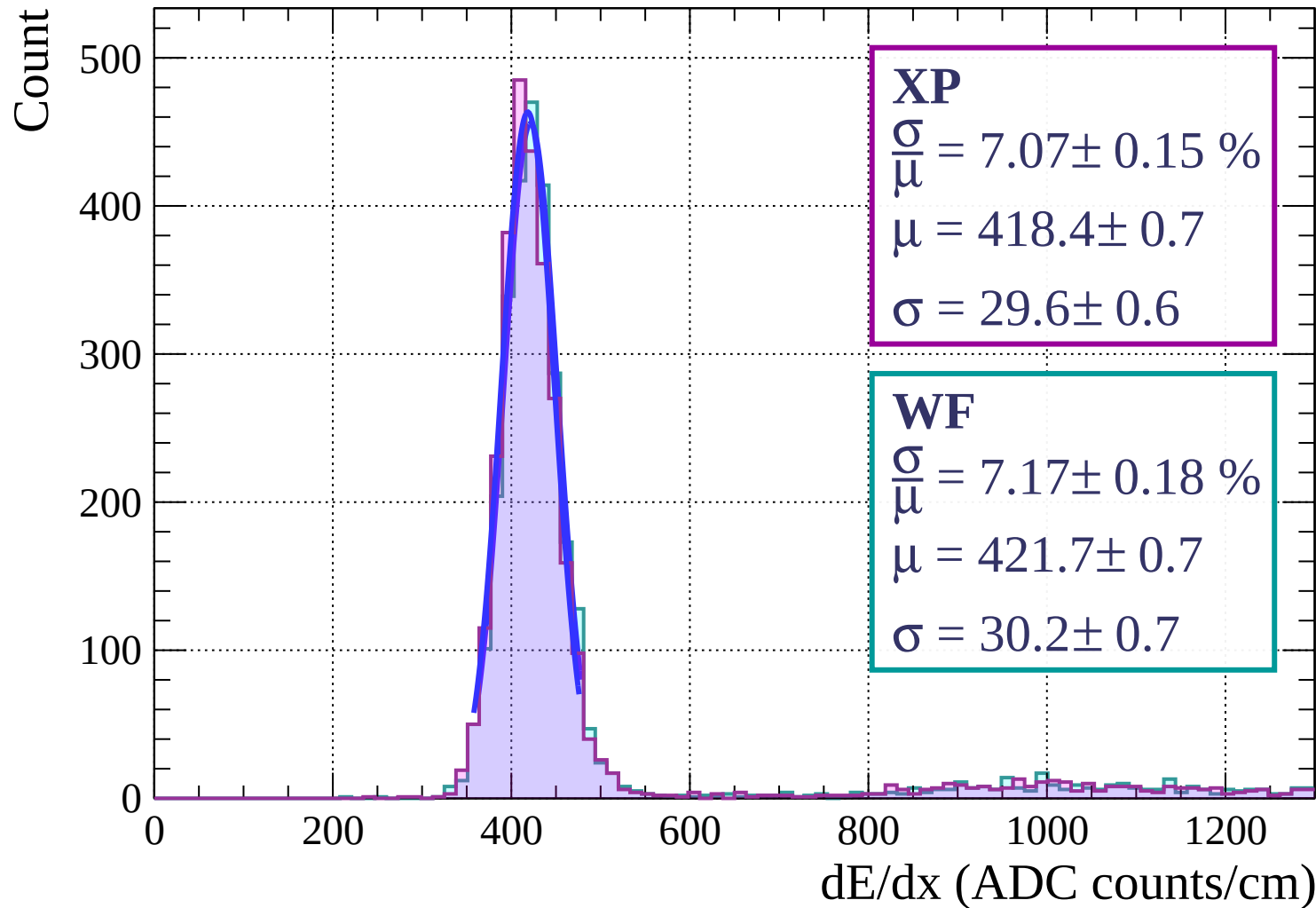
dE/dx (ADC counts/cm)

Energy loss | $3 < \theta < 6$

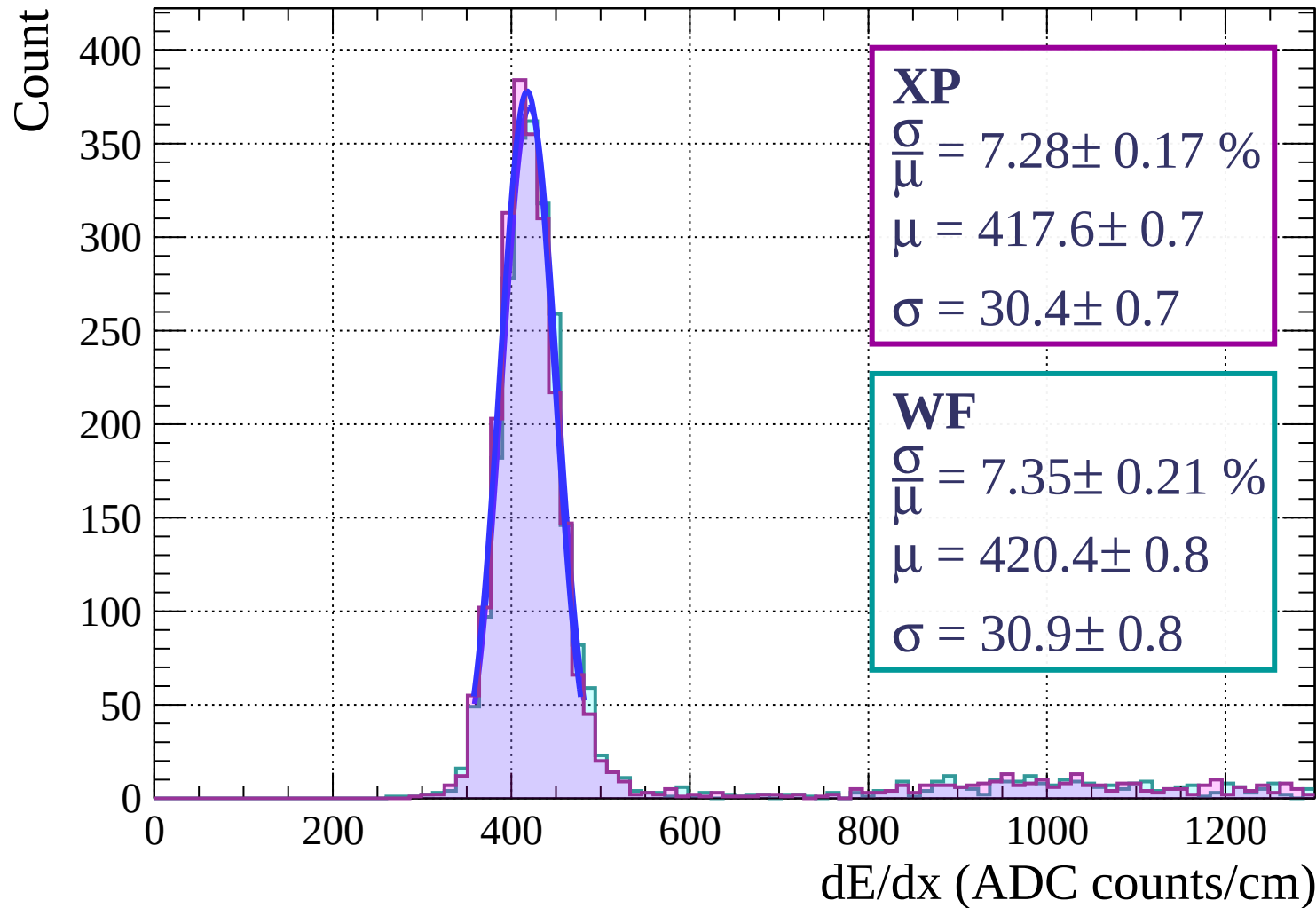
Count



Energy loss | $6 < \theta < 9$

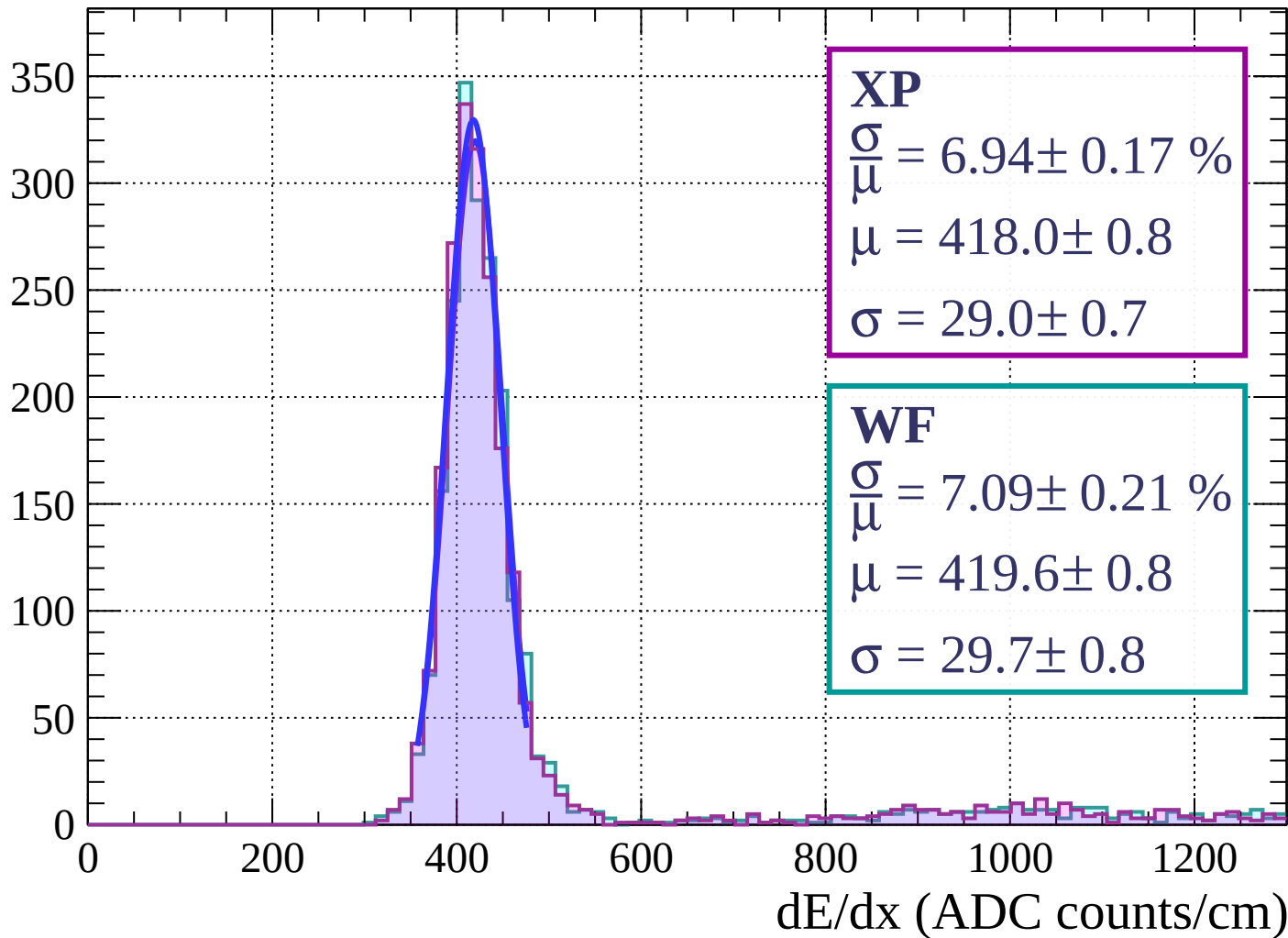


Energy loss | $9 < \theta < 12$



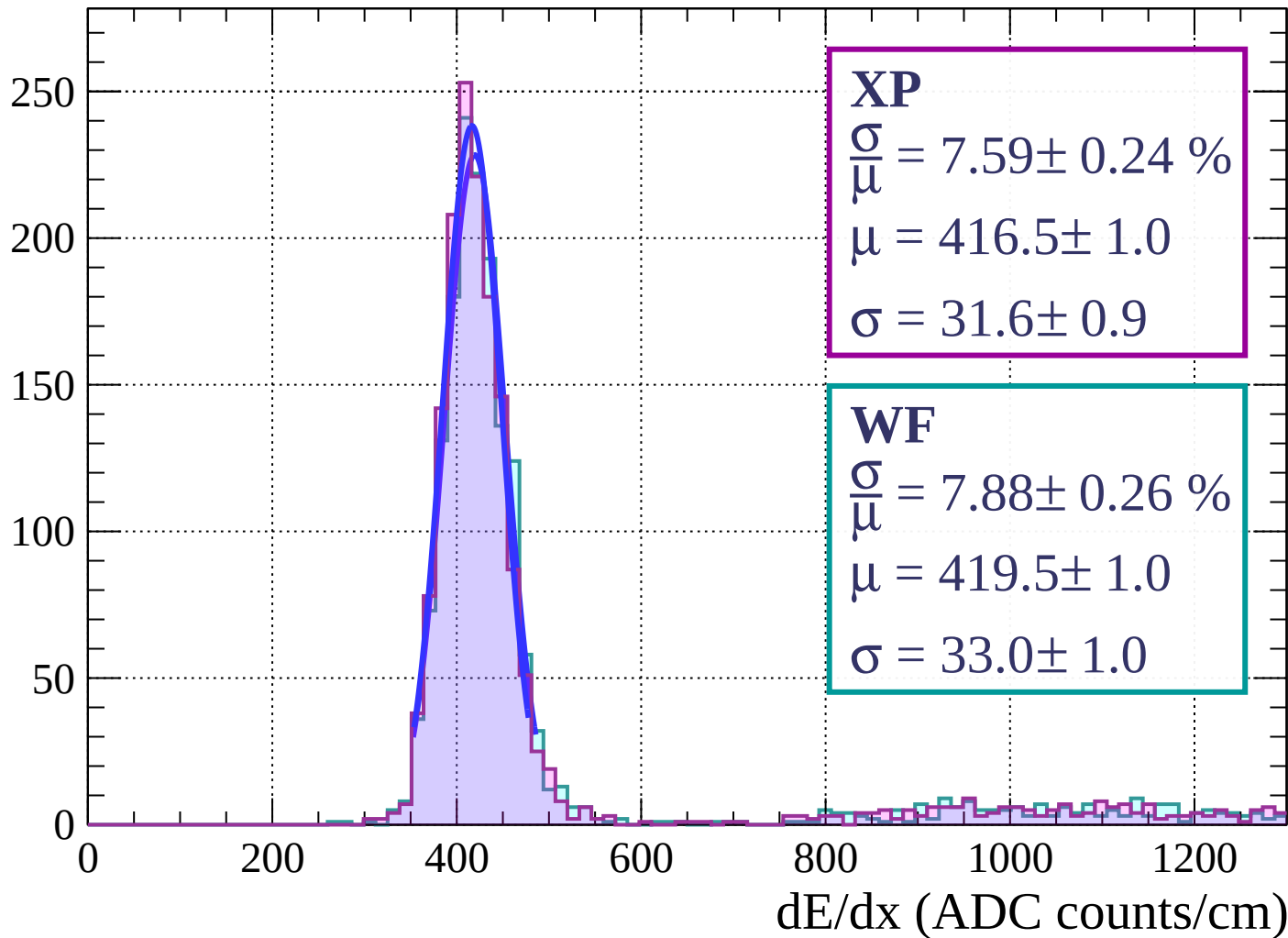
Energy loss | $12 < \theta < 15$

Count

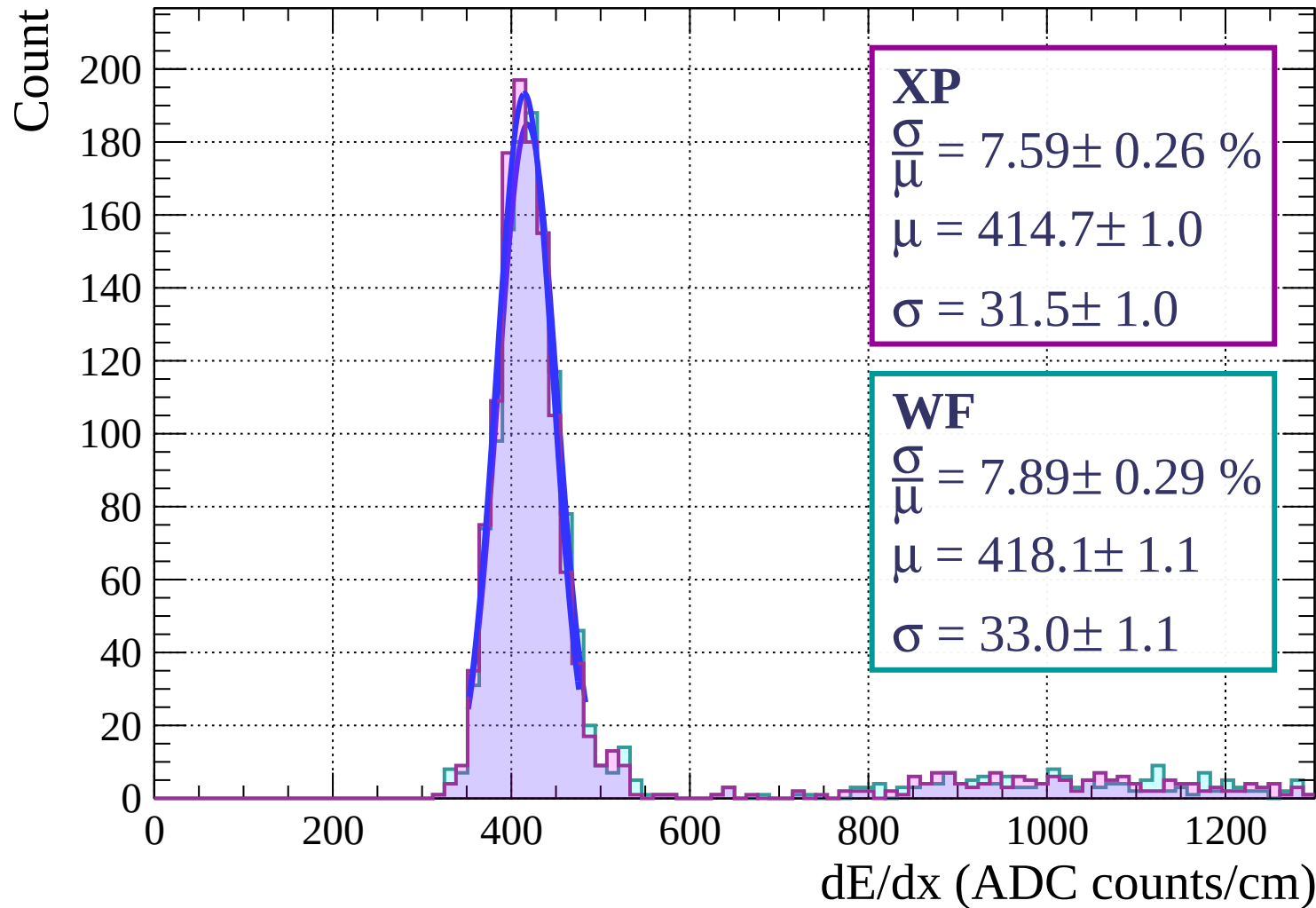


Energy loss | $15 < \theta < 18$

Count

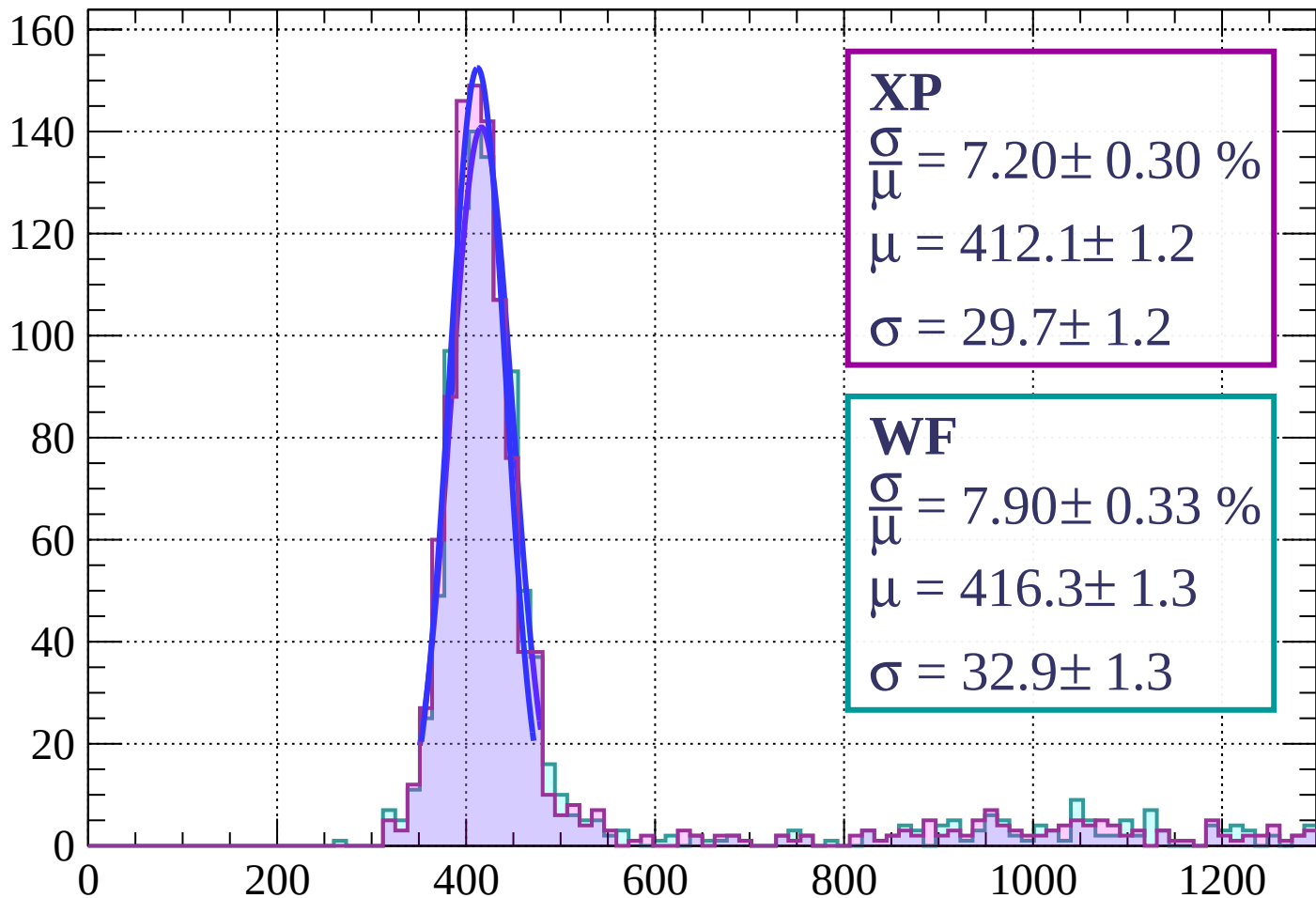


Energy loss | $18 < \theta < 21$



Energy loss | $21 < \theta < 24$

Count



XP

$$\frac{\sigma}{\mu} = 7.20 \pm 0.30 \%$$

$$\mu = 412.1 \pm 1.2$$

$$\sigma = 29.7 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 7.90 \pm 0.33 \%$$

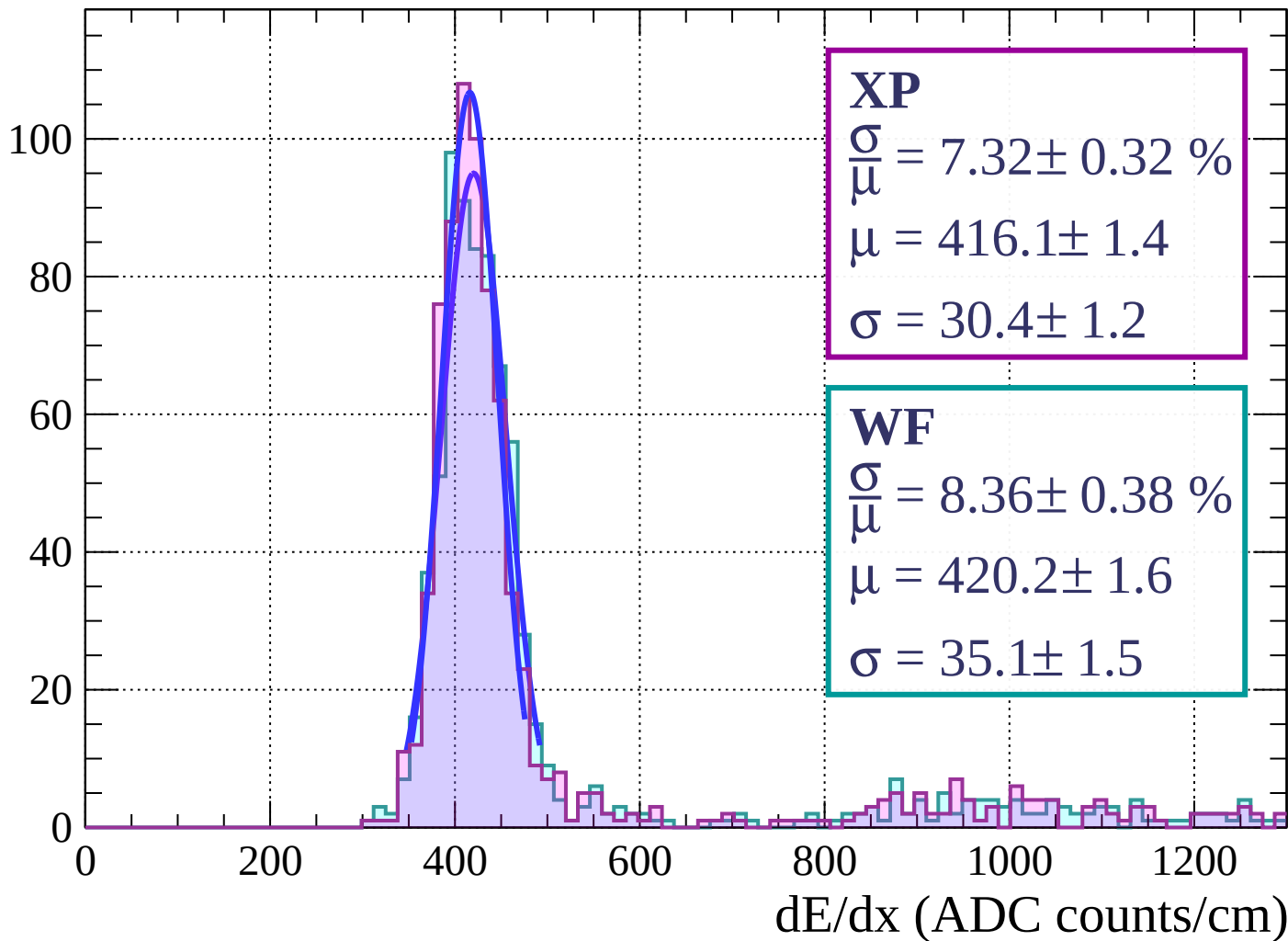
$$\mu = 416.3 \pm 1.3$$

$$\sigma = 32.9 \pm 1.3$$

dE/dx (ADC counts/cm)

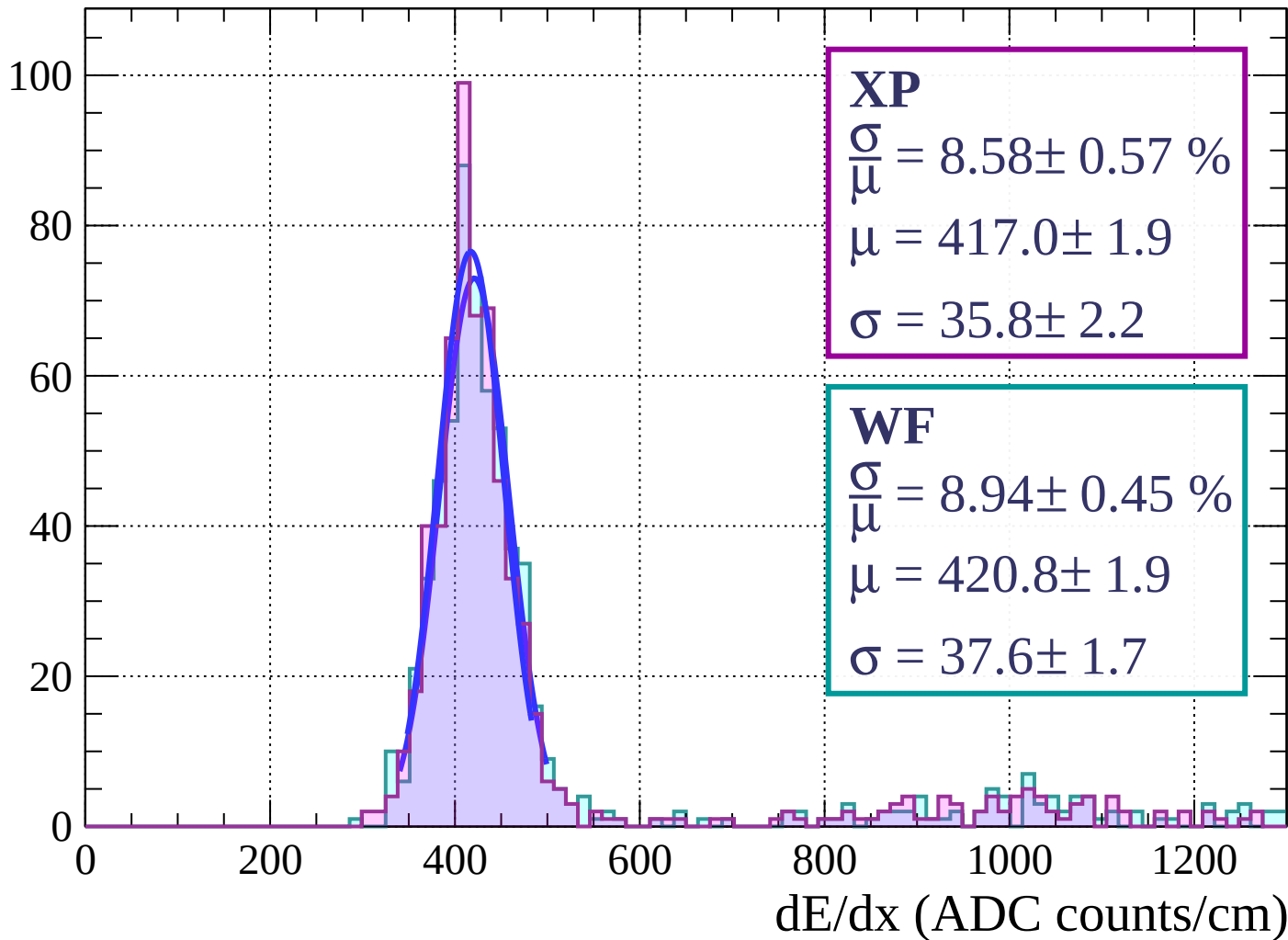
Energy loss | $24 < \theta < 27$

Count



Energy loss | $27 < \theta < 30$

Count



Energy loss | $30 < \theta < 33$

Count

XP

$$\frac{\sigma}{\mu} = 8.39 \pm 0.58 \%$$

$$\mu = 418.4 \pm 2.3$$

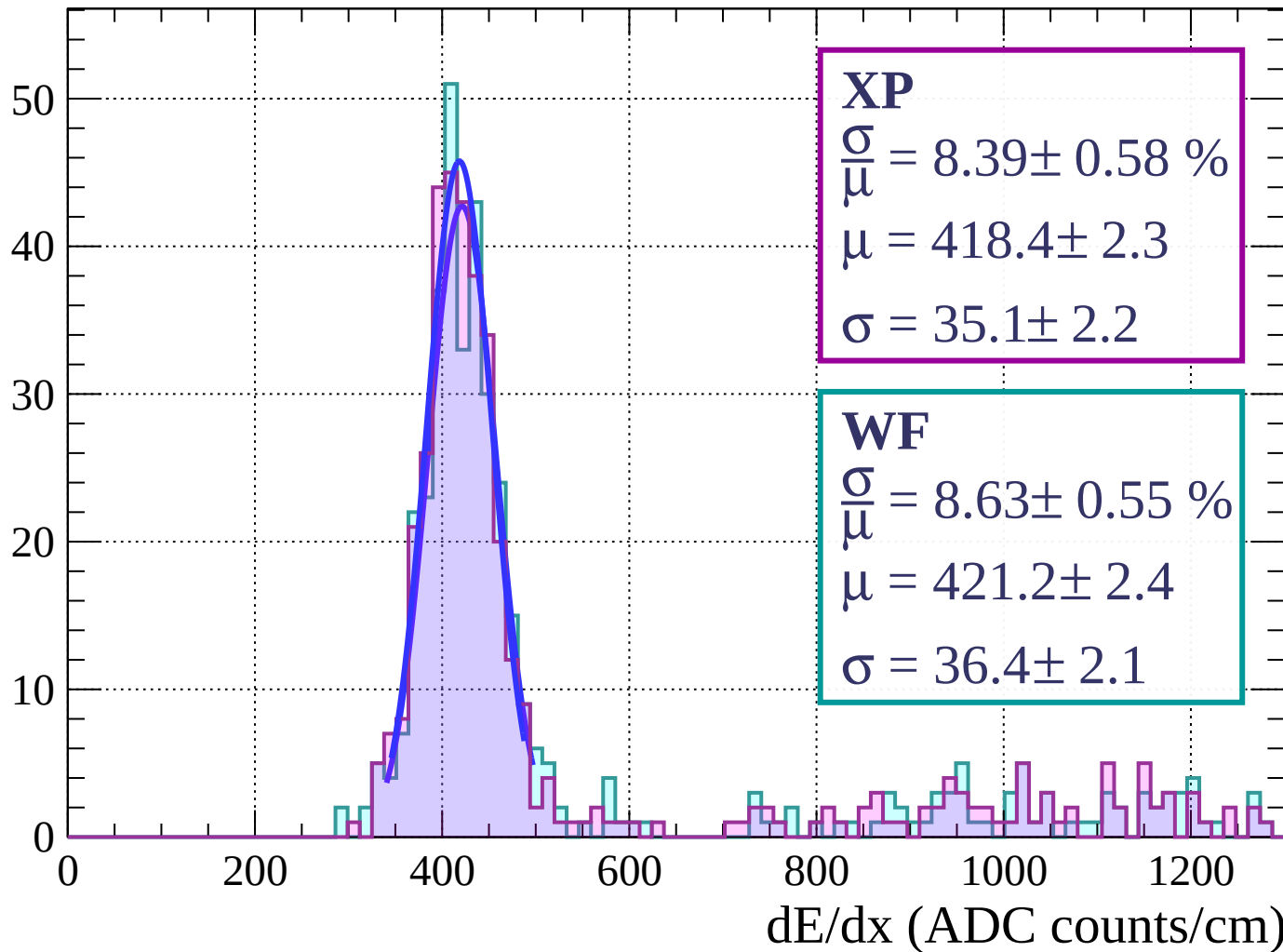
$$\sigma = 35.1 \pm 2.2$$

WF

$$\frac{\sigma}{\mu} = 8.63 \pm 0.55 \%$$

$$\mu = 421.2 \pm 2.4$$

$$\sigma = 36.4 \pm 2.1$$



Energy loss | $33 < \theta < 36$

Count

XP

$$\frac{\sigma}{\mu} = 8.54 \pm 0.72 \%$$

$$\mu = 421.9 \pm 2.8$$

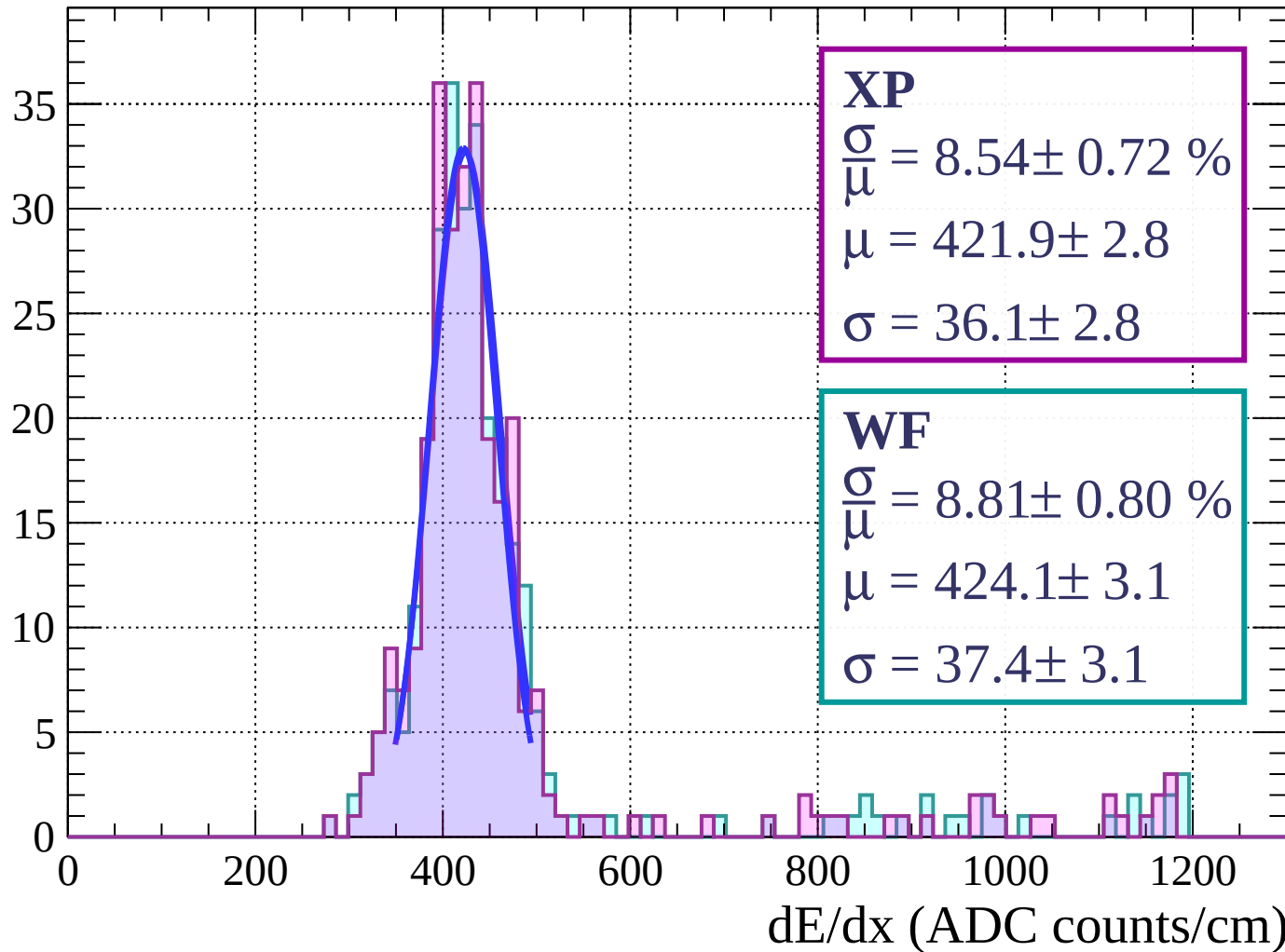
$$\sigma = 36.1 \pm 2.8$$

WF

$$\frac{\sigma}{\mu} = 8.81 \pm 0.80 \%$$

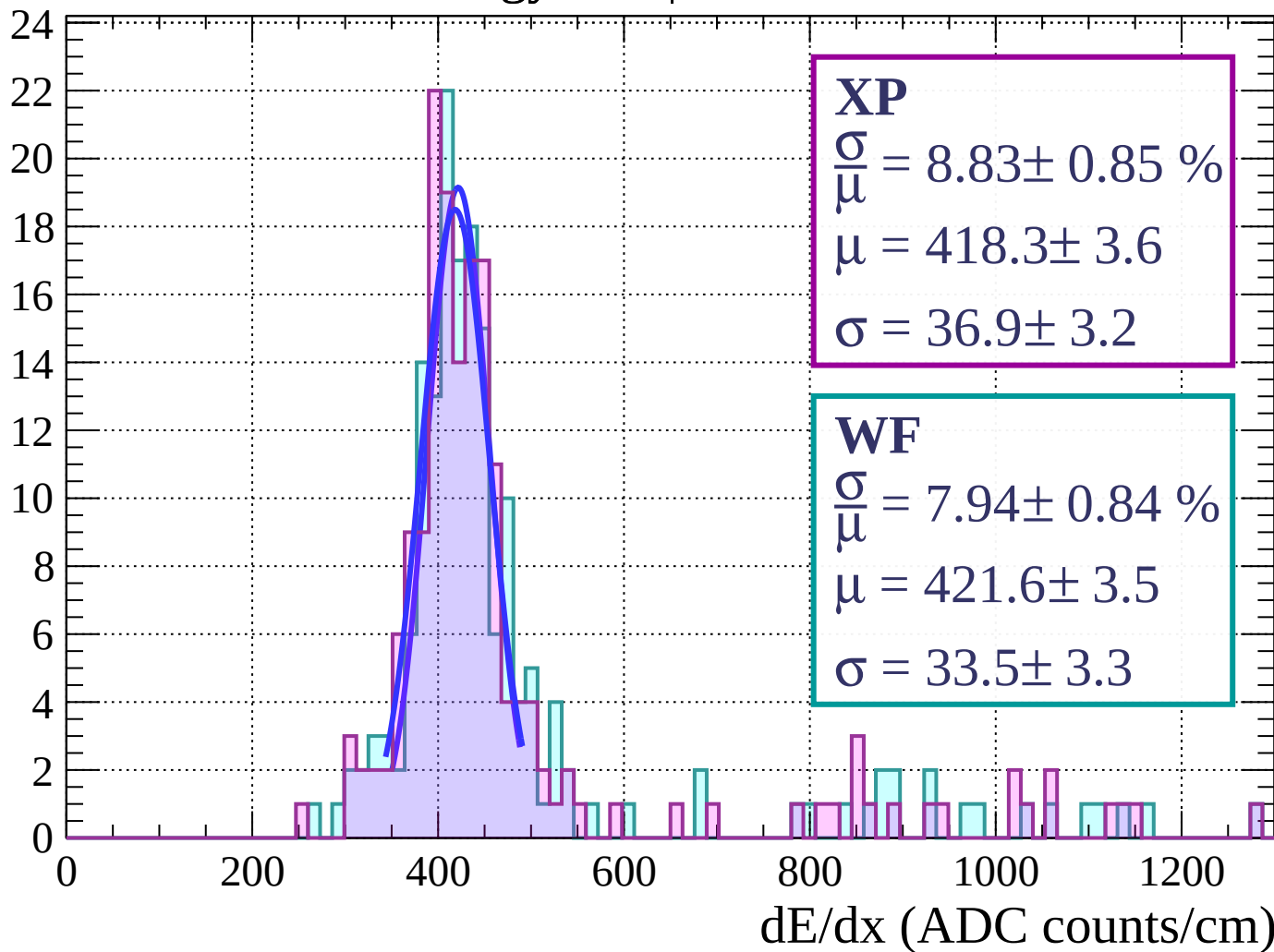
$$\mu = 424.1 \pm 3.1$$

$$\sigma = 37.4 \pm 3.1$$



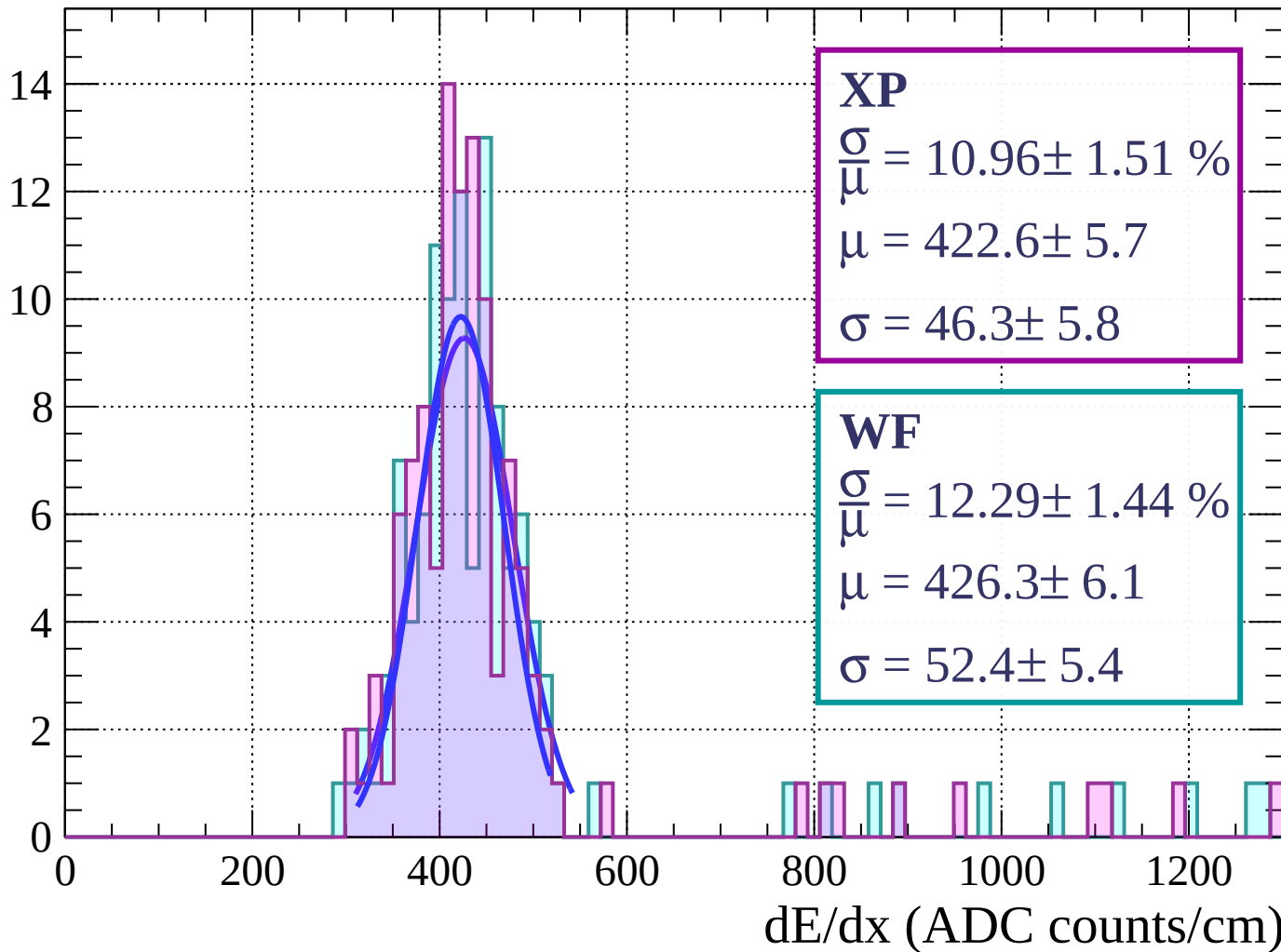
Energy loss | $36 < \theta < 39$

Count



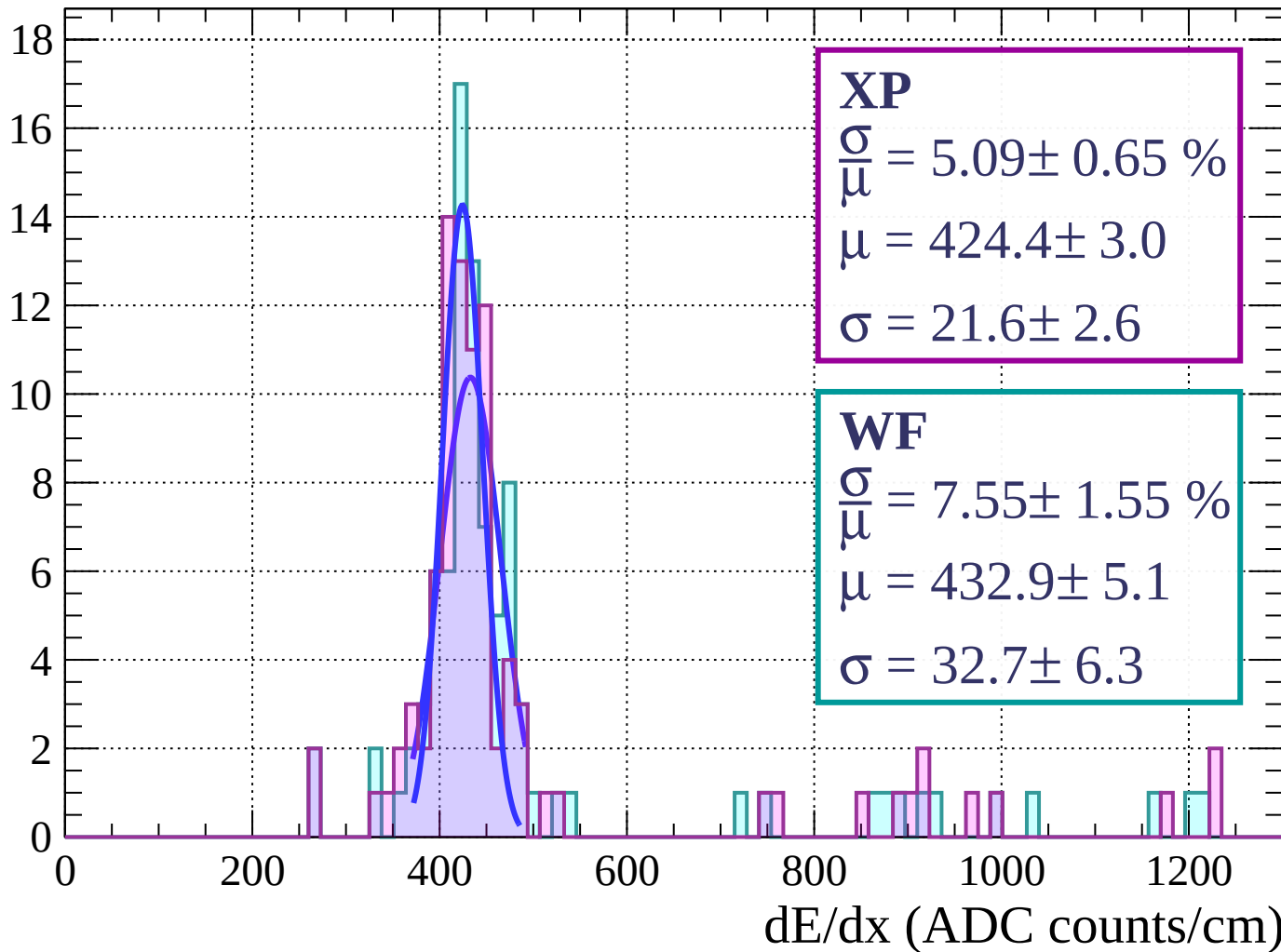
Energy loss | $39 < \theta < 42$

Count



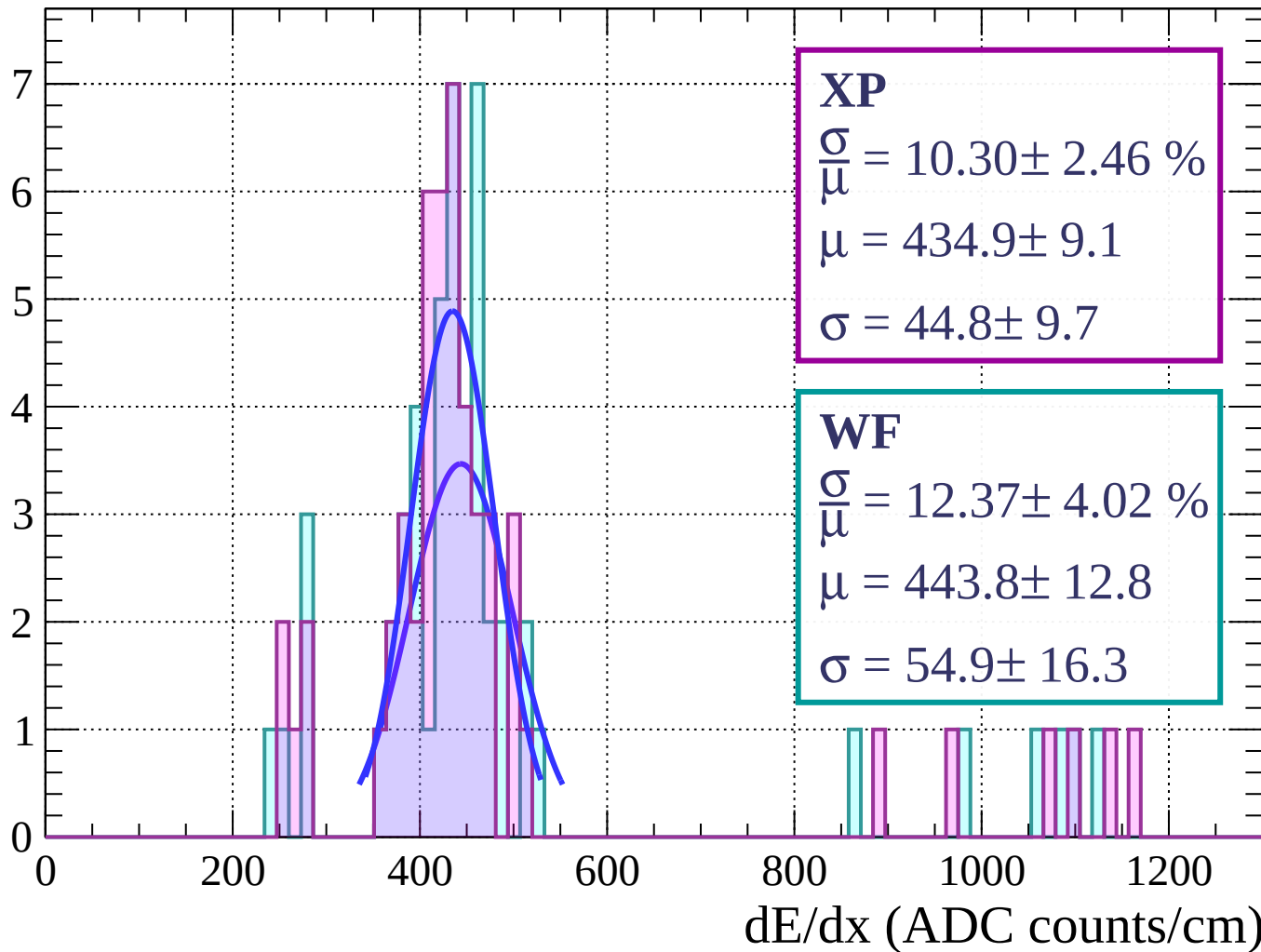
Energy loss | $42 < \theta < 45$

Count



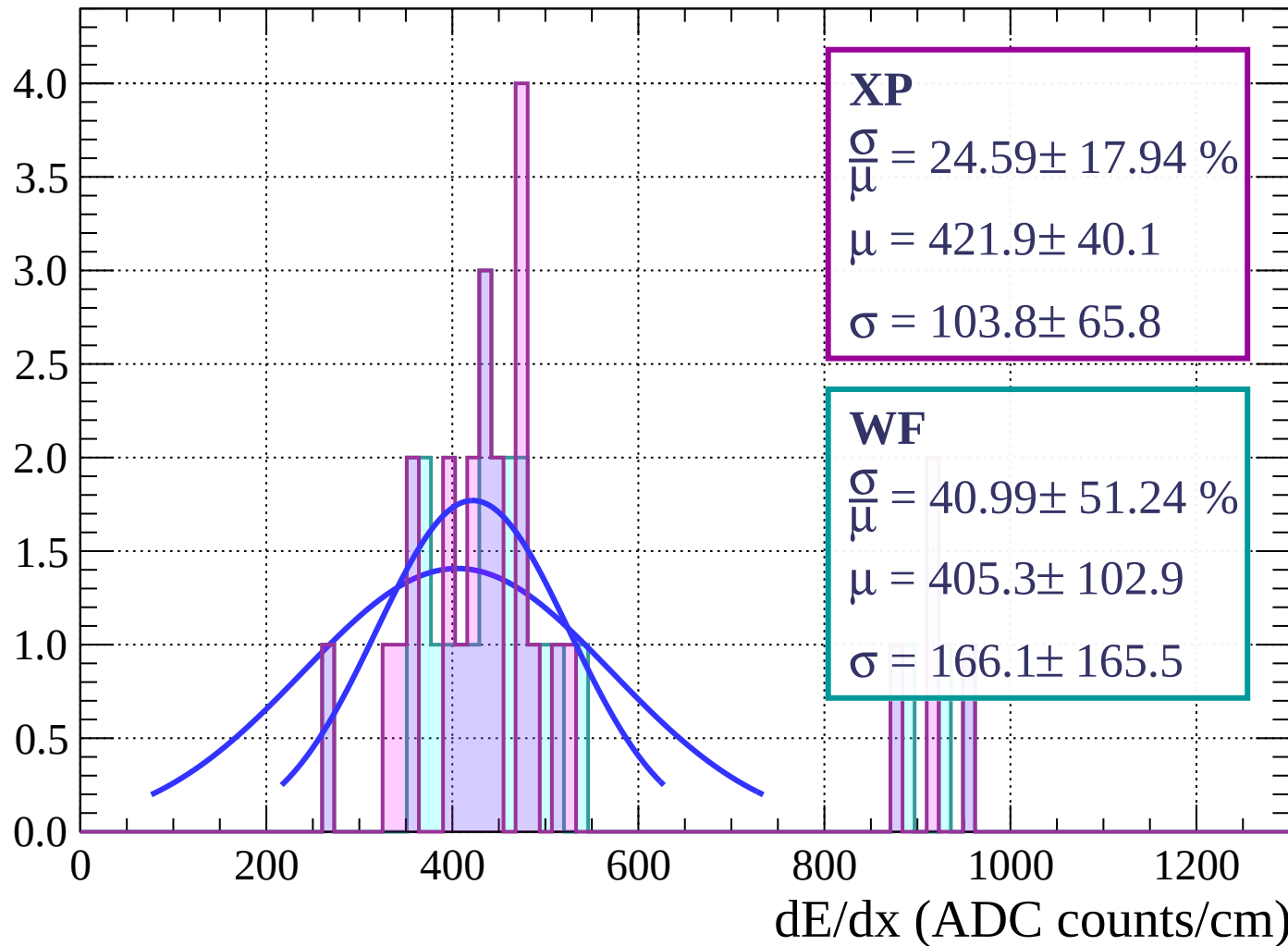
Energy loss | $45 < \theta < 48$

Count



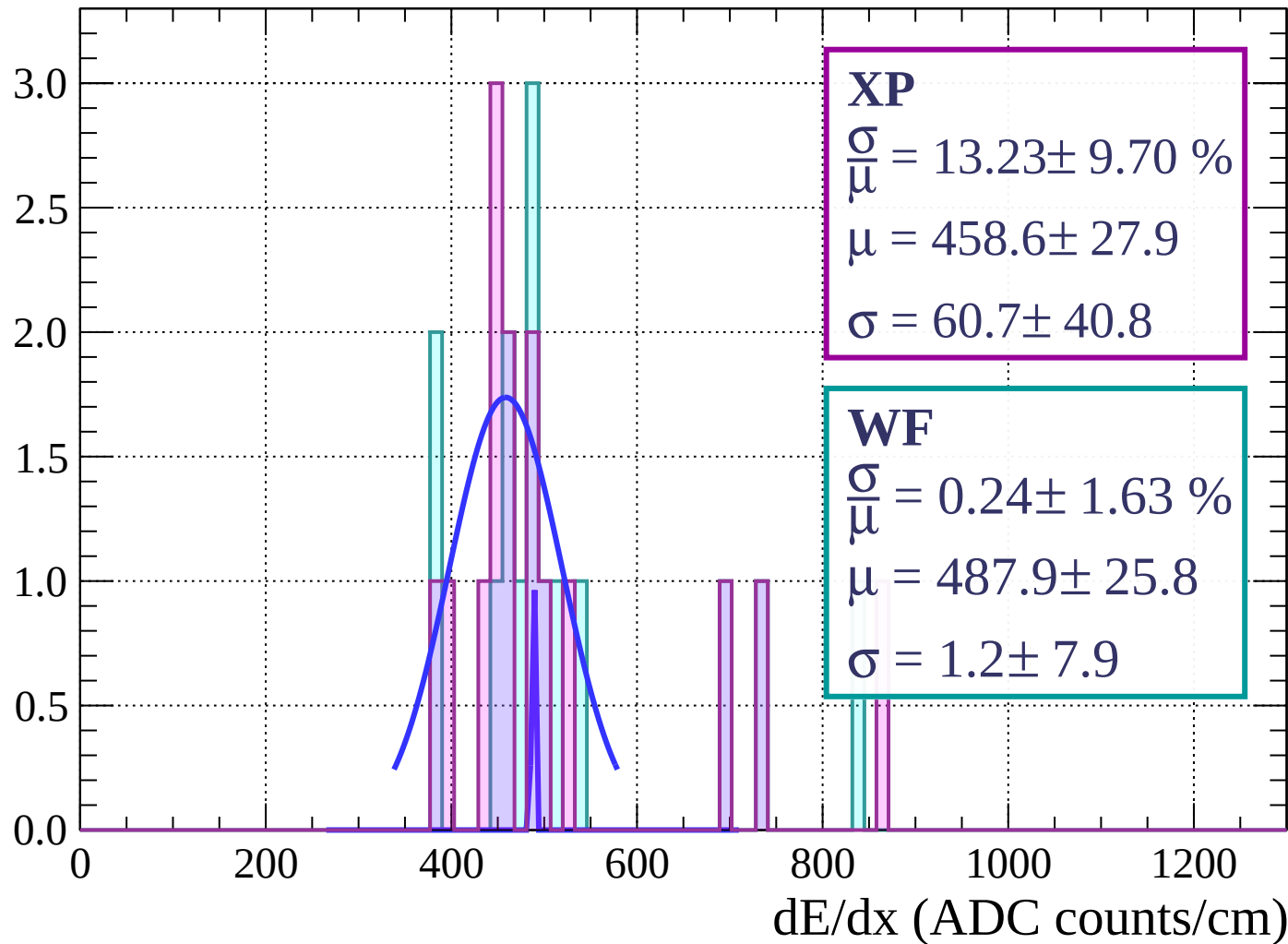
Energy loss | $48 < \theta < 51$

Count



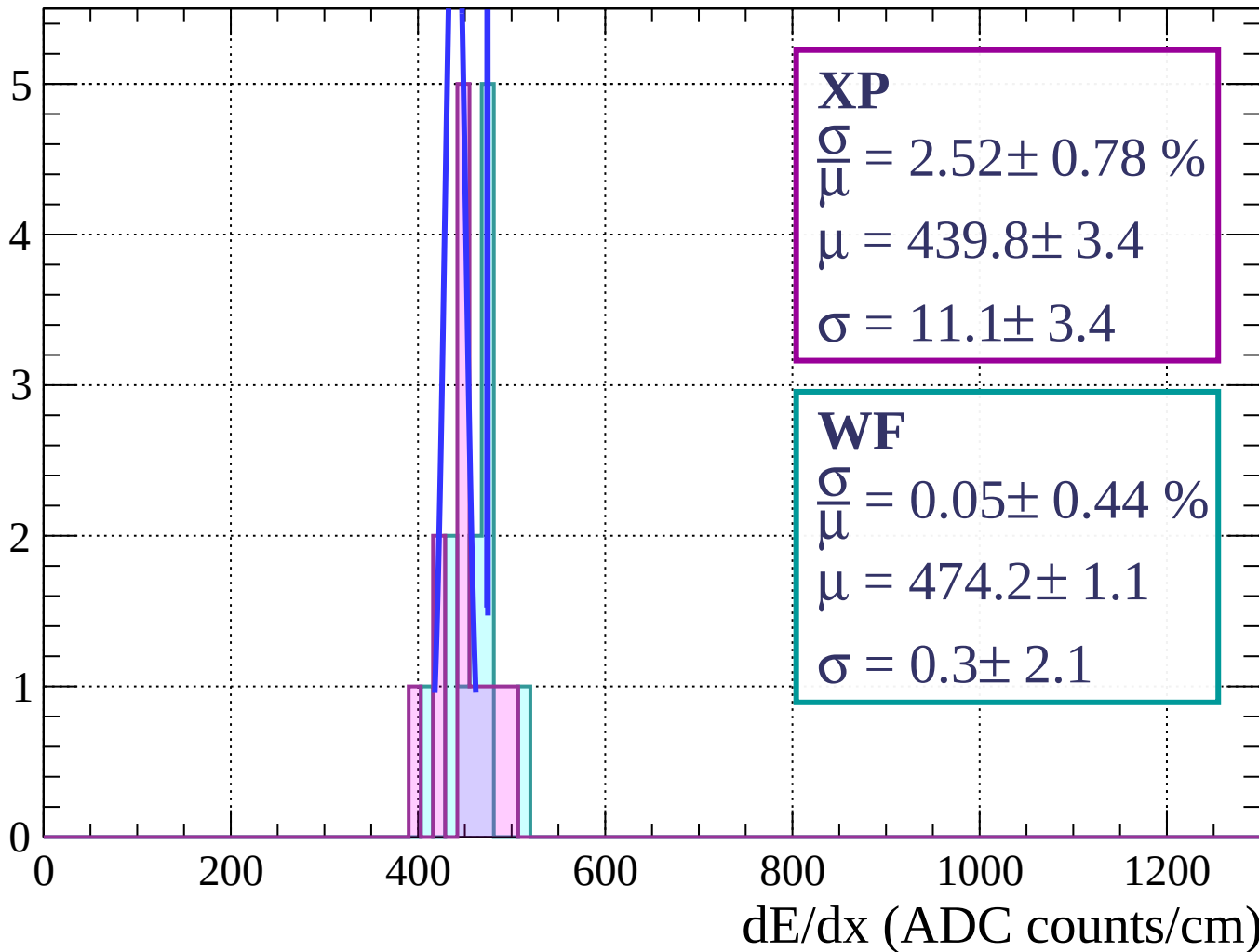
Energy loss | $51 < \theta < 54$

Count



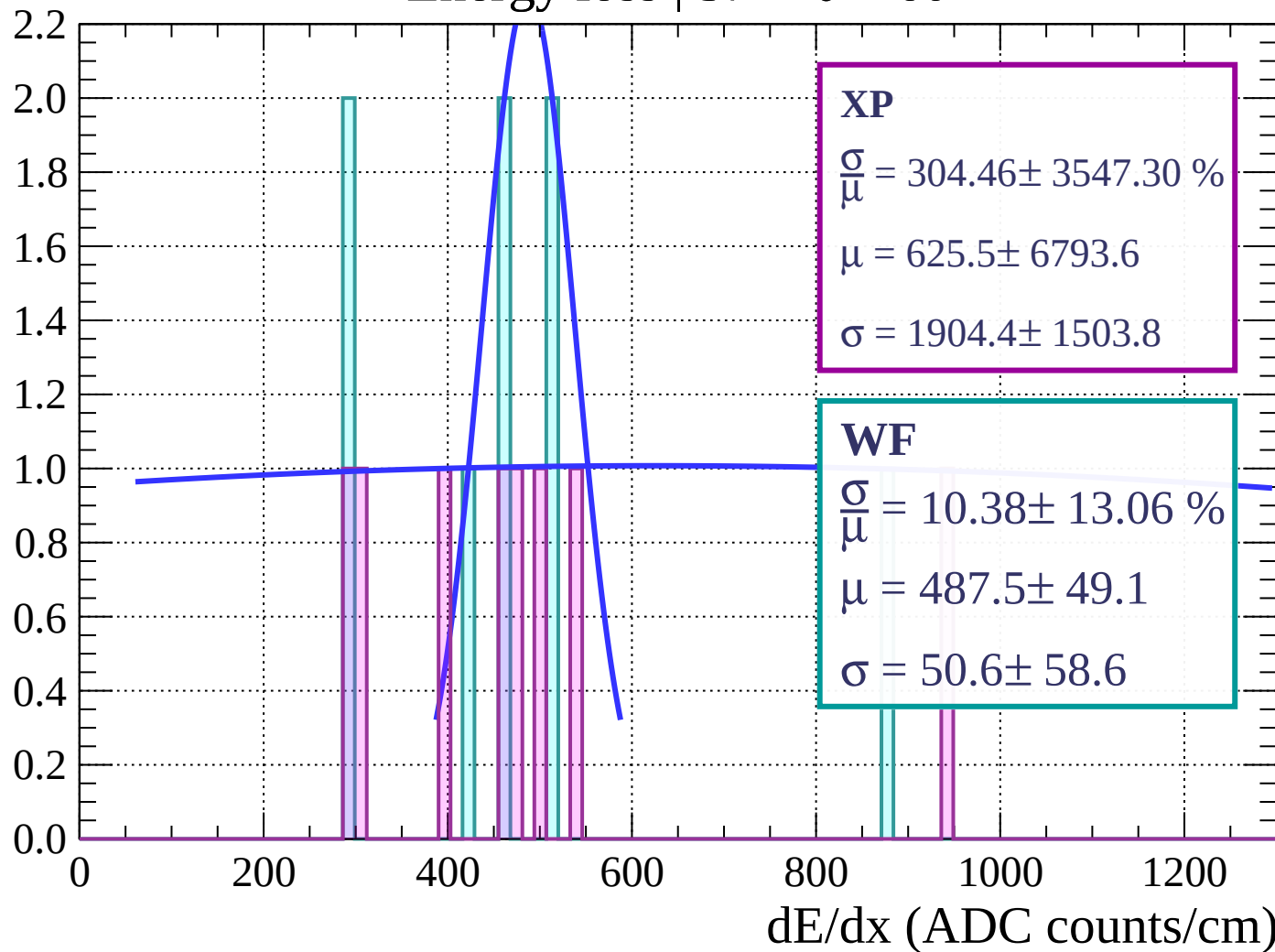
Energy loss | $54 < \theta < 57$

Count



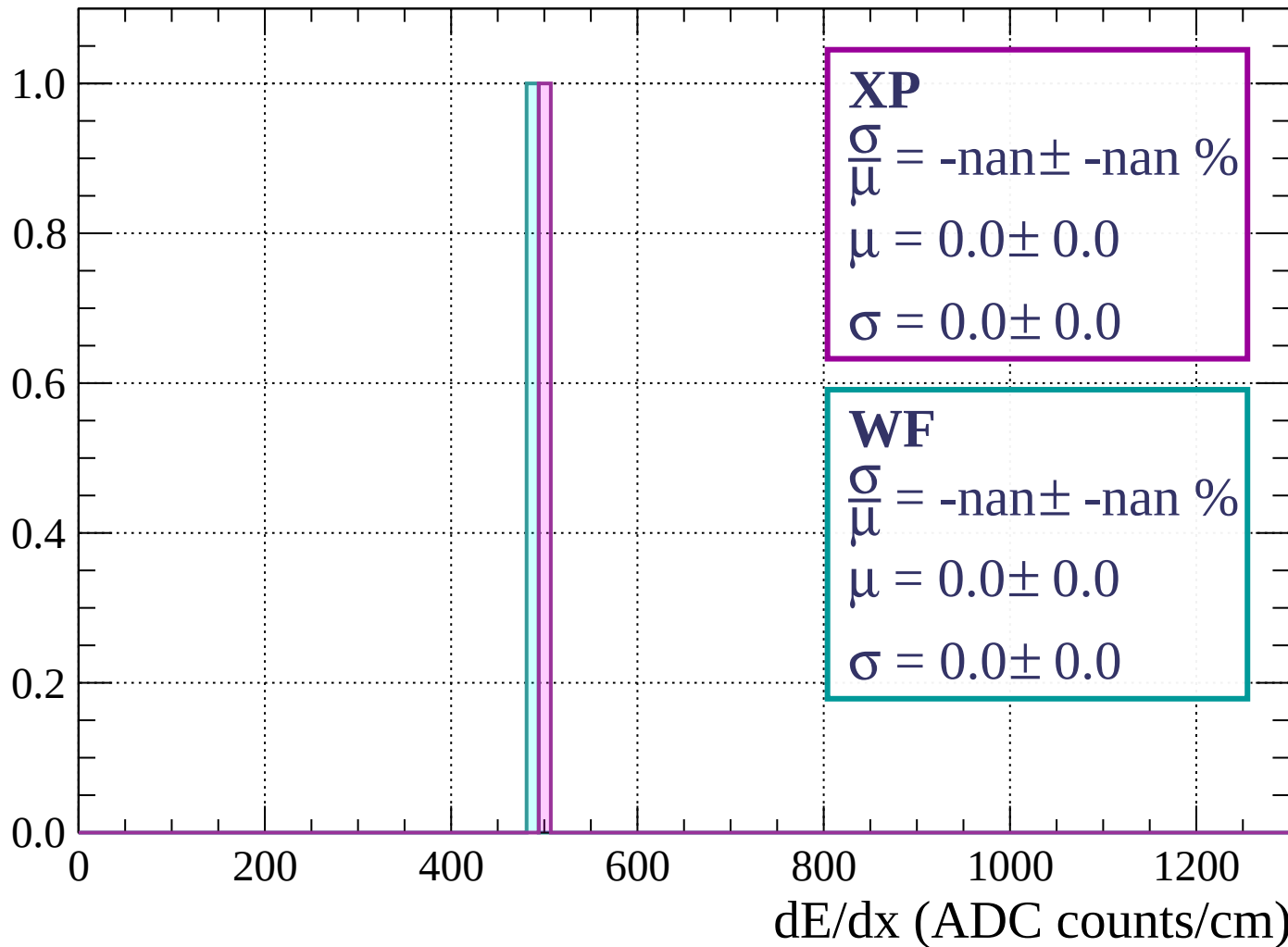
Energy loss | $57 < \theta < 60$

Count



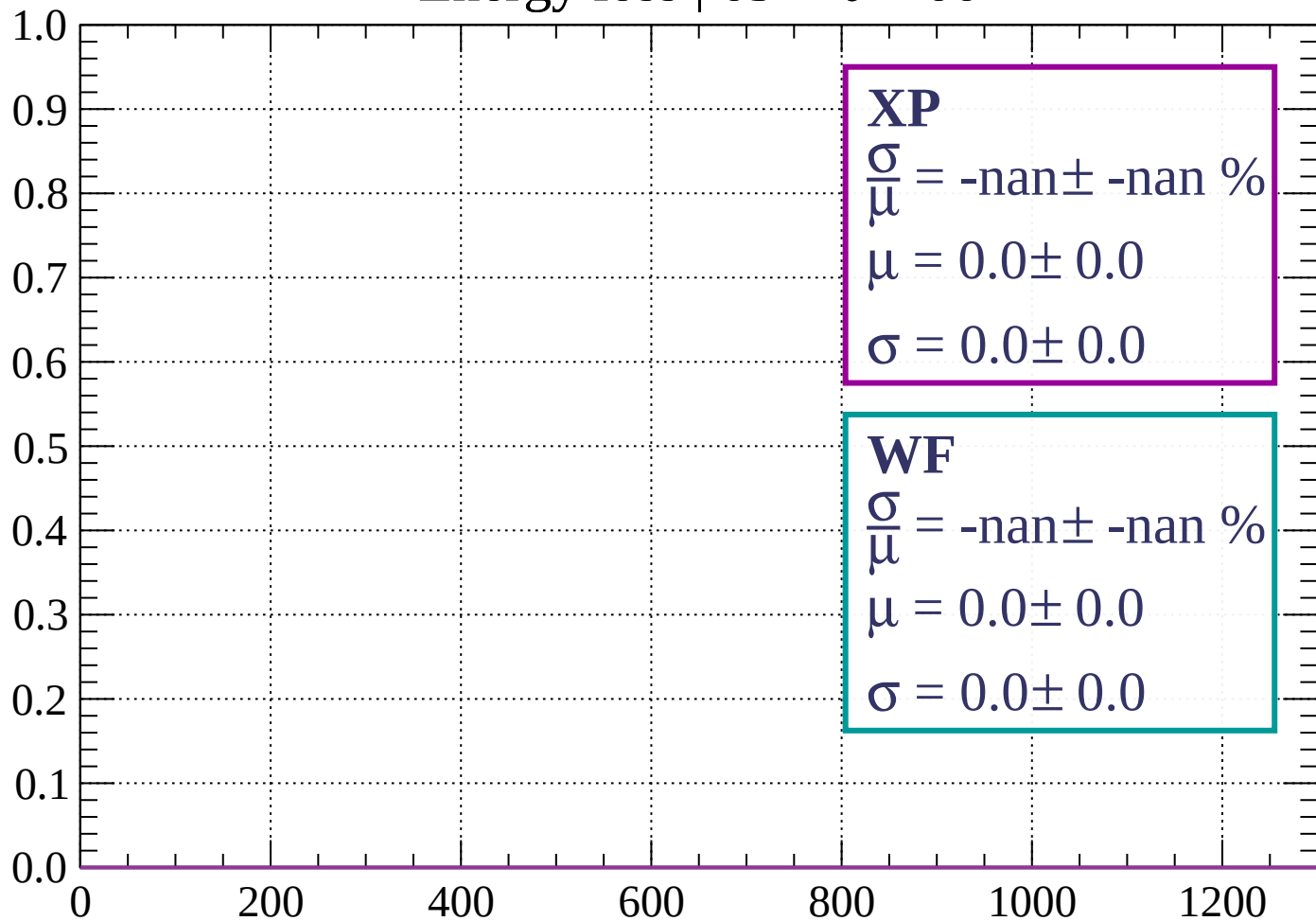
Energy loss | $60 < \theta < 63$

Count



Energy loss | $63 < \theta < 66$

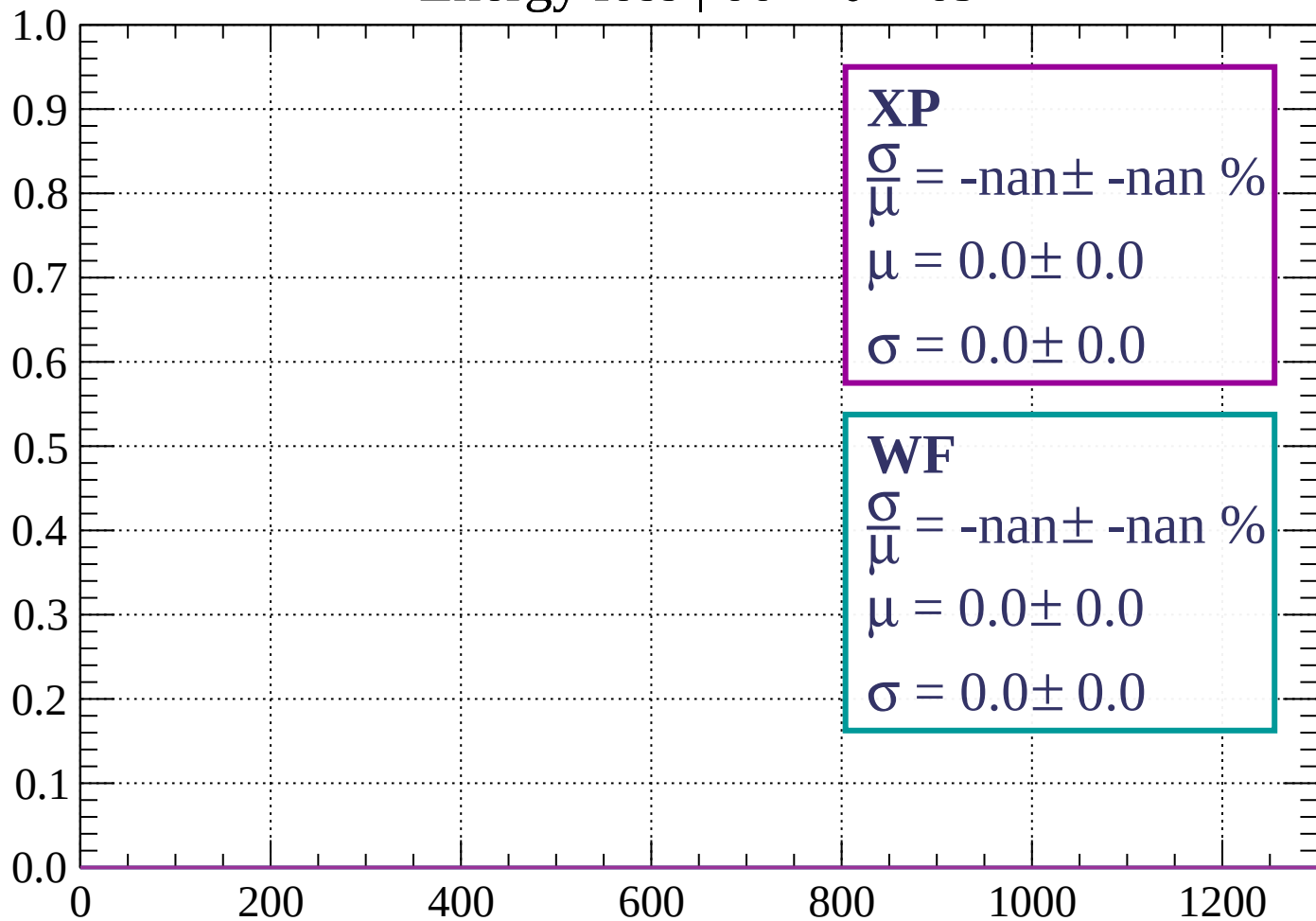
Count



dE/dx (ADC counts/cm)

Energy loss | $66 < \theta < 69$

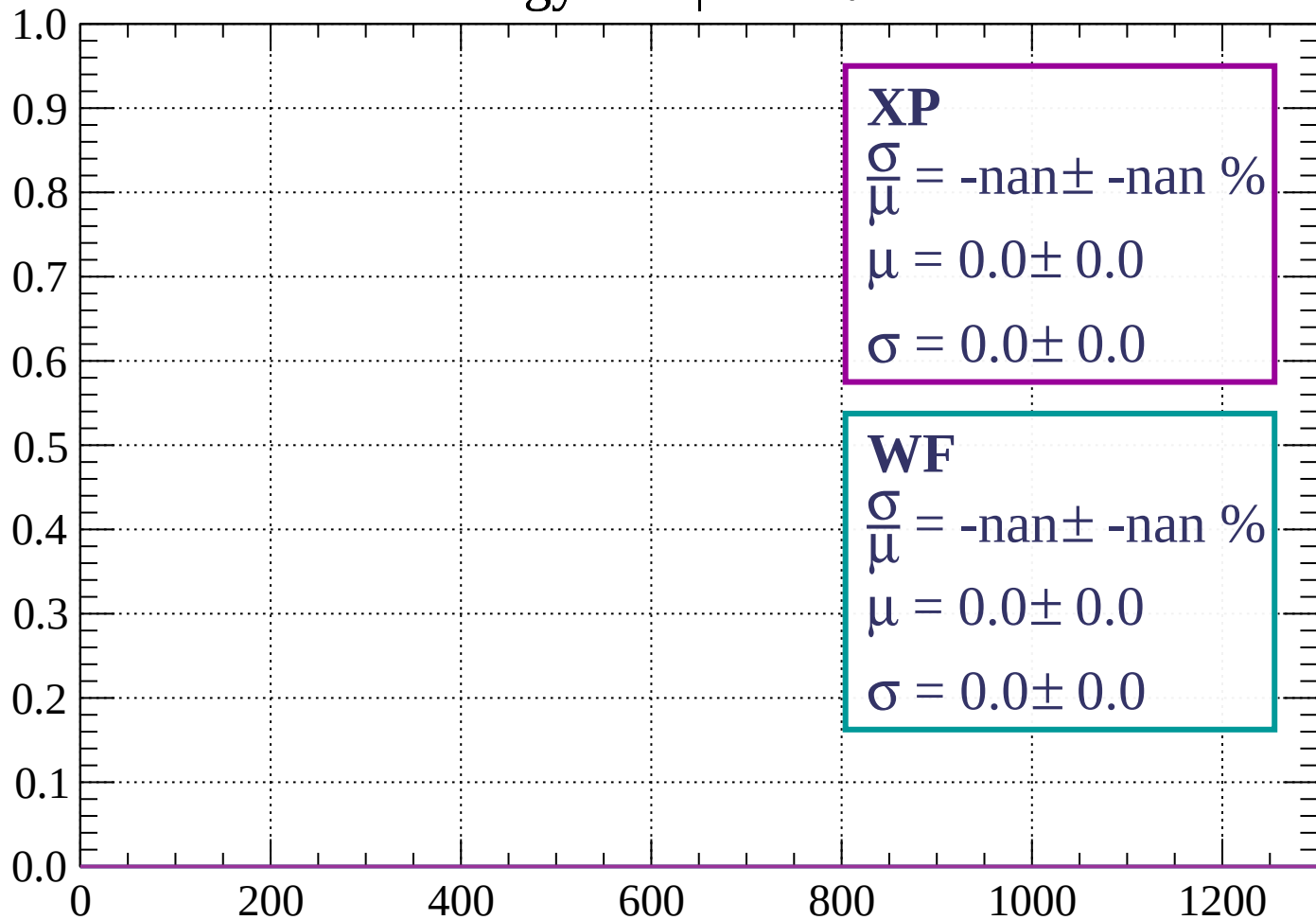
Count



dE/dx (ADC counts/cm)

Energy loss | $69 < \theta < 72$

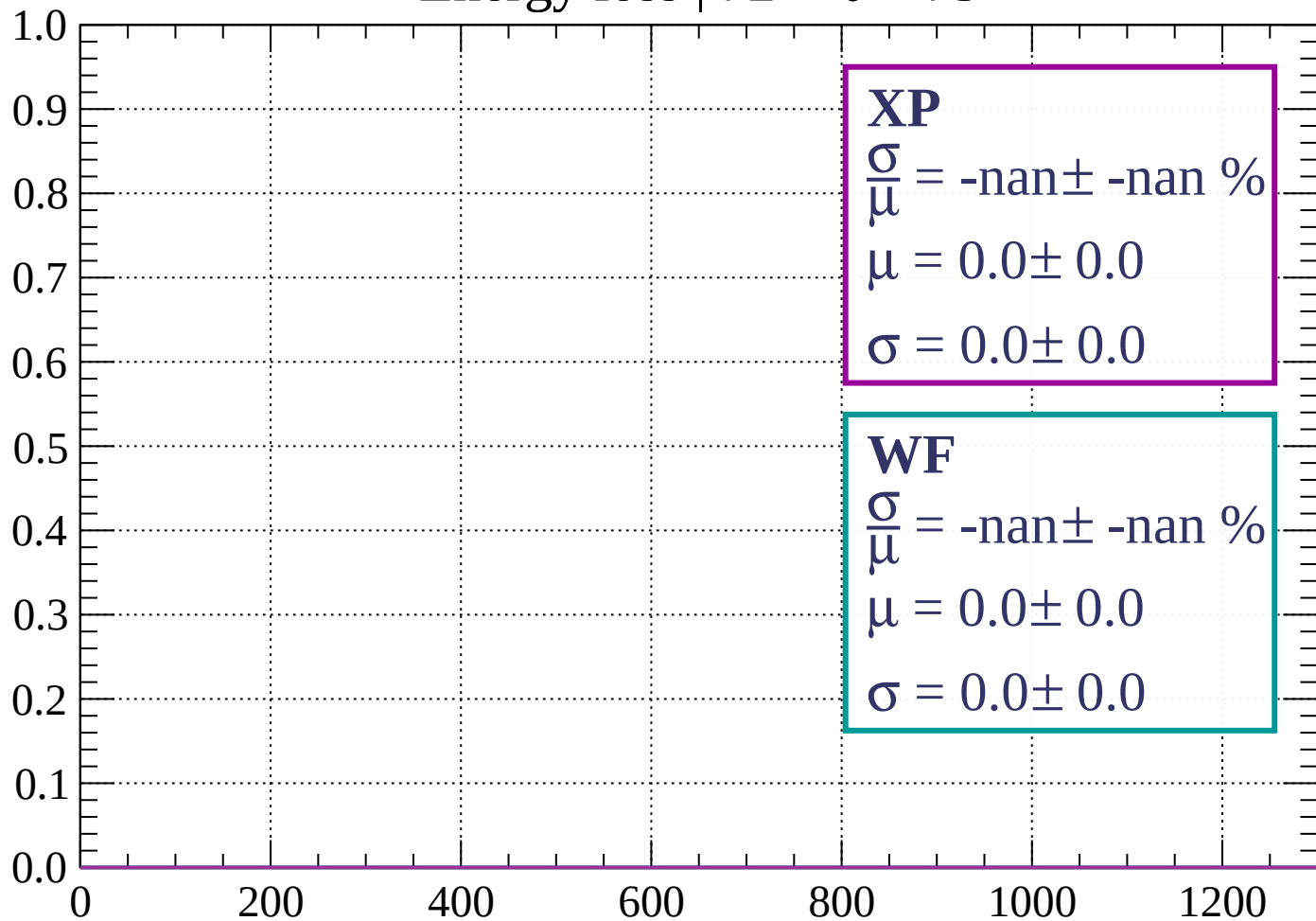
Count



dE/dx (ADC counts/cm)

Energy loss | $72 < \theta < 75$

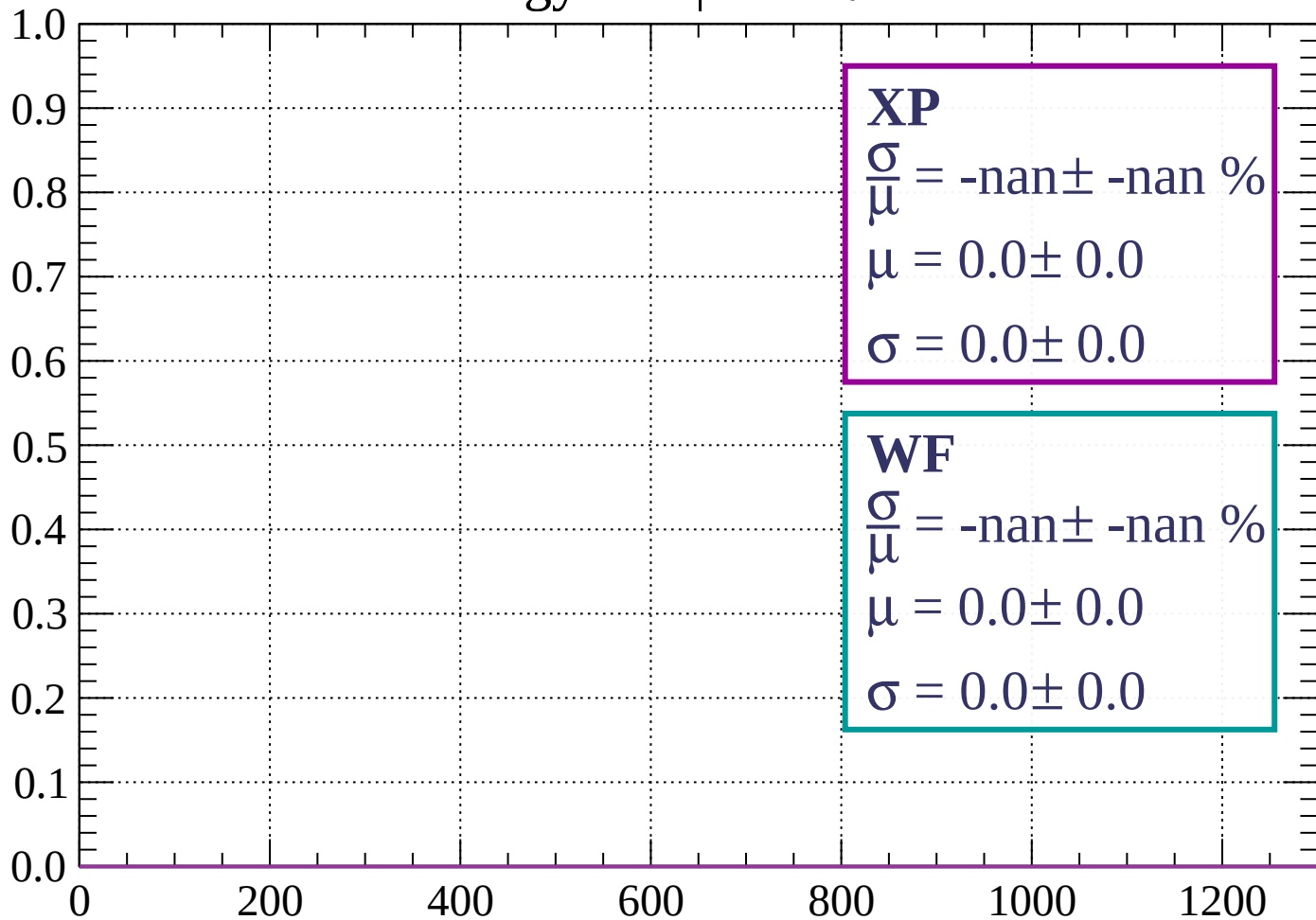
Count



dE/dx (ADC counts/cm)

Energy loss | $75 < \theta < 78$

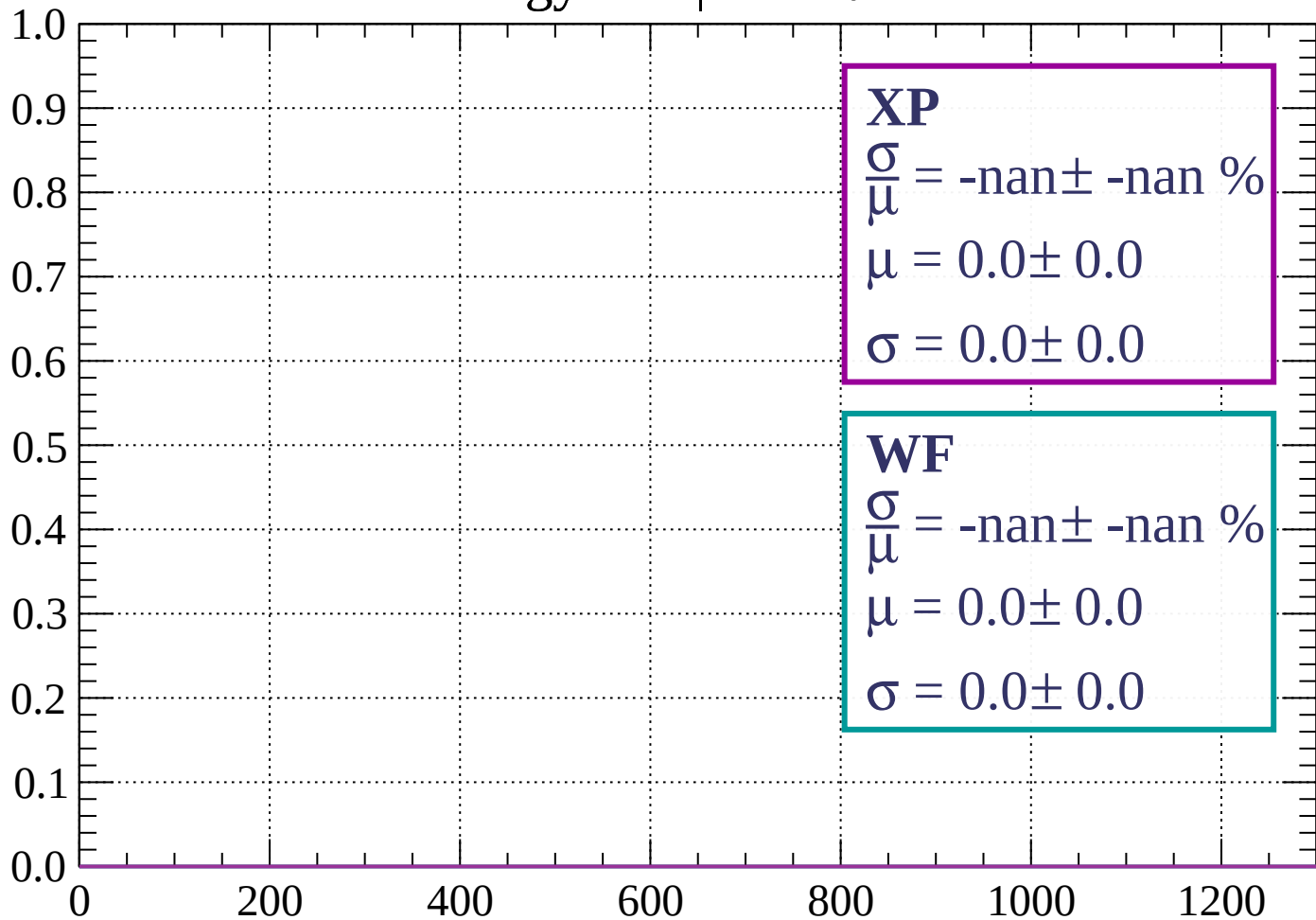
Count



dE/dx (ADC counts/cm)

Energy loss | $78 < \theta < 81$

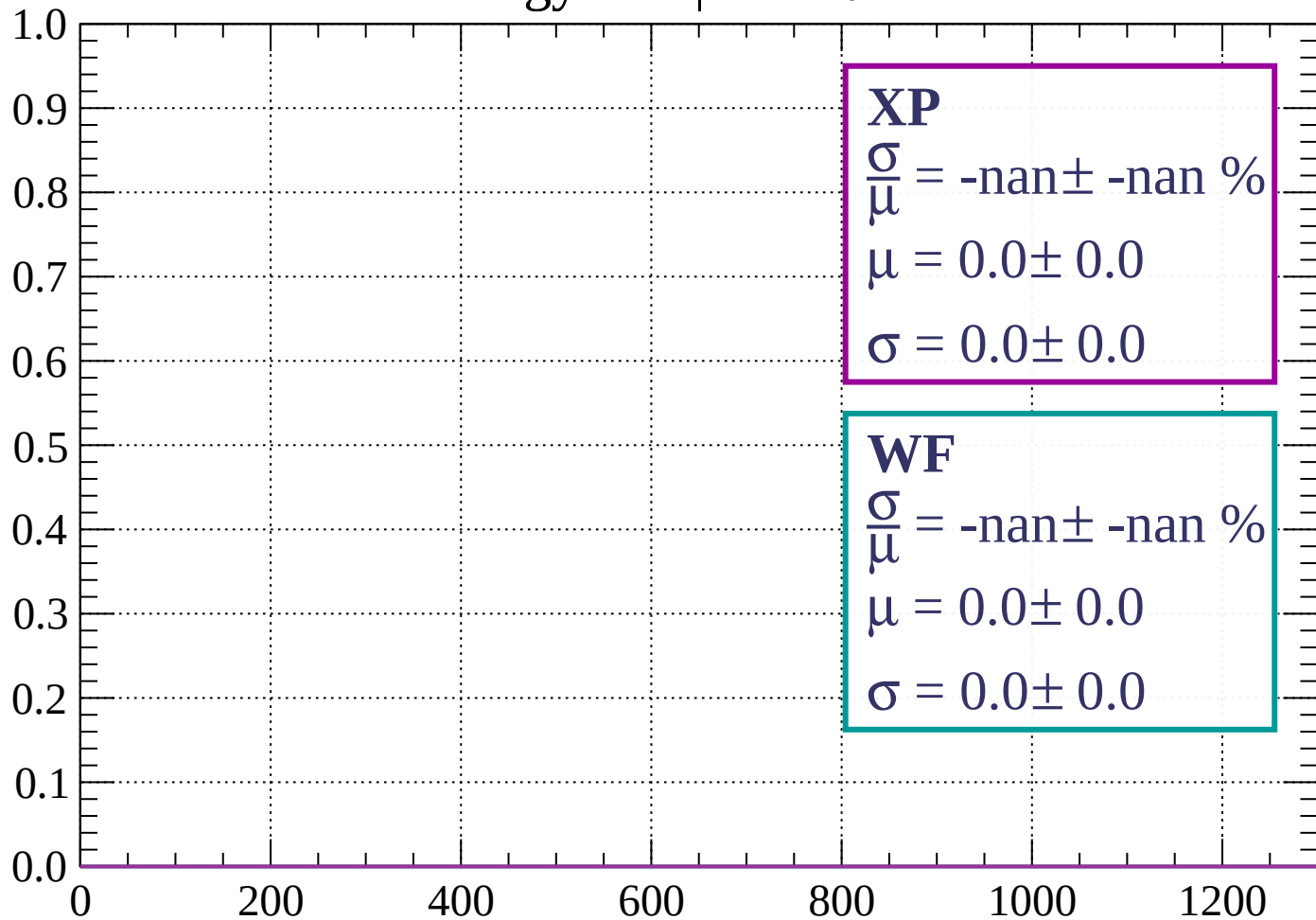
Count



dE/dx (ADC counts/cm)

Energy loss | $81 < \theta < 84$

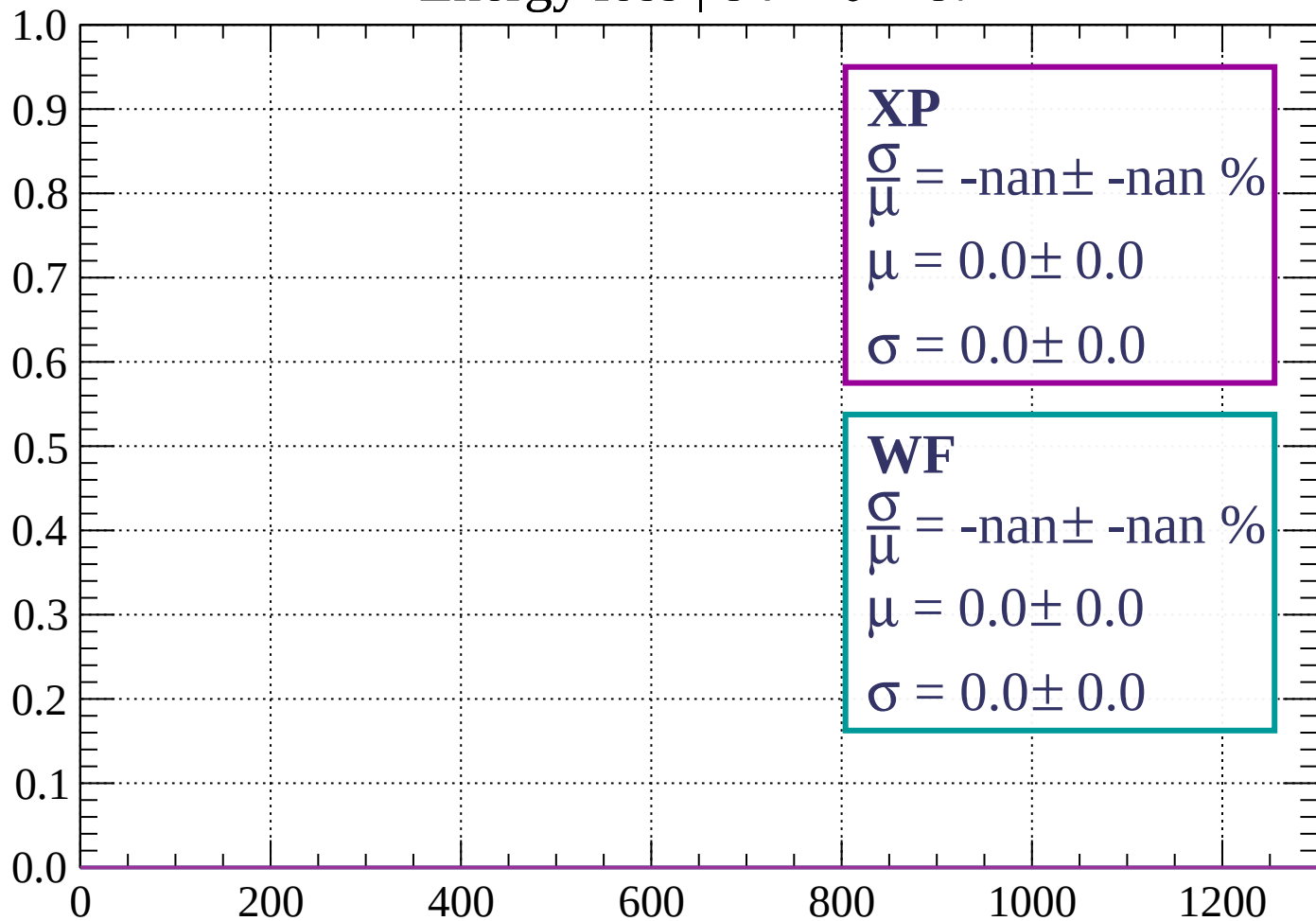
Count



dE/dx (ADC counts/cm)

Energy loss | $84 < \theta < 87$

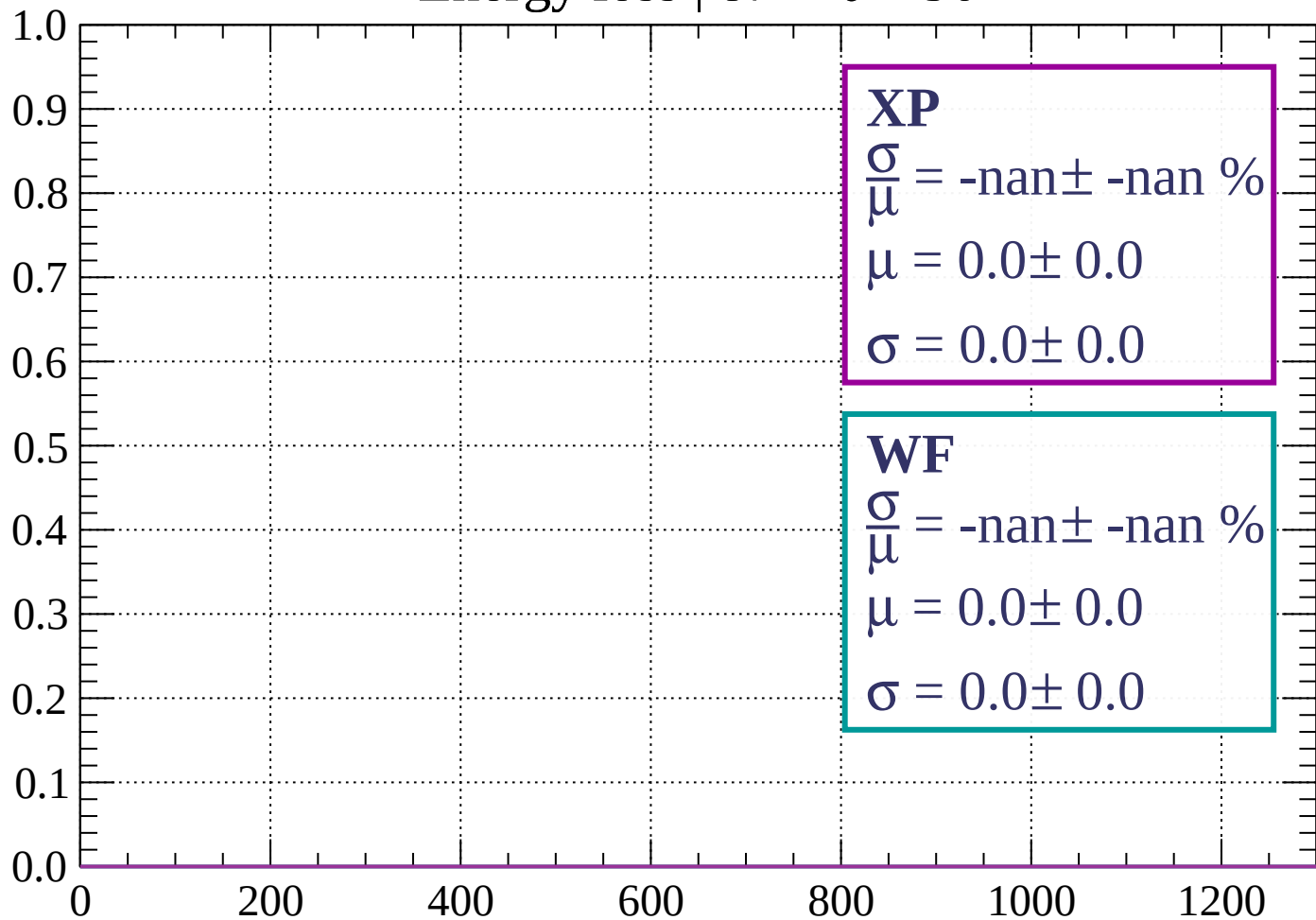
Count



dE/dx (ADC counts/cm)

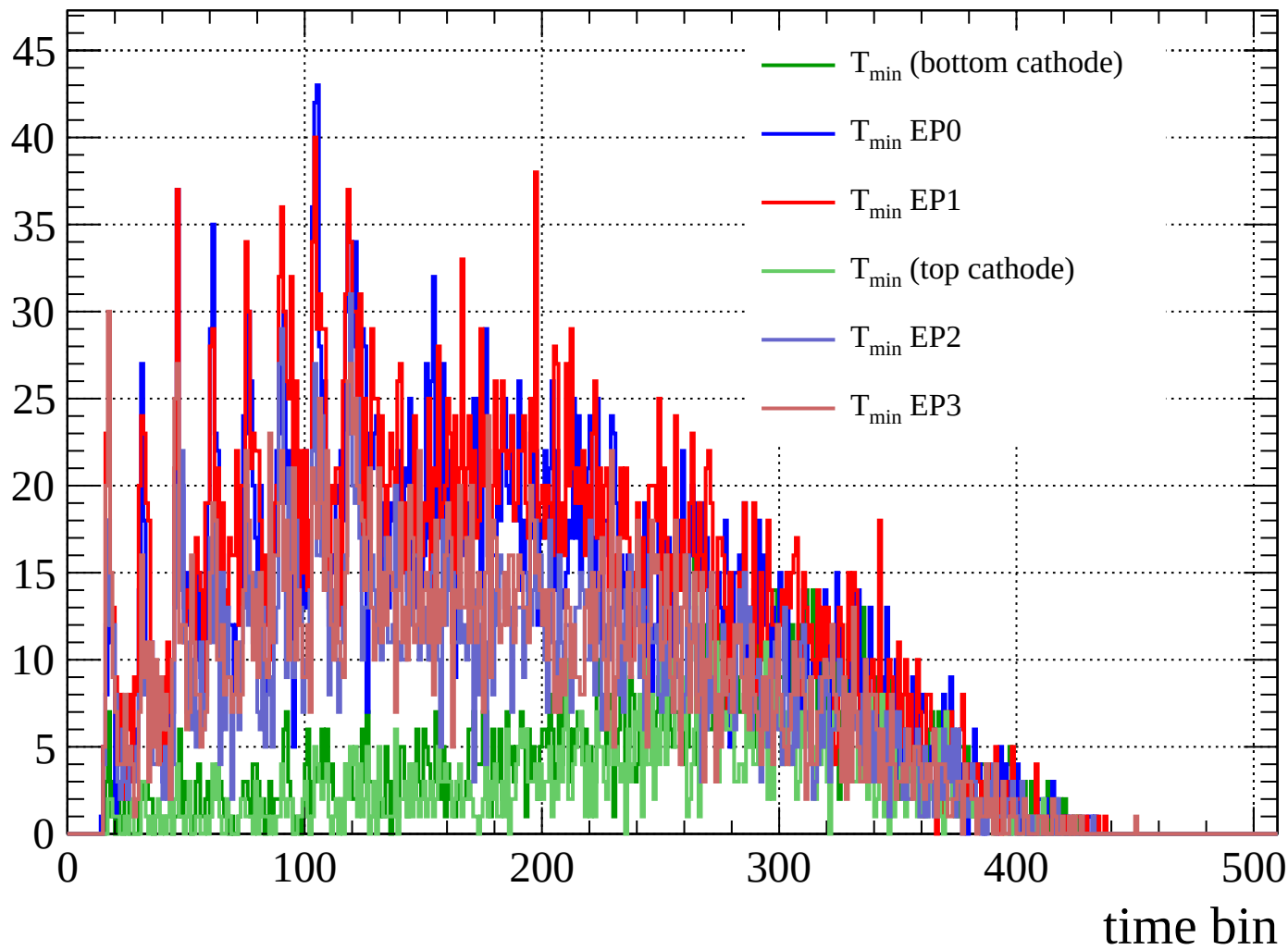
Energy loss | $87 < \theta < 90$

Count



dE/dx (ADC counts/cm)

Count



Count

